

Standard Bidding Document

327-Upgradation of Fire Fighting System at SBP Buildings, Head Office
Karachi
(Works)

National

Single Stage-Two Envelope



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REQUEST FOR BIDS

PROCUREMENT OF CIVIL WORKS

1. The **SBP BSC (SBP Banking Services Corporation (BSC))** has reserved Funds for the procurement planned for FY **2026-27**. The **SBP BSC (SBP Banking Services Corporation (BSC))** intends to apply part of the proceeds of this Fund to cover eligible payments under the contract for the "**327-Upgradation of Fire Fighting System at SBP Buildings, Head Office Karachi**" with the reference of "**P99841**".
2. The **SBP BSC (SBP Banking Services Corporation (BSC))** invites sealed Bids from eligible Bidders for procurement of Works (**327-Upgradation of Fire Fighting System at SBP Buildings, Head Office Karachi**) described in the bidding documents on **EPADS v2.0**.
3. **Single Stage-Two Envelope** will be used by adopting **Least Cost Based Selection (LCBS)** Technique for the subject procurement, in line with the Public Procurement Rules, 2004 and any Regulations, Regulatory Guides, Procurement Guidelines or Instructions issued by the Authority from time to time.
4. All Bids must be accompanied by a Bid Security amount described in Bid Security Section in Bidding Document in the form of **Pay Order, Banker's Cheque, Call at Deposit, Bank Guarantee, Demand Draft**. Or all bids must be accompanied by bid securing declaration in the format specified in the Bidding documents
5. E-Bidding documents, containing detailed terms & conditions, specifications and requirements etc. are available on **e-Pak Acquisition and Disposal System (EPADS)** at **<https://epads.gov.pk/opportunities/federal/procurements/99841>** for all the interested bidders registered on **EPADS v2.0**. Bidders are required to get themselves registered on **EPADS v2.0** to participate in Bidding process.
6. The e-bids, prepared in accordance with the instructions in the e-Bidding documents, must be submitted through **EPADS v2.0** on or before **Tuesday, October 20, 2026 12:00 PM**. E-bids will be opened by using **EPADS v2.0** on

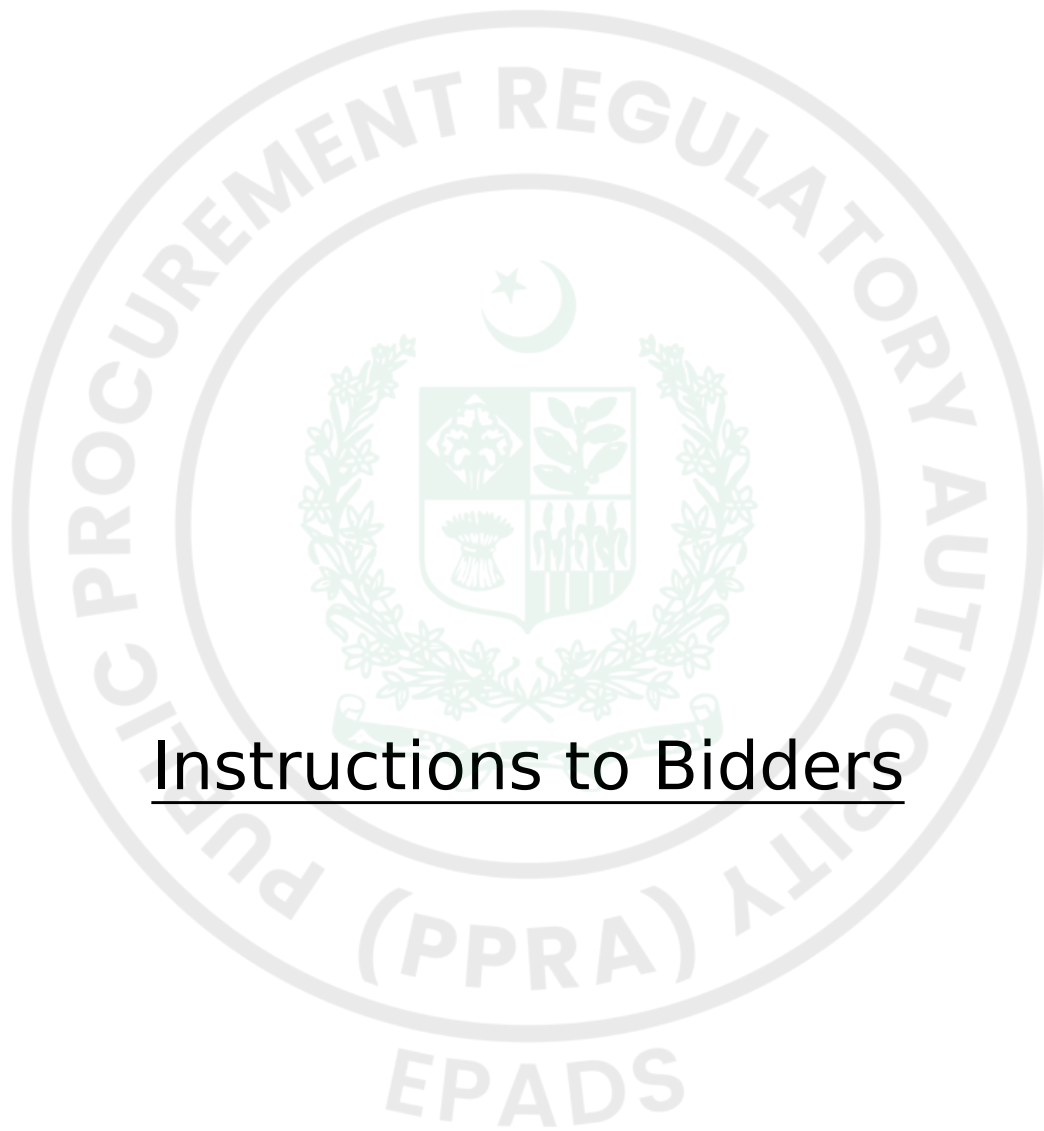
the same day at **Tuesday, October 20, 2026 12:30 PM**. Manual submission of Bids shall not be entertained. Those vendor who have not yet registered on the new version of **EPADS v2.0**, may register themselves on <https://vendors.epads.gov.pk/>. A tutorial to explain the registration process is available at <https://www.youtube.com/watch?v=MNW6T38v7tc>.

In terms of Rules 48 of Public Procurement Rules, 2004 Grievance Redressal Committee (GRC) is notified for the subject procurement and notification copy is available on the procuring agency's website and on Authority's website at (www.ppra.org.pk).

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Instructions to Bidders

A. INTRODUCTION

1. Scope of Bid

1.1. The Procuring agency/Employer (PA), as indicated in the Bid Data Sheet (BDS) invites Bids for the execution of Works as specified in the BDS and Section V- Works Requirements. The name, identification, and number of lots (contracts) of this National/ International Competitive Bidding process are specified in the BDS.

2. Source of Funds

2.1. Source of funds as referred in Clause 2 of Bid Data Sheet.

3. Eligible Bidders

3.1. A bidder may be natural person, company or firm or public or semi-public agency of Pakistan or any foreign country, or any combination of them with a formal existing agreement (on Judicial Papers) in the form of a joint venture or consortium. In the case of a joint venture or consortium, all members shall be jointly and severally liable for the execution of the Contract in accordance with the terms and conditions of the Contract. The joint venture or consortium shall nominate a Lead Member as nominated in the BDS, who shall have the authority to conduct all business for and on behalf of any and all the members of the joint venture or consortium during the Bidding process, and in case of award of contract, during the execution of contract. Verifiable copy of the agreement that forms a joint venture, consortium or association shall be required to be submitted as part of the Bid.

Any bid submitted by the joint venture, consortium or association shall indicate the part of proposed contract to be performed by each party and each party shall be evaluated (or post qualified if required) with respect to its contribution only, and the responsibilities of each party shall not be substantially altered without prior written approval of the Procuring Agency and in line with any instructions issued by the Authority.

(The limit on the number of members of JV or Consortium may be prescribed

in BDS, in accordance with the guidelines issued by the PPRA).

3.2. The invitation for bids is open to all prospective bidders subject to any provisions of incorporation or licensing by the respective national/international incorporating agency or statutory body established for that particular trade or business. Procuring agencies shall specify the registration/licensing requirements for the foreign bidder keeping in view the requirement of that business.

3.3. A Bidder shall not have a conflict of interest. All Bidders found to have a conflict of interest shall be disqualified. A Bidders may be considered to have a conflict of interest with one or more parties in this Bidding process, if they:

3.3.1. are associated or have been associated in the past, directly or indirectly with a firm or any of its affiliates which have been engaged by the Procuring agency/Employer to provide consulting services for the preparation of design or technical specifications of the works that are the subject of the bid; or

3.3.2. have controlling shareholders in common; or

3.3.3. receive or have received any direct or indirect subsidy from any of them; or

3.3.4. have the same legal representative for purposes of this Bid; or

3.3.5. have a relationship with each other, directly or through common third parties, that puts them in a position to have access to information about or influence on the bid of another bidder, or influence the decisions of the Procuring agency/Employer regarding this Bidding process; or

3.3.6. Submit more than one bid in this bidding process.

3.4. A Bidder may be ineligible if –

3.4.1. he is declared bankrupt or, in the case of company or firm, insolvent;

3.4.2. payments in favor of the bidder is suspended in accordance with the judgment of a court of law other than a judgment declaring bankruptcy and resulting (in accordance with the national laws) in the total or partial loss of the right to administer and dispose of its property;

3.4.3. the bidder is convicted, by a final judgment of a Court of Law or relevant Professional Statutory Body, of any offence involving professional conduct;

3.4.4. The bidder is debarred/ blacklisted by a national level Procuring agency/Employer and hence debarred due to involvement in corrupt and fraudulent practices, or performance failure or due to breach of bid securing declaration.

3.5. As and when required, bidders shall provide to the Procuring agency/Employer evidence of their eligibility, proof of compliance with the necessary legal requirements to carry out the contract effectively.

3.6. Bidders shall submit proposal relating to the nature, conditions and modalities of sub-contracting wherever the sub-contracting of any elements of the contract is envisaged.

4. Eligible Material and Equipment

4.1. All the material and equipment to be mobilized under the contract shall have their origin in eligible source countries, and all expenditures made under the contract will be limited to such materials and equipment. For this purpose, ineligible countries are stated in the section-IV titled as “Eligible Countries”.

B. BIDDING DOCUMENTS

5. Contents of Bidding Documents

5.1. The scope of Works, bidding procedures, and terms and conditions of the contract are prescribed in the bidding documents. In addition to the Invitation for Bids, the bidding documents which should be read in

conjunction with any addenda issued in accordance with ITB 7.1 include:

Section I -Invitation for Bids

Section II Instructions to Bidders (ITBs)

Section III Bid Data Sheet (BDS)

Section IV Eligible Countries

Section V Evaluation and Qualification Criteria

Section VI Works Requirements Technical Specifications & Schedule of Requirements

Section VII Standard Bidding Forms

Section VIII General Conditions of Contract (GCC)

Section IX Particular Conditions of Contract (PCC)

Section X Contract Forms

5.2. The bidder is expected to examine all instructions, forms, specifications, terms and conditions prescribed in the bidding documents. Failure to furnish all the information required in the bidding documents will be at the bidder's risk and may result in the rejection of his bid.

6. Clarification of Bidding Document, Pre-bid Meeting

6.1. A prospective bidder requiring any clarification of the bidding document may notify the Procuring agency/Employer through EPADS.

6.2. The Procuring agency/Employer shall respond to the request for clarification in accordance with Rule 31 of the Public Procurement Rules 2004.

6.3. Should the Procuring Agency deem it necessary to amend the BIDDING document as a result of a clarification, it shall do so following the procedure under ITB 7.

6.4. If indicated in the BDS, the bidder's designated representative is invited at the bidder's cost to attend a pre-bid meeting at the place, date and time mentioned in the BDS. During this pre-bid meeting, prospective bidders may request clarification of the schedule of requirement, the evaluation criteria or any other aspects of the bidding documents.

6.5. Minutes of the pre-bid meeting, if applicable, including the text of the questions asked by bidders, including those during the meeting (without identifying the source) and the responses given, together with any responses prepared after the meeting will be uploaded on EPADS. Any modification to the bidding documents that may become necessary as a result of the pre-bid meeting shall be made by the Procuring agency/Employer exclusively through the use of an Addendum pursuant to ITB 7. Non-attendance at the pre-bid meeting will not be a cause for disqualification of a bidder.

6.6. The bidder is advised to visit and examine the Site of Works and its surroundings and obtain for itself on its own responsibility all information that may be necessary for preparing the bid and entering into a contract for construction of the Works. The costs of visiting the Site shall be at the bidder's own expense.

6.7. The bidder and any of its authorized personnel will be granted permission by the Procuring agency/Employer to enter upon its premises and lands for the purpose of such visit, but only upon the express condition that the bidder and its personnel will release and indemnify the Procuring agency/Employer from and against all liability in respect thereof, and will be responsible for death or personal injury, loss of or damage to property, and any other loss, damage, costs, and expenses incurred as a result of the inspection.

7. Amendment of Bidding Documents

7.1. The procuring agency may issue notification of any change, addition, modification or deletion in accordance with Rule 23 of the Public Procurement Rules 2004 i.e. Bidding Documents.

7.2. To give prospective bidders reasonable time in which to take an addendum/corrigendum into account in preparing their bids, the Procuring agency/Employer may, at its discretion, extend the deadline for the submission of bids:

Provided that the Procuring agency/Employer shall extend the deadline for submission of bid in pursuance of Rule 27 of the Public Procurement Rules 2004, i.e. Extension of time for submission of bids, if such an addendum is

issued within last three (03) days of the bid submission deadline.

C. PREPARATION OF BIDS

8. Language of Bid

8.1. The bid prepared by the bidder, as well as all correspondence and documents relating to the bid exchanged by the bidder and the Procuring agency/Employer shall be written in the English language unless specified in the BDS. Supporting documents and printed literature furnished by the bidder may be in another language provided they are accompanied by an accurate translation of the relevant pages in the English language unless specified in the BDS, in which case, for purposes of interpretation of the bidder, the translation shall govern.

9. Documents Constituting the Bids

9.1. The Bids prepared by the Bidder shall constitute of all the documents required in the BDS.

10. Documents Establishing Eligibility of Material, Equipment and Works, their Conformity to Bidding Documents

10.1. The bid prepared by the bidder shall constitute the following components: -

10.1.1. Documentary evidence established in accordance with ITB 10 that the material and equipment to be utilized by the Bidder for the executions of works are eligible material and equipment and conform to the Bidding Documents;

10.1.2. Documentary evidence established in accordance with ITB 11 that the bidder has been authorized to carry out the Construction works;

10.1.3. Documentary evidence established in accordance with ITB 11 that the bidder is eligible and/or qualified for the subject bidding process;

10.1.4. Form of Bid and Bid Prices completed in accordance with ITB 12 and 13;

10.1.5. Completed schedules as required, including priced Bill of Quantities in accordance with ITB 13.

10.1.6. Technical Proposal completed in all aspects in accordance with ITB-15.

10.1.7. Bid security or Bid Securing Declaration furnished in accordance with ITB 17;

10.1.8. Any other document required in the BDS.

10.2. In addition to the requirements, bids submitted by a JV shall include a copy of the Joint Venture Agreement entered into by all members. Alternatively, a letter of intent to execute a Joint Venture Agreement in the event of a successful bid shall be signed by all members and submitted with the bid, together with a copy of the proposed Agreement.

10.3. The bidder shall furnish, as part of its bid, all those documents establishing the eligibility in conformity to the terms and conditions specified in the bidding documents for all material, equipment and works which the bidder proposes to execute.

10.4. The documentary evidence of conformity of the material, equipment and works to the Bidding Documents may be in the form of literature, drawings, and data, and shall consist of:

10.4.1. a detailed description of the work methodology, approach, schedule and resources to be mobilized at site;

10.4.2. an item-by-item commentary on the Procuring agency/Employer's Technical Specifications demonstrating substantial responsiveness of the material, equipment and works to those specifications, or a statement of deviations and exceptions to the provisions of the Technical Specifications;

10.4.3. any other procurement specific documentation requirement as stated in the BDS.

10.5. The required documents and other accompanying documents must be in English. In case any other language than English is used the pertinent translation into English shall be attached to the original version.

11. Documents Establishing Eligibility and Qualification of the Bidder

11.1. The bidder shall furnish, as part of its bid, all those documents establishing the bidder's eligibility to participate in the bidding process and/or its qualification to perform the contract if its bid is accepted.

11.2. The documentary evidence of the bidder's eligibility to bid shall establish to the satisfaction of the Procuring agency/Employer that the bidder, at the time of submission of its bid, is from an eligible country as defined in Section-IV titled as "Eligible Countries".

11.3. The documentary evidence of the bidder's qualification to perform the contract if its bid is accepted shall establish to the satisfaction of Procuring agency/Employer that:

11.3.1. The bidder has the financial and technical capability necessary to perform the Contract, meets the qualification criteria specified in Section-V, Evaluation and Qualification Criteria and BDS.

11.3.2. In the case of a bidder not doing business within Pakistan, the bidder is or will be (if awarded the contract) represented by a local bidder (Joint Venture) in accordance with the PEC works bylaws, and in case of award of works such foreign firm is required to participate in the execution of works to carry out its obligations as prescribed in the Conditions of Contract and /or Technical Specifications.

11.3.3. That the bidder meets the qualification criteria listed in Section-V, Evaluation and Qualification Criteria and BDS.

12. Forms of Bid

12.1. The Bidder shall fill the Form of Bids furnished in the bidding documents. The Bids Form must be completed without any alterations to its format and no substitute shall be accepted.

13. Bid Prices

13.1. The bid prices quoted by the bidder in the Standard bid Forms, Bill of Quantities and in the Price Schedules shall conform to the requirements specified below or exclusively mentioned hereafter in the bidding documents.

13.2. The bidder shall fill in rates and prices for all items of the Works described in the Bill of Quantities. If a Price Schedule shows items listed but not priced, their prices shall be construed to be included in the prices of other items in the Bill of Quantities and will not be paid for separately by the Procuring agency/Employer.

13.3. Items not listed in the Price Schedule shall be assumed not to be included in the bid, and provided that the bid is still substantially responsive in their absence or due to their nominal nature, the corresponding average price of the respective item(s) of the remaining substantially responsive bidder(s) shall be construed to be the price of those missing item(s):

Provided that:

13.3.1. where there is only one (substantially) responsive bidder, or

13.3.2. where there is provision for alternate proposals and the respective items are not listed in the other bids,

The Procuring agency/Employer may fix the price of missing items in accordance with market survey, and the same shall be considered as final price.

13.4. The Bid price to be quoted in the Form of Bid in accordance with ITB 12 shall be the total price of the bid.

13.5. Unless otherwise specified in the BDS and the Contract, the rates and prices quoted by the bidder are subject to adjustment during the performance of the Contract in accordance with the provisions of the

Conditions of Contract.

13.6. If so specified in ITB 1.1, bids may be invited for individual lots (contracts) or for any combination of lots (packages).

13.7. Prices quoted by the Bidder shall be fixed during the bidder's performance of the contract and not subject to variation on any account. A bid submitted with an adjustable price will be treated as non-responsive and shall be rejected, pursuant to ITB 27, unless otherwise price adjustment is permissible under Conditions of the Contract.

13.8. All duties, taxes, and other levies payable by the Contractor under the Contract, or for any other cause, as of the date twenty-eight (28) days prior to the deadline for submission of bids, shall be included in the rates and prices and the total bid price submitted by the bidder.

14. Currencies of Bid and Payment

14.1. Prices shall be quoted in Pakistani Rupees unless otherwise specified in the BDS. Comparison of bids and tie of bid shall be treated in accordance with the Rule 30(2) of Public Procurement Rules, 2004.

15. Documents Comprising the Technical Proposal

15.1. The bidder shall furnish a Technical Proposal including a statement of work methods, equipment, personnel, schedule and any other information as stipulated in Section VII – Standard Bid Forms, in sufficient detail to demonstrate the adequacy of the bidder's proposal to meet the work requirements and the completion time.

16. Bid Validity Period

16.1. Bids shall remain valid for the period specified in the BDS after the bid submission deadline prescribed by the Procuring agency/Employer. A bid valid for a shorter period shall be rejected by the Procuring agency/Employer as non-responsive. The period of bid validity will be determined from the complementary bid securing instrument i.e. the expiry period of bid security or bid securing declaration as the case may be.

16.2. Under exceptional circumstances, prior to the expiration of the initial Bids/Bid validity period, the Procuring Agency may request the Bidders' consent to an extension of the period of validity of their Bids/Bid. Such request for extension of the period of bid validity shall be carried out in accordance with Rule 26 of the Public Procurement Rules, 2004.

17. Bid Security or Bid Securing Declaration

17.1. Pursuant to ITB 11.1 unless otherwise specified in the BDS, the bidder shall furnish as part of its bid, a Bid Security in accordance with Rule 25 of the Public Procurement Rules, 2004 in the amount and currency specified in the BDS or Bid Securing Declaration as specified in the BDS in the format provided in Section VII (Standard Bidding Forms).

In case Procuring agency/Employer is inviting bids in lots / packages, the bidder shall be required to submit his bid security against the respective lot/package for which he is submitting his bid.

Until the development of functionality of auto verification of financial instrument in EPADS, the scanned copy of bid security or bid securing declaration, as the case may be, shall be uploaded on E-PADS whereas the original instrument to be submitted to the procuring agency before closing of bid submission deadline,

17.2. The Bid Security shall be denominated in the local currency or in another freely convertible currency, and it shall be in the form specified in the **BDS** which shall be in any of the following:

17.2.1. A bank guarantee, an irrevocable letter of credit issued by a Scheduled bank in the form provided in the Bidding Documents or another form acceptable to the Procuring agency/Employer and valid for twenty-eight (28) days beyond the end of the validity of the Bid. This shall also apply if the period for Bid Validity is extended. In either case, the form must include the complete name of the bidder;

17.2.2. A cashier's or certified cheque; or

17.2.3. Another security as indicated in the **BDS**.

17.3. The Bid Security or Bid Securing Declaration shall be in accordance with the Form of the Bid Security or Bid Securing Declaration included in Section VII (Standard Bidding Forms) or another form approved by the Procuring agency/Employer prior to the bid submission.

17.4. The Bid Security shall be payable promptly upon written demand by the Procuring agency/Employer in case any of the conditions listed in ITB 17.9 are invoked.

17.5. Any bid not accompanied by a Bid Security or Bid Securing Declaration in accordance with ITB 17.1 or 17.3 shall be rejected by the Procuring agency/Employer and shall be declared as non-responsive bid, pursuant to ITB 27.

17.6. Unsuccessful bidders' Bid Security will be discharged or returned as promptly as possible, however in no case later than thirty (30) days after the expiration of the period of Bid Validity prescribed by the Procuring agency/Employer pursuant to ITB 16. The Procuring agency/Employer shall make no claim to the amount of the Bid Security, and shall promptly return the Bid Security document, after whichever of the following that occurs earliest:

17.6.1. The expiry of the Bid Security;

17.6.2. The entry into force of a procurement contract and the provision of a performance security (or guarantee), for the performance of the contract if such a security (or guarantee), is required by the Bidding documents;

17.6.3. The rejection by the Procuring agency/Employer of all Bids;

17.6.4. The withdrawal of the bid prior to the deadline for the submission of bids, unless the bidding documents stipulate that no such withdrawal is permitted.

17.7. The successful bidder's Bid Security will be discharged upon the bidder signing the contract pursuant to ITB 40, or furnishing the performance security (or guarantee), pursuant to ITB 41.

17.8. The Bid Security may be forfeited or the Bid Securing Declaration executed:

17.8.1. if a Bidder:

17.8.1.1. Withdraws its Bid during the period of Bid Validity as specified by the Procuring agency/Employer, and referred by the bidder on the Form of Bid except as provided for in ITB 16.2; or

17.8.2. In the case of a successful bidder, if the bidder fails:

17.8.2.1. to sign the contract in accordance with ITB 40; or

17.8.2.2. to furnish performance security (or guarantee) in accordance with ITB 41.

17.9. In case of Bid Security issued by the foreign bank is allowed by the Procuring agency/Employer, the same should be counter guaranteed by a corresponding bank in Pakistan. Furthermore, in case of joint venture, it should be in the name of Joint venture to ensure joint responsibility. In case the JV is not legally constituted at the time of bid submission, the bid security or bid securing declaration shall be in the names of all future members as named in the letter of bid.

18. Withdrawal of Bids

18.1. Before bid submission deadline, any bidder may withdraw, substitute, or modify its bid after it has been submitted by sending a written notice, duly signed by an authorized representative, and the corresponding must accompany the respective written notice.

19. Format and Signing of Bid

19.1. The Bidder shall prepare and submit Bids through EPADS with due diligence after carefully reading all the terms and condition before bid submission deadline.

D. SUBMISSION OF BIDS

20. Submission of Bids through EPADS v2.0

20.1. All bids shall be submitted through EPADS v2.0.

21. Deadline for Submission of Bids

21.1. All bids shall be received through **EPADS v2.0** not later than bid submission deadline as specified in the **BDS**.

21.2. The Procuring agency/Employer may, under exceptional circumstances and at its discretion, extend the deadline for the submission of bids, pursuant to Rule 27 of the Public Procurement Rules, 2004. Extension of Time for submission of bid, by amending the Bidding Documents in accordance with ITB 7, in which case all rights and obligations of the Procuring agency/Employer and bidders previously subject to the deadline will thereafter be subject to the new deadline.

22. Substitution and Modification of bids

22.1. A bidder may substitute or modify his bid after it has been submitted, provided that written notice of the substitution or modification of the bid, is received by the Procuring agency/Employer prior to the deadline for submission of bids.

22.2. Revised bid may be submitted after the substitution or modification made in the original bid in accordance with the provisions referred in **ITB 18**.

E. OPENING AND EVALUATION OF BIDS

23. Opening of Bids

23.1. The Procuring Agency will open bids in accordance with Rule 28 of the Public Procurement Rules, 2004 and as specified in the BDS.

24. Confidentiality

24.1. Information relating to the examination, clarification, evaluation and comparison of bids and recommendation of contract award shall not be disclosed to bidders or any other persons not officially concerned with such process until the time of the announcement of the respective evaluation report.

25. Clarification of Bids

25.1. Clarification of Bidding Documents shall be carried out in accordance with Rule 31 of the Public Procurement Rules, 2004.

25.2. The alteration or modification in the bid which in any case affect the following parameters will be considered as a change in the substance of a bid:

- 25.2.1. evaluation & qualification criteria;
- 25.2.2. required scope of work;
- 25.2.3. contract price;
- 25.2.4. all securities requirements;
- 25.2.5. tax requirements;
- 25.2.6. terms and conditions of bidding documents.
- 25.2.7. change in the ranking of the bidder

26. Preliminary Examination of Bids

26.1. Prior to the detailed evaluation of bids, the Procuring agency/Employer will determine whether each bid:

- 26.1.1. meets the eligibility criteria defined in **ITB 3** and **ITB 4**;
- 26.1.2. has been prepared as per the format and contents defined by the Procuring agency/Employer in the bidding documents;
- 26.1.3. has been properly signed;

26.1.4. is accompanied by the required securities; and

26.1.5. is substantially responsive to the requirements of the bidding documents.

The Procuring agency/Employer's determination of a bid's substantial responsiveness will be based on the contents of the bid itself.

26.2. A substantially responsive Bid is one which conforms to all the terms, conditions, and specifications of the Bidding Documents, without material deviation or reservation. A material deviation or reservation is one that: -

26.2.1. affects in any substantial way the scope, quality, or performance of the Works;

26.2.2. limits in any substantial way, inconsistent with the bidding documents, the Procuring agency/Employer's rights or the bidders' obligations under the Contract; or

26.2.3. if rectified, would affect unfairly the competitive position of other bidders presenting substantially responsive bids.

26.3. The Procuring agency/Employer will confirm that the documents and information specified under ITB 9, 10 and 11 have been provided in the bid. If any of these documents or information is missing, or is not provided in accordance with the Instructions to Bidders, the bid shall be rejected.

26.4. The Procuring agency/Employer may waive-off any minor informality, nonconformity, or irregularity in a bid which does not constitute a material deviation, provided such waiver does not prejudice or affect the relative ranking of any Bidder.

Explanation: A minor informality, non-conformity or irregularity is one that is merely a matter of form and not of substance. It also pertains to some immaterial defect in a Bid or variation of a bid from the exact requirements of the invitation that can be corrected or waived without being prejudicial to other bidders. The defect or variation is immaterial when the effect on quantity, quality, or delivery is negligible when contrasted with the total cost or scope of the works. The Procuring agency/Employer either shall give the bidder an opportunity to cure any deficiency resulting from a minor

informality or irregularity in a bid or waive the deficiency, whichever is advantageous to the Procuring agency/Employer. Examples of minor informalities or irregularities include failure of a bidder to –

26.4.1. Submit the number of copies of signed bids required by the invitation;

26.4.2. Furnish required information concerning the number of its employees;

26.4.3. the firm submitting a bid has formally adopted or authorized, before the date set for opening of bids, the execution of documents by typewritten, printed, or stamped signature and submits evidence of such authorization and the bid carries such a signature.

26.5. Provided that a Technical Bid is substantially responsive, the Procuring agency/Employer may request the bidder to submit the necessary information or documentation, within a reasonable period of time, to rectify nonmaterial nonconformities or omissions in the Technical Bid related to documentation requirements. Requesting information or documentation on such nonconformities shall not be related to any such aspect of the technical Proposal linked with the ranking of the bidders. Failure of the bidder to comply with the request may result in the rejection of its bid.

26.6. Provided that a Technical Bid is substantially responsive, the Procuring agency/Employer shall rectify quantifiable nonmaterial nonconformities or omissions related to the Financial Proposal. To this effect, the Bid Price shall be adjusted, for comparison purposes only, to reflect the price of the missing or nonconforming item or component.

26.7. If a bid is not substantially responsive, it will be rejected by the Procuring agency/Employer and may not subsequently be evaluated for complete technical responsiveness.

27. Examination of Terms and Conditions; Technical Evaluation

27.1. The Procuring agency/Employer shall examine the bid to confirm that all terms and conditions specified in the **GCC** and the **PCC** have been

accepted by the bidder without any material deviation or reservation.

For this purpose:

“Deviation” means departure from the requirements specified in the Bidding Document.

“Reservation” means setting of limiting conditions or withholding from complete acceptance of the requirements specified in the Bidding Document.

27.2. The Procuring agency/Employer shall evaluate the technical aspects of the bid submitted in accordance with ITB 30, to confirm that all requirements specified in Section VI – Works Requirement, Technical Specifications of the Bidding Documents have been met without material deviation or reservation.

27.3. If after the examination of the terms and conditions and the technical evaluation, the Procuring agency/Employer determines that the bid is not substantially responsive in accordance with ITB 27, it shall reject the bid.

28. Correction of Arithmetic Errors

28.1. Bids determined to be substantially responsive will be checked for any arithmetic errors. Errors will be corrected as follows: -

28.1.1. if there is a discrepancy between unit prices and the sub-total price that is obtained by multiplying the unit price and quantity, the unit price shall prevail, and the sub-total price shall be corrected, unless in the opinion of the Procuring agency/Employer there is an obvious misplacement of the decimal point in the unit price, in which the total price as quoted shall govern and the unit price shall be corrected;

28.1.2. if there is an error in a total corresponding to the addition or subtraction of sub-totals, the sub-totals shall prevail, and the total shall be corrected; and

28.1.3. where there is a discrepancy between the amounts in figures and in words, the amount in words will govern.

28.1.4. Where there is discrepancy between grand total of price schedule and amount mentioned on the Form of Bid, the amount referred in Price Schedule shall be treated as correct subject to elimination of other errors.

28.2. The amount stated in the Bid will, be rectified by the Procuring agency/Employer in accordance with the above procedure for the correction of errors and, with, the concurrence of the bidder, shall be considered as binding upon the bidder. If the bidder does not accept the corrected amount, its bid shall be rejected after forfeiture of Bid Security or execution of the Bid Securing Declaration, as the case may be, in accordance with **ITB 41.3**.

29. Conversion to Single Currency

29.1. The unit rates and the prices shall be quoted by the bidder entirely in Pak rupees. A bidder expecting to incur expenditures in other currencies for inputs to the Works from outside the Procuring agency/Employer's country (referred to as the "Foreign Currency Requirements") shall indicate the same in the letter of bid-financial proposal. The proportion of the Bid Price (excluding Provisional Sums) needed by him for the payment of such Foreign Currency Requirements either (i) entirely in the currency of the Bidder's home country or, (ii) at the bidder's option, entirely in Pak rupees provided always that a bidder expecting to incur expenditures in a currency or currencies other than those stated in (i) and (ii) above for a portion of the foreign currency requirements, and wishing to be paid accordingly, shall indicate the respective portions in his bid. Comparison of bids quoted in different currencies and conversion of bids into a single currency shall be carried out in accordance with Rule 30(2) of the Public Procurement Rules, 2004.

30. Evaluation of Bids

30.1. The Procuring agency/Employer shall evaluate and compare only the bids determined to be substantially responsive, pursuant to **ITB 27**.

30.2. In evaluating the Technical Proposal of each Bid, the Procuring agency/Employer shall use the criteria and methodologies listed in the BDS

and in terms of works requirement. No other evaluation criteria or methodologies shall be permitted.

30.3. The Procuring agency/Employer's evaluation of a bid will take into account:

30.3.1. the bid price, excluding provisional sums and the provision, if any, for contingencies in the summary bill of quantities, but including day work items, where priced competitively;

30.3.2. converting the amount resulting from applying above, if relevant, to a single currency in accordance with ITB 29;

30.4. The estimated effect of the price adjustment provisions of the Conditions of Contract, applied over the period of execution of the Contract, shall not be taken into account in bid evaluation.

30.5. If these bidding documents allow bidders to quote separate prices for different lots, and the award to a single bidder of multiple lots, the methodology of evaluation to determine the lowest evaluated lot combinations in the Form of Bid, is specified in the **BDS**.

30.6. If the bid, which results in the Evaluated Bid Price (Successful Bid), is seriously unbalanced or front loaded in the opinion of the Employer, the Employer may require the Bidder to produce detailed price analyses for any or all items of the Bill of Quantities, to demonstrate the internal consistency of those prices with the construction methods and schedule proposed. After evaluation of the price analyses, taking into consideration the schedule of estimated Contract payments, the Employer may require that the amount of the performance security be increased at the expense of the Bidder to a level sufficient to protect the Employer against financial loss in the event of default of the successful Bidder under the Contract.

Explanation:

"Unbalanced" or "front-loaded" bids consist of deliberately submitting bids with artificially high prices or unit rates for the early stages of a construction project, offset by artificially low prices or unit rates for the later stages of the project, to improve the contractor's cash flow.

31. Domestic Preference

31.1. If the **BDS** so specifies, the Procuring agency/Employer will grant a margin of preference to the domestic contractor in line with the rules, regulations, regulatory guides or instructions issued by the Authority from time to time.

32. Determination of Successful Bid

32.1. The Procuring agency/Employer shall compare the evaluated bids in accordance with the predefined bidding procedure, of all substantially responsive bids to determine the Successful bidder.

33. Qualification of Bidder

33.1. The Procuring agency/Employer shall determine to its satisfaction whether the bidder is substantially responsive and whose bid is declared as Successful bid either continues to meet (if prequalification applies) or meets (if post-qualification applies) the qualifying criteria specified in Evaluation and Qualification Criteria.

Note: In case of international bidding, the parameters for incorporation or licensing within Pakistan may be fulfilled as part of post qualification.

33.2. The determination shall be based upon an examination of the documentary evidence of the bidder's qualifications submitted by the bidder, pursuant to **ITB 11**.

33.3. Prior to contract award, the Procuring agency/Employer will verify that the successful bidder (including each member of a JV) is not blacklisted/debarred. The Procuring agency/Employer will conduct the same verification for each sub-contractor proposed by the successful bidder.

34. Sub-Contractors

34.1. The bidder shall provide details regarding any specialized sub-contractor to the Procuring agency/Employer. In case change of sub-contractors, the bidder shall promptly notify the Procuring agency/Employer and obtain approval for replacement of sub-contractors.

34.2. Bidders may propose sub-contracting up to the percentage of total value of contracts or the volume of works as specified in the **BDS**.

35. **Abnormally Low Financial Bid**

35.1. A procuring Agency may reject abnormally low bids. The decision of the Procuring agency/Employer to reject a bid and reasons for the decision shall be recorded in the procurement proceedings and promptly communicated to the bidder concerned. Moreover, the Procuring agency/Employer shall not incur any liability solely by rejecting abnormally bid

Guidance for Procuring agency/Employer:

An abnormally low bid means, in the light of the Procuring agency/Employer's estimate and of all the bids submitted, the bid appears to be abnormally low by not providing a margin for normal levels of profit. In order to identify the Abnormally Low Bid (ALB) following approaches can be considered to minimize the scope of subjectivity:

35.1.1. Comparing the bid price with the cost estimate;

35.1.2. Comparing the bid price with the bids offered by other bidders submitting substantially responsive bids; and

35.1.3. Comparing the bid price with prices paid in similar contracts in the recent past either government- or development partner-funded.

35.2. The Procuring agency/Employer will determine to its satisfaction whether the bidder that is selected as having submitted the successful bid is qualified to perform the contract satisfactorily, in accordance with the criteria listed in **ITB 11**

35.3. The determination will take into account the bidder's financial and technical capabilities. It will be based upon an examination of the documentary evidence of the bidder's qualifications submitted by the bidder, pursuant to **ITB 11**, as well as such other information as the Procuring agency/Employer deems necessary and appropriate. Factors not included in these bidding documents shall not be used in the evaluation of the bidders' qualifications.

35.4. Procuring agency/Employer may seek “Certificate for Independent Price Determination” from the bidder and the results of reference checks may be used in determining award of contract.

Explanation: The Certificate shall be furnished by the bidder. The bidder shall certify that the price is determined keeping in view of all the essential aspects such as raw material, its processing, value addition, optimization of resources due to economy of scale, transportation, insurance and margin of profit etc.

35.5. An affirmative determination will be a prerequisite for award of the contract to the bidder. A negative determination will result in rejection of the bidder’s bid, in which event the Procuring agency/Employer will proceed to the next ranked bidder to make a similar determination of that bidder’s capabilities to perform satisfactorily.

F. AWARD OF CONTRACT

36. **Criteria of Award**

36.1. Subject to **ITB 36 and 37**, the Procuring agency/Employer will award the Contract to the bidder whose bid has been determined to be substantially responsive to the bidding documents and who has been declared as Successful Bidder, provided that such bidder has been determined to be:

36.1.1. eligible in accordance with the provisions of **ITB 3**;

36.1.2. is determined to be qualified to perform the Contract satisfactorily;
and

36.1.3. Successful negotiations have been concluded, if any.

37. **Negotiations**

37.1. The Committee of the Procuring agency/Employer may negotiate with the Most Advantageous Bidder relating to the following areas:

37.1.1. a minor alteration to the technical (drawings, design technical specifications) details of the statement of works;

37.1.2. Methodology, work plan, staffing in view to streamline the work;

37.1.3. a minor amendment to the Particular conditions of Contract;

37.1.4. finalizing payment arrangements;

37.1.5. clarifying details that were not apparent or could not be finalized at the time of Bidding;

37.2. Where negotiation fails to result into an agreement, the Procuring agency/Employer may invite the next ranked bidder for negotiations. Where negotiations are commenced with the next ranked bidder, the Procuring agency/Employer shall not reopen earlier negotiations.

38. Procuring agency's Right to reject All Bids

38.1. The procuring agency has the right to reject all bids in accordance with Rule 33 of the Public Procurement Rules, 2004. However, the Authority (i.e. **PPRA**) may call from the Procuring agency/Employer the justification of those grounds.

39. Notification of Award

39.1. The procuring agency shall announce and publish the evaluation result in accordance with Rule 35 of the Public Procurement Rules, 2004.

39.2. Where no complaints have been lodged, the bidder whose bid has been accepted will be notified of the award by the Procuring agency/Employer prior to expiration of the bid validity period through EPADS. However, the Procuring agency/Employer shall not award any procurement contract at least for five (05) days after the announcement of final evaluation report. The notification letter (herein after and in the condition of the contract and contract form called "Letter of Acceptance" will specify the sum that the Procuring agency/Employer will pay the successful bidder in consideration for the execution and completion of the

works as prescribed by the Contract (hereinafter and in the Contract called the "Contract Price).

39.3. The notification of award will constitute the formation of the Contract, subject to the bidder furnishing the Performance Security (or guarantee) in accordance with **ITB 41** and signing of the contract in accordance with **ITB 40**.

39.4. Upon the successful bidder's furnishing of the performance security (or guarantee) pursuant to **ITB 41**, the Procuring agency/Employer will promptly notify each unsuccessful bidder, the name of the successful bidder and the Contract amount and will discharge the Bid Security or Bid Securing Declaration of the bidder(s) pursuant to **ITB 17**.

40. Signing of Contract

40.1. Promptly after notification of award, Procuring agency/Employer shall send the successful bidder the draft agreement, incorporating all terms and conditions as agreed by the parties to the contract.

40.2. Immediately after the Redressal of grievance by the **GRC**, and after fulfillment of all conditions precedent of the Contract Form, the successful bidder and the Procuring agency/Employer shall sign the contract.

40.3. Where no formal signing of a contract is required, work order issued to the bidder shall be construed to be the contract.

41. Performance Security (or Guarantee)

41.1. After the receipt of the Letter of Acceptance, the successful bidder, within the specified time, shall deliver to the Procuring agency/Employer a Performance Guarantee in the amount and in the form stipulated in the BDS and PCC, denominated in the type and proportions of currencies in the Letter of Acceptance and in accordance with the Conditions of Contract.

41.2. If the Performance Guarantee is provided by the successful bidder and it shall be in the form specified in the BDS which shall be in any of the following:

41.2.1. certified cheque, cashier's or manager's cheque, or bank draft;

41.2.2. irrevocable letter of credit issued by a scheduled bank of Pakistan or in the case of an irrevocable letter of credit issued by a foreign bank, the letter shall be confirmed or authenticated by a scheduled bank of Pakistan;

41.2.3. bank guarantee confirmed by a reputable local bank or, in the case of a successful foreign bidder, bonded by a foreign bank; or

41.2.4. surety bond callable upon demand issued by any reputable surety or insurance company.

Any Performance Guarantee submitted shall be enforceable in Pakistan.

41.3. Failure of the Most Advantageous Bidder to comply with the requirement of **ITB 40** shall constitute sufficient grounds for the annulment of the award and forfeiture of the Bid Security or declare blacklisted (in case bid securing declaration is submitted) in which event the Procuring agency/Employer may make the award to the next most advantageous bidder or reinitiate the procurement process afresh (as a case may be).

42. **Advance Payment**

42.1. Advance payment will be provided to the bidder in percentage and in the manner as agreed by the both parties in terms of Conditions of the Contract.

42.2. The Procuring agency/Employer will provide an advance payment as stipulated in the Conditions of Contract, subject to a maximum amount, as stated and/or Conditions of the Contract. The advance payment request shall be accompanied by an advance payment security (guarantee) in the form provided in Section X. For the purpose of receiving the advance payment, the bidder shall make and estimate of, and include in its bid, the expenses that will relate to the purchase of equipment, machinery, materials, and on the engagement of labor during the first month beginning with the date of the Procuring agency/Employer's "**Notice to Commence**" as specified in the **PCC**.

43. General Performance of the Bidders

43.1. The Procuring agency/Employer reserves the right to obtain information regarding performance of the bidders on their previously awarded contracts / works. The Procuring agency/Employer may seek information / report from the previous employer for consideration. However, the Procuring agency/Employer shall incorporate such parameters in the evaluation criteria and accordingly decide the fate of the bid submitted.

44. Corrupt & Fraudulent Practices

44.1. Procuring agencies (including beneficiaries of Government funded projects and procurement) as well as Bidders/Suppliers/Contractors under Government financed contracts, observe the highest standard of ethics during the procurement and execution of such contracts, and will avoid to engage in any corrupt and fraudulent practices.

G. GRIEVANCE REDRESSAL & COMPLAINT REVIEW MECHANISM

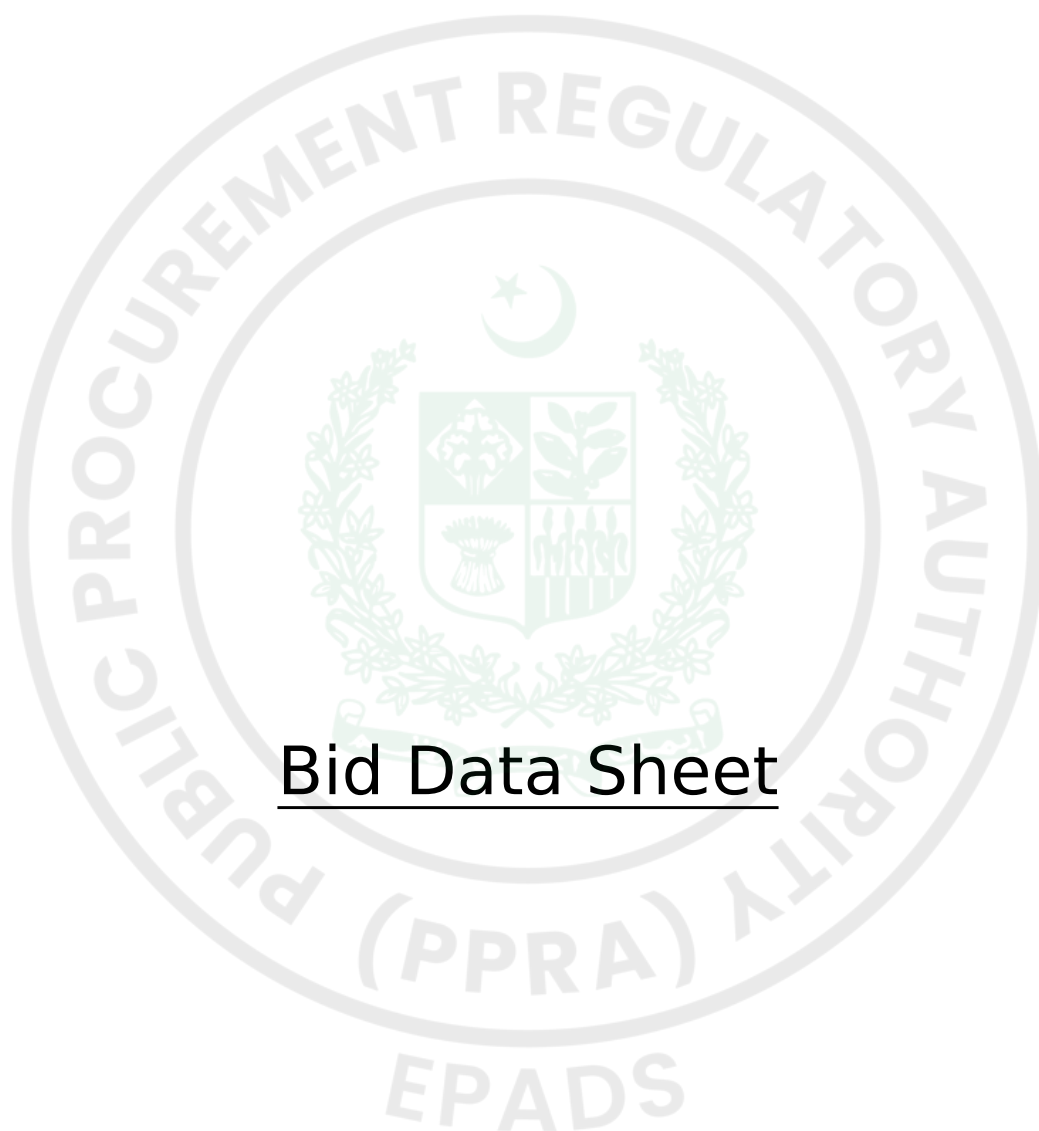
45. Grievance Redressal

45.1. Grievance Redressal shall be carried out in accordance with Rule 48 of the Public Procurement Rules, 2004 i.e. Redressal of grievances by the procuring agency and “Redressal of Grievances Regulations 2021”.

H. MECHANISM OF BLACKLISTING

46. Mechanism of Blacklisting

46.1. The Procuring agency/Employer shall proceed Blacklisting of Bidders/Contractors in accordance with Rule 19 of the Public Procurement Rules, 2004 i.e. Blacklisting and “Blacklisting and Debarment of Bidders or Contractors Regulations 2024”.



Bid Data Sheet

Bids Data Sheet (BDS)

The following specific data for the procurement of Goods to be procured shall complement, supplement, or amend the provisions in the Instructions to Bidders (ITB). Whenever there is a conflict, the provisions herein shall prevail over those in ITB.

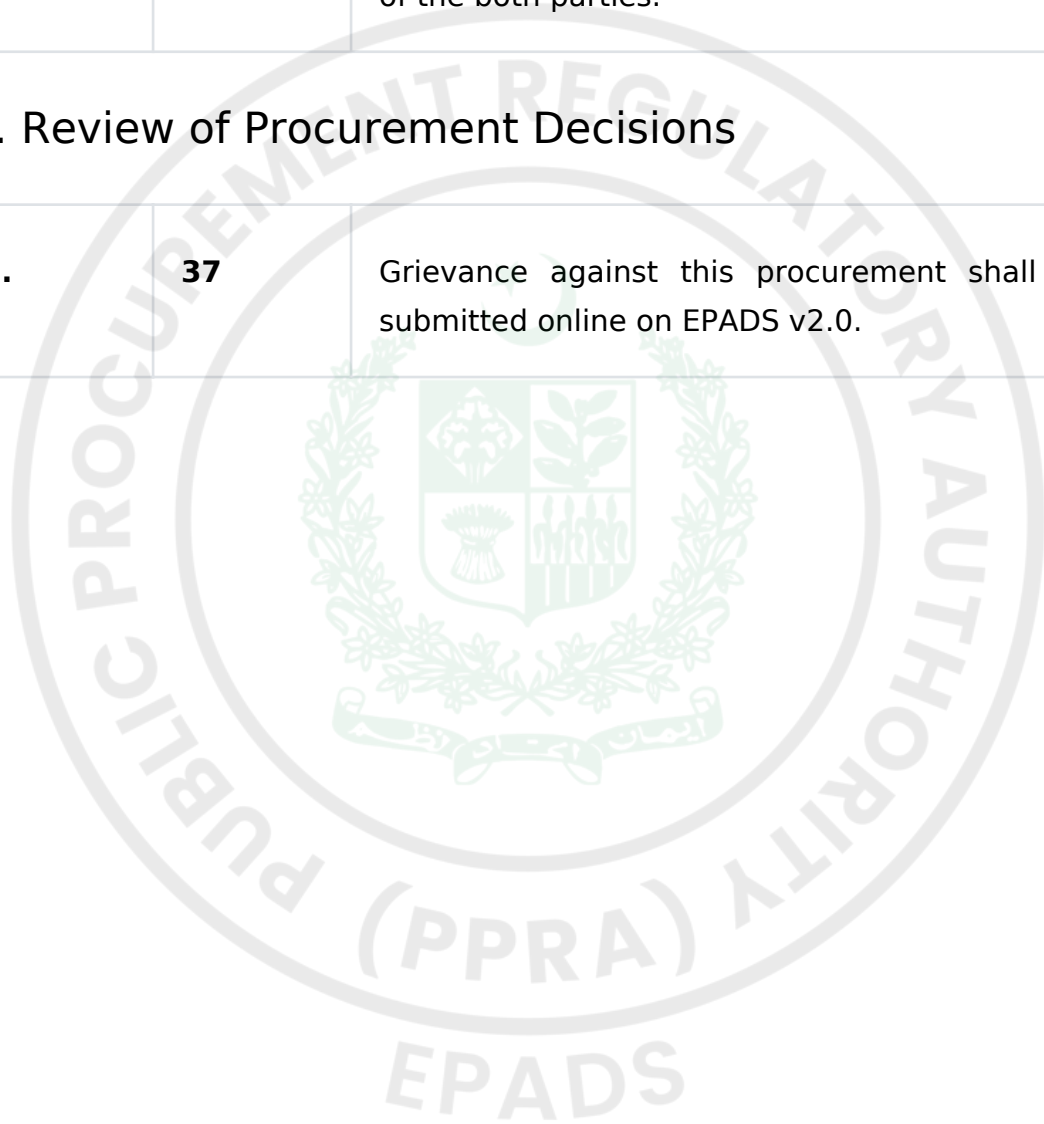
BDS Clause Number	ITB Number	Amendments of, and Supplements to, Clauses in the Instruction to Bidders
A. Introduction		
1	1.1	Name of Procuring Agency:SBP BSC (SBP Banking Services Corporation (BSC)) The subject of procurement is:327- Upgradation of Fire Fighting System at SBP Buildings, Head Office Karachi Expected commencement date: Wednesday, December 30, 2026
2.	2.1	Financial year for the operations of the Procuring Agency:2026-27 Name and identification number of the Contract: P99841
3.	3.1	JV/Consortium or Association Allowed: No Number of JV/Consortium Members: Nil
B. Bidding Documents		

4.	6.2 & 6.4	The Bidders may seek clarifications through EPADS v2.0: Clarification Date: Tuesday, October 6, 2026 Pre-Bid Meeting: Thursday, October 1, 2026 12:00 PM Venue: Engineering Department, 1st Flor, SBP Boulton Market Building, M.A Jinnah Road
5.	7.2	Any addendum, in case issued, shall be published on SBP BSC (SBP Banking Services Corporation (BSC)) website and on EPADS v2.0.
C. Preparation of Bids		
6.	8.1	List of documents required along with the bid: No
7.	9.1	The qualification criteria to establish the supply / production capability of the bidder. <i>see Eligibility Criteria</i>
8.	11.2	Works and Their related documents: <i>See section Required Scope of Work</i>
9.	11.1	Price schedule will be provided according to the format defined and acquired. <i>see section price schedule.</i>
10.	11.4	Specifications: <i>see section of specifications.</i>

11.	11.5 & 13.5	The price shall be Fixed . The bid price shall be adjusted in accordance with Appendix provided - Formula for Price Adjustment.
12.	14.1	Currency of the Bids shall be : PKR
13.	16.1	The Bids/Bid Validity period shall be: 180 Days
14.	17.1	The amount of Bid Security shall be as defined in Bid Security Section for items and lots given in BDS 6
15.	17.2	The Bid Security shall be in the form of: Pay Order, Banker's Cheque, Call at Deposit, Bank Guarantee, Demand Draft
16.	15.1	Alternative Bids to the requirements of the bidding documents willnot be permitted.
D. Submission of Bids		

17.	18.1 & 21.1	Bid shall be submitted online on EPADS v2.0 whereas hard copy of the bid security should be submitted to the following; Engineering Department, 1st Flor, SBP Boulton Market Building, M.A Jinnah Road, Saddar Sub-Division, Karachi South (District), Karachi (Division), Sindh (Province). Bids that are not submitted on EPADS v2.0 shall be disqualified. The deadline for Bids submission is: Tuesday, October 20, 2026 12:00 PM
E. Opening and Evaluation of Bids		
18.	24.1	The Bids opening shall take place on EPADS v2.0. Day : Tuesday Date: October 20, 2026 Time : 12:30 PM
19.	30.2	Selection technique adopted will be: Least Cost Based Selection (LCBS) <i>see Evaluation Criteria</i>
F. Award of Contract		

20.	41.1 & 41.2	The Performance guarantee shall: 10.00%. The Performance Guarantee shall be acceptable in the form of:Pay Order, Banker's Cheque, Call at Deposit, Bank Guarantee, Demand Draft
21.	45.1	Arbitrator shall be appointed by mutual consent of the both parties.
G. Review of Procurement Decisions		
22.	37	Grievance against this procurement shall be submitted online on EPADS v2.0.



Eligibility Criteria

Bidder's Type	Required Registration
Individual / Individual Consultant Sole Proprietorship Partnership Firm Company (Private Limited) Company (Public Limited) Company (Holding Company) Company (Limited by Guarantee) State Owned Enterprise (Private Limited) State Owned Enterprise (Public Limited)	FBR (NTN)

Eligibility Criteria	Document
ATL FBR: Bidder must be registered with FBR in Income Tax and must be active taxpayer as per Federal Board of Revenue (FBR's) Data base i-e Active Taxpayer List.	Yes
Revenue Authority Registration: The Bidder should be registered with relevant / concerned Provincial Revenue Authority. If not registered, the 100% tax shall be deducted at source as per applicable law and credited to relevant account.	No
Undertaking: The bidder is required to submit form of bid as per format attached in Annexure 08.	Yes
PEC Valid Registration: The Bidder shall be registered with the Pakistan Engineering Council (PEC) in Category C-2 or above, with the relevant specialization codes ME-02 and EE-04.	Yes

Evaluation Criteria

Least Cost Based Selection (LCBS)

Weightage

Technical Evaluation %	
100	
Technical Marks	100
Passing Marks	100
Technical Evaluation Criteria	
Experience: The Bidder must have projects pertaining to Firefighting Works, having gross value of completed / awarded / in-hand projects, at least Rs.200 million, during the last five (05) years. (Copies of work orders/completion certificates/ contract agreements or equivalent documents to be attached). (Quantitative)(Doc Required)	20
Financial Capability/Liquid Assets: Annual Sales volume or Gross Turnover available at least Rs.300.0 M in any of the last five years. Attach Copy(s) of Audited Financial Statements or Income Tax return filed in FBR --- OR --- Bank Credit Line facility available of Rs.100.0 M or above during last one year or equivalent Liquid Assets) ---OR --- Financial Capability Available of Rs. 50.0 M or above. (Submit Bank statement as evidence showing required balance at any one instant in the statement of last three months before date of publication of tender notice) (Quantitative)(Doc Required)	20
The bidder/firm must possess an OEM's trained engineer for installation and commissioning of Fire Fighting & Fire Detection Equipment. (Copies of training certificates/ appropriate documentary reference) (Quantitative)(Doc Required)	20

Main Equipment Manufacturer / Offered Brand 5 years' experience in manufacturing similar equipment/machinery conforming internationally recognized manufacturing and safety standards. (Appropriate documentary evidence/ Extracts from manufacturer's website/ reference). (Quantitative)(Doc Required)	20
Technical Literature and other details including manufacturers Catalogue in support of quoted equipment shall be submitted along with the bid for technical evaluation in light of specifications of tender (OEM provided Manual/ booklets/specs sheet/ appropriate documentary reference). (Quantitative)(Doc Required)	20

Jobs/Lots

Jobs Without Lots :

Job	Delivery Schedule	Quantity	Bid Security
Upgradation of Fire Fighting System at SBP Buildings, Head Office Karachi	Address: SBP Complex, I.I. Chundrigar road Karachi Schedule: 730 Days Quantity: 1/job	1/job	5000000 PKR

Related Services :

No



Work Specifications and Market Rates

Jobs Without Lots :

Job: Upgradation of Fire Fighting System at SBP Buildings, Head Office Karachi

Specifications / Requirements:

Sr. No	Item No.	Description	Unit	Qty	Rate	Amount
1	1	Upgradation of Fire Fighting System at SBP Buildings, Head Office Karachi	Job	1	564907971	564907971
					GRAND TOTAL:	Rs564,907,971.00

Scope of Work

See attachedAnnexure-04 - Bill of Quantities;Annexure-05 - Work Schedule & KPI;Annexure-06 Technical Specifications



Price Schedule

For Individual Jobs

#	Job Title	Quantity	Unit Price (PKR)	Total Price (PKR)	Delivery Location	Delivery Period / Year	Country of Origin
1							
2							

For Lots

#	Lot Title	Total Lot Price (PKR)	Country of Origin
1	[Lot 1 Title]		





General Conditions of Contract

A. General

1. Definitions

1.1. Unless the context otherwise requires, the following terms whenever used in this Contract shall have the same meaning and shall be interpreted as indicated

1.1.1. “Applicable Law” means the laws and any other instruments having the force of law in the Government’s Country, or in such other country as may be specified in the Special Conditions of the Contract (SC), as they may be issued and in force from time to time;

1.1.2. “The Contract” means an agreement enforceable by law;

1.1.3. “The Contract Price” means the price payable to the Contractor under the Contract for the full and proper performance of its contractual obligations;

1.1.4. “The Services” means the work to be performed by the Contractor pursuant to this Contract and as prescribed in the Specifications and Schedule of Activities included in the Contractor’s Bid;

1.1.5. “Ancillary Services” means those services ancillary to the provision of Services, such as transportation and insurance, and any other incidental services, such as installation, commissioning, provision of technical assistance, training, and other such obligations of the Contractor covered under the Contract;

1.1.6. “GCC” means the General Conditions of Contract contained in this section;

1.1.7. “SCC” means the Special Conditions of Contract by which the GCC may be amended or supplemented;

1.1.8. “Day” means calendar day unless indicated otherwise;

1.1.9. “Effective Date” means the date on which this Contract comes into force and effect;

1.1.10. “The Contractor” means the individual or corporate body whose Bids to provide the Services has been accepted by the Procuring Agency;

1.1.11. “The Project Site,” where applicable, means the place or places named in Bid Data Sheet and technical Specifications;

1.1.12. “Government” means the Government of Pakistan;

1.1.13. “Local Currency” means the currency of Pakistan;

1.1.14. “In Writing” means communicated in written form with proof of receipt;

1.1.15. “Completion Date” means the date of completion of the Services by the Contractor as certified by the Procuring Agency;

1.1.16. “Foreign Currency” means any currency other than the currency of the country of the Procuring Agency;

1.1.17. "Party" means the Procuring Agency or the Contractor, as the case may be, and "Parties" means both of them;

1.1.18. "Service" means any object of procurement other than goods or works;

1.1.19. "Subcontractor" means any entity to which the Bidder subcontracts any part of the Services.

2. Applicable Law

2.1. The contract shall be governed and interpreted in accordance with the laws of Pakistan, unless otherwise specified in SCC.

3. Language

3.1. The Contract as well as all correspondence and documents relating to the Contract exchanged between the Contractor and the Procuring Agency, shall be written in the **English language** unless otherwise stated in the SCC. Supporting documents and printed literature that are part of the Contract may be in another language provided these are accompanied by an accurate translation of the relevant passages in English, in which case, for purposes of interpretation of the Contract, this translation shall govern.

4. Notices

4.1. Any notice, request, or consent made pursuant to this Contract shall be in writing and shall be deemed to have been made when delivered in person to an authorized representative of the Party to whom the communication is addressed, or when sent by registered mail, telex, telegram, or facsimile to such Party at the address specified in the SCC.

5. Location

5.1. The Services shall be performed at such locations as the Procuring Agency may approve and as specified in SCC.

5.2. A {DOCUMENTS}

6. Authorized Representatives / Authority of Member in charge

6.1. Any action required or permitted to be taken, and any document required or permitted to be executed, under this Contract by the Procuring Agency or the Contractor may be taken or executed by the officials specified in the SCC.

B. Commencement, Completion, Modification, and Termination of Contract

7. Effectiveness of Contract

7.1. This Contract shall come into effect on the date the Contract is signed by both parties and such other later date as may be stated in the SCC.

8. Commencement of Services

8.1. The Contractor shall confirm availability of Key Experts and begin carrying out the Services not later than the number of days after the Effective Date specified in the SCC.

9. Program schedule

9.1. Before commencement of the Services, the Contractor shall submit to the Procuring Agency for approval a Program showing the general methods, arrangements, order and timing for all activities. The Services shall be carried out in accordance with the approved Program as updated.

10. Starting Date/Expiration Date

10.1. The Contractor shall start carrying out the Services Five (05) days after the date the Contract becomes effective, or at such other date as may be specified in the SCC.

10.2. Unless terminated earlier pursuant to Clause **GCC 14** hereof, this Contract shall expire at the end of such time period after the Effective Date as specified in the SCC.

11. Entire Agreement

11.1. This Contract contains all covenants, stipulations and provisions agreed by the Parties. No agent or representative of either Party has authority to make, and the Parties shall not be bound by or be liable for, any statement, representation, promise or agreement not set forth herein.

12. Modification

12.1. Any modification or variation of the terms and conditions of this Contract, including any modification or variation of the scope of the Services, may only be made by written agreement between the Parties. However, each Party shall give due consideration to any modification(s) or variation(s) made by the other Party.

12.2. In cases of any modification(s) or variation(s), the prior written consent of the Procuring Agency is required.

13. Force Majeure

13.1. Definition

For the purposes of this Contract, "Force Majeure" means an event which is beyond the reasonable control of a Contractor and which makes a Contractor's performance of its obligations under the Contract impossible or so impractical as to be considered impossible under the circumstances.

13.2. No Breach of Contract

The failure of a Party to fulfill any of its obligations under the contract shall not be considered to be a breach of, or default under, this Contract in so far as such inability arises from an event of Force Majeure, provided that the Party affected by such an event (a) has taken all reasonable precautions, due care and reasonable alternative

measures in order to carry out the terms and conditions of this Contract, and (b) has informed the other Party as soon as possible about the occurrence of such an event.

13.3. Extension of Time

Any period within which a Contractor shall, pursuant to this Contract, complete any action or task, shall be extended for a period equal to the time during which such Party was unable to perform such action as a result of Force Majeure.

13.4. Payments

During the period of their inability to perform the Services as a result of an event of Force Majeure, the Contractor shall be entitled to continue to be paid under the terms of this Contract, as well as to be reimbursed for additional costs reasonably and necessarily incurred by them during such period for the purposes of the Services and in reactivating the Service after the end of such period.

14. Termination

14.1. By the Procuring Agency

The Procuring Agency may terminate this Contract in case of the occurrence of any of the events specified in paragraphs (a) through (e) of this Clause. In such an occurrence the Procuring Agency shall give at least thirty (30) calendar days' written notice of termination to the Contractor in case of the events referred to in (a) through (d); at least sixty (60) calendar days' written notice in case of the event referred to in (e);

14.1.1. If the Contractor fails to remedy a failure in the performance of its obligations hereunder, as specified in a notice of suspension;

14.1.2. If the Contractor becomes (or, if the Contractor consists of more than one entity, if any of its members becomes) insolvent or bankrupt or enter into any agreements with their creditors for relief of debt or take advantage of any law for the benefit of debtors or go into liquidation or receivership whether compulsory or voluntary;

14.1.3. If the Contractor fails to comply with any final decision reached as a result of arbitration proceedings;

14.1.4. If, as the result of Force Majeure, the Contractor is unable to perform a material portion of the Services for a period of not less than sixty (60) calendar days;

14.1.5. If the Procuring Agency, in its sole discretion and for any reason whatsoever, decides to terminate this Contract;

14.2. By the Contractor

The Contractor may terminate this Contract, by not less than thirty (30) calendar days' written notice to the Procuring Agency, in case of the occurrence of any of the events specified in paragraphs (a) through (d) of this Clause.

14.2.1. If the Procuring Agency fails to pay any money due to the Contractor pursuant to this Contract and not subject to dispute within forty-five (45) calendar days after receiving written notice from the Contractor that such payment is overdue;

14.2.2. If, as the result of Force Majeure, the Contractor is unable to perform a material portion of the Services for a period of not less than sixty (60) calendar days;

14.2.3. If the Procuring Agency fails to comply with any final decision reached as a result of arbitration;

14.2.4. If the Procuring Agency is in material breach of its obligations pursuant to this Contract and has not remedied the same within forty-five (45) days (or such longer period as the Bidder may have subsequently approved in writing) following the receipt by the Procuring Agency of the Contractor's notice specifying such breach.

C. Obligations of the Contractor

15. General

15.1. Standard of Performance

15.1.1. The Contractor shall perform the Services and carry out the Services with all due diligence, efficiency and economy, in accordance with generally accepted professional standards and practices, and shall observe sound management practices, and employ appropriate technology and safe and effective equipment, machinery, materials and methods. The Contractor shall always act, in respect of any matter relating to this Contract or to the Services, as a faithful adviser to the Procuring Agency, and shall at all times support and safeguard the Procuring Agency's legitimate interests in any dealings with the third parties;

15.1.2. The Contractor shall employ and provide such qualified and experienced Experts and Sub-Contractors as are required to carry out the Services.

15.2. Law Applicable to Services

The Contractor shall perform the Services in accordance with the Contract and in accordance with the Law of Pakistan and shall take all practicable steps to ensure that any of its Experts and Sub-Bidders, comply with the Applicable Law.

16. Conflict of Interests

16.1. Contractor Not to Benefit from Commissions and Discounts

The remuneration of the Contractor shall constitute the Contractor's sole remuneration in connection with this Contract or the Services, and the Contractor shall not accept for their own benefit any trade commission, discount, or similar payment in connection with activities pursuant to this Contract or to the Services or in the discharge of their obligations under the Contract, and the Contractor shall use their best efforts to ensure that the Personnel, any Subcontractors, and agents of either of them similarly shall not receive any such additional remuneration.

16.2. Contractor and Affiliates Not to be Otherwise Interested in Project

The Contractor agree that, during the term of this Contract and after its termination, the Contractor and its affiliates, as well as any Subcontractor and any of its affiliates, shall be disqualified from providing Services (other than the Services and any continuation thereof) for any project resulting from or closely related to the Services.

16.3. Prohibition of Conflicting Activities

Neither the Bidder nor its Subcontractors nor the Personnel shall engage, either directly or indirectly, in any of the following activities:

- 16.3.1. during the term of this Contract, any business or professional activities in the Government's country which would conflict with the activities assigned to them under this Contract;
- 16.3.2. during the term of this Contract, neither the Contractor nor their Subcontractors shall hire public employees in active duty or on any type of leave, to perform any activity under this Contract;
- 16.3.3. after the termination of this Contract, such other activities as may be specified in the **SCC**.

17. Insurance to be Taken Out by the Contractor

17.1. The Contractor(a) shall take out and maintain, and shall cause any Subcontractors to take out and maintain, at its (or the Sub-contractors', as the case may be) own cost but on terms and conditions approved by the Procuring Agency, insurance against the risks, and for the coverage, as shall be specified in the **SCC**; and (b) at the Procuring Agency's request, shall provide evidence to the Procuring Agency showing that such insurance has been taken out and maintained and that the current premiums have been paid.

18. Contractor's Actions Requiring Procuring Agency's Prior Approval

18.1. The Contractor shall obtain the Procuring Agency's prior approval in writing before taking any of the following actions:

- 18.1.1. appointing such members of the Personnel not provided by the Contractor;
- 18.1.2. changing the Program of activities; and
- 18.1.3. any other action that may be specified in the **SCC**.

19. Reporting Obligations

19.1. The Contractor shall submit to the Procuring Agency the reports and documents in the numbers, and within the periods as prescribed by the Procuring Agency.

20. Liquidated Damages

20.1. Payments of Liquidated Damages

The Contractor shall pay liquidated damages to the Procuring Agency at the rate per day stated in the **SCC** for each day that the Completion Date is later than the Intended Completion Date. The total amount of liquidated damages shall not exceed the amount defined in the **SCC**. The Procuring Agency may deduct liquidated damages from payments due to the Contractor. Payment of liquidated damages shall not affect the Contractor's liabilities.

20.2. Correction for Over-payment

If the Intended Completion Date is extended after liquidated damages have been paid, the Procuring Agency shall correct any overpayment of liquidated damages by the Contractor by adjusting the next payment certificate. The

Contractor shall be paid interest on the overpayment, calculated from the date of payment to the date of repayment, at the rates specified in **SCC**.

20.3. Lack of performance penalty

If the Contractor has not corrected a Defect within the time specified in the Procuring Agency's notice, a penalty for Lack of performance will be paid by the Contractor. The amount to be paid will be calculated as a percentage of the cost of having the Defect corrected, assessed as specified in the Contractor

21. Performance Guarantee

21.1. Within the time stipulated in the acceptance letter from the Procuring Agency, the successful Bidder shall furnish the Performance Guarantee in shape and amount **specified in SCC**.

21.2. The proceeds of the Performance Guarantee shall be payable to the Procuring agency as compensation for any loss resulting from the Supplier's failure to complete its obligations under the Contract.

21.3. The Performance Guarantee shall be denominated in the currency of the Contract, or in a freely convertible currency acceptable to the Procuring agency and shall be in the acceptable form as specified in **SCC**.

21.4. The Performance Guarantee will be discharged by the Procuring agency and returned to the Supplier not later than thirty (30) days following the date of completion of the Supplier's performance obligations under the Contract, including any warranty obligations, unless otherwise **specified in SCC**.

22. Sustainable Procurement

22.1. The Contractor shall conform to the sustainable procurement contractual provisions, if and as specified in the **SCC**.

D. Contractor's Personnel

23. Description of Personnel

23.1. The titles, agreed job descriptions, minimum qualifications, and estimated periods of engagement in the carrying out of the Services of the Contractor's Key Personnel. The Key Personnel listed by title as well as by name are hereby approved by the Procuring Agency.

24. Removal and / or Replacement of Personnel

24.1. Except as the Procuring Agency may otherwise agree, no changes shall be made in the Key Personnel. If, for any reason beyond the reasonable control of the Contractor, it becomes necessary to replace any of the Key Personnel, the Contractor shall provide as a replacement a person of equivalent or better qualifications.

24.2. If the Procuring Agency finds that any of the Personnel have (i) committed serious misconduct or have been charged with having committed a criminal action, or (ii) have reasonable cause to be dissatisfied with the performance of any of the Personnel, then the Contractor shall, at the Procuring Agency's written request specifying the grounds thereof, provide as a replacement a person with qualifications and experience acceptable to the Procuring Agency.

24.3. The Contractor shall have no claim for additional costs arising out of or incidental to any removal and/or replacement of Personnel.

E. Obligations of the Procuring Agency

25. Change in the Applicable Law

25.1. If, after the date of this Contract, there is any change in the Applicable Law with respect to taxes and duties which increases or decreases the cost of the Services rendered by the Contractor, then the remuneration and reimbursable expenses otherwise payable to the Contractor under this Contract shall be increased or decreased accordingly by agreement between the Parties, and corresponding adjustments shall be made to the amounts referred in the SCC.

26. Services and Facilities

26.1. The Procuring Agency shall make available to the Contractor and the Experts, for the purposes of the Services and free of any charge, the services, facilities and property described in the Terms of Reference, at the times and in the manner specified in the Terms of Reference.

26.2. In case that such services, facilities and property shall not be made available to the Contractor, the Parties shall agree on (i) any time extension that it may be appropriate to grant to the Contractor for the performance of the Services, (ii) the manner in which the Contractor shall procure any such services, facilities and property from other sources, and (iii) the additional payments, if any, to be made to the Contractor as a result thereof.

F. Payments to the Contractor

27. Contract Price

27.1. The price payable shall be in Pakistani Rupees unless otherwise specified in the SCC.

28. Terms and Conditions of Payment

28.1. Payments will be made to the Contractor according to the payment schedule stated in the SCC and as per actual invoice submitted by the Contractor.

28.2. Unless otherwise stated in the SCC, the advance payment shall be made against the provision by the Contractor of a bank guarantee for the same amount, and shall be valid for the period stated in the SCC. Any other payment shall be made after the conditions listed in the SCC for such payment have been met, and the Contractor have submitted an invoice to the Procuring Agency specifying the amount due.

29. Quality Control Identifying Defects

29.1. The principle and modalities of Inspection of the Services by the Procuring Agency shall be as indicated in the SCC. The Procuring Agency shall check the Contractor's performance and notify him of any Defects that are

found. Such checking shall not affect the Contractor's responsibilities. The Procuring Agency may instruct the Contractor to search for a Defect and to uncover and test any service that the Procuring Agency considers may have a Defect. Defect Liability Period is as defined in the SCC.

29.2. A {INSPECTION}

30. Correction of Defects, and Lack of Performance Penalty

30.1. The Procuring Agency shall give notice to the contractor of any Defects before the end of the Contract. The Defects liability period shall be extended for as long as Defects remain to be corrected.

30.2. Every time notice a Defect is given; the contractor shall correct the notified Defect within the length of time specified by the Procuring Agency's notice.

30.3. If the contractor has not corrected a Defect within the time specified in the Procuring Agency's notice, the Procuring Agency will assess the cost of having the Defect corrected, the contractor will pay this amount, and a Penalty for Lack of Performance.

31. Settlement of Disputes Amicable Settlement

31.1. The Parties shall use their best efforts to settle amicably all disputes arising out of or in connection with this Contract or its interpretation.

32. Dispute Settlement

32.1. Arbitration

If any dispute of any kind whatsoever shall arise between the procuring agency and the contractor in connection with or arising out of the Contract, including without prejudice to the generality of the foregoing, any question regarding its existence, validity or termination, or the execution of the contract, the parties shall seek to resolve any such dispute or difference by mutual consultation. If the parties fail to resolve such a dispute or difference even after negotiations or mediation, then the dispute shall be referred within fourteen (14) days in writing by either party to the Arbitrator, with a copy to the other party.

Any dispute in respect of which a notice of intention to commence arbitration has been given, in accordance with **GCC sub-clause 32.1**, shall be finally settled by arbitration. Arbitration may be commenced prior to or after completion of the Contract. Arbitration proceedings shall be conducted in accordance with Arbitration Act 1940. Notwithstanding any reference to arbitration herein, the parties shall continue to perform their respective obligations under the Contract unless otherwise agreed. The Procuring Agency shall continue to pay the Contractor any undisputed amounts due under the Contract during the resolution of any dispute.



Special Conditions of Contract

SECTION VIII. SPECIAL CONDITIONS OF CONTRACT

The following Special Conditions of Contract shall supplement the General Conditions of Contract. Whenever there is a conflict, the provisions herein shall prevail over those in the Conditions of Contract. The corresponding clause number of the GCC is indicated in parentheses.

Number of GC Clause

Amendments of, and Supplements to, Clauses in the General Conditions of Contract

Number of GC Clause 1

Definitions

The Procuring Agency is: SBP BSC (SBP Banking Services Corporation (BSC)), Executive Engineer Engineering Department, 1st Flor, SBP Boulton Market Building, M.A Jinnah Road, Saddar Sub-Division, Karachi South (District), Karachi (Division), Sindh (Province).

The Supplier is:

The title of the subject procurement is: 327-Upgradation of Fire Fighting System at SBP Buildings, Head Office Karachi

Number of GC Clause 2

Applicable/Governing Law:

The Contract shall be interpreted in accordance with the laws of Islamic Republic of Pakistan

Number of GC Clause 3

Language:

The language of the Contract, all correspondence and communications to be given, and all other documentation to be prepared and supplied under the Contract shall be in **English**.

Number of GC Clause 4

Notices:

The addresses for the notices are:

Procuring Agency:

SBP BSC (SBP Banking Services Corporation (BSC)), Executive Engineer Engineering Department, 1st Flor, SBP Boulton Market Building, M.A Jinnah Road, Saddar Sub-Division, Karachi South (District), Karachi (Division), Sindh (Province).

+92-213-450-4122

shehzad.memon@sbp.org.pk

Contractor/ Bidder:

[Name, address and telephone number]

The Contractor/ Bidder's Representative(s)

[Name, address, telephone number and e-mail address]

Number of GC Clause 6.1

The Authorized Representatives are:

For the Procuring Agency:

SBP BSC (SBP Banking Services Corporation (BSC)), Executive Engineer
Engineering Department, 1st Flor, SBP Boulton Market Building, M.A Jinnah Road, Saddar Sub-Division,
Karachi South (District), Karachi (Division), Sindh (Province).
+92-213-450-4122
shehzad.memon@sbp.org.pk

For the Bidder:

Name:

Designation:

Address:

Number of GC Clause 7

Effectiveness of the contract

The Contractor/Bidder shall be effective within days from the date of signature of the Contract by both parties

Number of GC Clause 8

Commencement of Contract:

The Contractor/Bidder shall provide Non-Consultancy Services from the effective date of contract.

Number of GC Clause 10.2

Expiration of Contract:

The time period shall be

Number of GC Clause 14

Termination

In the event of termination of the contract due to any reason as already defined in the General Conditions of Contract, the Bidder shall be responsible for providing to the Authority the Goods till the time of alternate arrangements.

Number of GC Clause 16

Conflict of Interest:

The Procuring Agency reserves the right to determine on a case-by-case basis whether the Bidder should be disqualified from providing goods or services due to a conflict of a nature described in Clause GCC 17.

Number of GC Clause 20

Liquidated Damages

If the Bidder fails to provide services as required under the contract or in case of any data loss/data breach or any incident compromising the data security or other such failures related to any services, the Bidder shall pay to the Procuring Agency as Liquidated Damages at a rate of **0.10% to 10.00%** of the Contract value, in accordance with the extent of performance failure & the cost of investigating such incidents as judged by the Authority.

Number of GC Clause 21

Performance Guarantee:

The amount of performance guarantee shall be **10.00%** of the contract price in acceptable form of **Pay Order, Banker's Cheque, Call at Deposit, Bank Guarantee, Demand Draft**

Number of GC Clause 27

Currency of Payment:

All the payment to be released to the contractor/Bidder shall be in Pakistani Rupees.

Number of GC Clause 28

Payment terms:

Payment will be made to the Bidder against the procured Goods and services according to the actual invoice or running bills submitted by the Bidder against the services provided within the time given in the conditions of the contract.

Number of GC Clause 29

Identifying Defects:

The Authority reserves the right at any time to inspect the premises of the provider to inspect the goods and monitor the goods being provided.

Number of GC Clause 31

Following is the guidance for Dispute Resolution

- i. If any dispute of any kind whatsoever shall arise between the Authority and the Bidder in connection with or arising out of the Contract... (full clause unchanged)
- ii. At future of negotiation the dispute shall be resolved through mediation...
- iii. At failure of mediation, arbitration under Arbitration Act 1940...
- iv. Cost sharing equally...

v. Proceedings may commence before/after completion...

Notwithstanding any reference to the arbitration herein, the parties shall continue to perform their respective obligations under the Contract.

Rules of procedure for arbitration proceedings:

Any dispute between the Authority and a Bidder shall be referred to arbitration under the laws of Pakistan including Arbitration Act 1940.

Place of Arbitration and Award:

The arbitration shall be conducted in English language and place shall be Islamabad. The award shall be final and binding.





Bid Securing Declaration

Form 9: Bid Securing Declaration

Date: *[insert date (as day, month and year)]*

Bid No.: **P99841**

To: **SBP BSC (SBP Banking Services Corporation (BSC)), Executive Engineer Engineering Department, 1st Flor, SBP Boulton Market Building, M.A Jinnah Road, Saddar Sub-Division, Karachi South (District), Karachi (Division), Sindh (Province).**

We, the undersigned, declare that:

We understand that, according to your conditions, Bids must be supported by a Bid Securing Declaration.

We accept that we will be blacklisted and henceforth cross debarred for participating in respective category of public procurement proceedings for a period of (not more than) six months, if fail to abide with a bid securing declaration, however without indulging in corrupt and fraudulent practices, if we are in breach of our obligation(s) under the Bid conditions, because we:

1. have withdrawn or modified our Bid during the period of Bid Validity specified in the Form of Bid;
2. Disagreement to arithmetical correction made to the Bid price; or
3. having been notified of the acceptance of our Bid by the Procuring Agency during the period of Bid Validity, (i) failure to sign the contract if required by Procuring Agency to do so or (ii) fail or refuse to furnish the Performance Security or to comply with any other condition precedent to signing the contract specified in the Bidding Documents.

We understand this Bid Securing Declaration shall expire if we are not the successful

Bidder, upon the earlier of (i) our receipt of your notification to us of the name of the successful Bidder; or (ii) twenty-eight (28) days after the expiration of our Bid.



Contract Form

SECTION IX: CONTRACT FORMS

THIS AGREEMENT made the ____ day of _____ 20____ between **SBP BSC (SBP Banking Services Corporation (BSC)), Executive Engineer Engineering Department, 1st Flor, SBP Boulton Market Building, M.A Jinnah Road, Saddar Sub-Division, Karachi South (District), Karachi (Division), Sindh (Province).**

(hereinafter called “the Procuring Agency”) of the one part and [name of Bidder] of [city and country of Bidder] (hereinafter called “the Bidder”) of the other part:

WHEREAS the Procuring Agency invited Bids for provision of goods, viz., **327-Upgradation of Fire Fighting System at SBP Buildings, Head Office Karachi (P99841)** and has accepted a Bids by the Bidder for the provision of Goods in the sum of [contract price in words and figures] (hereinafter called “the Contract Price”).

NOW THIS CONTRACT WITNESSETH AS FOLLOWS:

1. In this Contract words and expressions shall have the same meanings as are respectively assigned to them in the Conditions of Contract referred to.

2. The following documents shall be deemed to form and be read and construed as part of this Contract, In the event of any ambiguity or conflict between the Contract Documents listed below, the order of precedence shall be the order in which the Contract Documents are listed below:-

1. This form of Contract;
2. the Form of Bids and the Price Schedule submitted by the Bidder;
3. the Schedule of Requirements;
4. the Technical Specifications;
5. the Special Conditions of Contract;
6. the General Conditions of the Contract;
7. the Procuring Agency’s Letter of Acceptance; and

8. [add here: any other documents]

3. In consideration of the payments to be made by the Procuring Agency to the Bidder as hereinafter mentioned, the Bidder hereby covenants with the Procuring Agency to provide the Goods related services and to remedy defects therein in conformity in all respects with the provisions of the Contract.

4. The Procuring Agency hereby covenants to pay the Bidder in consideration of the provision of Goods and the remedying of defects therein, the Contract Price or such other sum as may become payable under the provisions of the contract at the times and in the manner prescribed by the contract.

IN WITNESS whereof the parties hereto have caused this Contract to be executed in accordance with their respective laws the day and year first above written.

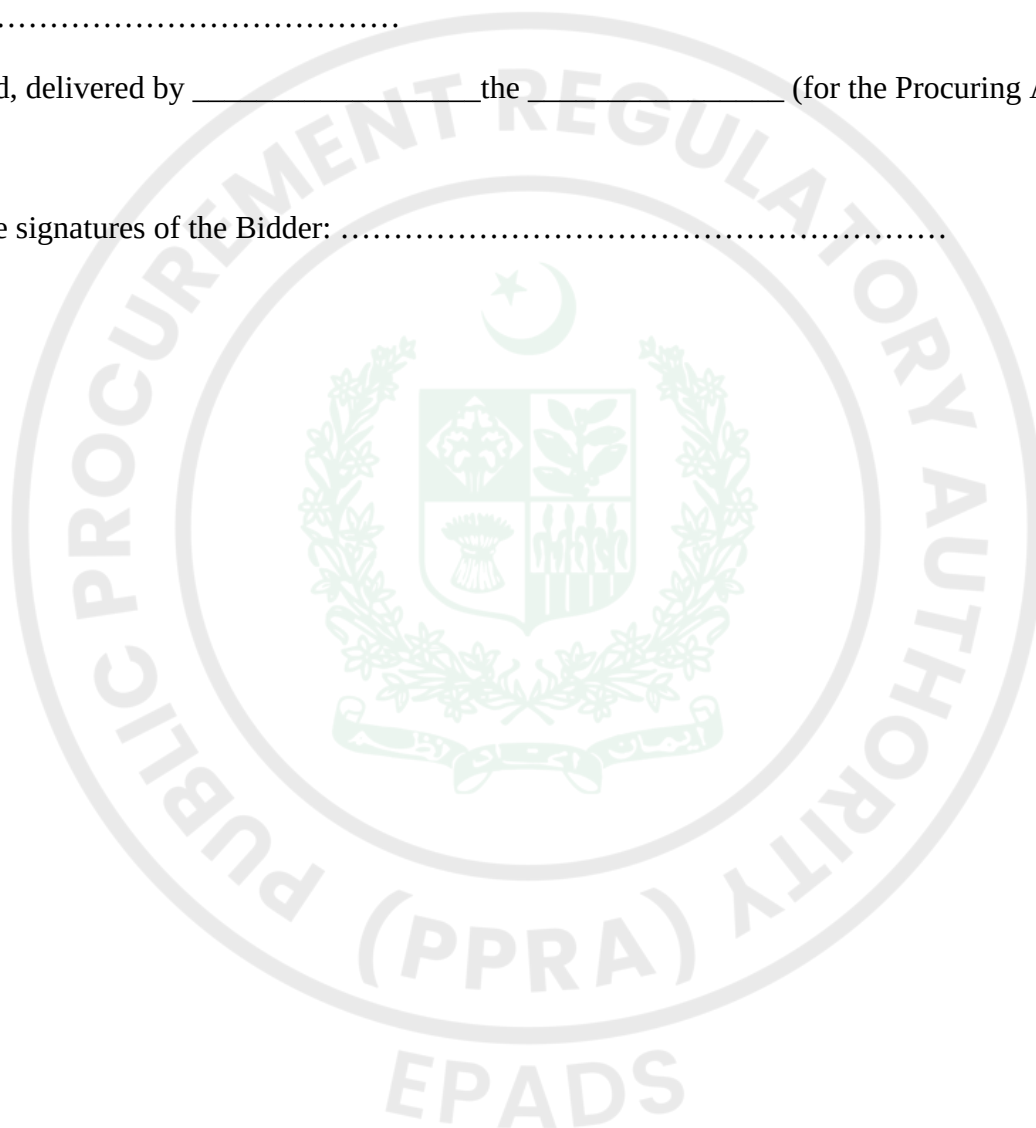
Signed, sealed, delivered by _____ the _____ (for the Procuring Agency)

Witness to the signatures of the Procuring Agency:

.....

Signed, sealed, delivered by _____ the _____ (for the Procuring Agency)

Witness to the signatures of the Bidder:





Integrity Pact

Integrity Pact

DECLARATION OF FEES, COMMISSION AND BROKERAGE ETC. PAYABLE BY THE SUPPLIERS OF GOODS, SERVICES & WORKS IN CONTRACTS WORTH RS.10.00 MILLION OR MORE

Contract

Number: Contract

Value: Contract Title:

Dated:

[Name of Supplier] hereby declares that it has not obtained or induced the procurement of any contract, right, interest, privilege or other obligation or benefit from Government of Pakistan or any administrative subdivision or agency thereof or any other entity owned or controlled by it (GoP) through any corrupt business practice.

Without limiting the generality of the foregoing [Name of Supplier] represents and warrants that it has fully declared the brokerage, commission, fee etc. paid or payable to anyone and not given or agreed to give and shall not give or agree to give to anyone within or outside Pakistan either directly or indirectly through any natural or juridical person, including its affiliate, agent, associate, broker, consultant, director, promoter, shareholder, sponsor or subsidiary, any commission, gratification, bribe, finder's fee or kickback, whether described as consultations fee or otherwise, with the object of obtaining or inducing the procurement of a contract, right, interest, privilege or other obligation or benefit in whatsoever form from GoP, except that which has been expressly declared pursuant hereto.

[Name of Supplier] certifies that it has made and will make full disclosure of all agreements and arrangements with all persons in respect of or related to the transaction with GoP and has not taken any action or will not take any action to circumvent the above declaration, representative or warranty.

[Name of Supplier] accepts full responsibility and strict liability for making and false declaration, not making full disclosure, misrepresenting fact or taking any action likely to defeat the purpose of this declaration, representation and warranty. It agrees that any contract, right interest, privilege or other obligation or benefit obtained or procured as aforesaid shall, without prejudice to any other right and remedies available to GoP under any law, contract or other instrument, be voidable at the option of GoP.

Notwithstanding any rights and remedies exercised by GoP in this regard, [Name of Supplier] agrees to indemnify GoP for any loss or damage incurred by it on account of its corrupt business practices and further pay compensation to GoP in an amount equivalent to ten times the sum of any commission, gratification, bribe, finder's fee or kickback given by [Name of Supplier] as aforesaid for the purpose of obtaining or inducing the procurement of any contract, right, interest, privilege or other obligation or benefit in whatsoever form from GoP.



Performance Guarantee Form

Performance Guarantee Form

To: **SBP BSC (SBP Banking Services Corporation (BSC)), Executive Engineer Engineering Department, 1st Flor, SBP Boulton Market Building, M.A Jinnah Road, Saddar Sub-Division, Karachi South (District), Karachi (Division), Sindh (Province).**

WHEREAS *[name of Bidder]* (hereinafter called “the Bidder”) has undertaken, in pursuance of Contract No. *[reference number of the contract]* dated *[insert date]* for provision of Goods (hereinafter called “the Contract”).

AND WHEREAS it has been stipulated by you in the said Contract that the Bidder shall furnish you with a Bank Guarantee by a reputable bank for the sum specified therein as security for compliance with the Bidder’s performance obligations in accordance with the Contract.

AND WHEREAS we have agreed to give the Bidders guarantee:

THEREFORE, WE hereby affirm that we are Guarantors and responsible to you, on behalf of the Bidder, up to a total of *[amount of the guarantee in words and figures]*, and we undertake to pay you, upon your first written demand declaring the Bidder to be in default under the Contract and without cavil or argument, any sum or sums within the limits of *[amount of guarantee]* as aforesaid, without your needing to prove or to show grounds or reasons for your demand or the sum specified therein.

This guarantee is valid until the: *[insert date]*

Signature and seal of the Guarantors

[name of bank or financial institution]

[address]

[date]





Annexure

Annexure-01 - Bidding Data Sheet (BDS) – Part-B

Annexure-01 - Bidding Data Sheet (BDS) – Part-B

Information (Read-Only)

See Form Under Additional Forms and Documents: **Annexure-01 - Bidding Data Sheet (BDS) – Part-B** (page number: 78)

Annexure-02 - Section VIII - SCC (Part-B)

Annexure-02 - Section VIII - SCC (Part-B)

Information (Read-Only)

See Form Under Additional Forms and Documents: **Annexure-02 - Section VIII - SCC (Part-B)** (page number: 79)

Annexure-04 - Bill of Quantities

Annexure-04 - Bill of Quantities

Information (Read-Only)

See Form Under Additional Forms and Documents: **Annexure-04 - Bill of Quantities** (page number: 88)

Annexure-05 - Work Schedule & KPI

Annexure-05 - Work Schedule & KPI

Information (Read-Only)

See Form Under Additional Forms and Documents: **Annexure-05 - Work Schedule & KPI** (page number: 98)

Annexure-06 - Technical Specifications

Annexure-06 Technical Specifications

Information (Read-Only)

See Form Under Additional Forms and Documents: **Annexure-06 - Technical Specifications** (page number: 101)

Annexure-08 - Forms

Annexure-08 - Forms

Technical Submission (Vendor)

Document Required

See Form Under Additional Forms and Documents: **Annexure-08 - Forms** (page number: 165)

Annexure-09 - Supplier Account Form

Annexure-09 - Supplier Account Form

Technical Submission (Vendor)

Document Required

See Form Under Additional Forms and Documents: **Annexure-09 - Supplier Account Form** (page number: 167)

Annexure-7.1 - Drawings (1/2)

Annexure-7.1 - Drawings (1/2)

Information (Read-Only)

See Form Under Additional Forms and Documents: **Annexure-7.1 - Drawings (1/2)** (page number: 168)

Annexure-7.2 - Drawings (2/2)

Annexure-7.2 - Drawings (2/2)

Information (Read-Only)





Procurement Forms







Additional Forms and Documents

Bidding Data Sheet (BDS) – Part-B

The following specific data shall complement, supplement, or amend the provisions in the Instructions to Bidders (ITB) and Bidding Data Sheet. Whenever there is a conflict, the provisions herein shall prevail over those in ITB and Bidding Data Sheet.

ITB No.	Amendments of, and Supplements
Preparing of Bids	
6.2 & 6.4	Add the following Pre-Bid Meeting: Date & Time: Thursday, October 01, 2026 at 12:00 PM Venue: Engineering Department, 1st Flor, SBP Boulton Market Building, M.A Jinnah Road Interested bidders attendees may be required to intimate at (proc.engg.hok@sbp.org.pk). In case pre-bid meeting is falling on public holiday, the same will be conducted on next working day at the same time and venue.
11.5 & 13.5	The bid price shall be fixed and no price adjustment will be made.
13.3.2	Alternative Bids to the requirements of the bidding documents will not be permitted.
18.1	Add the following The Bid Security shall be in favor of “State Bank of Pakistan (SBP BSC)” in the form of Payment Order / Banker’s Cheque / Bank Draft and shall be valid till Bid Validity period.
28.	Determination of Unit Rates The total quoted Bid Price shall be apportioned among the individual BOQ items in proportion to their respective employer’s / engineer’s estimated unit rates, and the corresponding individual unit rates shall be determined accordingly, without altering the total quoted Bid Price. The Bidder shall be deemed to have accepted such determination and adjustment of the individual unit rates. In the event that the Bidder fails or refuses to accept the aforesaid determination or adjustment, the Bid shall be treated as non-responsive, and the Bid Security shall be forfeited.
41.1	Performance Security (or Guarantee) Add the following at the end Performance Guarantee shall be valid till successful completion of Defect Liability Period
42.2	Advance Payment The Procuring agency/Employer will provide an advance payment as stipulated in the Conditions of Contract.

Section VIII – Special Conditions of Contract (SCC) – Part-B

The following Special Conditions of Contract (SCC) shall supplement the GCC. Whenever there is a conflict, the provisions herein shall prevail over those in the GCC and SCC.

GCC Clause No.	Amendments of, and Supplements to, Clauses in the GCC
6.1	Authorized Representatives / Authority of Member in charge The Authorized representative will be mentioned in Notice to Commence / Work Order
8.1	Commencement of Services The Commencement date will be mentioned in Notice to Commence / Work Order
9.1	Program schedule Contract Duration is: 730 days , from start of commencement of Contract. Refer Annexure-04 - Scope of Works & KPI
12	VARIATIONS AND CLAIMS 12.1 Right to Vary The Employer/Engineer may issue Variation Order(s) in writing. Where for any reason it has not been possible for the Employer/Engineer to issue such Variations Order(s), the Contractor may confirm any verbal orders given by the Employer/Engineer in writing and if the same are not refuted/denied by the Employer/Engineer within seven (7) days of the receipt of such confirmation the same shall be deemed to be a Variation Orders for the purposes of this Sub-Clause. 12.2 Valuation of Variations Variations shall be valued as follows: a) at a lump sum price agreed between the Parties, or b) where appropriate, at rates in the Contract, or c) rates notified by the Government Departments/agencies, or d) in the absence of appropriate rates, the rates in the Contract shall be used as the basis for valuation, or failing which e) at appropriate new rates, as may be agreed or which the Engineer/Employer considers appropriate Contractor's profit & overheads shall be up to 15% of the cost of labour & materials, cartage, etc. Taxes shall be considered as per actual rate while analyzing rates for any extra items. For the valuation of variations and determination of rates for non-BOQ/extra items, the approval of the Head/Director Engineering shall be obtained. The decision of the Head/Director Engineering in this regard shall be final and binding upon both parties. The approval / finalization of rates of all variations shall not relieve the Contractor of his obligations under the Contract. The Contractor shall neither stop the work nor slow down the progress of the Works in awaiting the approval of rates of all variations. 12.3 Early Warning The Contractor shall notify the Engineer/Employer in writing as soon as he is aware of any circumstance which may delay or disrupt the Works, or which may give rise for Extension in time.

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	To the extent of the Contractor's failure to notify, which results to the Engineer/Employer being unable to keep all relevant records or not taking steps to minimize any delay, disruption, the Contractor's entitlement to extension of the Time for Completion may be reduced/rejected. 12.4 Variation Procedure The Contractor shall submit to the Engineer/Employer an itemized make-up of the value of variations within twenty-eight (28) days of the instruction. The Engineer/Employer shall check and if possible agree the value. In the absence of agreement, the Employer shall determine the value.
13.3	Extension of Time The Contractor shall, within such time as may be reasonable under the circumstances, notify the Employer of any event(s) falling within the scope of Employer's Risks, or Early Warnings, or adverse ground conditions, force majeure not leading to Contract termination, or any instruction of the Employer to slow down the progress of works or change in the scope of Works by the Employer requiring additional time for completion of the Works as per the revised scope of Works, may request the Employer to extend the Completion Time reasonably. The Contractor must have to provide complete record of the circumstances on account of which extension of time is being applied along with an updated program/schedule in bar chart form for completion of the balance works. The Employer may evaluate the request lodged by the Contractor and extend the Completion Time. The decision of the Director/Head Engineering will be final & binding upon the Contractor. It is being understood and agreed by the Contractor that the sole remedy for delay shall be extension in the time for performance of the Contractor's work, which extension shall not be subject to any valuation.
13.4	Force Majeure & Payment upon Termination In the event the project cannot be materialized for any reason, or if the works are suspended or discontinued at any stage, the Contractor shall only be entitled to payment for the works executed. No other payments claim in this regard shall be admissible. The decision of the Director/Head of Engineering shall be final and binding in this regard.
17	Insurance to be taken out by the Contractor From AA rated Insurance Company of PACRA/ JCR. The Works Amount of Cover: The sum stated in Letter of Acceptance plus fifteen percent 15%. Contractor 's equipment Amount of Cover: Full replacement Cost Injury to Person and Damaged to Property including Third Party Insurance Amount of Cover: I) As per workmen compensation act II) Contractor's all Risk including Third party III) damages to the Structure, stores if supplied by the Bank
20.	Liquidated Damages If the Contractor fails to complete the Works, or any Section where a separate Time for Completion is specified, within the applicable Time for Completion, the Contractor shall

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	pay Delay Damages to the Employer for each day of delay from the expiry of the applicable Time for Completion until the date of completion of the Works or relevant Section. The rate of Delay Damages shall be 0.1% of the Contract Value at the expiry of the applicable Time for Completion per day of delay, subject to a maximum aggregate amount of 10% of the Contract Price. The Employer may, without prejudice to any other right or remedy available to it under the Contract and/or applicable law, recover such Delay Damages by deduction from any amount due or becoming due to the Contractor, including amounts payable under interim or final payment certificates. The payment or deduction of Delay Damages shall not relieve the Contractor from its obligation to complete the Works or from any other obligations, liabilities or responsibilities under the Contract.
20.2	Correction for Over-payment Not applicable
28	Terms and Conditions of Payment Add the following 28.2.1 The Contractor shall submit to the Employer/Engineer, from time to time, a Statement, Bill or Invoice setting out in detail all amounts to which the Contractor considers himself entitled under the Contract, together with all supporting documents and records (if required) to substantiate such amounts, in such form and manner as may be prescribed by the Employer/Engineer from time to time. 28.2.2 The Employer/Engineer shall have the right, by any Bill or Interim Payment Certificate, to make any correction, adjustment or modification to any amount previously certified in any Interim Payment Certificate. Furthermore, if any part of the Works is not being executed to the satisfaction of the Employer/Engineer or is not in accordance with the Contract, the Employer/Engineer shall have the authority to omit, withhold or reduce the value of such work in any Interim Payment Certificate until the Contractor has satisfactorily rectified the relevant deficiencies. 28.2.3 The Contractor shall pay all taxes, duties and fees required to be paid by him under the Contract, and the Contract Price shall not be adjusted for any of this cost except where otherwise expressly provided for in the Contract. 28.2.4 The quantities given in the Bill of Quantities are estimated and provisional and are given to provide a common basis for bidding. The basis of payment will be the actual quantities of work executed and measured by the Contractor and verified by the Engineer. 28.2.5 Upon issuance of Final Completion Certificate, the payments pertaining to the Works completed during Defects Liability Period and / or money as Retention Money shall be released as per sub-Clause-28.3 and the accounts for the contract shall be closed. The Employer shall release payments within sixty (60) days. 28.3 Retention Money 10% of the net payable amount from each bill (except Mobilization Advance) of the Contractor will be deducted as retention money. The retention money would be released as follows:

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- a. 5% on Completion of the whole work,
- b. Balance 5% after successful completion of the Defects Liability Period / Warranty Period.

28.4 Advance Payment:

If requested by the Contractor, an interest-free Advance up to 15 % of the Contract Price stated in the Letter of Acceptance shall be paid by the Employer to the Contractor in two equal parts upon submission by the Contractor of an Advance Bank Guarantee for the full amount of the Advance in the specified form from a Scheduled Bank in Pakistan subject to approval of Director/Head Engineering Payments shall be made as per the following criteria;

First part of the Mobilization Advance within twenty (20) days after verification of the approved/ legible Bank Guarantee from the issuing bank.

Second part of the mobilization advance on arrival of required tools & plants and commencement of Works at site to the entire satisfaction of the Employer.

28.6 Recovery of Advance:

Advance paid to the Contractor shall be recovered from the interim bills of the Contractor @ 15% of total amount of work done at site for the bill being processed till the time that whole of the amount of Mobilization Advance has been recovered

28.7 Secured Advance

The Employer at his sole discretion may provide to Contractor Secured Advance against an indemnity bond acceptable to the Employer of such sum as the Engineer in charge may consider proper in respect of non-perishable materials brought at the Site but not yet incorporated in the permanent Works subject to approval of Director/Head of Engineering, provided that:

- (a) The materials are in accordance with the Specifications for the permanent works;
- (b) Such materials have been delivered to the Site and are properly stored and protected against loss or damage or deterioration to the satisfaction of the Engineer in charge but at the risk to cost of the Contractor;
- (c) The Contractor's records of the requirements, orders, receipts and use of materials are kept in a form approved by the Engineer in charge, and such records shall be available for inspection by the Engineer in charge;
- (d) The Contractor shall submit with his monthly statement the estimated value of the materials on Site together with such documents as may be required by the Engineer in charge for the purpose of valuation of materials and providing evidence of ownership and payment therefore;
- (e) Ownership of such materials shall be deemed to vest in the Employer and these materials shall not be removed from the Site or otherwise disposed of without written permission of the Employer.
- f) 60% of the purchase price of the item/material or 50% of tender price of the item/material whichever is lesser, after measuring the quantity & verification of the quality of materials at site by the Employer may be paid

The amount of Secured Advance against any item(s)/materials shall be recovered from the next three (03) bill of the Contractor in the form of three equal installments. The Employer reserves the right to recover all outstanding amount of the Secured Advance from very 1st bill of the Contractor submitted after release of the Secured Advance.

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29.1	Defects Liability Period Defects Liability Period (DLP) shall be 365 days
30.1	Correction of Defects, and Lack of Performance Penalty Remedying Defects The Contractor shall from the date of issue of the Taking Over Certificate carry out, at no cost to the Employer, repair and rectification work which is necessitated by the earlier execution of poor quality of work or use of below specifications material in the execution of Works and which is so identified by the Employer/Engineer in writing within the said period. Upon expiry of the said period, and subject to the Contractor's faithfully performing his aforesaid obligations, the Employer/Engineer shall issue a Defect Liability Certificate Failure to remedy any such defects or complete outstanding work within a reasonable time shall entitle the Employer to carry out all necessary works at the Contractor's cost. Uncovering and Testing The Contractor shall give not less than five days' notice in writing to the Engineer-in-charge of work before covering up or otherwise placing beyond the reach of measurement any work in order that same may be measured, and correct dimensions thereof be taken before same is covered or placed beyond the reach of measurement. The Engineer-in-Charge shall check the Contractor's Work and notify the Contractor of any Defects if found. Such checking shall not affect the Contractor's responsibilities. The Employer may instruct the Contractor to search for a defect and to uncover/ or testing any work that the he considers may have a Defect Failure to remedying Defects Failure to remedy any defect(s) or complete the outstanding Works except the defects not attributable to the Contractor, to the entire satisfaction of the Employer within a notified time shall entitle the Employer to extend the defect liability period and / or carry out all necessary works at the Contractor's risk to cost.
31.1	Settlement of Disputes Amicable Settlement For Notices Procuring agency/Employer/ Engineer's Authorized representatives name and address: Director / Head Engineering, State Bank of Pakistan
32.1	Dispute Settlement 28.2.1 Any complaint or grievance relating to the procurement process arising before the Contract comes into force shall be dealt with in accordance with the grievance mechanism prescribed under Rule 48 of the Public Procurement Rules, 2004. 28.2.2 Any dispute arising out of or in connection with this Contract after it comes into force, which cannot be resolved amicably between the parties, shall be referred to arbitration in accordance with Rule 49 of the Public Procurement Rules, 2004 and the applicable laws of Pakistan. 28.2.3 The arbitration shall be conducted by a sole arbitrator mutually appointed by the parties. In the event of failure to agree on the appointment, the arbitrator shall be appointed in accordance with the applicable provisions of the Arbitration Act, 1940. 28.2.4 The place of arbitration shall be Karachi, Pakistan. The arbitration shall be conducted in English.

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	28.2.5 This provision shall prevail over any inconsistent provision concerning arbitration contained in the document.
33	Add the following clauses Attendance of Meetings The Contractor shall attend all meetings along with his authorized representative(s) when called by the Employer to discuss the quality and progress of Works, site matters and other matters related to the Contract, without any compensation from the Employer
34	Progress Report Monthly progress report in the format acceptable to the Engineer shall be prepared by the contractor and submitted to the Engineer in soft/hard format
35	Lighting Work at Night In the event of work being carried out at night, the Contractor shall, at his own cost, provide and maintain such good and sufficient light as will enable the work to proceed satisfactorily and without danger. The approaches to the Site and the Works where the night-work is being carried out shall be sufficiently lighted. All arrangements adopted for such lighting shall be to the satisfaction of the Engineer's Representative.
36	Right of way and Facilities/utilities The Contractor shall bear all costs and charges for special and or temporary right of way which may be required for the purposes of the work including those for access to the site The Contractor shall also obtain/arrange at the contractor's risk and cost such facilities /utilities which may be required for the purpose of the work including but not limited to material storage areas, labor accommodation, site offices, temporary access roads, water, electricity etc. All costs, charges, rents, licenses, permissions, utilities and other expenses associated with such facilities shall be borne/obtained by the Contractor.
37	Health, Safety, Environment and Security (HSE&S) 37.1 General Compliance The Contractor shall, at its own cost and responsibility, comply with all applicable laws, regulations, codes, standards and statutory requirements relating to Health, Safety, Environment and Security (HSE&S), together with the Employer's applicable HSE&S policies, procedures, rules and instructions communicated to the Contractor from time to time. The Contractor shall take all necessary measures to protect the health, safety and security of its personnel and all other persons affected by the Works and to prevent damage to property and adverse environmental impacts arising from the performance of the Works. All costs associated with compliance with the requirements of this Clause shall be deemed included in the Contract Price, and no separate or additional payment shall be made to the Contractor on this account. 37.2 HSE&S Management, Inspection and Compliance The Employer/Engineer may, at any time, conduct inspections, audits and assessments of the Contractor's compliance with HSE&S requirements. The Contractor shall provide all necessary access, information, records and

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cooperation and shall promptly rectify any deficiency or non-compliance identified by the Employer/Engineer.

The Contractor shall immediately comply with any reasonable instruction issued by the Employer/Engineer for the prevention or rectification of any unsafe condition, security risk or environmental hazard.

The Employer/Engineer may direct the Contractor to suspend any activity that is being performed in an unsafe manner or in violation of applicable HSE&S requirements. The Contractor shall not be entitled to any extension of time, additional payment or compensation arising from any suspension caused by its failure to comply with its HSE&S obligations.

37.3 Personnel Competence and Training

The Contractor shall ensure that all personnel engaged in the Works are adequately trained, competent and qualified to perform their assigned duties safely.

The Contractor shall ensure that its personnel understand the risks and hazards associated with their work and shall maintain appropriate records of HSE&S training, inductions and certifications. Such records shall be provided to the Employer/Engineer upon request.

37.4 Safety, Security and Incident Reporting

The Contractor shall perform the Works in a manner that does not compromise the safety, health, security or environmental standards applicable to the Site.

The Contractor shall promptly notify the Employer/Engineer of:

- a. any accident, injury or dangerous occurrence;
- b. any actual or potential security threat relating to the Site or the Works;
- c. any environmental incident, pollution or release of hazardous substances; and
- d. any other condition that may materially affect the health, safety, security or environment of the Site.

The Contractor shall provide all relevant information regarding its working practices, materials, plant and equipment as may reasonably be required by the Employer/Engineer.

37.5 Environmental Protection

The Contractor shall take all reasonable and necessary measures to prevent pollution and minimize adverse environmental impacts arising from the Works, including, where applicable:

- a. control of air emissions, dust and other pollutants;
- b. control of noise and vibration;
- c. recovery, reuse and recycling of usable materials;
- d. proper handling, storage, transportation and disposal of hazardous materials and chemicals;

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- e. preservation of natural land, vegetation and other environmental resources to the extent practicable; and
- f. protection of archaeological, historical or culturally significant sites encountered during the execution of the Works.

37.6 Personal Protective Equipment (PPE)

The Contractor shall, at its own cost and responsibility, provide, maintain and ensure the mandatory use of appropriate Personal Protective Equipment (PPE) by all persons engaged in or affected by the Works under its control, including its employees, subcontractors and labourers.

The Contractor shall ensure that PPE appropriate to the nature of the work and associated hazards is used at all times. Such PPE shall include, where applicable:

- a. safety helmets;
- b. safety shoes or boots;
- c. high-visibility jackets or vests;
- d. safety gloves;
- e. safety goggles or face shields;
- f. hearing protection;
- g. respiratory protection; and
- h. safety harnesses and fall-arrest systems for work at height.

All PPE shall comply with applicable standards and shall be properly maintained, regularly inspected and promptly replaced when damaged, defective or unsuitable for use.

No person shall be permitted to undertake any activity without the PPE required for that activity.

37.7 First Aid

The Contractor shall, at its own cost, provide and maintain an adequately stocked first-aid box at the Site and ensure that appropriate arrangements are available for providing prompt first aid and medical assistance in the event of injury or emergency.

The first-aid box shall be readily accessible, and the Contractor shall ensure that at least one suitably trained person is available to provide first aid during the execution of the Works.

37.8 Non-Compliance and Remedies

Any failure by the Contractor to comply with the requirements of this Clause shall constitute a breach of its contractual obligations.

Without prejudice to any other rights or remedies available under the Contract or applicable law, the Employer/Engineer may:

- a. require the Contractor to immediately rectify any non-compliance;
- b. stop or suspend any unsafe activity until the required corrective measures have been satisfactorily implemented;

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	c. arrange for the required safety measures or Personal Protective Equipment (PPE) to be provided where the Contractor fails to do so, and recover the full cost thereof from the Contractor by deduction from any amounts due or becoming due to the Contractor; d. where any person under the Contractor's control is found undertaking work without the required PPE, impose a penalty of Rs. 5,000/- (Rupees Five Thousand only) per person per occurrence; e. in case of repeated or serious PPE violations, including failure to use mandatory fall-protection equipment while working at height, impose a penalty of Rs. 10,000/- (Rupees Ten Thousand only) per person per occurrence, in addition to stopping or suspending the relevant activity until the required corrective measures have been implemented; and/or f. in the event of serious, material or repeated non-compliance with HSE&S requirements, take such further action, including suspension or termination of the Contract. Any delay, loss, damage, cost or consequence arising from the Contractor's failure to comply with its HSE&S obligations shall be borne solely by the Contractor. All penalties and recoverable costs under this Clause may be deducted from the Contractor's bills or from any other amounts due or becoming due to the Contractor.
38	Site Clearance Upon completion of the Works, or at such earlier stage as may be directed by the Employer, the Contractor shall remove from the Site all temporary structures, camps, dumpsites, debris, surplus materials, waste and other obstructions arising from the execution of the Works, and shall fill and properly reinstate all trenches, excavations and other areas disturbed by the Contractor. The Contractor shall obtain a Site Clearance Certificate from the Employer and submit the same with each Running Bill or the Final Payment Application, as the case may be. In the event that the Contractor fails or refuses to carry out the required site clearance within the period specified by the Employer, the Employer may, without prejudice to any other rights or remedies available under the Contract, undertake or arrange for such work to be carried out through other sources or contractors. The Employer shall be entitled to recover all costs and expenses so incurred from any amount due or becoming due to the Contractor, including retention money and/or any other sums payable under the Contract. If such amounts are insufficient, the Employer may recover the balance from the Performance Security, subject to the terms and conditions of the Contract. In this regard, the decision of the Director/Head of Engineering shall be final and conclusive, subject to the dispute resolution provisions of the Contract

BILL OF QUANTITIES

1. The Bill of Quantities shall be read in conjunction with the Conditions of Contract, Specifications and Drawings.
2. The quantities given in the Bill of Quantities are estimated and provisional, and are given to provide a common basis for bidding. The basis of payment will be the actual quantities of work executed and measured by the Contractor and verified by the Engineer.
3. The bid amount shall, except insofar as it is otherwise provided under the Contract include all costs of Contractor's labour, supervision, materials, execution, insurance, profit, taxes and duties, together with all general risks, liabilities and obligations set out or implied in the Contract. Furthermore, all duties, taxes, GST, PST/SST and other levies payable by the Contractor under the Contract, or for any other cause, as on the date 28 days prior to deadline for submission of Bids, shall be included in the rates and prices and the total Bid Price submitted by the Bidder.
4. Prices quoted by the Supplier will remain fixed during the contact period. Except otherwise mentioned in the Contract, the Supplier shall NOT be entitled for price adjustment on account of any change in wages, material prices, POL prices, tax rates etc. whatsoever on any other account.
5. The Supplier hereby irrevocably and unconditionally waives all claims, direct, indirect or consequential arising out of price escalations, decrease or increase in quantities etc. in connection with the Agreement.
6. General directions and description of work and materials are not necessarily repeated nor summarized in the Bill of Quantities. References to the relevant sections of the Bidding Documents shall be made before entering prices against each item in the priced Bill of Quantities.
7. The brand names have been provided in order to establish a standard of performance and reliability. However, it does not indicate a preference for a particular brand. The bidder may propose other brands which can be accepted subject to necessary tests to establish equivalency.
8. The total quoted Bid Price shall be apportioned among the individual BOQ items in proportion to their respective employer's / engineer's estimated unit rates, and the corresponding individual unit rates shall be determined accordingly, without altering the total quoted Bid Price.
9. The Bidder shall be deemed to have accepted such determination and adjustment of the individual unit rates. In the event that the Bidder fails or refuses to accept the aforesaid determination or adjustment, the Bid shall be treated as non-responsive, and the Bid Security shall be forfeited.

BILL OF QUANTITIES

Item No.	Description	Unit	Qty
1	Supply and installation of Fire Pump Set and controller including testing and commissioning as per drawings and specifications		
a	Electric motor driven vertical turbine fire pump-motor set (UL Listed/FM approved) 1250 GPM @ 14 Bar	Set	1
b	Diesel engine driven vertical turbine fire pump and engine (UL Listed/FM approved) 1250 GPM @ 14 Bar	Set	1
c	Electric motor driven vertical turbine jockey pump-motor set (UL Listed/FM approved) 25 GPM @ 14 Bar	Set	1
d	Fire pumps controller, Fire pump controller shall be integrated with centralized monitoring software through SNMP/Modbus etc. or least have remote monitoring capability installed in control room.	Set	1
2	Supply and installation of MCC including power cables, materials and accessories complete in all respect	No.	1
3	Supply of 400 Amp T.P Adjustable MCCB 25kA, Make: Schneider/GE/ABB or approved equivalent, manufactured on IEC Standard, with all the mounting material, complete in all respect as directed by Engineer In-Charge	No.	2
4	Installation, Testing & Commissioning of 400 Amp T.P Adjustable MCCB 25kA mentioned above at serial # 1, one breaker in existing panel in Substation and second in Pump room along with Copper Busbar links for connecting existing cables, The cost includes the safe de-termination of existing cables, Dismantling of old circuit breakers, installation of new circuit breakers with termination including lugs, Busbar links, Glands, Distribution box for breaker in Pump room, as per specifications and complying with standards, complete in all respect as directed by Engineer In-Charge	No.	2
5	Supply of 4 x 1-C 240 sq.mm Al/XLPE/PVC + 1-C 90 sq.mm Al/XLPE/PVC cables. The job includes cost of all accessories including cost of lugs (metallic, bi metallic, copper etc.), glands, lugs punching, cable ties, nut bolts, taping etc. complete in all respects as directed by Engineer in charge	RM	130
6	Laying, end to end termination, testing and commissioning of 4 x 1-C 240 sq.mm Al/XLPE/PVC + 1-C 90 sq.mm Al/XLPE/PVC power Cables in 3 ft./ 1 Mtr underground, Schedule-40 UPVC pipe as per site requirements including minimum 03 mainholes (suitable size), all fitting, cutting, chipping, perfect back filling and repairing of flooring as per original. Complete in all respect as per standards and as directed by Engineer In-charge.	RM	100
7	Supply, Installation, testing and commissioning of LED Battern type light, Length: 4 feet, 18-22 Watts, 6500K (Make: Fast or approved equivalent) complete in all respects as per direction of Engineer In-charge.	Nos.	6
8	Supply, Installation, testing and commissioning of Multi socket/international socket 13A of make Clipsal, Legrand or equivalent along with back box complete in all respects as directed by Engineer in charge	Nos.	4
9	Supply, Installation, testing and commissioning of 6 Gang switch sheet of make Clipsal, Legrand or equivalent along with back box complete in all respects as directed by Engineer in charge	No	1
10	Supply, laying, testing & commissioning of circuit wiring for light points mentioned at serial # 5 by using PVC/Cu wire (2x1.5 sqmm + 1.5 sqmm as ECC). The cost includes laying, and installation of wires in PVC conduit pipe with all necessary installation material and dressing as per site condition, complete in all respects as per standard industry practice to the satisfaction of the Engineer In-Charge.	Jobs	6

BILL OF QUANTITIES

Item No.	Description	Unit	Qty
11	Supply, laying, testing & commissioning of circuit wiring for power points mentioned at serial # 6 by using PVC/Cu wire (2x2.5 sqmm + 1.5 sqmm as ECC). The cost includes laying, and installation of wires in PVC conduit pipe with all necessary installation material and dressing as per site condition, complete in all respects as per standard industry practice to the satisfaction of the Engineer In-Charge.	Jobs	4
12	Supply, installation, termination, testing and commissioning of wall mounted breaker box including: 01 Nos. 32 Amp MCB (Incoming), 04 Nos. of 6 amp MCB(Outgoing), 04 Nos. 20 amp MCB(Outgoing). Cost includes all the mounting and installation material and complete in all respects as per site requirement.	Job	1
13	Supply, Installation, termination, testing & commissioning of metallic body exhaust fan (size: 12 Inches) make: Pak Fans or approved equivalent. Complete in all respects and as directed by Engineer In-charge.	No.	1
14	Supply, Installation, termination, testing & commissioning of wall mounted bracket fan (size: 18 Inches) make: Pak Fans or approved equivalent. Complete in all respects and as directed by Engineer In-charge.	Nos.	4
15	Supply, laying/pulling & fluke testing of CAT-6 UTP voice cable in appropriate size of PVC Pipe/PVC Channel, including labeling at both ends. Make Vivanco, Clipsal, Systemax, or approved equivalent, Insulation Solid PVC color coded in accordance with telephone industry standards, The cost includes supply, laying, and installation of cables, PVC conduit/channel with all necessary installation material, chiseling, running on walls and make good as per the original condition, complete in all respects as per standard industry practice to the satisfaction of the Engineer In-charge.	Mtr	80
16	Supply, installation & termination of dual telephone face plates with RJ-11 socket outlets. Make: Clipsal / Systemax, or approved equivalent complete in all respects as per site requirements and as directed by the Engineer In-charge.	No.	1
17	Making and finishing of RCC foundation pads for fire pumps along with vibration isolators, rubber pads and accessories etc., complete in all respect	Job	1
18	FIRE PUMP ACCESSORIES Supply and installation of OS&Y Gate Valves Class 125 Flanged, 200 PSI working UL/FM Approved, W/Tamper Switch,UL/FM Approved		
a	10" Dia	Nos.	2
b	2" Dia	No.	1
c	6" Dia	Nos.	4
19	Supply and installation of Check Valve Class 125 Flanged, 300 PSI working UL/FM Approved		
a	10" Dia	Nos.	2
b	2" Dia	No.	1
20	Supply and installation of 4"X2" way breeching inlet (with matching flanges) Breeching inlet cabinet: single door, surface mounted, complete stainless steel, brush finish W/Wired glass door check valve 4" class 125 flanged, 175 psi working UL/FM approved.	Set	1
21	Supply and installation of Pressure gauge 0-250 psi, 3-1/2" dia, 1/4" BSPT end connection, UL listed, with isolation valve (1/4" ball valve) double female threaded.	Nos.	13
22	Supply and installation of "Automatic air release valve 1" FNPT inlet & outlet , 175 psi water, UL/FM.	Nos.	13
23	Supply and installation of OS&Y gate valve 2" Bronze, 175 psi, UL listed.	Nos.	8

BILL OF QUANTITIES

Item No.	Description	Unit	Qty
24	Supply and installation OS&Y Gate Valve 6" Class 125 Flanged, 200 PSI working UL/FM Approved, w/ Tamper Switch, UL/FM Approved	Nos.	6
25	FIRE FIGHTING SYSTEM: Supply and installation of Double vertical fire hose reel cabinet, surface mounted type, complete stainless steel, solid door, brush finish. Upper Compartment: 1"X30 mtr fire hose reel, cabinet mounted, swinging automatic type, kitemark approved with 1" jet spray nozzle, 1" lock shield valve, LPCB approved, 1" pressure reducing valve. Lower Compartment: Fire hose reel single jacket 2.5"X30 mtrs. UL listed with AL coupling; Branch pipe 2-1/2" kitemark approved; Pressure reducing angle 2-1/2" with built in PRV-UL/FM; CO2 extinguisher portable type, capacity:5kg, red color, fire rating:55B, CE/KITEMARK/MED/LPCB Approved; Dry powder extinguisher, portable pressure type, capacity: 6kg, fire rating 24A & 113B, CE/Kitemark/MED/LPCB approved.	Nos.	120
26	Supply and installation of CO2 extinguisher, single cylinder, trolley type, capacity:30kg, fire rating: 113B, red color, BSI kitemark approved	Nos.	3
27	Supply and installation of water fire extinguisher, portable pressure type, capacity: 9 ltr, fire rating 21A, CE/kitemark/MED/LPCB approved	Nos.	20
28	Supply and installation of CO2 extinguisher portable type, capacity:2kg, red color, fire rating:34B, CE/KITEMARK/MED/LPCB Approved	Nos.	55
29	Supply and installation of CO2 extinguisher portable type, capacity:6kg, red color, fire rating:55B, CE/KITEMARK/MED/LPCB Approved	Nos.	40
30	Supply and installation of Dry powder extinguisher, portable pressure type, capacity: 6kg, fire rating 21A & 113B, CE/kitemark/MED/LPCB approved	Nos.	55
31	Supply and installation of Dry powder extinguisher, mobile pressure type, capacity: 25kg, fire rating: IIB 113B, red color, LPCB/BSI kitemark approved	Nos.	40
32	Supply and installation of CO2 extinguisher, mobile pressure type, capacity:50kg, red color, fire rating:113B, CE/KITEMARK/MED/LPCB Approved. Substation Ground & 1st Floor = 2 Nos., Canteen Building Lift machine room (Medical service Department) LXBXH = 15ftX15ftX16ft = 1 No. Lift machine room (Cargo Canteen Building) 19ftX14ftX16ft = 1 No. Lift machine room (Capsule MD side Canteen Building) 16ftX8ftX8ft = 1 No. Main Building Lift machine room (for executives 6Nos. Of lifts) 29ftX35ftX10ft = 1 No. Lift machine room (Cargo) LXBXH =24ftX24ftX16ft = 1 No.	Nos.	7
33	Supply and installation of Dry powder extinguisher, mobile pressure type, capacity: 50kg, fire rating: IIB 113B, red color, LPCB/BSI kitemark approved. Substation Ground and 1st Floor = 2 Nos.	Nos.	2
34	Supply and installation of CO2 extinguisher portable type, capacity:6kg, red color, fire rating:55B, CE/KITEMARK/MED/LPCB Approved. Substation Ground and 1st Floor = 2 Nos. Control Room Ground Floor = 3 Nos. Gate#01 Entrance = 4 Nos.	Nos.	17

BILL OF QUANTITIES

Item No.	Description	Unit	Qty
	Gate#07 Guard Room = 2 Nos. A/C Plant Room = 6 Nos.		
35	Supply and installation of Dry powder extinguisher, portable pressure type, capacity :6kg, fire rating 21A & 113B,CE/Kitemark/MED/LPCB approved Substation Ground and 1st Floor = 2 Nos. Control Room Ground Floor = 3 Nos. Gate#01 Entrance = 4 Nos. Gate#07 Guard Room = 2 Nos. A/C Plant Room = 6 Nos.	Nos.	17
36	Supply and installation of CO2 extinguisher ,mobile pressure type, capacity:25kg,red color, fire rating:113B,CE/KITEMARK/MED/LPCB Approved Control Room Ground Floor = 1 Nos. Lift Machine Room (Roof) = 8 Nos. Pump Room = 1 No. Roof UPS room LXBXH = 11ftX15ftX12ft = 1 No. Main Building AHU rooms = 10 Nos. LXBXH =11ftX19ftX14ft Main Building (Data Controls) LXBXH =15ftX11ftX10ft = 1 No. Data Center - 9th floor LXBXH =16ftX13ftX10ft = 1 No. Data Center - 8th floor LXBXH =14ftX8ftX10ft = 1 No. Data Center - 7th floor LXBXH =15ftX15ftX13ft = 1 No. Data Center - 6th floor LXBXH =16ftX12ftX10ft = 1 No. Data Center - 5th floor LXBXH =10ftX11ftX10ft = 1 No. Telephone Exchange - Room 1 LXBXH =7ftX9ftX15ft = 1 No. Telephone Exchange - Room 2 LXBXH =15ftX16ftX10ft = 1 No. Data Center - 1st floor LXBXH =17ftX10ftX8ft = 1 No. Bolton Market Building = 15 No.	Nos.	45
37	Supply and installation of Dry powder extinguisher, mobile pressure type, capacity: 25kg, fire rating: 113B, red color, LPCB/BSI kitemark approved. Control Room Ground Floor = 1 Nos. Lift Machine Room (Roof) = 8 Nos. Pump Room = 1 No. Bolton Market Building = 15 No.	No.	25
38	Supply and installation of Staircase Pressurization fan 8000 cfm 3" ESP with fan motor and controls, complete in all respects	Nos.	2
39	A/C Plant Room		
a	Supply and installation of 50 Gallons Capacity Vertical Bladder Tank complete with all accessories and trims. Tank Shell: Carbon Steel, SA 516 Gr. 70. Bladder: Vinyl Based Polymer. Pressure Rating: 175 PSI UL Listed with AFFF Foam concentrate, 3%, 250 litres of Foam (20% for T & C) UL Listed	Set	1

BILL OF QUANTITIES

Item No.	Description	Unit	Qty
b	Supply and installation of Fire Blanket 1.2X1.2 meter (4ftX4ft) glass fiber with twill weaving yarn, silicon coated in PVC flat hard container red, LPCB approved	Nos.	46
c	Supply and installation of Pressure gauge 0-250 psi, 3-1/2" dia, 1/4" BSPT end connection, UL listed, model:SD-P-shield with isolation valve (1/4" ball valve) double female threaded.	Nos.	10
d	Supply and installation of "automatic air release valve 1" FNPT inlet & outlet, 175 psig water	Nos.	10
e	Supply and installation of OS&Y gate valve 6" class 125 flanged, 200psi working UL/FM approved, D-shield W/tamper switch, UL/FM approved	Nos.	6
f	Supply and installation of OS&Y gate valve 4" class 125 flanged, 200psi working UL/FM approved, Shield W/tamper switch UL/FM approved	Nos.	6
40	Hydrant System Supply and installation of Wet type fire hydrant 4" (barrel type) with 2X2 1/2" control outlet valve and 6" HDPE pipe - 15Rft with all accessories, materials and etc, including the civil works, complete in all respects as per drawings and specifications	Nos.	12
41	Supply and installation of 4" lever operated butterfly valve, wafer type	Nos.	12
42	Supply and installation of Gate valve, non rising stem, EPDM coated resilient wedge with handle, flanged ends to ANSI B 16.1, class 125, WP 200psi, red RAL3000, UL/FM approved shield with post indicator, suitable for 3"-12" vertical type, UL/FM approved - 8" Dia		
a	8" Dia	Nos.	11
b	6" Dia	Nos.	3
43	Supply and installation of Straight through landing valve 2-1/2" brass painted red RAL3000, inlet flanged outlet BS female with PVC cap and chain	Nos.	2
44	Supply and installation of Self standing type fire hydrant cabinet, solid door, complete mild steel, C/W: 2 Nos.-synthetic single jacket fire hose 2.5"x 30 mtr. UL listed with AL Coupling; 2 Nos-2-1/2" Branch pipe-Kite mark approved, NF SB 202A 1 Nos-hydrant wrench 1 Nos-spanner	Set	12
45	Supply and installation of "COMBINATION AIR RELEASE VALVE DN80, ductile iron body, PN-16, with OS &Y gate valve 3", rising stem with slot, bronze trim, flanged ends class 125 to ANSI B16.1, WP 175 Psi.	Nos.	12
46	Fire Alarm System Supply and installation of Addressable Multi detector (smoke + heat) with base, UL LISTED	Nos.	900
47	Supply and installation of fire resistant cables (Prysmian FP 200 Gold/BS-7629-1 or equivalent) and conduiting of Addressable multi detector complete in all respects as per specifications.	Rft	36,300
48	Supply and installation of Addressable dual action manual pull station with wiring and conduiting, UL Listed	Nos.	61
49	Supply and installation of Addressable indoor/open area sounder red (weather proof), UL listed	Nos.	74
50	Supply and installation of wiring and conduiting of Addressable open area sounder red complete in all respects as per specifications, UL listed	Rft	9,300
51	Supply and installation of Addressable multi loop control panel, UL/FM approved	Nos.	6
52 a	Supply and installation of NETWORK CARD with third party BMS integration	Nos.	6
52b	Supply and installation of centralized monitoring and management software installed on dedicated server/PC having large display. Integrated with fire detection and suppression system as mentioned in specifications	Nos.	1

BILL OF QUANTITIES

Item No.	Description	Unit	Qty
53	Supply and installation of ADDRESSABLE SOUNDER CIRCUIT MODULE, UL LISTED	Nos.	30
54	Supply and installation of Addressable switch monitor input/output module UL LISTED	Nos.	25
55	Supply and installation of Addressable DUAL priority switch monitor module UL LISTED	Nos.	53
56	Supply and installation of POWER SUPPLY UNIT, 3A 24VDC	Nos.	7
57	Supply and installation of Addressable isolator, UL LISTED	Nos.	92
58	Emergency & Exit Light Supply and installation of Non maintained, surface mounted emergency luminaire, 3 hrs duration with wiring and conduiting as per specifications	Nos.	30
59	Supply and installation of Non maintained recessed mounted emergency luminaire, 3 hrs duration with wiring and conduiting as per specifications	Nos.	1,200
60	Supply and installation of addressable Non maintained EXIT box single sided, surface mounted emergency luminaire with wiring and conduiting as per specifications	Nos.	4
61	Supply and installation of Non maintained exit light, single sided, hanging type complete with wiring and conduiting as per specifications	Nos.	150
62	Supply and installation of Addressable Non maintained exit light, DOUBLE sided, system relevant, hanging type emergency luminaire complete with wiring and conduiting as per specifications	Nos.	325
63	Supply and installation of Non emergency lighting management systems monitoring panel	Nos.	17
64	Supply and installation of Digital addressable line interference power supply	Nos.	36
65	Foam System GENSet Room-1 Supply and installation of "50 Gallons Capacity Vertical Bladder Tank complete with all accessories and trims. Tank Shell: Carbon Steel, SA 516 Gr. 70. Bladder: Vinyl Based Polymer. Pressure Rating:175 PSI. UL Listed with AFFF Foam concentrate, 3%, 250 litres of Foam (20% for T & C) UL Listed	Set	1
66	Vertical 2" DELUGE VALVE ASSEMBLY C/W Supply and installation of Deluge Valve Flanged, 90 Deg Pattern Vertical Mounted, Pressure Rating: 175 PSI, Solenoid Valve, 24 V DC, Water Motor Alarm Gong, INT 3/4" BSPT, Epoxy Red powder coated, Pressure Switch: SD-P1 with Isolation Valve & Test and Drain Valve 1" UL Listed / FM Approved	No.	1
67	Supply and installation of 2" Bronze Butterfly Valve, Grooved, 175 PSI UL/FM Approved Part with Tamper switch, UL/FM Approved	No.	1
68	Supply and installation of 1-1/2" dia SS ball valve	Set	1
69	Supply and installation of 1-1/2" dia SS Non- return valve	Set	2
70	Supply and installation of 1-1/2" dia hydraulic operated ball valve	Set	2
71	Supply and installation of Foam Sprinkler Head - Open and Air Aspirating Type, K-Factor 42	Nos.	8
72	Supply and installation of Automatic Air Release Valve 1" FNPT inlet & outlet, 175 PSIG water, UL/FM Approved	Set	1
73	Supply and installation of Extinguishing Control Panel AX-T With 2 Detection Zone, 1 Release Area, C/W 24 V DC Back-Up Battery, CRM 4 Relay Module	Set	1
74	Supply and installation of Combo Detector W / Base, UL Listed as per specifications	Nos.	4
75	Supply and installation of Abort Switch UL Listed.	No.	1
76	Supply and installation of Bell weather proof, 6" UL Listed	No.	1
77	Supply and installation of Strobe Horn W/Back Box weather proof, UL listed	No.	1

BILL OF QUANTITIES

Item No.	Description	Unit	Qty
78	Supply and installation of Xenon/Strobe light W/Back Box weather proof, UL Listed	No.	1
79	Supply and installation of Warning Sign (Set of 3)	No.	1
80	FOAM SYSTEM GENSet ROOM-2 Supply and installation of "50 Gallons Capacity Vertical Bladder Tank complete with all accessories and trims. Tank Shell: Carbon Steel, SA 516 Gr. 70. Bladder: Vinyl Based Polymer. Pressure Rating: 175 PSI UL Listed with AFFF Foam concentrate, 3%, 250 litres of Foam (20% for T & C) UL Listed	Set	1
81	Supply and installation of Vertical 2" Deluge Valve Assembly C/W Deluge Valve Flanged, 90 Deg Pattern Vertical Mounted, Pressure Rating: 175 PSI. Solenoid Valve, 24 V DC, Water Motor Alarm Gong, INT 3/4" BSPT, Epoxy Red Coated, 2 Nos-Pressure Gauge Connection 1/4 BSPT with Isolation Valve & Test and Drain Valve 1" UL Listed / FM Approved	No.	1
82	Supply and installation of 2" Bronze Butterfly Valve, Grooved, 175 PSI UL/FM Approved Part, Tamper switch Potter, UL/FM Approved	Set	2
83	Supply and installation of 1-1/2" mm dia SS ball valve	Set	1
84	Supply and installation of 1-1/2" mm dia SS Non- return valve	Set	2
85	Supply and installation of 1-1/2" mm dia hydraulic operated ball valve	Set	2
86	Supply and installation of Foam Sprinkler Head - Open and Air Aspirating Type, K-Factor 42	Nos.	8
87	Supply and installation of "Automatic Air Release Valve 1" FNPT inlet & outlet, 175 PSIG water	Set	1
88	Supply and installation of Extinguishing Control Panel AX-T With 2 Detection Zone, 1 Release Area, C/W 24 V DC Back-Up Battery, CRM 4 Relay	Set	1
89	Supply and installation of Combo Detector W / Base, UL Listed as per specifications	Nos.	4
90	Supply and installation of Abort Switch UL Listed	No.	1
91	Supply and installation of Bell weather proof, 6" UL Listed	No.	1
92	Supply and installation of Strobe Horn weather proof W/Back Box, UL listed	No.	1
93	Supply and installation of Xenon/Strobe light W/Back Box, UL Listed	No.	1
94	Supply and installation of Warning Sign weather proof (Set of 3)	No.	1
95	Supply and installation of Schedule 40 MS piping with fittings and specials for fire suppression system, including all cutting, fixing and laying, complete in all respect		
a	10 inch dia	Rft	100
b	8 inch dia	Rft	170
c	6 inch dia	Rft	540
d	4 inch dia	Rft	3,630
e	2-1/2 inch dia	Rft	5,810
96	Supply and installation of HDPE PE 4710 pipe with fittings and specials for external fire suppression system, including all cutting, fixing and laying, complete in all respect		
a	10 inch dia	Rft	3,800
97	Main AC Plant Room		
a	Supply and installation of Fire Fighting Works , including all civil works required to complete the Job, complete in all respects	Job	1
98	Making of a sump for fire pumps, inner size 4'x4' and 12' deep, with 6" thick RCC walls all around and 12" thick RCC base (all RCC not leaner than 1:2:4 with 1/2" dia. steel bars @ 6" c/c bothways in double layer), including the cost of required steel reinforcement, formwork, required excavation in any type of soil and removal of concrete pavers, dismantling of existing PCC flooring/sub-flooring, disposal of debris etc, making of connection with	Job	1

BILL OF QUANTITIES

Item No.	Description	Unit	Qty
	existing UG water tank with about 10 Rft long two HDPE pipes of 12" dia. and making of hole in wall of of existing tank and sealing the same after fixing of pipe, complete in all respects		
99	Making of pump room of about 10ft x 12ft inner size and 10ft ceiling height with 6" thick block masonry set in 1:4 CS mortar on inner and outer sides, 6" high PCC (1:2:4) parapet, one Aluminium sliding window (1.6mm thick) of 4'x7' size of approved design, girder-slab type pre-cast roofing of approved make, including water proofing with 4mm thick membrane (Hy-Grip or approved equivalent) and average 2-1/2" thick cement screed(1:2:4), laid according to proper slopes etc, one 4" dia rainwater pipe with shoe bend, with smooth travel finished cement concrete flooring 6" high from outside level and any other works required, complete in all respects	Job	1
100	Cutting of core in existing RCC slabs/Projections of any thickness with the help of core cutter, including refinishing of holes after laying of pipes		
a	6" dia	Nos.	32
b	4" dia	Nos.	90
101	Cutting of core in existing masonry wall of any thickness with the help of core cutter, including refinishing of holes after laying of pipes		
a	6" dia	Nos.	10
b	4" dia	Nos.	92
c	3" dia	Nos.	30
d	2 1/2" dia	Nos.	45
102	Safe sound dismantling of existing false ceiling/ceiling and reinstalling after execution of works	Job	1
103	Providing & fixing in position false ceiling comprising of 600 mm x 600mm size 7.5mm thick Gypsum sheets of DFB or other approved equivalent make with PVC film at front side and metallic foil pasted at back side, with approved quality powder coated suspension system of CKM or other approved equivalent make, with 14SWG GI wires at every junction of the main Tee runner and lateral Tee sections, hanging with the RCC slab with the help of approved quality rawal bolts/fixing anchors (3" long) provided @ approximately 4ft both ways & long GI I-hooks with approved quality PVC plug provided @ approximately 4ft both ways, including cutting of sheets, etc. where required, making of holes for fixtures, complete in all respects as per site requirements and as directed by the Engineer In-Charge.	Sft	43,000
104	Dismantling of existing concrete pavers including stacking at designated location and refixing the same after laying of pipes etc	Sft	1,500
105	Dismantling of existing concrete flooring/Sub-flooring of any thickness including disposal of debris away from the bank premises	Cu.ft	11,000
106	Excavation and backfilling of any type of soil including stone sub base and backfilling the suitable fine excavated materials in trenches after laying of pipes and compacting the same including disposal of debris away from the bank premises complete in all respects	Cu.ft	136,800
107	Providing, mixing, laying, vibrating and curing of machine mixed PCC 1:2:4 in flooring, complete in all respects	Cu.ft	11,250
108	Same as above but in 1:4:8 ratio, 4" thick layer	Cu.ft	5,000
109	Making of Shop Drawings with sectional details complete in all respect for complete Fire suppression Systems as per specifications	Job	1
110	Making of As Built drawings with sectional details complete in all respect for complete Fire suppression Systems as per Specifications	Job	1

BILL OF QUANTITIES

Item No.	Description	Unit	Qty
111	Operation and Maintenance of the Complete System including Provisioning of One no. Technician and One no. helper, who will be responsible for the Operation & Maintenance of Complete Firefighting System	Months	12
112	4 years service level agreement (SLA) with parts after completion of 12 months operation and maintenance. Vendor shall include cost of consumable parts for 48 months. Contractor shall submit the maintenance schedule and will submit the monthly inspection/fitness/maintenance report to the Client.	Months	48



SCHEDULE OF REQUIREMENT

Completion / Delivery Time: 730 Days

Bidder shall provide a program in a bar-chart showing the sequence of items by which the bidder proposes to complete the Works of the entire Contract. The program should indicate the sequence of work items and the period of time during which he proposes to complete the Works including the activities like designing, schedule of submittal of drawings, ordering and procurement of materials, manufacturing, delivering, construction of civil works, erection, testing and commissioning of Works to be supplied under the Contract. The proposed sequence of execution of various sections of the project shall be according to the given Key Performance Indicators (KPIs) as following:

Work Schedule

The below Key Performance Indicators should be taken into consideration by the bidder in order to execute the works.

If the KPI is not acceptable to any bidder, the bidder will prepare its own Work Program along with detailed justification for its change. Bids without above given Work Program or bidder's own proposed Work Program without justification shall not be considered responsive.

UPGRADATION OF FIREFIGHTING SYSTEM AT SBP BUILDINGS HEAD OFFICE KARACHI

S.No	Duration (weeks)	Start Week	End Week	Activity	month 1	month 2	month 3	month 4	month 5	month 6	month 7	month 8	month 9	month 10	month 11	month 12	month 13	month 14	month 15	month 16	month 17	month 18	month 19	month 20	month 21	month 22	month 23	month 24
1	2	1	2	Site Mobilization																								
2	4	1	4	Technical Submittal																								
3	6	1	6	Shop Drawings																								
4	4	5	8	Review and approval																								
5	4	9	12	Placement of order																								
6				Addressable fire alarm																								
6.1	12	13	24	Site works																								
6.2	4	25	28	Equipment delivery																								
6.3	44	29	72	Equipment installation																								
6.4	4	73	76	commissioning of system																								
6.5	4	77	80	One month test run																								
7				Fire fighting System																								
7.1	28	13	40	Site works																								
7.2	4	37	40	Equipment delivery																								
7.3	48	41	88	Equipment installation																								
7.4	5	57	61	commissioning of system																								
7.5	4	62	65	One month test run																								
8				Pump Room																								
8.1	16	21	36	Site works																								
8.2	4	37	40	Equipment delivery																								
8.3	24	41	64	Equipment installation																								
8.4	4	89	92	commissioning of system																								
8.5	4	93	96	One month test run																								
9				Emergency lighting																								
9.1	12	13	24	Site works																								
9.2	4	25	28	Equipment delivery																								
9.3	44	29	72	Equipment installation																								
9.4	4	73	76	commissioning of system																								
9.5	4	77	80	One month test run																								
10	4	62	65	As built Drawings																								
11	4	93	96	Operations Manual																								
12	8	85	92	Training of Clients personal																								
13	4	93	96	Handling over of system																								

Contractor's Representative and Key Personnel

MINIMUM MANDATORY STAFF:

The Contractor shall arrange all requisite resources for timely completion of project as per provisions given in the Bidding Documents.

Following is the list of Minimum Mandatory Staff Requirement & Equipment to be deployed at site by the Contractor as and when required by the Engineer till completion of the Project. In case Contractor fails to deploy any of the following personnel, the Engineer will deduct the amount mentioned at column (E) for the respective personnel. However, the deduction of the amount does not relieve the Contractor to deploy the minimum staff at site and shall not prejudice Engineer / Employer from any other remedy defined in the Contract.

<i>Designation</i>	<i>Nos.</i>	<i>Minimum Qualification</i>	<i>Min. Relevant Working Experience</i>	<i>Minimum Amount to be deducted per month per person in case the personnel is not deployed at site (Rs.)</i>
<i>(A)</i>	<i>(B)</i>	<i>(C)</i>	<i>(D)</i>	<i>(E)</i>
Engineer (Mechanical/Electrical)	01	B.E/ BSC (Registered with PEC)	03 years	250,000/-
Site Supervisors (Mechanical/Electrical)	01	DAE Mechanical/Electrical/ B-Tech	08 years	120,000/-

TECHNICAL SPECIFICATIONS

1. Brand Names

Equipment and materials specified with brand names have been provided to establish a standard of performance and do not necessarily indicate a preference for a particular manufacturer or material.

The names of manufacturers given below are to indicate the level of quality and performance anticipated by the Procuring Agency. Other makes may be accepted provided that the quality and performance of such equipment, in the sole opinion of the Procuring Agency, are at least equal to or better than the equipment offered by the recommended manufacturer listed hereunder.

2. List of Manufacturers Recommended

The acceptance of equipment/ materials offered by these manufacturers will be subject to compliance of offered models/ materials with the specifications, capacity and/ or performance requirements.

All other items not included in the list hereunder will be approved by the Procuring Agency in accordance to their compliance with the specifications, capacity and/ or performance requirements.

Onus lies with the Contractor / supplier for establishing the genuineness of any material/ product.

Following are the specification of material. Bidder can also offer equivalent products, however, adequate documentation in the form of technical data sheets and type test certificates from independent laboratories stating the standards to which the product complies with shall be furnished in the support of their being equivalent. The Employer/ Consultant reserves the right to ask the bidder of additional documentation in the regard.

Item	Recommended Manufacturer
Fire/Smoke Damper	a) Aerovac b) TUF c) ISONE d) Simplex
Seamless / Galvanized Steel Pipes - UL Listed	a) Sumitomo b) Shield c) Nippon
Galvanized threaded fittings (UL listed and or FM Approved).	a) Grinell b) Shield c) Central
Grooved Fittings (UL listed and or FM Approved).	a) Grinell b) Shield c) Central
Hangers & Accessories	a) PHD b) Flamco c) Shield d) Fischer e) Sikla
HDPE pipes & fittings	a) SHIELD b) ISCO c) IPF

Fire Hydrant – UL Listed	a) NAFFCO b) Kennedy c) Mueller d) SRI e) Sffico
Fire Department connection (LPCB, BSI/Kitemark/ UL listed and or FM approved).	a) Naffco b) Shield c) Chubb d) Sffeco
Fire Hose Reel, Fire Hose, Nozzle, Landing Valve, Cabinets and accessories (LPCB, BSI/Kitemark/ UL listed and or FM Approved).	a) Naffco b) Shield c) Chubb d) SRI e) Sffico f) Macron (UK)
Special Valves (Alarm valve, pre-action valve)	a) Automatic Sprinkler Corporation b) Central / Spraysafe c) Shield d) Tyco e) Lifeco

LIST OF STANDARDS

Product	Standards
MS seamless schedule 40 pipes	- Must conform to ANSI B36.10 and ASTM 106
Poly Ethylene (PE) pipe	- Must conform to NFPA 24 - Must conform to ASTM D2239
SS (Stainless Steel) Twin Compartment Hose Reel Cabinet	- Must be CE certified to EN 671-1:2012 - At least 1.2 mm thick galvanize steel sheet - Rated to Maximum working pressure 12 bar - Jetting range ≥ 10 m - Material for Hose Valve & Adapter : Brass
Fire hydrant	- U/L & FM approved - Rated to Maximum working pressure 12 bar - Operating temperature range -10-49 °centigrade - Manufacture to BS-5041 and BS 5154
Valves	- U/L & FM approved - Rated to Maximum working pressure 12 bar - Operating temperature range 0-49 °centigrade - Manufacture to BS-5041 and BS 5154
Fire Extinguishers	- Must comply to NFPA - 10
Sprinklers	- capable of operating up-to 180 degree C - Made of corrosion resistant copper Alloy

Product	Standards
Fire Pump	- UL & FM Approved - Must comply to NFPA-20 - CE certified to EN:1028-1:2002 + A1:2008 & EN:1028-2:2002 + A1:2008 - Single stage, end suction , top discharge - NEMA 2 (IP-54) rated protection - Diesel pump engine NFPA-25 and SAE J 1349 testing requirement
Fire Retardant Cables	- Must conform to BS 7629-1 and BS 5839-1
Electrical Conduits	- As per National Electric Code
Fire detection System	- Heat / smoke detectors must be CE certified to EN-671-1: 2012 - Heat / Smoke Detectors Must conform to NFPA 72 - Manual call point must be reusable (Non glass breaking type) semi-flush or surface mounted with LED indicator - Addressable horn strobe sound to be ≥ 100 dB at 1 m

FIRE SUPPRESSION SYSTEM SPECIFICATIONS

GENERAL REQUIREMENTS FOR FIRE SUPPRESSION SYSTEM

1.1. SCOPE OF WORKS

The work covered by this section consists of providing all material and equipment and performing all the work necessary for the complete execution and completion, including testing and commissioning of all systems of Fire Suppression works as specified in Division 210000 and as shown on the Drawings or as directed by the Engineer.

All the above mentioned systems complete in all respects including connection of the internal systems with the external systems are included in the scope of work.

All the materials and equipment shall be of the specifications mentioned herein and shall be limited to products regularly produced and recommended by the manufacturers for the service intended. Contractor shall submit samples, necessary catalogue, sketches, the name of manufacturer and guarantee, if necessary, before installation for approval by the Engineer. All material and equipment shall be new and unused.

All other equipment and material shall be installed in strict accordance with manufacturer's instructions. These instructions shall be considered as part of specifications.

It is specifically intended and shall be agreed to by the Contractor that any material or labor which is usually furnished as a part of such equipment and which is necessary for its proper completion and best operation shall be furnished as a part of this contract without any additional cost whether or not shown in detail on the drawings or described in detail in the specifications.

The Contractor shall be responsible for his work until its completion and final acceptance and shall replace any of the same which may be damaged, lost or stolen without any additional cost to the Employer.

All openings left in floor for passage of ducts, pipes and cables etc. shall be covered and protected.

All open ends of pipes shall be properly plugged off to prevent any foreign material from entering the pipe during installation.

The Contractor shall allow in his bid for the cost of all cutting, making of holes and subsequently making good to the desired finish as approved by the Engineer. No separate payment shall be made for this item. Also the Contractor shall allow in his bid for the cost of providing protective painting or coating as specified in the relevant sections and no claim shall be entertained for this item.

The arrangement, positions and connections of pipe fittings and appurtenances shall be as shown on the drawings but the Engineer reserves the rights to change the locations. Special precautions shall be taken for the installation of concealed pipes as shown on the drawings and/or as required. Should it be necessary to correct piping so installed, the Contractor shall be held liable for any injury caused to other work in the correction of piping? For any expedient change, the Contractor will correctly show it in shop drawings. A minimum distance between different services shall be maintained as shown in the contract documents or as approved by the Engineer.

All exterior openings provided for the passage of piping and ducting shall be properly sealed with snugly fittings collars of metal of other approved rat-proof material securely fastened in to place.

Joints at the roof, around ducts and pipes, shall be made water-tight by the use of lead, copper, galvanized iron or other approved flashings of flashing material. Exterior wall openings shall be made water-tight.

Piping in ground shall be laid on a firm bed for its entire length. No separate payment shall be made for any earth work required in connection with laying of piping in ground. Piping shall be installed without undue strains and stresses. Vertical piping shall be securely held to keep the pipe in alignment and carry the weight of the pipe and contents. Horizontal piping shall be supported to keep it in alignment and prevent sagging. Hangers and anchors shall be of metal of sufficient strength to maintain their proportional share of pipe alignments and prevent ratting. Hangers and anchors shall be securely attached to the building construction.

The work shall conform in all respects to local Municipality, and any other authorities having jurisdiction.

The systems shall be in accordance with the requirements of the Civil Defense Department and National Fire Protection Association (NFPA).

These specifications are supplementary to the requirements above. All required approvals of shop drawings and materials shall be obtained by the Contractor from the appropriate authorities before proceeding with work on site.

All connections to/from city points of utilities supply/discharge shall be the mechanical contractor's responsibility.

1.2. DEFINITIONS

Reviewed: Reviewed by the Engineer; normally a design or piece of equipment must be shown to have operated successfully for not less than two years under conditions generally similar to those required by this Contract. Facilities shall be given at the tender stage, if required by the Engineer, to enable him to evaluate the equipment's features and performance. The review of a design or piece of equipment shall be without prejudice to the acceptance tests required after installation.

Size: When related to pipe work means nominal size which generally approximates to the bore of steel tubes and the outside diameter of copper tubes.

"Complete installation" shall mean not only the major items of plant and equipment conveyed by this specification, but all the incidental sundry components necessary for the complete execution of the works and for the proper operation of the installation, with their labor charges, whether or not these sundry components are mentioned in detail in the tender documents issued in connection with the contract.

"As indicated", "where indicated", and "unless otherwise indicated", refer to items or requirements which are, or may be given elsewhere in the tender documents issued in connection with the contract, (e.g. on a drawing, in a supplementary specification or in a schedule to this specification).

Provide: Supply, install and test.

Remove: Remove complete with all ancillary devices and equipment. Dispose of off- site in an environmentally safe manner. Ensure that equipment is completely isolated from any energy sources prior to testing.

Removal: All piping and wiring shall be removed back to the nearest isolating point.

1.3. REGULATIONS

The work shall accord strictly with all rules, regulations, By-laws and requirements of all authorities having jurisdiction.

Drawings and specifications should not conflict with the above regulations but where there are apparent discrepancies the Contractor shall notify the Engineer in writing and obtain clarification before proceeding with the Work.

1.4. CODES AND STANDARDS

The installation shall comply with all relevant statutory instruments and regulations and in particular with the following:

1. The IEE regulations for electrical equipment in buildings.
2. Any Regulations under Electricity.
3. Local Health and safety at work regulations.
4. Local Control of Pollution Regulations.
5. ASHRAE standards.
6. ASME standards.
7. British Standard.
8. National Plumbing Code
9. Standards mentioned elsewhere in these documents

The Tender shall be based on regulations and standards current on the date of return of tenders. If these regulations are amended or new regulations are enacted after that date, the Engineer shall be notified immediately.

The materials, equipment and installations detailed in this specification are based on the standards and codes of practice indicated in the Contract Documents. In the event of a contradiction between this specification and any applicable standard or Code of Practice, this specification shall govern and the Engineer shall be notified immediately.

Where material and equipment are specified under a particular standard and supplied under an equivalent standard, a certificate of compliance shall be provided to the Engineer at his request.

1.5. ABBREVIATIONS

Generally, the following abbreviations are used in this Division

a)	ANSI	:	American National Standards Institute
b)	ASHRAE	:	ASHRAE
c)	ASME :		American Society of Mechanical Engineers.
d)	ASTM	:	American Society for Testing and Materials
e)	AWWA	:	American Water Works Association
f)	bhp	:	boiler horsepower or brake horsepower
g)	BS	:	British Standard
h)	Btu/hr	:	British thermal units per hour
i)	BWG	:	British Wire Gauge
j)	CFM	:	cubic feet per minute
k)	CI	:	cast iron
l)	CSA	:	Canadian Standards Association
m)	Dia	:	diameter
n)	EEMAC	:	The Electrical and Electronics Manufacturer's Association of Canada
o)	° F	:	degree Fahrenheit
p)	° K	:	degree Kelvin
q)	ft	:	feet
r)	gpm	:	gallons per minute
s)	hp	:	high pressure or motor horsepower
t)	lbs	:	Pounds
u)	lbs/ft ³	:	pounds per cubic feet
v)	NBS	:	National Bureau of Standards
w)	N.C.	:	Noise criterion as define by graph in ASHRAE.
x)	NFPA	:	National Fire Protection Association
y)	NPSH	:	Net positive Suction Head
z)	No.	:	Number
aa)	O.S. & Y	:	outside screw and yoke
ab)	ph	:	hydrogen ion concentration
ac)	ppm	:	parts per million
ad)	psi	:	pounds per square inch
ae)	rpm	:	revolutions per minute

af)	SMACNA	:	Sheet Metal and Air Conditioning .Contractors National Association
ag)	SS	:	Stainless steel
ah)	UL	:	Underwriters' Laboratories
ai)	ULC	:	Underwriters' Laboratories of Canada
aj)	W	:	Watts
ak)	KW	:	Kilo Watts
al)	w.g.	:	water gauge
am)	w.p.	:	working pressure
an)	Lit	:	Liter
ao)	m3	:	Cubic Meter
ap)	L/s	:	Liters per sec

1.6. DRAWINGS, CHANGES AND INSTALLATION

The drawings shall be considered to show the general character and scope of the work and not the exact details of the installation. The installation shall be complete with all accessories required for a complete and operative installation.

The location, arrangement and connection of equipment and material as shown on the drawings represent a close approximation to the intent and requirements of the Contract. The right is reserved by the Engineer to make reasonable changes required to accommodate conditions arising during the progress of the work, at no extra cost to the Contract.

All piping and duct work in finished areas shall be concealed in ceiling spaces and shafts or chased into walls. No exposed piping or duct work shall be installed in such areas unless specifically accepted by the Engineer. Any pipe or mechanical equipment mounted on roof, or housing for such equipment, shall not be closer to the edge of roof than a distance equal to the height of the pipe, hood or equipment; unless specifically accepted by the Engineer.

The actual location of sensing devices, thermostats, switches, etc., shall be reviewed by the Engineer before installation.

The location and size of existing services shown on the drawings are based on the best available information. The actual location of existing services shall be verified in the field before work is commenced.

Changes and modifications necessary to ensure coordination and to avoid interference and conflicts with other trades, or to accommodate existing conditions, shall be made at no extra cost to the Contract.

The Contractor shall ensure that all plant to be supplied by him can be installed in the available space and that there is adequate access to admit all plant to its position and enable maintenance to be carried. Special care shall be taken in areas where pour-gaps take place. Coordinate the work schedule with the Contractor.

1.7. RECORD DRAWINGS

The Contractor shall clearly mark, as the job progresses, all changes and deviations from that SHOWN On contract drawings on two sets of prints. After inspection and acceptance of service lines in trenches, the Contractor shall take 'as-built' measurements. Definite measurements shall be taken for each service line. Drawings shall be kept up-to-date during construction. On completion of the building, the Contractor shall forward to the Engineer, drawings (quantities as specified elsewhere) indicating all such changes and deviations. This contractor shall include all existing information on the final record drawings.

The cost of furnishing above prints and preparing these record drawings shall be borne by the Contractor and shall be included in the Contract Price. When all revisions showing the work as finally installed are made, the corrected sepia's shall be submitted for review by the Engineer, and delivered to the Consultants before final payment for the completed work will be made.

1.8. SHOP DRAWING

Shop Drawings shall indicate clearly all services and the materials and/or equipment actually being supplied, all details of construction, accurate dimensions, capacity operating characteristics and performance.

Each shop drawings for non-catalogue items shall be prepared specifically for this project. Material submissions and brochures for catalogue items shall be marked clearly to show the Items being supplied.

This contractor shall prepare all shop drawings related to the particular material submittal and shall submit both the shop drawings and the material submittal simultaneously.

Ensure that electrical coordination is complete before submitting drawings for review.

Installation of any equipment shall not be commenced until after the shop drawings have been reviewed by the Engineer.

When requested, shop drawings shall be supplemented by data explaining the theory of operation. The Engineer may also request that this information be added to the maintenance and operating manual.

1.9. DRAWING SUBMISSIONS

Builder's work drawings shall show fully dimensioned, all foundation, bases plinths, sumps, holes and sleeves details required and the overall size and weights of the plant concerned.

With the agreement of the Engineer, smaller holes, built-in fixings, etc., other than in plant rooms, may be marked out on site instead of on drawings.

Fully dimension plant room drawings shall detail for each plant room the location of each unit, pipe routes and connections with valves and fittings, duct routes and connection with accessories, drain connections, electrical connections and controls. All drawings shall include explanatory notes and shall show the required sizes of pipes and ducts with and without insulation as applicable.

If abbreviations are employed for the designation of components, an integral schedule shall be provided on the drawings to explain the meanings of the abbreviations.

Individual equipment drawings from the various manufacturers will not be accepted in lieu of these composite plant room drawings.

1.10. COORDINATION, INSTALLATION, INTERFERENCE AND SETTING DRAWINGS

Coordination, Installation, interference and setting drawings dimensioned and to scale, shall be submitted for the Engineer's review to make clear the work intended and to, show its relation to adjacent work and to the work of other trades. Three copies of such drawings shall be submitted for review, of which one will be retained by the Engineer.

The drawings must be comprehensive showing all details, dimensions, equipment, supports for services and sections through critical areas to properly coordinate all services and work of other trades. Unless accepted by the Engineer, all drawings must be to 1/8" : 1' scales for IP Units or 1:100 scales for SI Units.

Site services drawing shall be prepared to show all existing services, modification to existing services and all new services within the entire construction area. Location of all existing services shall be carefully detailed and dimensioned on the drawings showing all sizes and invert elevations.

Drawings shall be prepared with due regard to the construction schedule and shall be submitted to the Engineer, allowing reasonable time for examination and review.

Work shall not proceed in areas involved until after final approval of all such drawings has been obtained.

These drawings shall include all existing information.

1.11. EXAMINATION OF SITE

Before submitting tenders, each trade shall examine the site to determine the conditions which may affect the proposed work. No claims for extra payment will be considered, because of failure to fulfill this condition.

1.12. CONTRACTOR'S TECHNICAL RESPONSIBILITIES

This Contractor shall provide detailed calculations for flow and head requirements for all Pipes, pumps and etc. These calculations shall be done prior to ordering the equipment and shall be submitted to the Engineers for review prior to ordering such equipment. This Contractor shall bear full responsibility for the final equipment sizing and selection.

The cost of furnishing above calculations shall be borne by the Contractor and shall be included in the Contract Price.

1.13. EXCAVATIONS AND BACK FILLING

1.13.1 Excavations

Excavation for the drains in open trenches shall be to the line and depths indicated on the drawings or as directed. Great care shall be taken to excavate only to such depths as are correctly required for regular gradients. Grips for joint as required shall be formed, trench bottom shall have sufficient width to allow adequate working space for the pipe jointed and should in no case be less than 24 inches or the internal diameter of the pipe plus 18 inches. Trenches are to be kept clear of water including any necessary pumping. Perform all tunneling for pipe when required. Wherever soft places in excavated areas are encountered, the Contractor shall excavate such soil to hard foundation and replace with granular material before any drain are laid. In the event of the excavation being made deeper than necessary by error they shall be filled to proper level with cement concrete 1:4:8 at the Contractor's expense. Cement concrete 1:4:8 is to be laid before the pipes are placed in position.

1.13.2 Back Filling

No filling or back filling shall be carried out until debris, trash and other foreign material from trenches have been removed and drains, manholes, etc. have been tested and approved by the Engineer-In-Charge. Fill material shall be free from timber, rocks 3 inches or large, organic material and other unsuitable material as determine by the Engineer. Earth filling to the bottom of the trenches and to a height of 12 inches above the top of the pipes shall be of selected materials, hand backed, watered if necessary and well rammed on either side of the pipe. Special care shall be exercised where pipes are laid direct on the earth or on beds without benching or covers. The remainder of the earth filling shall be in 6 inches layers, each layer watered if necessary, and well rammed. Mechanical rammers are not to be used until atleast 24" of consolidated material has been returned over the pipes. Where vegetable soil has been removed, it shall be replaced at top of the trench and similarly consolidated. Heading shall be filled with the excavated materials, packed tightly in layers and rammed by hand (or concrete of mix. 1:3:6) or broken rock or hard core kept to a stepped face as the work proceeds. Surplus earth shall be disposed of as directed.

The trenches under pavings shall be back filled by local sand, well compacted. Existing pavements, roadways, walkways, curbs and landscaped areas disturbed during the programme of the excavation and backfill work shall be restored to original condition.

1.13.3 Planking Strutting and Staging

Excavated materials shall not be deposited within 1'-6" of the edge of the trench and sides of the excavation shall be supported by planking and strutting if necessary to ensure a proper and speedy execution of the work.

1.13.4 Blasting

No blasting shall be allowed near the rocks. The rocks shall be excavated by chiseling.

1.13.5 Buried Services

All pipes, ducts, cables, main and other service exposed by all excavation shall be effectively supported by timber or other means.

1.14. BEDDING AND ENCASTING PIPES

The concrete beds shall be of ratio 1:3:6 (6 inches thick for pipes more than 8" dia) and not less than 12 inches wider than the external diameter of the pipe. These shall be provided at all buried bends.

Where cast iron or steel (or concrete) pipes are specified to be haunched, the concrete shall be carried by from the outside edge of the bed to meet the pipe barrel tangentially, where the pipes are specified to be surrounded, the concrete shall be carried up from the beds in square section with a minimum of 6 inches in thickness over the barrel of the pipe.

The rest bend in the ground at the set of all soil, rain water, ventilating and other vertical pipes, to be embedded in solid concrete 6 inches thickness.

1.15. PLUMBING AND DRAINAGE SERVICES CONNECTION

The cost of all service connections required for the plumbing and drainage system shall, be the responsibility of the contractor. This includes the supply and installation of the water meter and all associated fitting, piping, etc.

Make any adjustments to piping systems necessary to suit the local requirements.

Coordinate routing and final points of connection with the external services layout and satisfaction of authorities having jurisdiction. Manholes and covers shall be installed to the municipality requirements.

Install water meters in meter cabinets and isolation valves to the local authority standards.



OPERATION AND MAINTENANCE OF FIRE SUPPRESSION SYSTEM**1.1. MAINTENANCE**

The Contractor shall be responsible without additional charge to the Employer for maintenance and servicing of the complete plant during the period of maintenance after the issue of the Certificate of Substantial Completion by the Engineer.

The Contractor shall be responsible for arranging all tools, instruments and Technical Staff including Specialist Technicians/Engineers required for the work.

The Contractor shall service the complete plant regularly according to the Schedule of Servicing and Maintenance as approved or amended by the Consultant but not less than once a month during the operational seasons. The servicing and maintenance shall be carried out by competent skilled labor under supervision of a qualified Engineer.

The Contractor shall take a certificate of satisfactory completion of monthly servicing from the Employer's Representative.

The Contractor shall carry out annual servicing, maintenance and overhauling of the complete plant at the end of the operational season and make the plant ready for operation in all respects well before the commencement of the next operational season. On receiving notification from the Contractor that annual servicing, etc. is nearing completion, the Consultant shall check the work carried out and give directions to the Contractor for completion of outstanding work, if any.

On satisfactory completion of annual servicing, maintenance and overhauling of the complete plant, the Engineer shall issue a Certificate of satisfactory completion to the Contractor.

1.2. SPECIAL TOOLS AND INSTRUMENTS FOR MAINTENANCE

The Contractor shall supply to the Employer all necessary special tools and instruments required for proper operation, servicing and maintenance of the complete plant.

MOTOR CONTROL CENTRE (MCC)**1.1. CODE & COMPLIANCE**

All materials and workmanship described herein shall be in accordance with the latest editions and addenda of the codes and standards listed below and all State, and local codes, and/or specifications, the more stringent shall be govern.

ASHRAE, BS, IEC, NFPA

1.2. QUALITY ASSURANCE

All materials and equipment to be used for this contract shall be standard products of a reputable manufacturer regularly engaged in the production of same. Where more than one unit of an item is required, all shall be provided by the same manufacturer using identical components. The manufacturer's nameplate shall include model number, serial number and performance data and shall be permanently affixed to all equipment furnished under this contract. Any work or materials found to be defective or which do not meet the requirements of this specifications, shall be replaced by the Vendor at his own expense. All materials shall be new, unused and free of defects and imperfections.

1.3. COORDINATION

In selecting the units, the tenderer shall carefully check and confirm that the units can be installed and conveniently serviced and maintained within the respective spaces indicated on the drawings for unit installation.

1.4. SUBMITTALS

Submit manufacturer's technical product data, installation instructions and dimensioned drawings and single line diagram.

1.5. SPARE PARTS

The tenderer shall include the price of complete set of spare parts for all items supplied by him required for two years operation for the specified operating hours.

1.6. DELIVERY, STORAGE & HANDLING

MCC shall be delivered factory assembled with protective crating and covering and their delivery coordinated in sufficient time to allow movement into building.

Preparation for shipment shall be in accordance with Vendor's standards. Vendor shall be solely responsible for the adequacy of the preparation for shipment provisions with respect to materials and applications. Adequate protection shall be provided against mechanical damage and atmospheric corrosion in transit and for a 6 month period of outdoor storage at job site prior to installation. Control panel shall be protected from moisture penetration. Miscellaneous parts shall be tagged or marked with the equipment item number for which they are intended.

1.7. WARRANTY

Vendor shall guarantee and provide a written statement as proof that the unit shall perform as specified on the Data Sheets.

All equipment shall be guaranteed for satisfactory performance and shall be warranted against faulty design, defective or improper materials.

The Material and Workmanship Guarantees shall be extended for a period of eighteen months from the date of shipment or one year from the date the equipment is placed in service whichever occurs first.

1.8. FACTORY INSPECTION AND TESTING

Each unit shall be assembled at the factory, inspected and determined to be in compliance with the specifications. Any deviations found shall be corrected.

1.9. PRODUCT

The scope of work shall cover the supply, design, manufacture, testing in the factory, packing, insurance, delivery to site, unloading, installation, testing and commissioning of all components with all the necessary accessories in accordance with this technical specification.

Climatic conditions are as follows:

Maximum ambient temperature: 45°C

Maximum relative humidity: 100%

Electric motors shall be of the sizes and types as specified for driving all plant and equipment. The motors shall be of at least the horsepower specified but shall be of proper horse-power and speed to suit the specific plant and equipment offered by the tenderer. Any adjustment in motor horsepower or speed must be included in the tender and no additional cost will be allowed on this account. The motors and starters shall be heavy duty quiet running type suitable for continuous operation under the site conditions. The minimum motor efficiency and power factor shall be 0.85.

The motors shall be designed for 50 cycle AC supply of following voltage characteristics:

	<u>Imported</u>	<u>Indigenous</u>
Fractional HP, 1-phase	220±10%	220±10%
1 HP & above, 3-phase	400±10%	400+5%-10%

All motors shall be constant speed type unless otherwise specified. 3-phase motors, 150 HP and below shall be squirrel cage type and above 150 HP slip ring type. Squirrel cage motors 10HP and above shall have 6 winding leads brought to motor terminal block for star-delta starting. Single phase motors shall be split phase type or capacitor start induction run type.

The motors and starters shall be tropicalized and fungus proof. Unless otherwise specified, drip proof ventilated or totally enclosed fan cooled construction with Class IP44 Protection for indoor installation, totally enclosed fan cooled weather proof construction with Class IP54 Protection for outdoor installation or where coming in contact with high humidity air. The motors and starters shall be suitable for operation under site conditions as specified; minimum ambient temperature 45°C and altitude 600m amsl. Where required, motors installed outside shall be provided with sheet metal cover to protect from direct sun.

The motors to have cast iron frame with cast in cooling ribs, integral feet and cast in end shields, protective cover for cooling fan, quiet running ball or roller bearings to meet the specified duty, terminal box and grease nipples in upper part of end shields to be accessible while motor is running. Motors to be suitable for direct or pulley drive.

All motors should be arranged for quiet operation and guaranteed to give the required output and fulfill the requirements of the driven machinery without producing any sound audible outside the machine room.

Motors driving pumps shall be directly connected through flexible couplings, while motors driving fans, compressors, etc shall be belt connected unless otherwise specified. All belt connected motors should have sturdy adjustable bases with arrangement to maintain proper belt alignment and tension and complete with belt guards. All belt connected fan driving motors shall have variable pitch pulley for adjusting the fan speed +5-10% of design selected speed.

The slip ring motors shall have inspection windows for access to brushes and slip rings. Rotor connections shall be made thru a cable gland or terminal box. There shall be a safety control to ensure that the motor cannot be started without the brushes in position.

Single phase motors 0.5 HP and above shall have automatic magnetic direct-on-line starters with hand/off/auto switch, two adjustable overload cutouts, low voltage cutout, and atleast one auxiliary contact for electric interlocking circuit. Motors below 0.5 HP rating may be provided with a heavy duty MCB.

Squirrel cage motors 10 HP and below shall have automatic magnetic direct-on-line starters with hand/off/auto switch, three adjustable overload cutouts, low voltage cutout, single phasing preventer, atleast two auxiliary contacts for electric interlocking circuits and one ammeter for motors 5 HP and above.

Squirrel cage motors above 10 HP shall have automatic magnetic star-delta type reduced voltage starters with hand/off/ auto switch, three adjustable overload cutouts, low voltage cutout, single phasing preventer, atleast two auxiliary contacts for electric interlocking circuits and one ammeter. All three contactors of the starter shall be identical.

Slip ring motors shall have automatic magnetic stator-rotor type starters with hand/off/auto switch, three adjustable overload cutouts, low voltage cutout, single phasing preventer, atleast two auxiliary contacts for electric interlocking circuits and one ammeter. The rotor Rheostat, natural air ventilation cooled, should be suitable for atleast 8 starts per hour. Electric interlocking control shall ensure that the motor cannot be energized without the Rheostat in starting position. At power supply contactor tripping, the Rheostat shall automatically return to the starting position. The Rheostat operating system shall be step by step ensuring smooth starting to full speed and short circuiting of rotor when full speed achieved.

All starters control circuit and magnetic coils to be suitable for 230 volts 50 cycles 1-phase AC. For motors requiring electric interlocking or remote control or sequence starting control or any other such feature, starters should have necessary auxiliary contacts (normally open and normally closed) providing the desired control arrangement. A separate set of terminals is required for each control circuit. Single phase starters shall have off-on green-red indicator lights and 3-phase starter's off-on-overload green-red-yellow indicator lights.

All starters for equipment not visible from the starters and for remote control operation shall be provided with a pilot light. An off-on switch in the starter control circuit shall be installed near the motor from which the starter is not visible to ensure that the motor cannot be started or electric circuit energized by error. Alternatively a disconnect switch can be provided near the motor.

All motors and starters provided under this contract should be of one manufacturer except for the equipment where special motors and starters are provided as standard component. The Contractor shall submit manufacturers' technical bulletins of motors and starters to the Consultant for approval before supply.

Where single phasing preventer or any other safety devices or controls specified above are not part of standard control panel and/or starters supplied by the equipment manufacturer as standard component, the Contractor shall provide all such safety devices and controls.

All motor starters in one Plant Room shall be installed in one Motor Control Center of type and construction as specified elsewhere, except where these are installed in the equipment control panel by the manufacturer. Special fire resistant, explosion proof, 2-speed or multi speed motors and starters shall be supplied if specified elsewhere in the documents.

The Motor Control Center (MCC) shall be factory fabricated of weather proof construction. The MCC shall have main incoming MCCB and a MCCB for each outgoing. 3 phase indicating lights and voltmeter with selector switch for incoming supply and on-off green-red indication lights for each outgoing. MCCB shall be heavy continuous duty type with 10KVA rupture rating and shall have magnetic short circuit trip, thermal overload trip single phasing preventer and under voltage release.

Four sets of plastic coated complete detail wiring diagram for the MCC shall be supplied. One set kept in a pocket in MCC and 3 sets handed over to the Employer.

Sub-switch boards shall be manufactured BS5486 Part 1 with Degree of Protection to IP 42, BS 5420 and IEC529.

Refer to single line diagram for rating and type of feeders required.

Equipment shall be indoor type 380 volt, 3 phase and neutral (TP/N), 4 wire, 50 Hz earth wire system with neutral solidly earthed and a withstand rating of 25 kA RMS symmetrical for one second or as specified.

Sub-switchboards shall be wall mounted unless noted otherwise on the drawings, metal enclosed, tamperproof, dead front type and separation to IEC439.1 form 2, Type 2.

Sections shall be 2mm zinc coated corrosion resistant treat after all forming and welding have been completed and all holes have been punched. The finish coat shall be epoxy powder paint deposited by an electrically charged procedure both inside and outside light blue/grey color RAL 7032.

Main incoming section shall contain a system service circuit breaker with thermal magnetic protection.

Bus bars shall be copper (size as indicated on the drawings).

Provide a copper earth bus extending the full width of the cubicle, and located at the bottom, with lugs at each end for earth cable connections.

Switchboards shall be factory assembled at the manufacturer's factory.

Ensure sufficient glanding and termination facilities are available for the main incoming cables.

Removable gland plates shall be provided at the top and bottom of all panel boards enclosures for cable gland termination.

Name plates for the sub main switchboards and for each outgoing to be fixed and shall be engraved lamincoid, with white letters on a black background.

1.10. EXECUTION

Sub-switchboards shall be wall mounted securely fixed to the structure by built-in fixings or Rawl bolts. Connect main feeders and branch feeders as shown on drawings and schedules. Provide 50mm x 6mm earth bar within the sub switchboard at low level. Check all factory made connections for mechanical security and electrical continuity. Provide a 2 meter by 1 meter rubber mat 1000 volt rating under all sub-switchboards.

WIRES, CABLES, CONDUITES AND CABLE TRAY

1.0. GENERAL

1.1 CODE & COMPLIANCE

All materials and workmanship described herein shall be in accordance with the latest editions and addenda of the codes and standards listed below and all State, and local codes, and/or specifications, the more stringent shall be govern.

ASHRAE, BS, IEC, NFPA

1.2 QUALITY ASSURANCE

All materials and equipment to be used for this contract shall be standard products of a reputable manufacturer regularly engaged in the production of same. Where more than one unit of an item is required, all shall be provided by the same manufacturer using identical components. The manufacturer's nameplate shall include model number, serial number and performance data and shall be permanently affixed to all equipment furnished under this contract.

Any work or materials found to be defective or which do not meet the requirements of these specifications, shall be replaced by the Vendor at his own expense.

All materials shall be new, unused and free of defects and imperfections.

1.3 COORDINATION

In selecting the units, the tenderer shall carefully check and confirm that the units can be installed and conveniently serviced and maintained within the respective spaces indicated on the drawings for unit installation.

1.4 SUBMITTALS

Submit manufacturer's technical product data, installation instructions and dimensioned drawings and single line diagram.

1.5 SPARE PARTS

The tenderer shall include the price of complete set of spare parts for all items supplied by him required for two years operation for the specified operating hours.

1.6 DELIVERY, STORAGE & HANDLING

Adequate protection shall be provided against mechanical damage and atmospheric corrosion in transit and for a 6 month period of outdoor storage at job site prior to installation. Control panel shall be protected from moisture penetration.

Miscellaneous parts shall be tagged or marked with the equipment item number for which they are intended.

1.7 WARRANTY

Vendor shall guarantee and provide a written statement as proof that the unit shall perform as specified on the Data Sheets.

All equipment shall be guaranteed for satisfactory performance and shall be warranted against faulty design, defective or improper materials.

The Material and Workmanship Guarantees shall be extended for a period of eighteen months from the date of shipment or one year from the date the equipment is placed in service whichever occurs first.

1.8 FACTORY INSPECTION AND TESTING

Each unit shall be assembled at the factory, inspected and determined to be in compliance with the specifications. Any deviations found shall be corrected.

2.0 PRODUCT

The scope of work shall cover the supply, design, manufacture, testing in the factory, packing, insurance, delivery to site, unloading, installation, testing and commissioning of all components with all the necessary accessories in accordance with this technical specification.

Climatic conditions are as follows:

Maximum ambient temperature: 45°C

Maximum relative humidity: 100%

The Contractor will be responsible for complete electric wiring and earthing of the equipment and controls. The Employer shall only provide 3 phase and natural 4 wire supply with two earthing wire up to the Motor Control Center as shown on drawing.

The terminal connections at the units shall be made in flexible conduits.

Color code conductors as follows:

3 phase - **red, yellow, blue**

1 phase - **red**

Control wiring - **orange or red with cable markers to match the**

diagrams at each point of connection and termination.

Neutrals	-	black
Earth	-	green/yellow striped

Conductors shall have the color impregnated into the insulation at the time of manufacture.

Painting of conductor insulation will not be accepted.

3.0 EXECUTION

Wiring of multi-point circuits shall utilize a 'looping-in' system. Joints and connections other than those required for the connection of switches, fuses, socket outlets, motors and the like shall not be allowed.

PVC cables shall not be in direct contact with any form of polystyrene used in the building.

PVC cables shall not be installed when manufacturers recommended installation temperature. If cable has been exposed to a temperature below recommended value a warming up cable is installed.

PVC cables shall not enter any Luminaire or heat-producing equipment. For tungsten luminaires, heat-resisting cables shall be installed from the luminaires to the lighting switches or equivalent.

For fluorescent luminaries high temperature PVC cables shall be installed from the lighting switch or equivalent. For recessed tungsten and fluorescent luminaries and heat-producing/emitting equipment, final connections shall be made using heat-resistant flexible cables. Where cables are permitted to traverse channel-ways or similar on continuously mounted fluorescent luminaries, heat-resistant cables shall be used throughout.

Soldered connections or lugs shall not be permitted. Conductors requiring bolted connection shall be terminated with compression lugs using an automatic compression crimp tool which will only release after the correct crimp depth has been obtained. All crimped connections shall be contained within an appropriately sized conduit box. Bolted connections shall have spring washers. Pinch screw terminals may not be used for conductors greater than 6.0mm.

Single strand cables shall be doubled back on themselves when terminations are made.

PVC cables shall not be used for final connections to any appliances containing a heating element or any appliance emitting heat. Where flexible conduit is used as a final connection wiring medium, heat-resistant cables shall be used and these shall commence at the solid conduit end of the flexible conduit provided it is not in a heated area. If this is not possible, heat-resistant cables shall be run back to the first switch outside the hot area.

PVC cables shall not exceed the carrying capacity of the conduit or trunking. The installations shall comply strictly with the IEE Wiring Regulations regarding capacity of conduits and trunking for 450/750V cables and a space fill factor of 35% shall not be exceeded for trunking.

Circuit protective conductors shall be used throughout the installation and shall be the same grade and temperature rating of the live conductors of the circuit.

Metal conduit and/or the trunking system enclosure shall not be used exclusively as a circuit protective medium. Separate circuit protective conductors shall be installed and shall be colored green and yellow.

No cable installation work shall be carried out in temperatures at which the cable being installed is likely to suffer damage.

Each drum length of cable shall be allotted a distinctive and separate reference number. This number shall appear on the test sheet covering the respective length of cable and shall also be clearly marked on the cable drum. The Contractor shall advise the Engineer upon delivery to site of each drum length, quoting this reference number, which shall also appear on the invoices.

All cables shall be delivered to site with the manufacturer's seals, labels or other proof of origin intact. These labels and seals shall not be removed until the cable is required for use and shall be retained for inspection by the Engineer.

The arrangement of cables and all methods of installation shall be approved by the Engineer.

Cables shall be installed from terminal point to terminal point and straight through joints shall not be made unless approved. Where a run of MICC cable is of a length which exceeds the maximum to which the cable can be manufactured, a through joint of approved pattern will be permitted.

The radius of each bend or change in direction of the route of any cable shall not be less than that laid down in the relevant table of current issues of IEE Regulations and the relevant British Standard Specifications, and shall generally not be less than eight times the overall diameter of the cable.

The spacing of cable supports shall be those laid down for the relevant size and type of cable in the current issue of the IEE Regulations and the manufactures recommendations, but the horizontal longitudinal distance between wire armored support centers of unarmored cable support centers shall not exceed 1 meter, the horizontal distance between support centers of unarmored cables shall not exceed 800 mm and the distance between support centers for either type of cable on vertical runs is not to exceed 1 meter. Cables of and less than 50 mm external diameter shall be supported at not more than 800 mm centers.

The cables may be installed in a number of different ways:

- (a) Laid direct in the ground
- (b) Laid in ducts
- (c) Installed in covered concrete trenches
- (d) Installed on cable ladders
- (e) Installed on perforated steel trays
- (f) Fixed to concrete and brickwork
- (g) Fixed to steelwork

Single core cables forming a three phase group shall be fixed in trefoil cleats of a British Insulated Cables Limited alloy pattern or similar. Flat formation may be used where specified or approved.

Unarmored PVC cables and wires shall be enclosed in trunking or conduit unless otherwise stated.

Home runs exceeding 25m in length to distribution boards shall be minimum 4mm² unless otherwise stated.

The cross-sectional area of circuit protective conductors shall be in strict accordance with the requirement of the IEE Wiring Regulations.

Conductors shall be pulled into conduit in a careful and workmanlike manner without overstressing or exceeding the pulling limitations of the conductors.

Conductors shall be 'combed' as pulling proceeds. The neutral and circuit protective conductor or each circuit shall be pulled with the phase conductors.

Conductors of a circuit run in trunking shall be 'grouped' together with approved plastic binding clips. Tape shall not be used.

Clip cables neatly to tray with cable separation and support spacing as recommended by the manufacturer.

Where more than two power cables are run together, provide a cable tray. Secure cables with cleats, saddles or ties as appropriate to location and use spacers where more than one layer is required.

Install cables in accordance with the manufacturer's instructions, using manufacturer's approved terminating devices.

Terminate single conductor cables entering steel cabinets through a non-ferrous plate.

Where cables pass through a floor or fire barrier provide cable transit or equivalent fire stop with openings properly sized for the cables.

Run surface cables similar to exposed conduit installations. Run cables concealed above ceilings in finished areas. Where exposed, run parallel to building lines and avoid proximity to water.



VIBRATION AND SEISMIC CONTROLS FOR FIRE SUPPRESSION PIPING AND EQUIPMENT**1.0. GENERAL****1.1. CODE & COMPLIANCE**

All materials and workmanship described herein shall be in accordance with the latest editions and addenda of the codes and standards listed below and all State, and local codes, and/or specifications, the more stringent shall be govern.

ASHRAE, ASME, ASPE, ASTM, BS, IPC, NFPA

1.2. QUALITY ASSURANCE

All materials and equipment to be used for this contract shall be standard products of a reputable manufacturer regularly engaged in the production of same. Where more than one unit of an item is required, all shall be provided by the same manufacturer using identical components. The manufacturer's nameplate shall include model number, serial number and performance data and shall be permanently affixed to all equipment furnished under this contract.

Any work or materials found to be defective or which do not meet the requirements of these specifications, shall be replaced by the Vendor at his own expense.

All materials shall be new, unused and free of defects and imperfections.

1.3. COORDINATION

In selecting the units, the tenderer shall carefully check and confirm that the units can be installed and conveniently serviced and maintained within the respective spaces indicated on the drawings for unit installation.

1.4. SUBMITTALS

Shop drawings and submittal data shall be provided for review by the successful vendor within fifteen (15) calendar days of notice of the award. Submit two copies of the drawings and data in English Language. The submittals and shop drawings shall be complete in all respects including the following information:

Product Data:

- A. Submit seismic force level (F_p) calculations from applicable building code. Submit pre-approved restraint selections and installation details from Cooper B-Line's Seismic Restraints catalog (SRS-02). [or engineer approved equal]
- B. Restraint selection and installation details shall be pre-approved by a professionally licensed engineer experienced in seismic restraint design.
- C. Submit manufacturer's product data on strut channels including, but not limited to, types, materials, finishes, gauge thickness, and hole patterns. For each different strut cross-section, submit cross sectional properties including Section Modulus (S_x) and Moment of Inertia (I_x).

Shop Drawings: Submit assembly-type shop drawings indicating dimensions, weight loadings, required clearances, and methods of assembly of components.

Maintenance Data: Submit maintenance data and parts list for each type of equipment and material including "trouble shooting" maintenance guide. Include this data, product data, and shop drawings, in maintenance manuals.

Vendor shall resubmit five sets of submittals and drawings, which have been returned for correction within ten (10) calendar days of receipt.

1.5. SPARE PARTS

The tenderer shall include the price of complete set of spare parts for all items supplied by him required for two years operation for the specified operating hours.

1.6. DELIVERY, STORAGE & HANDLING

All vibration and seismic control material shall be supplied to the site in manner to prevent corrosion while under storage and installation. Where possible, store product inside and protected from weather. Where necessary to store outside, elevate above grade and enclose with durable, or by packaging with durable, water proof wrapping. Protect product and material from moisture and dirt by inside storage and enclosure, or by packaging with durable, water proof wrapping.

1.7. WARRANTY

Vendor shall guarantee and provide a written statement as proof that the unit shall perform as specified on the Data Sheets. All material shall be guaranteed for satisfactory performance and shall be warranted against faulty design, defective or improper materials. The Material and Workmanship Guarantees shall be extended for a period of eighteen months from the date of shipment or one year from the date the equipment is placed in service whichever occurs first.

1.8. FACTORY INSPECTION AND TESTING

Each product shall be inspected and determined to be in compliance with the specifications. Any deviations found shall be corrected.

2.0. PRODUCT**2.1 GENERAL**

The scope of work shall cover the supply, design, manufacture, testing in the factory, packing, insurance, delivery to site, unloading, installation, testing and commissioning of all components with all the necessary accessories in accordance with this technical specification.

The contractor shall provide pre-engineered seismic restraint systems to meet total design lateral force requirements for support and restraint of piping, conduit, cable trays and other similar Systems and equipment where required by the applicable building code including the following.

Seismic Requirements for single rod hanger supports for conduit, pipe and other similar systems. Seismic Requirements for trapeze type supports for cable tray, conduit, pipe and other similar systems.

2.2 SEISMIC BRACING COMPONENTS

Steel strut shall be 1-5/8 wide in varying heights and mig-welded combinations as required to meet load capacities and designs indicated. A material heat code, part number, and manufacturer's name shall be stamped on all strut and fittings to maintain traceability to material test reports.

- a. Material for epoxy painted strut: ASTM A1011, SS, Grade 33
- b. Material for pre-galvanized strut: ASTM A653, SS, Gr. 33
- c. Material for Hot-Dip Galvanized strut: ASTM A1011, SS, Grade 33 and hot-dip galvanized after fabrication in accordance with ASTM A123.
- d. Material for fittings and accessories: ASTM A907 Gr. 33, Structural Quality or ASTM A1011, SS, Gr.33.
- e. Fittings and accessories: Products shall be of the same manufacturer as strut and designed for use with that product.

2.3 SEISMIC BRACING AND SUPPORT OF SYSTEMS AND COMPONENTS

General:

- a. Seismic restraint designer shall coordinate all attachments with the structural engineer of record.
- b. Design analysis shall include calculated dead loads, static seismic loads, and capacity of materials utilized for the connection of the equipment or system to the structure.
- c. Analysis shall detail anchoring methods, bolt diameter, and embedment depth.
- d. All seismic restraint devices shall be designed to accept without failure the forces calculated per the applicable building code and as summarized in Section 3.01.

Friction from gravity loads shall not be considered resistance to seismic forces.

Fire protection systems shall meet the requirements of NFPA-13 and NFPA-14 for the building seismic requirements.

3.0 EXECUTION**3.1 Seismic Restraint of Piping:**

- a. All seismic restraint systems shall be installed in strict accordance with the manufacturer's seismic restraint guidelines manual and all certified submittal data.
- b. Transverse piping restraints shall be at 40-foot maximum spacing for all pipe sizes, except where lesser spacing is required to limit anchorage loads.
- c. Longitudinal restraints shall be at 80-foot maximum spacing for all pipe sizes, except where lesser spacing is required to limit anchorage loads.
- d. Transverse restraint for one pipe section may also act as a longitudinal restraint for a pipe section of the same size connected perpendicular to it if the restraint is installed within 24-inches of the elbow or tee or combined stresses are within allowable limits at longer distances.
- e. Hold down clamps must be used to attach pipe to all trapeze members before applying restraints. Branch lines may not be used to restrain main lines. Provide reinforced clevis bolts when required.
- f. Piping crossing building seismic or expansion joints, passing from building to building, or supported from different portions of the building shall be installed to allow differential support displacements without damaging the pipe, equipment connections, or support connections. Pipe offsets, loops, anchors, and guides shall be installed as required to provide specified motion capability and limit motion of adjacent piping.
- g. Do not brace a system to two independent structures such as ceiling and wall.
- h. Provide appropriately sized openings in walls, floors, and ceilings for anticipated seismic movement.
- i. Provide fire seal systems in fire-rated walls.

3.2 Seismic Restraint of Electrical Services

- a. All seismic restraint systems shall be installed in strict accordance with the manufacturer's seismic restraint guidelines manual and all certified submittal data.
- b. Installation of seismic restraints shall not cause any change in position of equipment or piping, resulting in stresses or misalignment.
- c. No rigid connections between equipment and the building structure shall be made that degrade the noise and vibration-isolation system specified.
- d. Do not install any equipment, piping, duct, or conduit that makes rigid connections with the building unless isolation is not specified.
- e. Prior to installation, bring to the architect's/engineer's attention any discrepancies between the specifications and the field conditions, or changes required due to specific equipment selection.
- f. Bracing may occur from flanges of structural beams, upper truss cords of bar joists, cast in place inserts, or wedge-type concrete anchors. Consult structural engineer of record.

- g. Overstressing of the building structure shall not occur from overhead support of equipment. Bracing attached to structural members may present additional stresses. The contractor shall submit loads to the structural engineer of record for approval in this event.
- h. Brace support rods when necessary to accept compressive loads. Welding of compressive braces to the vertical support rods is not acceptable.
- i. Provide reinforced clevis bolts where required.
- j. Seismic restraints shall be mechanically attached to the system. Looping restraints around the system is not acceptable.
- k. Do not brace a system to two independent structures such as a ceiling and wall.
- l. Provide appropriately sized openings in walls, floors, and ceilings for anticipated seismic movement. Provide fire seal systems in fire-rated walls.



PIPES, FITTINGS AND SUPPORT

01. Piping used for Fire systems shall be as follows:

SERVICE	MATERIAL	TYPE	WEIGHT
FIRE	SEAMLESS	BLACK	SCHEDULE 40

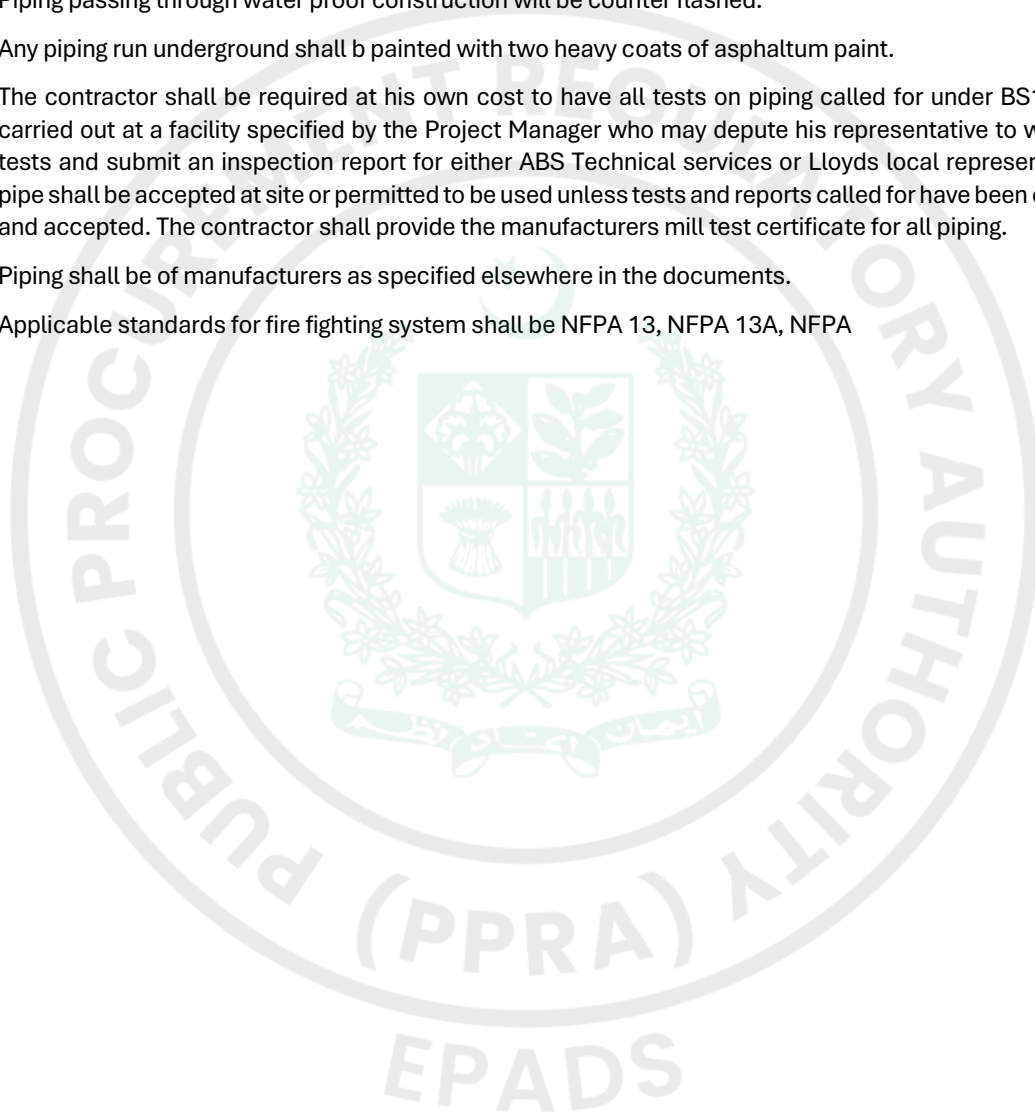
02. *All black piping shall be supplied lacquered from the factory to prevent corrosion while under storage and installation.*

03. Piping shall be jointed as follows:

Black steel piping upto 2 inch	Screwed
Black Steel piping above 2 inch	Welded

04. Threaded fittings shall comply with the requirements of dimensional standards BS-142 and BS-143 and material/physical standards BS-309 and BS-310 in the case of “whiteheart” and “blackheart” fittings. All fittings shall be of the “banded” or “collar” type. The contractor shall not deliver at site any fittings unless prior approval of submittals verifying compliance with standards has been approved.
05. Screw threads shall be cut as per BS21:1957 clean and true, screw joints made tight without caulking and without red or white lead. No bushing shall be used. Reducing fittings shall be used to change pipe size. Reductions to be made with eccentric reducers or eccentric fittings. All pipes, 2 inches and less, shall be reamed out after cutting to remove all burrs.
06. Welded fittings shall be of weight equal to that specified for the piping system and shall be factory made complying with the BS equivalent of ANSI 16.9 and ASTM A-105. Elbows shall be of the long radius type.
07. The drawings indicate generally the size and location of piping as designed for space conditions, ceiling heights, etc. and may not be changed until coordinated with other contractors.
08. Piping shall be properly graded to secure easy circulation and prevent noise and water hammer. As much pitch as space conditions permit, to point where vent relief, drip and drain connections are provided. Capped dirt pockets at all risers, bells, low points and other places where dirt and scale may accumulate. Proper provision for expansion and contraction in all portions of pipe work to prevent under strains on piping or apparatus connected therewith. Double swing at riser transfer wherever possible to take up expansion. Expansion joints and loops where indicated. In general all piping shall conform to ANSI/ASME B31.9-1982 or equivalent British standard with regard to installation details.
09. Approved screw unions, with steel or bronze bodies and ground brass taper or spherical joints, shall be installed at traps, instruments, etc. and wherever else directed to permit easy connection and disconnection.
10. All pipe work shall be so arranged and put together as to prevent undue strains or leaks caused by expansion and contraction. Riser branches and other offsets made up with 4 elbow swings wherever possible. Riser branches arranged to take up motion of risers and mains.
11. To meet job conditions, water supply and return mains shall be set up and down where directed. In any such case, provide drain cocks at low points and vent traps at high points.
12. All piping passing through floor construction shall have No. 20 US gauge galvanized iron sleeves the full depth of the floor construction. Any sleeves in wet areas, such as equipment rooms, shall extend 1 inch above floor, and space between pipe and sleeve shall be caulked watertight.

13. All visible piping passing through finished floors and ceilings shall be provided with escutcheons. Escutcheons shall be bell shaped, with plated finish and shall be fastened to pipe.
14. Sleeves for exposed horizontal pipes passing through partitions shall be galvanized steel pipe with heavy spun metal, and set-screwed to the pipe on each side of partition.
15. All sleeves shall be of sufficient size to include pipe covering.
16. All sleeves shall be furnished, located and set by the Contractor and he shall be responsible for the accuracy of their location when concrete is poured.
17. Piping passing through water proof construction will be counter flashed.
18. Any piping run underground shall be painted with two heavy coats of asphaltum paint.
19. The contractor shall be required at his own cost to have all tests on piping called for under BS1387:1967 carried out at a facility specified by the Project Manager who may depute his representative to witness the tests and submit an inspection report for either ABS Technical services or Lloyds local representation. No pipe shall be accepted at site or permitted to be used unless tests and reports called for have been completed and accepted. The contractor shall provide the manufacturers mill test certificate for all piping.
20. Piping shall be of manufacturers as specified elsewhere in the documents.
21. Applicable standards for fire fighting system shall be NFPA 13, NFPA 13A, NFPA



HDPE PIPES AND FITTINGS**1.1 HDPE PIPE & FITTINGS****I) UNDER GROUND PIPES & FITTINGS (UPTO 13.7 BAR) FM approved**

1. High density Polyethylene Pipes and fittings (**HDPE**) PE100 Manufactured for underground fire protection lines, **FM** approved and listed to be used for fire underground main, Class 200, confirm to **AWWA C 906** and **NFPA 24**.
2. Pipes and fittings to be from one manufacturer to meet above standards.
3. Welding to be butt fusion only, suitable for pipe pressure ratings and each joint should have separate report showing full details on joint and welding process and result.

II) UNDER GROUND PIPES (UPTO 16 BAR) FM approved

1. High density Polyethylene Pipes & fittings (**HDPE**) PE 100 **FM approved** manufactured for underground fire protection lines, FM approved and listed to ISO 4427 – SDR 7.4 to be used for fire underground main.
2. Pipes and fittings to be from one Manufacturer to meet above standards.
3. Welding to be butt fusion only, suitable for pipe pressure ratings and each joint should have separate report showing full details on joint and welding process and result.

FIRE HYDRANTS

1.1 DRY BARREL FIRE HYDRANT

Fire hydrants shall be manufactured to comply with AWWA Standard C502. Fire hydrants shall meet all requirements of **LPCB/UL listed/FM approved**.

1. Fire hydrants shall be rated for a minimum working pressure of 150 Psi (1034 kPa).
2. Fire hydrants shall have a main valve which shall provide complete closing of the drains after 4 to 5 turns of the operating nut in the opening direction.
3. The drain ring assembly shall be replaceable without removing the hydrant from the connecting pipe or having to dig.
4. Fire hydrants shall be three-way in design, having one pumper nozzle (size as specified) and two 2-1/2" hose nozzle (side outlet). Nozzle thread type shall be as specified by the end user.
5. The operating nut shall be a one piece design, manufactured of Bronze. It shall be pentagon in shape and the nut dimensions shall be as specified by the end user.
6. An arrow shall be cast on the bonnet flange to indicate the specified opening direction.
7. Hydrants shall be a "traffic-model" having upper and lower barrels joined at the ground line by a separate and breakable "swivel" flange providing 3600 rotation of upper barrel for proper nozzle facing.
8. The lower barrel shall be an integrally cast unit. The use of threaded on or mechanically attached flanges is deemed unacceptable.
9. All ferrous part shall be epoxy coated, the lower barrel shall be bitumen coated.
10. Hydrants shall be internally coated by Epoxy to avoid corrosion.
11. A bury line shall be marked on the lower barrel below the break flange to indicate proper installation depth.
12. Hydrant Inlet shall flange and the size shall be minimum 6".

1.2 WET BARREL FIRE HYDRANT

General

Fire hydrants shall be manufactured to comply with AWWA Standard C503. Fire hydrants shall meet all requirements UL/FM. **Coupling must be quick coupling as per Civil Defense requirements where coupling must be Listed & Certified.**

1. Fire hydrants shall be rated for a working pressure of 200 Psig.
2. Hydrant Body shall be Ductile Iron, Stainless Steel, or Bronze as specified.
3. All ferrous part shall be Internally & Externally Epoxy coated.

1.3 UNDER GROUND HYDRANTS

Under ground fire hydrant valves shall be of screw down type and are suitable for a maximum working pressure of 16 bars. The Inlet shall be flange type and the Outlet of this hydrant shall be round threaded 2 1/2" size to BS 336 Standard.

Under ground fire hydrants are manufactured to comply BS 750:2006 & EN14339:2005 and shall be **BSI Kitemark Approved**.

1.4 MATERIALS

- Body material shall be made of Grey cast iron or Ductile Iron as per BS standard.
- Material of spindle shall be made of Stainless steel to BS EN 10088-1.

- Material for the screwed outlet is made of High tensile brass to BS 12163.
- Spindle sealing (two seals) is of the toroidal sealing ring ('O' ring) type.
- Design of the spindle cap in accordance with BS 5163.
- A Wiper ring is positioned above the spindle sealing to prevent the ingress of foreign matter.

1.5 DESIGN FEATURES

- The Hydrant shall be compactable against resistance to disinfection products.

1.6 RESISTANCE TO OPERATING LOADS

- Maximum operating torque value (MOT) 105 N-m.
- Minimum strength torque value (mST) 210 N-m.

1.7 CORROSION PROTECTION

- a. All Ferrous components liable to corrosion shall be shot blasted (inside & outside) to SA 2-1/2 standard.
- b. Each hydrant should be painted from inside a 3 Coat epoxy paint system including primer, intermediate & Top coat.

1.8 DRAINING

- a. Hydrant draining system is by self operating frost valve.
- b. Volume of retained water 80 ml.
- c. Time of draining~3 min. 20 Sec.

1.9 HYDRAULIC CHARACTERISTICS

Flow co-efficient Kv equal 120 m³/hr @ 1.7 bars.

FIRE SUPPRESSION STAND PIPES AND HOSE SYSTEMS

FIRE HOSE CABINETS

1. FIRE HOSE CABINET

1.1 General

- 1.1.1 The scope of work shall cover the supply, design, manufacture, testing in the factory, packing, insurance, delivery to site, unloading, installation, testing and commissioning of Fire Department Connection pipes and fittings with all the necessary accessories in accordance with this technical specification.
- 1.1.2 Fire hose cabinet, valves, hose and accessories shall be approved by and shall be in compliance with the requirements of local Civil Defense Authorities.
- 1.1.3 All cabinets shall be 12 swg stainless steel with hair line finish internally. Mirror finished stainless steel doors shall be provided for the cabinets subject to approval of Consultant.
- 1.1.4 FHC shall have two compartments. Upper compartment shall have permanent die cut lettering label "FIRE HOSE REEL", the lower compartment shall have permanent die cut lettering label " FIRE HOSE AND EXTINGUISHERS".
- 1.1.5 Cabinets in unconditioned areas and parking levels shall have similar configuration but suitable for surface or recess mounting with stainless steel body, door and architrave. Cabinets located at roof shall be suitable for weather exposed installation.
- 1.1.6 Cabinet shall contain a chrome plated swing hose reel complete with 30 meter of 25 mm double red braided rubber fire hose to BS 316SA with chrome plated brass shutoff nozzle clamped ends, automatic valve to open water flow after 2-1/2 turns of the reel and 25 mm key operated shut-off valve. All metal parts shall be polished and chrome plated. Where necessary to limit water pressure to hose to 65 psi (450 kPa), a suitable pressure reducing valve shall be provided. The cabinet shall contain a 65 mm chrome plated, cast brass landing valve connected to the stand pipe riser as well as 30 m hose canvas with hose rack.

1.2 FIRE BLANKETS (FIRST AID EQUIPMENT)

Provide fire blankets in Kitchens. Fire blanket shall be good quality with silicon coating provided on both sides. The size shall be 4 feet x 4 feet, 4 Feet x 6 feet and 6 feet x 6 feet. **The product should have at least been certified by two independent Certification Organizations from International Testing Laboratories & Accredited Certification Bodies as LPCB/ Kitemark approved.**

1.4 STANDPIPE AND HOSE SYSTEM

1.4.1 FIRE HOSE & NOZZLE: (FIRST AID EQUIPMENT)

Fire Hose and Nozzles: **The product should have at least been certified by two independent Certification Organizations from International Testing Laboratories & Accredited Certification Bodies as LPCB approval/BSI Kitemark approval and/or UL listed/FM Approved.**

Fire hose shall be of synthetic re-enforced rubber lined, 65 mm diameter and 30 m long equipped with brass chrome plated male coupling for the connection with the landing valve, and female coupling for the connection 65 mm chrome plated adjustable diffuser nozzle. The fire hose unit shall include the landing valve specified above, and shall have minimum working pressure of 15 Bar and bursting pressure of minimum 45 Bar.

1.4.2 LANDING VALVES (PRESSURE REDUCING & RESISTING DEVICES)

Landing valves shall be of pressure reducing and restricting type devices acceptable to the Fire Brigade use inside buildings. This device shall have 65mm inlet diameter bore fitted with 65 mm. instantaneous female coupling conforming to BS 336 and a blank cap secured by a suitable length of chain. **The product should have at least been certified by two independent Certification Organizations from International Testing**

Laboratories & Accredited Certification Bodies as LPCB approval/BSI Kitemark approval and or UL listed/FM Approved.

The valve shall be of brass / Gunmetal construction with approved finish. If the landing valve is UL listed then the outlet coupling must be quick instantaneous with local fire services. The installation of the coupling should be done by an authorized installer.

Landing valves shall be installed inside the cabinet, at a height above floor level between 750 mm and 1100 mm. The outlet shall be installed to give clearances of not less than 150 mm on both sides and below the valve and not less than 200 mm clearance above the hand wheel.

1.4.3 HOSE REEL CABINETS

Hose reel cabinets: **The product should have at least been certified by two independent Certification Organizations from International Testing Laboratories & Accredited Certification Bodies as LPCB/BSI Kitemark approval.**

The hose reel cabinet shall be either wall-mounted type or recessed type. The cabinet can be single or double to accommodate fire hose reel in the upper compartment and other fire equipments including hose rack in the bottom compartment. The cabinet has to be made of minimum 1.2mm thickness electro-galvanized steel sheet or stainless steel as per design requirement. The door can be different type like wood to steel finish, marble finish, MDF wooden door, clear glass or wired glass. The stainless steel cabinets shall be polished finish type, the finish shall be decided upon the approval of the architect.

The Double fire hose reel cabinet in different locations of the building shall have following specified fire equipments:

- a) Automatic or manual Fire Hose Reel
- b) Landing Valves, Fire Hose and Nozzle or fire hose rack assembly
- c) 2 kg CO2 Fire Extinguisher
- d) 6 kg DCP Fire Extinguisher

1.4.4 FIRE BREECHING INLET VALVE

The unit shall be of **BSI Kitemark/LPCB approved.**

The inlet breeching piece shall be of horizontal pattern having a 150 mm flanged outlet and four 65 mm instantaneous male inlets complete with blank caps and chains. A nonreturn (check) valve shall protect the inlet. The coupling shall comply with BS 336. The inlet connections shall be of suitable type of local fire brigade department.

The breeching piece shall be equipped with a 25 mm drain valve to drain the riser.

All parts including valve body, chain and caps shall be brass / gunmetal if it is UL Listed and valves body shall be made of DI body with Brass Inlet Valves polished chrome plated if it is BSI Kitemark/LPCB approved.

The product should have at least been certified by two independent Certification Organizations from International Testing Laboratories & Accredited Certification Bodies.

1.4.5 FIRE BREECHING INLET VALVE CABINET

The unit shall be **LPCB approved. The product should have certified by independent Certification Organizations from International Testing Laboratories & Accredited Certification Bodies as LPCB approved.**

The inlet breeching unit shall be housed in an steel flush box painted by polyester powder coated with wired glass glazed front panel as per BS 5041-5 standard and:

- a) Conspicuously indicated by the words: “Dry Riser Inlet, Sprinkler Inlet, Wet Riser Inlet” in block letters on the inner face of the glass, depending on the project type.
- b) Fastened only by means of a spring lock, which can also be operated from the inside without the aid of a key after the glass has been broken.
- c) Made large enough for hose to be connected to inlets even if the door cannot be opened and the only means of access is by breaking the glass.

1.4.6 EXECUTION/ INSTALLATION

Standpipe and hose systems shall be installed in accordance with NFPA-14 and Authority having jurisdiction requirements and as shown on the drawing.

The Fire protection contractor shall be responsible to co-ordinate his work with all other trades and shall be completely familiar with and shall make allowance for conditions which affect the fire protection installation.

The Fire hose cabinets shall be mounted securely in the concrete block walls, shimmying the back box and mortaring tightly in place.

Provide a pressure gauge with a isolation cock with inspectors test plug on top and bottom of the each stand pipe risers.

Chrome plated nipples shall be provided in between the landing valve and hose connection. Chrome plated escutcheons shall be provided where pipes penetrate the cabinet.

A waterflow switch shall be provided in each standpipe riser and interfaced to fire alarm system for monitoring and to locate the section of waterflow during fire fighting operation.

A drain valve and piping, located at the lowest point of the standpipe piping downstream of the isolation valve, shall be arranged to discharge water at an approved location.

1.4.7 CLEANING AND TESTING

Tubes and all items of equipment shall be delivered, stored and maintained in closed storage with their open ends effectively plugged, capped or sealed. All fittings, valves and sundry items shall be stored in clean bins or bagged and stowed in suitable racks. All such stored items shall be maintained under weatherproofed cover to be supplied by the MEP sub-contractor until they are ready for incorporation in the works. Particular care shall be taken to ensure that electrical equipment and components are kept clean and dry.

Before installations are handed over or subjected to the inspection and tests, the entire installation shall be thoroughly cleaned, both internally and externally.

All fire protection installations shall be flushed out with clean water. During the flushing out, provision shall be made to exclude any items of plant, which could be damaged by the cleaning operation. The entire operation shall be carried out to the satisfaction of the Engineer.

The fire protection system shall be tested in accordance with the NFPA codes and the relevant standards, with particular reference the requirements set forth in this section:

All testing shall be performed under the work of this section. All services required for testing shall be a responsibility of the work of this section. The MEP sub-Contractor shall notify the Engineer of all tests, 48 hours prior to testing.

If the systems are tested in sections, the connection to the previously tested section shall be included.

All tests that may be required by the local fire authorities shall be performed under the work of this section, in the presence of their representative, if so required.

All leaks shall be corrected and the system re-tested until no leaks are found, at no cost to the client.

Functional tests of all water flow alarms are a responsibility of the works of this Section.

When the various systems are completed, operation tests shall be run on all equipment to demonstrate proper operating conditions. These tests shall be run under the observation of the inspector and the client's operating engineer. The client's operators shall be instructed during this period. Operation tests shall be performed under the work of this section.

Should any piece of equipment or apparatus of any material or work fail in any of these tests, it shall be immediately removed and replaced by a perfect material, and the portion of the work replaces shall again be tested under the work of this section.



FIRE EXTINGUISHERS

1.0 GENERAL

- Fire Extinguishers shall be manufactured to comply with BS EN 3 (portable), BS EN 1866 (Mobile) standards.
- Fire Extinguishers shall comply with NFPA-10 or BS 5306 for the installation.
- The classification of fire extinguishers shall consist of a letter that indicates the class of fire on which a fire extinguisher has been found to be effective, preceded by a rating number (Class A, Class B and Class F only) that indicates the relative extinguishing effectiveness.
- The distribution of fire extinguishers shall comply with the prescribed requirements of NFPA-10, as shown on the drawings and the specific requirements of the local Civil Defense Authorities.
- Cabinets housing fire extinguishers shall not be locked.
- Portable fire extinguishers shall be maintained in a fully charged and operable condition, and kept in their designated places at all times when they are not being used.

1.1 DEFINITIONS

The type of extinguisher needed depends on the right type of fire that may occur in the room or building.

According to NFPA 10, several classes of fire have been defined:

- I. **Class A Fires:** Fires in ordinary combustible materials, such as wood, cloth, paper, rubber, and many plastics.
 - II. **Class B Fires:** Fires in flammable liquids, combustible liquids, petroleum greases, tars, oils, oil-based paints, solvents, lacquers, alcohols, and flammable gases.
 - III. **Class C Fires:** fires that involve energized electrical equipment where the electrical non-conductivity of the extinguishing media is of importance. (Where electrical equipment is de-energized, fire extinguishers for class A or B fires can be used safely).
 - IV. **Class D Fires:** Fires in combustible metals, such as magnesium, titanium, zirconium, sodium, lithium, and potassium.
 - V. **Class K or Class F Fires:** Fires in cooking appliances that involve combustible cooking media (vegetable or animal oils and fats).
- a. Building presenting class B or C fire hazard shall be equipped with a regular number of class A extinguishers dedicated to the building protection, and shall also be equipped with adding class B or C extinguishers.
 - b. According to these definitions, all parts of the buildings shall be equipped with class ABC type extinguishers. In the meantime, rooms such as Storages, Workshops and other rooms housing electrical equipments, shall be equipped with adding class C extinguishers.

1.2 ABC DRY POWDER FIRE EXTINGUISHER (FIRST AID EQUIPMENT)

ABC Dry powder fire extinguishers shall be manufactured in accordance with BS EN3 for portable extinguishers and BS1866 for mobile extinguishers and shall be operated by means of a lever/squeeze handle operated valve provided with a safety pin which shall avoid inadvertent operation and the valve shall be capable for controlling partial discharge. The cylinder shall have minimum working pressure of 14bar. The temperature shall range from -20° to 60° C. The extinguisher must have a plastic base (Only for portable). It must be coated externally by electrostatic polyester powder and oven baked for corrosion resistance. The cylinder shall be made special cold rolled steel sheet with Argon/CO2 automatic welding process and the suction tube shall be made of PVC material. All extinguishers shall carry instructions for operation and maintenance in English and Arabic.

The Dry powder fire extinguishers: **The product should have at least been certified by two independent Certification Organizations from International Testing Laboratories & Accredited Certification Bodies as LPCB Approval / BSI Kitemark approval and or UL listed.**

Multipurpose Dry Chemical Extinguishers (Class A, Class B, and Class C Fires and Electrical Fires).

The extinguishers are recommended with the capacities and minimum fire rating as follows:

SL #	Capacity	Fire Rating
1	1Kg ABC Dry Powder	8A & 34B
2	21Kg ABC Dry Powder	13A & 70B
3	4 Kg ABC Dry Powder	21A & 113B
4	6 Kg ABC Dry Powder	34A & 183B
5	9 Kg ABC Dry Powder	43A & 233B
6	12 Kg ABC Dry Powder	55A & 233B
7	25 Kg Dry Powder Mobile	34A IB
8	50 Kg Dry Powder Mobile	34A IIB
9	100 Kg Dry Powder Mobile	34A IIIB
10	150 Kg Dry Powder Mobile	34A IVB

1.3 CARBON DIOXIDE FIRE EXTINGUISHER (FIRST AID EQUIPMENT)

Carbon dioxide fire extinguishers shall be manufactured in accordance with BS EN 3 (portable) and BS EN 1866 (mobile) and shall be operated by means of a lever /squeeze handle operated valve provided with a safety pin which shall avoid inadvertent operation and the valve shall be capable for controlling partial discharge. The cylinder shall be made of high grade seamless carbon steel pipe or aluminum alloy and it must be coated externally by electrostatic polyester powder quating and oven baking to resist corrosion. The cylinders shall be free from leakage during hydrostatically pressure tested to 250bar. The temperature shall range from -20° to 60° C. The suction tube shall be Aluminum. All extinguishers shall carry instructions for operation and maintenance in English and Arabic.

Typical applications are electrical & electronic equipments, food preparation areas, laboratories, and printing or duplicating areas, telephone equipment Rooms, Engine Rooms, Control Rooms, Paint spray Booths, Flammable liquid storage areas etc,. Since the agent is discharged in the form of a gas/snow cloud, it has a relatively short range of 3 ft to 8 ft (1 m to 2.4 m). This type of fire extinguisher is not recommended for outdoor use where windy conditions prevail, or for indoor use in locations that are subject to strong air currents because the agent can rapidly dissipate and prevent extinguishment. The concentration needed for fire extinguishment reduces the amount of oxygen (air) needed for life safety when the discharge is in a confined area (space).

The Carbon dioxide fire extinguishers: **The product should have at least been certified by two independent Certification Organizations from International Testing Laboratories & Accredited Certification Bodies as LPCB approval/ Kitemark approval and/or UL listed.**

The CO2 extinguishers are recommended with the capacities and minimum fire rating as follows:

SL #	Capacity	Fire Rating
1	2 Kg CO2	55B
2	5 Kg CO2	55B
3	10 Kg CO2 Mobile	Min 70B
4	20 Kg CO2 Mobile	Min 89B
5	30 Kg CO2 Mobile	Min 113B
6	50 Kg CO2 Mobile	Min 144B

1.4 WATER FIRE EXTINGUISHER (FIRST AID EQUIPMENT)

Water type Fire extinguisher shall manufactured in accordance to BS EN 3 (portable) and BS EN 1866 (mobile) and shall be operated by means of a lever operated self closing control valve to enable the discharge to be interrupted temporarily and provided with a safety pin. The cylinder shall have minimum working pressure of 14bar. The temperature shall range from +5° to 60°C. The extinguisher must have a plastic base. It must be coated inside by electrostatic PVC resin to resist corrosion. The cylinder shall be made special cold rolled steel sheet with Argon/CO2 welding process and the suction tube shall be made of Rigid PVC with filter. All extinguishers shall carry instructions for use in English and Arabic.

The water extinguishers are recommended with the capacities and minimum fire rating as follows:

SL#	Capacity	Fire Rating
1	6 ltr Water	13A
2	9 ltr Water	21A
3	20 Litre Mobile	Min 13A
4	25 Litre Mobile	Min 13A
5	45 Litre Mobile	Min 13A
6	50 Litre Mobile	Min 13A
7	90 Litre Mobile	Min 13A
8	100 Litre Mobile	Min 13A
9	135 Litre Mobile	Min 13A
10	150 Litre Mobile	Min 13A

The water type fire extinguishers: **The product should have at least been certified by two independent Certification Organizations from International Testing Laboratories & Accredited Certification Bodies as LPCB approval/ Kitemark approval and/or UL listed.**

1.5 FOAM FIRE EXTINGUISHER (FIRST AID EQUIPMENT)

Foam type Fire extinguisher shall manufactured in accordance to BS EN 3 (portable) and BS EN 1866 (mobile) and shall be operated by means of a lever operated self closing control valve to enable the discharge to be interrupted temporarily and provided with a safety pin. The cylinder shall have minimum working pressure of 14bar. The temperature shall range from +5° to 60°C. The extinguisher must have a plastic base. It must be coated inside by electrostatic PVC resin to resist corrosion. The cylinder shall be made special cold rolled

steel sheet with Argon/CO2 welding process and the suction tube shall be made of Rigid PVC with filter. All extinguishers shall carry instructions for use in English and Arabic.

The Foam extinguishers are recommended with the capacities and fire rating as follows:

SL #	Capacity	Fire Rating
1	6 ltr Foam	21A & 144B
2	9 ltr Foam	21A & 183B
3	20 Litre Foam Mobile	Min 13A IB
4	25 Litre Foam Mobile	Min 13A IB
5	45 Litre Foam Mobile	Min 13A IIB
6	50 Litre Foam Mobile	Min 13A IIB
7	90 Litre Foam Mobile	Min 13A IIIB
8	100 Litre Foam Mobile	Min 13A IIIB
9	135 Litre Foam Mobile	Min 13A IVB

The Foam type fire extinguishers: **The product should have at least been certified by two independent Certification Organizations from International Testing Laboratories & Accredited Certification Bodies as LPCB approval/ Kitemark approval and/or UL listed.**

1.6 WET CHEMICAL FIRE EXTINGUISHER (FIRST AID EQUIPMENT)

The Wet Chemical Fire Extinguishers shall have **LPCB/BSI KITE mark approved and comply with BS EN3 standards**. The cylinder shall be tested to 30 bar and shall have minimum pressure of 10 bar. The capacity shall be of 3 liters 25F/13A & 6 liters with 75F/13A ratings. They shall be provided in all the kitchens. The temperature shall range from +5° to 60°C.

The Wet Chemical extinguishers are recommended with the capacities and fire rating as follows:

SL#	Capacity	FireRating
1	3 ltr Wet Chemical	13A & 25F
2	6 ltr Wet Chemical	13A & 75F

The Wet chemical type fire extinguishers: **The product should have at least been certified by two independent Certification Organizations from International Testing Laboratories & Accredited Certification Bodies as LPCB approval/ Kitemark approval and/or UL listed.**

1.7 INSTALLATION

1. Fire extinguishers shall be conspicuously located where they will be readily accessible and immediately available in the event of fire. Preferably they shall be located along normal paths of travel, including exits from areas.
2. Fire extinguishers shall not be obstructed or obscured from view.
3. Portable fire extinguishers other than wheeled types shall be securely installed on the hanger or in the bracket supplied or placed in cabinets or wall recesses. The hanger or bracket shall be securely and properly anchored to the mounting surface in accordance with the manufacturer's instructions. Wheeled-type fire extinguishers shall be located in a designated location.
4. Fire extinguishers having a gross weight not exceeding 40 lb (18.14 kg) shall be installed so that the top of the fire extinguisher is not more than 5 ft (1.53 m) above the floor. Fire extinguishers having a

gross weight greater than 40 lb (18.14 kg) (except wheeled types) shall be so installed that the top of the fire extinguisher is not more than 3 1/2 ft (1.07 m) above the floor. In no case shall the clearance between the bottom of the fire extinguisher and the floor be less than 4 in. (10.2 cm).

5. Fire extinguishers shall be installed in a 1.2 mm thick, stainless steel, polished finish cabinets where required in finished rooms / office areas and as shown on the drawings and shall be confirmed with the Architect.



FIRE PUMPS

1.0 GENERAL

Fire pumps shall be listed for fire protection use as per NFPA 20.

Fire pump manufacturer shall bear the unit responsibility for the total packaged fire pump set. Major components of fire pump set such as fire pumps, engines and controller shall be manufactured by a single manufacturer.

Each pump with its engine or motor shall be assembled at factory. Factory performance test shall be conducted for fire pumps using actual job motor and actual job engine. Factory test shall not be conducted using suction pressure created by other systems. It shall be done at true suction pressure i.e. suction pressure based on water tank level. A certificate showing the results of this performance test shall be submitted for review and approval along with factory test curve. In addition to the performance test a string test should also be conducted at factory with actual job pumps and actual job drivers along with their actual job controllers as a packaged-unit prior to dispatch. A certificate showing the successful completion of this packaged-unit string test shall also be submitted for review & approval.

Fire Pumps should be covered by a warranty of 1 year from the date of testing and commissioning. 10 years spare parts availability should be assured by the fire pump manufacturer.

The scope of work includes design, manufacturing, shop testing and supply of the Fire Pump Set.

The fire pump shall consist of Electric Motor driven pump to act as duty pump, and Diesel Engine driven pump to act as standby pump and Electric motor driven Jockey pump.

2. ELECTRIC-DRIVE, MULTISTAGE FIRE PUMP

PART 1 -GENERAL

2.1 CODE & COMPLIANCE

- 2.1.1 All materials and workmanship described herein shall be in accordance with the latest editions and addenda of the codes and standards listed below and all State, and local codes, and/or specifications, the more stringent shall govern.

NFPA	National Fire Protection Association
ANSI	American National Standards Institute
ASTM	American Society for Testing and Materials
FM	Factory Mutual Research Corporation and manufacturer's printed recommendations.
BS	British Standard

2.2 QUALITY ASSURANCE

- 2.2.1 Firms regularly engaged in manufacture of fire pump and allied equipment of the types and sizes required, whose products have been in satisfactory use in similar service for not less than five (5) years.
- 2.2.2 The manufacturer shall supply all necessary pump accessories (controller, driver and accessories, gauges, etc.) to provide a complete pump installation, as detailed in FM Global Property Loss Prevention Data Sheets. The pump manufacturer shall be held accountable for the complete pump package and installation.
- 2.2.3 Any work or materials found to be defective or which do not meet the requirements of this specifications, shall be replaced by the Vendor at his own expense.
- 2.2.4 All materials shall be new, unused and free of defects and imperfections.
- 2.2.5 Pumps and components shall be permanently marked with the model number, type and manufacturer name or logo and FM approval mark.
- 2.2.6 Pumps shall be UL listed and FM approved.

2.3 **COORDINATION**

- 2.3.1 In selecting the units, the tenderer shall carefully check and confirm that the units can be installed and conveniently serviced and maintained within the respective spaces indicated on the drawings for unit installation.

2.4 **SUBMITTALS**

- 2.4.1 Submit manufacturer's technical product data, installation instructions and dimensioned drawings for fire pump and allied equipment.
- 2.4.2 Test Certificate shall be submitted as per NFPA Standard 20.

2.5 **DELIVERY, STORAGE & HANDLING**

- 2.5.1 Unit shall be delivered with protective crating and covering and their delivery coordinated in sufficient time to allow movement into building.
- 2.5.2 Preparation for shipment shall be in accordance with Vendor's standards. Vendor shall be solely responsible for the adequacy of the preparation for shipment provisions with respect to materials and applications.
- 2.5.3 Adequate protection shall be provided against mechanical damage and atmospheric corrosion in transit and for a 6 month period of outdoor storage at job site prior to installation. Control panel shall be protected from moisture penetration.
- 2.5.4 Miscellaneous parts shall be tagged or marked with the equipment item number for which they are intended.

2.6 **WARRANTY**

- 2.6.1 Vendor shall guarantee and provide a written statement as proof that the unit shall perform as specified on the Data Sheets.
- 2.6.2 All equipment shall be guaranteed for satisfactory performance and shall be warranted against faulty design, defective or improper materials.
- 2.6.3 The Material and Workmanship Guarantees shall be extended for a period of eighteen months from the date of shipment or one year from the date the equipment is placed in service whichever occurs first.

2.7 **FACTORY INSPECTION AND TESTING**

- 2.7.1 Each unit shall be assembled at the factory, inspected and determined to be in compliance with the specifications. Any deviations found shall be corrected. Test Certificate shall be submitted as per NFPA Standards.
- 2.7.2 Each individual pump shall be hydrostatically tested and run tested prior to shipment. The pump shall be hydrostatically tested at a pressure of not less than one and one-half times the no flow (shut off) head of the pump's maximum diameter impeller plus the maximum allowable suction head but in no case less than 250 psig.

2.8 **SPARE PARTS**

- 2.8.1 The tenderer shall include the price of complete set of spare parts for all items supplied by him required for two years operation for the specified operating hours.
- 2.8.2 Amongst other spare parts, the tenderer shall include the following without limitation.
Provide one complete set of gaskets, screens and seals for each pump type and model supplied.

PART 2 -PRODUCT

- 2.1 Pump shall be of the vertical turbine type with suction case, intermediate bowls, top bowl and discharge case made of cast iron, free of foundry imperfections and other detrimental defects.
- 2.2 Impellers shall be of bronze, of the enclosed type, statically balanced. They shall be adjustable vertically by means of an adjustable nut located on top of the driver.
- 2.3 Pump shaft shall be of polished stainless steel and shall be supported by bearings above and below each impeller. The suction case bearing shall be of bronze, grease packaged and protected against entrance of sand and other abrasive.
- 2.4 Column pipe size shall be such that the friction loss shall not exceed 5 ft./100 ft. of column. The pipe shall be furnished in interchangeable sections having a nominal length not exceed 3 meters.
- 2.5 A thrust bearings of ample capacity to carry the weight of the rotating parts plus the hydraulic thrust shall be incorporated as an integral part of the driver. The bearings shall have an average life rating of not less than 5 years continuous operation.

- 2.6 Motor – shall be drip proof, 2900 RPM and shall be especially selected for quiet operation. The horse power of the motor shall be of such a size as to insure non overloading of the motor through the capacity range of the pump.
- 2.7 Installation – A concrete base 8 inches higher than the surrounding floor shall be provided. Construct base and install as shown by the detail on the drawings and as per manufacturer's recommendations.
- 2.8 Controls – The pump shall be controlled by a combined manual and automatic across the line type fire pump control panel. The control panel shall have an externally operated circuit breaker, a pilot light to indicate when power is on, switches or push button for manual or automatic operation, a pressure switch control for automatic operation and a remote alarm signal when pump is operating. The entire control panel shall be factory assembled in a steel cabinet. The panel shall have provision for connection to the BMS. Controllers to have auto-program for start/stop for future periodic maintenance.

PART 3 -EXECUTION

2.9 Examination

- 2.9.1 Examine areas, equipment foundations, and conditions with installer present for compliance with requirements for installation and other conditions affecting pump performance. Do not proceed with installation until unsatisfactory conditions have been corrected.
- 2.9.2 Examine the installed fire fighting system, verify actual locations of piping connections prior to installation.

2.10 Concrete equipment bases

- 2.10.1 Wherever necessary, install concrete equipment bases of dimensions required for fire pumps, and controllers.

2.11 Installation

- 2.11.1 Comply with pump manufacturers written installation and alignment instructions.
- 2.11.2 Install pumps in locations indicated and arrange to provide access for periodic maintenance, including removal of motors impellers, couplings and accessories.
- 2.11.3 Set base-mounted pumps on concrete equipment bases as indicated on drawings. Disconnect couplings halves before setting. Do not reconnect couplings until alignment operations have been completed.
- a. Support pump base plate on rectangular metal blocks and shims or on metal wedges having small taper, at points near foundation bolts to provide a gap of 19 to 38 mm between pump base and foundation for grouting.
- b. Adjust metal supports or wedges until pump and driver shafts are level. Check couplings faces and pump suctions and discharge flanges to verify that they are level.
- 2.11.4. Install valves of types and at locations indicated that are same size as the piping connecting fire pump, bypasses, test headers and other piping systems.
- 2.11.5. Install suction pressure gages on pump suction and discharge at integral pressure gages, and equipment supports as indicated for complete installation.
- 2.11.6. Support pumps and piping separately so that weight of piping system does not rest on pumps.
- 2.11.7 Install piping accessories, hangers and supports, anchors, valves meters and gages, and equipment support as indicated for complete installation.
- 2.11.8 Install flow meter and sensing elements where indicated, install flow-measuring system components and make connections according to manufacturers written installation instructions.
- 2.11.9 Electrical Wiring: Install electrical devices furnished by equipment manufacturers but not specified to be factory mounted. Furnish copy of manufacturers wiring diagram submittal.
- 2.11.10 Verify that electrical wiring is installed according to manufacturer's submittal and installation requirements. Do not proceed with equipment start up until wiring installation is acceptable.

2.12 Alignment

- 2.12.1 Align pump and driver shafts after complete unit has been leveled on foundation and after grout has set and foundation bolts have been tightened.
- 2.12.2 After alignment is correct, tighten foundation bolts evenly but not too firmly. Fill base plate completely with non shrink, non metallic grout, with metal blocks and shims or wedges in place. After grout has hardened, fully tighten foundation bolts. Check alignment and take corrective measures required.

2.13 Connections

- 2.13.1 Connect suction and discharge side of the piping system to the pumps.

- 2.13.2 Connect flow-measuring and other hydronic system components according to manufacturer's written installation instructions.

3.0 DIESEL-DRIVE, VERTICAL FIRE PUMP

PART 1 -GENERAL

3.1 CODE & COMPLIANCE

- 3.1.1 All materials and workmanship described herein shall be in accordance with the latest editions and addenda of the codes and standards listed below and all State, and local codes, and/or specifications, the more stringent shall govern.

NFPA	National Fire Protection Association
ANSI	American National Standards Institute
ASTM	American Society for Testing and Materials
FM	Factory Mutual Research Corporation and manufacturer's printed recommendations.
BS	British Standard

3.2 QUALITY ASSURANCE

- 3.2.1 Firms regularly engaged in manufacture of fire pump and allied equipment of the types and sizes required, whose products have been in satisfactory use in similar service for not less than five (5) years.
- 3.2.2 The manufacturer shall supply all necessary pump accessories (controller, driver and accessories, gauges, etc.) to provide a complete pump installation, as detailed in FM Global Property Loss Prevention Data Sheets. The pump manufacturer shall be held accountable for the complete pump package and installation.
- 3.2.3 Any work or materials found to be defective or which do not meet the requirements of this specifications, shall be replaced by the Vendor at his own expense.
- 3.2.4 All materials shall be new, unused and free of defects and imperfections.
- 3.2.5 Pumps and components shall be permanently marked with the model number, type and manufacturer name or logo and FM approval mark.
- 3.2.6 Pumps shall be UL listed and FM approved.

3.3 COORDINATION

- 3.3.1 In selecting the units, the tenderer shall carefully check and confirm that the units can be installed and conveniently serviced and maintained within the respective spaces indicated on the drawings for unit installation.

3.4 SUBMITTALS

- 3.4.1 Submit manufacturer's technical product data, installation instructions and dimensioned drawings for fire pump and allied equipment.
- 3.4.2 Test Certificate shall be submitted as per NFPA Standard 20.

3.5 DELIVERY, STORAGE & HANDLING

- 3.5.1 Unit shall be delivered with protective crating and covering and their delivery coordinated in sufficient time to allow movement into building.
- 3.5.2 Preparation for shipment shall be in accordance with Vendor's standards. Vendor shall be solely responsible for the adequacy of the preparation for shipment provisions with respect to materials and applications.
- 3.5.3 Adequate protection shall be provided against mechanical damage and atmospheric corrosion in transit and for a 6 month period of outdoor storage at job site prior to installation. Control panel shall be protected from moisture penetration.
- 3.5.4 Miscellaneous parts shall be tagged or marked with the equipment item number for which they are intended.

3.6 WARRANTY

- 3.6.1 Vendor shall guarantee and provide a written statement as proof that the unit shall perform as specified on the Data Sheets.
- 3.6.1 All equipment shall be guaranteed for satisfactory performance and shall be warranted against faulty design, defective or improper materials.

- 3.6.2 The Material and Workmanship Guarantees shall be extended for a period of eighteen months from the date of shipment or one year from the date the equipment is placed in service whichever occurs first.

3.7 FACTORY INSPECTION AND TESTING

- 3.7.1 Each unit shall be assembled at the factory, inspected and determined to be in compliance with the specifications. Any deviations found shall be corrected. Test Certificate shall be submitted as per NFPA Standards.
- 3.7.2 Each individual pump shall be hydrostatically tested and run tested prior to shipment. The pump shall be hydrostatically tested at a pressure of not less than one and one-half times the no flow (shut off) head of the pump's maximum diameter impeller plus the maximum allowable suction head but in no case less than 250 psig.

3.8 SPARE PARTS

- 3.8.1 The tenderer shall include the price of complete set of spare parts for all items supplied by him required for two years operation for the specified operating hours.
- 3.8.2 Amongst other spare parts, the tenderer shall include the following without limitation. Provide one complete set of gaskets, screens and seals for each pump type and model supplied.

3.9 PRODUCT

- 3.9.1 Fire pump shall constitute a diesel engine and fire pump of vertical turbine design.
- 3.9.2 Pump shall be of the vertical turbine type with suction case, intermediate bowls, top bowl and discharge case made of cast iron, free of foundry imperfections and other detrimental defects.
- 3.9.3 Impellers shall be of bronze, of the enclosed type, statically balanced. They shall be adjustable vertically by means of an adjustable nut located on top of the driver.
- 3.9.4 Pump shaft shall be of polished stainless steel and shall be supported by bearings above and below each impeller. The suction case bearing shall be of bronze, grease packaged and protected against entrance of sand and other abrasive.
- 3.9.5 Column pipe size shall be such that the friction loss shall not exceed 5 ft./100 ft. of column. The pipe shall be furnished in interchangeable sections having a nominal length not exceed 3 meters.
- 3.9.6 A thrust bearings of ample capacity to carry the weight of the rotating parts plus the hydraulic thrust shall be incorporated as an integral part of the driver. The bearings shall have an average life rating of not less than 5 years continuous operation.
- 3.9.7 Pump speed shall be according to the diesel engine speed and shall be especially selected for quiet operation. The engine shall be of such a size as to insure non overloading through the capacity range of the pump.
- 3.9.8 Installation – A concrete base 8 inches higher than the surrounding floor shall be provided. Construct base and install as shown by the detail on the drawings and as per manufacturer's recommendations.
- 3.9.9 Controls – The pump shall be controlled by a combined manual and automatic across the line type fire pump control panel. The control panel shall have an externally operated circuit breaker, a pilot light to indicate when power is on, switches or push button for manual or automatic operation, a pressure switch control for automatic operation and a remote alarm signal when pump is operating. The entire control panel shall be factory assembled in a steel cabinet. The panel shall have provision for connection to the BMS. Controllers to have auto-program for start/stop for future periodic maintenance.
- 3.9.10 Rating of diesel engine shall be chosen at the prevailing conditions at ambient temperature of 120 deg F. and a humidity ratio of more than 90%.
- 3.9.11 The diesel engine for driving the fire pump shall be UL listed, FM approved specifically designed for fire pump service. It shall develop sufficient horse power to drive the pump and shall have bare engine brake horse power not less than 20 percent greater than the maximum horsepower required to drive the pump at its rated RPM.
- 3.9.12 The engine shall be an open type self contained power unit including the following accessories :-
- 3.9.12.1 Stub shaft
- 3.9.12.2 Fuel pumps and filters.
- 3.9.12.3 Air Cleaner.
- 3.9.12.4 Adjustable governor capable of regulating engine speed within a range of 10 percent between shut-off and maximum load conditions. The governor shall be set to maintain rated pump speed at maximum pump load.

- 3.9.12.5 Lube oil pump.
- 3.9.12.6 Lube oil filter.
- 3.9.12.7 Overspeed shut down device arranged to shut down engine at a speed approximately 20 percent above rated engine speed and with manual reset. The overspeed shut down device shall be so supervised that the automatic engine controller will continue to show an overspeed trouble signal until the device is manually reset to normal operating position.
- 3.9.12.8 Electric Starter, generator or alternator and voltage regulator.
- 3.9.12.9 Dual storage battery units. Each unit shall have capacity to maintain cranking speed through a 6 minute cycle (15 seconds cranking and 15 seconds rest, in 12 consecutive cycles). Provide certification that the battery unit complies with this requirement. Batteries shall be furnished in a dry charge condition, with electrolyte liquid in a separate container. Electrolyte shall be added at the time the engine is put in service, and the battery is given a conditioning charge.
- 3.9.12.10 Dual battery chargers, automatically controlled.
- (i) Chargers shall be specifically designed for fire pump service.
 - (ii) The rectifier shall be a semiconductor type.
 - (iii) The charger for a lead acid battery shall be a type which automatically reduces the charging rate to less than 500 milliamperes when the battery reaches a full charge condition.
 - (iv) The control equipment incorporated in an "off-on" type of charger for a lead acid battery shall start the rectifier hourly. It shall shut off automatically when the battery has been fully charged.
 - (v) The battery charger at its rated voltage shall be capable of so delivering energy into a full discharged battery in such a manner that it will not damage the battery. It shall restore to the battery 100 percent of the battery's ampere hour rating within 24 hours.
 - (vi) An ammeter with an accuracy of five percent of the normal charging rate shall be furnished to indicate the operation of the charger.
 - (vii) The charger shall be designed so that it will not be damaged or blow fuses during the cranking cycle of the engine when operated by an automatic or manual controller.
 - (viii) A manual charge switch with indicator light shall be provided or, in lieu thereof, the charger shall automatically charge at the maximum rate whenever required by the state of charge of the battery.
 - (ix) When not connected through a control panel it shall be arranged to indicate loss of current output on the load side of the d-c over-current protective device.
 - (x) Storage batteries shall be substantially supported, secured against displacement, and located where they shall be readily accessible for servicing.
- 3.9.12.11 Engine cooling system shall be by unit mounted radiator.
- 3.9.12.12 Engine jacket temperature regulating device.
- 3.9.12.13 Engine jacket water pump.
- 3.9.12.14 Instrument panel with water temperature gauge, oil pressure gauge, ammeter, tachometer of the totalizing type or an hour meter.
- 3.9.12.15 Exhaust mufflers and flexible connectors, stainless steel non-sparking.
- 3.9.12.16 The engine shall be arranged for automatic operation with all wiring terminating in a junction box suitable for field hook-up.
- 3.9.12.17 Exhaust piping from flexible connection at engine to muffler and from muffler to point of discharge to atmosphere shall be schedule 40 black steel pipe and fittings, welded. A flap valve shall be provided at the external end of piping.
- 3.9.12.18 Pump and diesel engine shall be mounted on a common base plate, direct connected through a flexible coupling, and equipped with a coupling guard. Coupling shall be a spacer type to permit pump shaft and impellers to be removed without moving diesel engine or pump bottom casing.
- 3.9.13 Automatic Diesel Engine Control Panel**
- 3.9.13.1 The automatic engine control panel shall be UL listed, FM approved specifically designed for fire pump service. The panel shall be enclosed in a floor mounted duct and moisture resistant housing. The controller shall have possibility to program the start/stop of equipment for periodic maintenance.
- 3.9.13.2 The controller shall be the combined manual and automatic type including a "Manual-Off-Automatic selector switch.

3.9.13.3 A pressure switch and single phase power failure relay with local visual indication and auxiliary contact for remote indication of 240 volt AC power failure shall be provided which will, upon drop in system pressure or loss of single phase AC power, cause the controller to open or close all circuits necessary to automatically start the engine by performing the following functions :-

- (a) Turn on fuel supply.
- (b) Crank the engine in a series of crank-rest cycles automatically alternating between the dual batteries on each cycle.
- (c) Disconnect cranking motor upon engine start.
- (d) Once the engine is started, it shall remain in operation until shut down manually. A signal in the Synoptic Panel shall indicate that the Diesel Engine is operating.
- (e) If the engine should fail to start after several cranking attempts, the controller shall disconnect all starting circuits and energize an audible and visual alarm to indicate "overcrank" in the synoptic panel as well as in the engine control panel.

3.9.13.4 Alarm and Signal Devices on Controller

- (a) A pilot lamp(s) shall be provided to indicate that the controller is in the "automatic" position. The lamp shall be accessible for replacement.
- (b) Separate pilot lamps and a common bell shall be provided to indicate trouble caused by:-
- (c) Low oil pressure in the lubrication system. The controller shall provide means for testing the position of the pressure switch contacts without causing trouble alarms.
- (d) Failure of engine to start automatically.
- (e) Shutdown from overspeed.
- (f) Battery failure. Each battery shall be provided with separate lamps which shall be lighted or extinguished on battery failure on attempt to start.
- (g) High engine jacket water temperature.

3.9.13.5 Water Pressure Control : A pressure actuated switch having high and low calibrated adjustments and responsive to water pressure in the fire system shall be provided in the controller circuit. Suitable provision shall be made for relieving pressure to the pressure switch to test the operation of the controller and the pump.

3.9.13.6 Enclosure : The structure or panel shall be securely mounted in an enclosure(s) which will protect the equipment against mechanical injury and falling drops of water striking the enclosure from the downward vertical. All switches required to keep the controller in the "automatic" position shall be within the locked cabinets having to break glass panels.

3.9.13.7 A wiring diagram shall be provided and permanently attached to the inside of the enclosure showing exact wiring for the controller, including a legend of identifying numbers of individual components. All wiring terminals shall be plainly marked to correspond with the wiring diagram furnished.

3.9.13.8 Wiring Elements : Wiring elements of the controller shall be designed on a continuous-duty basis.

3.9.13.9 Field Wiring : All wiring leading from the panel to the engine and to the batteries shall have adequate carrying capacity. Such wiring shall be protected against mechanical injury. Controller manufacturer's specifications for distance and wire size shall be followed. Provision for connection to BMS shall be provided.

3.9.13.10 Marking : Each operating component of the controller shall be marked to plainly indicate an identifying number referenced to the wiring diagram. The markings shall be located so as to be visible after installation.

3.9.13.11 Instructions : Complete instructions covering the operation of the controller shall be provided and conspicuously mounted on the controller.

3.9.14 Fuel Storage Tank

3.9.14.1 The fuel storage tank shall be an integral part of the unit and shall be mounted within the support base of the unit. The tank shall have a capacity of 8 hours operation. Provide all piping, fittings and valves required.

3.9.14.2 The unit support base shall be designed to allow space for oil pan clearance.

3.9.15 Tests

3.9.15.1 Prior to shipment, the pump and engine shall be thoroughly shop-tested as a unit by the pump manufacturer. A characteristic curve showing the pump performance based upon the results of the shop test shall be submitted to the consultant prior to shipment, and shall include the performance pump and

engine at the set-governor's speed. Absence of these test reports, indicating that the complete unit has been tested in the factory by the driver, will be ample grounds for the equipment rejection.

3.9.15.2 Each control panel shall be fully assembled and factory tested by the control manufacturer prior to their shipment, and evidence of this test shall be furnished with the controller.

3.9.15.3 Field Acceptance Tests

- (a) The field acceptance test results shall be as good as the manufacturer's certified shop test characteristic curve for the pump being tested within the accuracy limits of the test equipment.
- (b) Overheating : The pump shall operate at peak load conditions without objectionable heating of the bearings or of the prime mover. The operating pump speed shall be the speed at which the pumping unit would be expected to operate during a fire.
- (c) The engine shall not show signs of overload or stress and its governor shall properly regulate the speed.
- (d) Controllers:
 - (i) The automatic controller shall be put through not less than ten (10) automatic and ten (10) manual operations.
 - (ii) A running interval of at least five minutes at full speed shall be allowed before repeating the starting cycle.
 - (iii) In an automatic operation, the controller shall start the pump from all the provided starting features, such as pressure switches, valves, etc.
 - (iv) Length of Test. The pump shall be in operation not less than one hour (total time) during the foregoing tests.

3.9.16 Single Source Responsibility :

3.9.16.1 The pump, driver, and all controls and necessary attachments, specified herein, shall be purchased under a unit contract. The pump manufacturer shall assume unit responsibility and shall provide the services of a qualified Engineer to supervise the installation and alignment of equipment.

3.10 EXECUTION

3.10.1 Examination

- 3.10.1.1 Examine areas, equipment foundations, and conditions with installer present for compliance with requirements for installation and other conditions affecting pump performance. Do not proceed with installation until unsatisfactory conditions have been corrected.
- 3.10.1.2 Examine the installed fire fighting system, verify actual locations of piping connections prior to installation.

3.10.2 Concrete equipment bases

3.10.2.1 Wherever necessary, install concrete equipment bases of dimensions required for fire pumps, and controllers.

3.10.3 Installation

- 3.10.3.1 Comply with pump manufacturers written installation and alignment instructions.
- 3.10.3.2 Install pumps in locations indicated and arrange to provide access for periodic maintenance, including removal of motors impellers, couplings and accessories.
- 3.10.3.3 Set base-mounted pumps on concrete equipment bases as indicated on drawings. Disconnect couplings halves before setting. Do not reconnect couplings until alignment operations have been completed.
- (a) Support pump base plate on rectangular metal blocks and shims or on metal wedges having small taper, at points near foundation bolts to provide a gap of 19 to 38 mm between pump base and foundation for grouting.
- (b) Adjust metal supports or wedges until pump and driver shafts are level. Check couplings faces and pump suctions and discharge flanges to verify that they are level.
- 3.10.3.4 Install valves of types and at locations indicated that are same size as the piping connecting fire pump, bypasses, test headers and other piping systems.
- 3.10.3.5 Install suction pressure gages on pump suction and discharge at integral pressure gages, and equipment supports as indicated for complete installation.
- 3.10.3.6 Support pumps and piping separately so that weight of piping system does not rest on pumps.
- 3.10.3.7 Install piping accessories, hangers and supports, anchors, valves meters and gages, and equipment support as indicated for complete installation.
- 3.10.3.8 Install flow meter and sensing elements where indicated, install flow-measuring system components and make connections according to manufacturers written installation instructions.
- 3.10.3.9 Electrical Wiring: Install electrical devices furnished by equipment manufacturers but not specified to be factory mounted. Furnish copy of manufacturers wiring diagram submittal to the consultant.

3.10.3.10 Verify that electrical wiring is installed according to manufacturer's submittal and installation requirements. Do not proceed with equipment start up until wiring installation is acceptable.

3.10.4 Alignment

3.10.4.1 Align pump and driver shafts after complete unit has been leveled on foundation and after grout has set and foundation bolts have been tightened.

3.10.4.2 After alignment is correct, tighten foundation bolts evenly but not too firmly. Fill base plate completely with non shrink, non metallic grout, with metal blocks and shims or wedges in place. After grout has hardened, fully tighten foundation bolts. Check alignment and take corrective measures required.

4. JOCKEY PUMP

PART 1 -GENERAL

4.1 CODE & COMPLIANCE

4.1.1 All materials and workmanship described herein shall be in accordance with the latest editions and addenda of the codes and standards listed below and all State, and local codes, and/or specifications, the more stringent shall govern.

NFPA	National Fire Protection Association
ANSI	American National Standards Institute
ASTM	American Society for Testing and Materials
BS	British Standard

4.2 QUALITY ASSURANCE

4.2.1 Firms regularly engaged in manufacture of fire pump and allied equipment of the types and sizes required, whose products have been in satisfactory use in similar service for not less than five (5) years.

4.2.2 The manufacturer shall supply all necessary pump accessories (controller, driver and accessories, gauges, etc.) to provide a complete pump installation. The pump manufacturer shall be held accountable for the complete pump package and installation.

4.2.3 Any work or materials found to be defective or which do not meet the requirements of this specifications, shall be replaced by the Vendor at his own expense.

4.2.4 All materials shall be new, unused and free of defects and imperfections.

4.2.5 Pumps and components shall be permanently marked with the model number, type and manufacturer name or logo.

4.3 COORDINATION

4.3.1 In selecting the units, the tenderer shall carefully check and confirm that the units can be installed and conveniently serviced and maintained within the respective spaces indicated on the drawings for unit installation.

4.4 SUBMITTALS

4.4.1 Submit manufacturer's technical product data, installation instructions and dimensioned drawings for fire pump and allied equipment.

4.4.2 Test Certificate shall be submitted as per NFPA Standard 20.

4.5 DELIVERY, STORAGE & HANDLING

4.5.1 Unit shall be delivered with protective crating and covering and their delivery coordinated in sufficient time to allow movement into building.

4.5.2 Preparation for shipment shall be in accordance with Vendor's standards. Vendor shall be solely responsible for the adequacy of the preparation for shipment provisions with respect to materials and applications.

4.5.3 Adequate protection shall be provided against mechanical damage and atmospheric corrosion in transit and for a 6 month period of outdoor storage at job site prior to installation. Control panel shall be protected from moisture penetration.

4.5.4 Miscellaneous parts shall be tagged or marked with the equipment item number for which they are intended.

4.6 WARRANTY

4.6.1 Vendor shall guarantee and provide a written statement as proof that the unit shall perform as specified on the Data Sheets.

- 4.6.2 All equipment shall be guaranteed for satisfactory performance and shall be warranted against faulty design, defective or improper materials.
- 4.6.3 The Material and Workmanship Guarantees shall be extended for a period of eighteen months from the date of shipment or one year from the date the equipment is placed in service whichever occurs first.
- 4.7 FACTORY INSPECTION AND TESTING**
- 4.7.1 Each unit shall be assembled at the factory, inspected and determined to be in compliance with the specifications. Any deviations found shall be corrected. Test Certificate shall be submitted as per NFPA Standards.
- 4.7.2 Each individual pump shall be hydrostatically tested and run tested prior to shipment. The pump shall be hydrostatically tested at a pressure of not less than one and one-half times the no flow (shut off) head of the pump's maximum diameter impeller plus the maximum allowable suction head but in no case less than 250 psig.
- 4.8 SPARE PARTS**
- 4.8.1 The tenderer shall include the price of complete set of spare parts for all items supplied by him required for two years operation for the specified operating hours.
- 4.8.2 Amongst other spare parts, the tenderer shall include the following without limitation.
Provide one complete set of gaskets, screens and seals for each pump type and model supplied.
- 4.9 PRODUCT**
- 4.9.1 The scope of work shall cover the supply, design, manufacture, testing in the factory, packing, insurance, delivery to site, unloading, installation, testing and commissioning of **electric motor driven vertical inline jockey pump** with all the necessary accessories in accordance with this technical specification.
- 4.9.2 Type - The jockey pump shall be vertical inline, especially designed and constructed for quiet operation.
- 4.9.3 Motor - Shall be drip proof, 2900 RPM and shall be especially selected for quiet operation. The horse-power of the motor shall be of such a size as to insure non-overloading of the motor throughout the capacity range of the pump.
- 4.9.4 Controls - The jockey pump shall have a relief valve set for the proper pressure and a pressure switch for automatic operation. There shall be a magnetic across-the-line starter with "Hand-Off-Automatic" selector switch in cover provided to start and stop the pump. Provision for connection to BMS shall be provided.
- 4.9.5 Single Source Responsibility :
- 4.9.5.1 The pump, driver, and all controls and necessary attachments, specified herein, shall be purchased under a unit contract. The pump manufacturer shall assume unit responsibility and shall provide the services of a qualified Engineer to supervise the installation and alignment of equipment.
- 4.9.5.2 Field tests shall be conducted in the field to satisfy the NFPA requirements and Local Authorities having jurisdiction of satisfactory operation of pump, driver and controller. The pump manufacturer's engineer shall be at the job site, supervising the installation and testing. The control manufacturer shall have their representative at the job site to train operators in the use of the controls. All costs related to the services shall be borne by the Contractor.
- 4.9.6 Provide pump lifting arrangement at the slab level above pumps, as per the recommendation of the manufacturer.
- 4.10 EXECUTION**
- 4.10.1 Examine areas, equipment foundations, and conditions with installer present for compliance with requirements for installation and other conditions affecting pump performance. Do not proceed with installation until unsatisfactory conditions have been corrected.
- 4.10.2 Examine the installed fire fighting system, verify actual locations of piping connections prior to installation.
- 4.10.3 Wherever necessary, install concrete equipment bases of dimensions required for fire pumps, and controllers.
- 4.10.4 Comply with pump manufacturers written installation and alignment instructions.
- 4.10.5 Install pumps in locations indicated and arrange to provide access for periodic maintenance, including removal of motors impellers, couplings and accessories.
- 4.10.6 Install valves of types and at locations indicated that are same size as the piping connecting fire pump, bypasses, test headers and other piping systems.
- 4.10.7 Install suction pressure gages on pump suction and discharge at integral pressure gages, and equipment supports as indicated for complete installation.

- 4.10.8 Support pumps and piping separately so that weight of piping system does not rest on pumps.
- 4.10.9 Install piping accessories, hangers and supports, anchors, valves meters and gages, and equipment support as indicated for complete installation.
- 4.10.10 Install flow meter and sensing elements where indicated, install flow-measuring system components and make connections according to manufacturers written installation instructions.
- 4.10.11 Electrical Wiring: Install electrical devices furnished by equipment manufacturers but not specified to be factory mounted. Furnish copy of manufacturers wiring diagram submittal to the Consultant.
- 4.10.12 Verify that electrical wiring is installed according to manufacturer's submittal and installation requirements. Do not proceed with equipment start up until wiring installation is acceptable.
- 4.10.13 Connect suction and discharge side of the piping system to the pumps.
- 4.10.14 Connect flow-measuring and other hydronic system components according to manufacturer's written installation instructions.
- 4.10.15 Contractor to ensure services of pump manufacturer representative to supervise field assembly of components, installation of fire fighting pump units, including piping and electrical connections and field acceptance tests. Report test results in writing.

2.1.1 FIRE PUMP ACCESSORIES

The Fire Pumps shall be provided with the following accessories, as required by NFPA 20:

1. Flow Metering Device (One for complete fire pump package unit). It shall be of the Venturi type.
2. Air release valve (for each Horizontal Split Case fire pumps only).
3. Suction pressure gauge (for each fire pumps).
4. Discharge pressure gauge (for each fire pumps).
5. Main relief valve (for diesel engine driven fire pumps only).
6. Casing circulation relief valve (for electric motor driven fire pumps only).

2.2 ELECTRIC MOTOR DRIVE FOR FIRE PUMP

On installations where the Fire Pump is to be operated electrically, the electric motor shall be **UL listed**, wound for 415 V, 3 phase, 50 Hz (cycle) current. The motor capacity in horsepower shall be such that the maximum motor current in any phase under any condition of pump load and voltage unbalance shall not exceed the motor-rated full-load current multiplied by the service factor.

Fire pump electric motor ambient temperature rating shall be 40 deg. C and it shall be capable of operating continuously with an overload of 15% without stress or excessive rise in temperature. The locked rotor current shall not exceed the values specified in NFPA 20.

Bearings shall be anti-friction ball or roller type.

2.3 FIRE PUMP CONTROLLER FOR ELECTRIC MOTOR DRIVE

Enclosure rating shall be NEMA 2.

The Main Electric Fire Pump Controller shall be listed for fire protection use as per NFPA 20 and shall be microprocessor based with star-delta open transition type starter.

It shall have the following features:

- Voltage Surge Protector.

- Main disconnecting switch with rotary handle sized for disconnecting motor horsepower and voltage.
- Circuit breaker with shunt trip coil and rotary handle.
- Rated motor contactors.
- Emergency run mechanism to mechanically close motor contactor contacts to start motor in case of emergency.
- Manual start and stop push buttons.
- 20x4 LCD display showing all system parameters and variables.
- Software which would enable user to program all related parameters like timers, pressure, inputs and outputs.
- 3 phase line voltages indication.
- 3 phase line motor currents indication.
- Should indicate real value of discharge pressure.
- Should indicate motor running hours.
- Should indicate ambient temperature.
- Starting delay timer (should be programmable) for sequence multiple pumps starting.
- Automatic or manual shutdown after automatic start (should be selectable).
- Automatic shutdown mode enabled/disabled indicator.
- Power ON/Healthy indicator.
- Automatic mode indicator.
- Manual mode indicator.
- Motor Run indicator/free contact.
- Phase Loss/Failure alarm / indicator / free contact.
- Phase sequence error alarm/indicator / free contact.
- System error alarm / indicator / free contact.
- Fail to start alarm / indicator.
- System error alarm / indicator.
- Push button for alarm silence (should be only for additional alarms).

- Motor locked rotor trip at 600% of FLA, after programmable time delay.
- Programmable automatic weekly test, which should be able to program to start at any desired day, hour and minute in the week.
- Manual test push button.
- 10 additional programmable indicators (LED's).
- 10 additional programmable auxiliary digital inputs.
- 7 additional programmable output relays (free contacts).
- Pressure transducer with analog voltage output.
- Data logging system for pressure and events.
- USB port for saving recorded pressure and events on USB memory, and can be viewed with MS word and Excel.
- Electrically actuated built in discharge solenoid valve.
- Electrical alarm bell.

Controller shall be rated for 415 VAC, 3 Phase, and 50 Hz power supply.

Controller shall be completely wired, assembled, programmed and tested at the factory before shipment, and ready for immediate installation.

2.3.1 FUEL SYSTEM

A fuel tank with fittings shall be furnished in accordance with NFPA 20.

It shall have sufficient capacity for 8 hours running at full load; equal to 1 gallon/horse power + 5 % volume for expansion + 5% volume for sump.

2.4 FIRE PUMP CONTROLLER FOR DIESEL ENGINE DRIVE

Enclosure rating shall be NEMA 2.

The standby diesel fire pump controller shall be listed for fire protection use as per NFPA 20 and shall be microprocessor based. It shall have the following features:

- System error alarm / indicator.
- AC power healthy indicator.
- DC power healthy indicator / free contact.
- Speed switch error or failed while running alarm / indicator.

- Automatic shutdown enabled/disabled indicator.
- 10 additional programmable indicators.
- 11 additional programmable digital inputs.
- 8 additional programmable output relays (free contacts).
- Common alarm free contact.
- Electrical alarm bell.
- Push button for alarm silence (should be only for additional alarms).
- Pressure transducer with analog voltage output.
- Data logging system for pressure and events.
- USB port for saving recorded pressure and events on USB memory, and can be viewed with MS word and Excel.
- Electrically actuated built in discharge solenoid valve.
- 220 VAC main line system power.
- 12 or 24 VDC system control voltage.
- Automatic / Manual / OFF mode (should be selectable).
- Two 12 / 24 VDC, 10 Amps smart automatic microprocessor battery chargers.
- Protection circuit breakers before and after battery chargers.
- 20x4 LCD display showing all system parameters and variables.
- Software which would enable user to program all related parameters like timers, pressure, inputs and outputs.
- Battery 1 and 2 voltages indication.
- Battery 1 and 2 charging currents indication.
- Should indicate real value of discharge pressure.
- Should indicate engine running hours.
- Should indicate ambient temperature.
- Automatic cranking from two different batteries.
- Two push buttons for manual cranking from two different batteries.
- Manual stop push button.

- Programmable automatic weekly test, which should be able to program to start at any desired day, hour and minute in the week.
- Manual test push button.
- Starting delay timer (programmable) for sequence multiple pumps starting.
- Automatic or manual shutdown after automatic start (should be selectable).
- Should be compatible with fuel solenoid engines and stopper solenoid engines.
- Engine run indicator / free contact.
- Test running indicator / free contact.
- Automatic mode indicator / free contact.
- Manual mode indicator.
- Battery Charger1 error indicator.
- Battery Charger2 error indicator.
- Battery1 failure alarm / indicator.
- Battery2 failure alarm / indicator.
- Low oil pressure alarm / indicator.
- High water temperature alarm / indicator.
- Over speed shut down alarm / indicator.
- Engine failed to start alarm / indicator.
- Cranking coil 1 failure alarm / indicator.
- Cranking coil 2 failure alarm / indicator.
- Two storage battery units. Each battery unit shall have twice the capacity sufficient to maintain the cranking speed recommended by the engine manufacturer through a 3-minute attempt-to-start cycle, which is six consecutive cycles of 15 seconds of cranking and 15 seconds of rest.

Controller shall be rated for 12 or 24 volts DC operating voltage, 220 volts AC input voltage and 50 Hz power supply.

Controller shall be completely wired, assembled, programmed and tested at the factory before shipment, and ready for immediate installation.

2.5 JOCKEY PUMP CONTROLLER

Enclosure rating shall be NEMA 2.

The Jockey Pump Controller shall be listed and shall be relay logic based. Starting type shall be direct on line. Jockey controller shall have the following features:

- Main disconnected switch with rotary handle, sized for disconnecting motor horsepower and voltage.
- Motor Starter rated to motor's horse power, with thermal and short circuit protection.
- Rated motor contactors.
- Circuit breaker for protection of control circuit.
- Selector switch for Hand—Off—Automatic operation.
- Star—Delta starting timer (for controllers above 4 kW).
- Power ON/Healthy indicator/free contacts.
- Pump Run indicator/free contact.
- Pump Trip indicator/free contact.
- Adjustable Pressure Switch or Pressure Transducer.

Controller shall be rated for 415 VAC, 3 Phase, and 50 Hz power supply.

Controller shall be completely wired, assembled, programmed and tested at the factory before shipment, and ready for immediate installation.

Jockey Pump shall run either manually (by turning the selector switch to hand mode) or automatically. In auto mode, the Jockey Pump shall run automatically due to pressure fall in discharge line and it shall stop automatically when the pressure reaches to the set point. Pressure Switch or Pressure Transducer provided inside the jockey controller should be set to the required points for this

EMERGENCY LIGHTING SYSTEM

1.1 SCOPE OF WORK

Design, supply, install, test and commission Emergency lighting systems. The systems shall provide emergency supply to emergency mains luminaries during mains supply or local lighting circuit failure. Design and installation of the system shall be as per the relevant British Standards, and the local Civil Defense authority. It is the contractor's responsibility to produce all necessary submittals and Civil Defense approved / stamped shop drawings before ordering of equipment.

1.2 SYSTEM DESIGN

The Emergency lighting equipment shall be designed and manufactured as per relevant BS, EN, and ICEL standards in an ISO9001 / 9002 certified facility in the UK. All the equipments and accessories shall be in accordance with **SGS/Kitemark/CE standards**.

The contractor shall ensure that the final system shall include the following design requirements in addition to that shown on the tender drawings:

1. Escape routes up to 2m wide and permanently unobstructed shall be illuminated to a minimum of 0.2 lux.
2. Escape routes up to 2m wide with risk of being obstructed shall be illuminated to a minimum of 1 lux along the center line.
3. Escape routes wider than 2m wide shall considered as open areas.
4. Open areas shall be illuminated to a minimum of 0.5 lux anywhere in the area.
5. High risk work areas shall be illuminated to 10% of the normal lighting level.
6. In addition emergency luminaries shall be sited as follows
 1. Near changes of level
 2. Near each change of direction
 3. To illuminate Exit doors and safety signs
 4. Near firefighting equipment and call points
 5. Near every final Exit
 6. Near each first aid point
 7. Near each intersection of corridors

All Exit and converted luminaries shall be maintained type, and Emergency / Safety luminaries shall of non-maintained type. The battery capacity shall be calculated to provide a minimum of 3 hours operation at 100% light output. Where indicated on the drawings, local lighting circuits shall be monitored for failure via sub-circuit monitoring / changeover relays. Refer to the tender drawings for a list of luminaries which need to be converted for emergency operation using appropriate conversion modules.

1.3 PHILOSOPHY OF OPERATION

1) CENTRAL BATTERY

The central battery units shall constantly monitor their mains charging input for failure. In the case of maintained luminaries, these shall be fed via a step down maintaining transformer in during mains healthy condition. In the event of mains failure, both the maintained and non-maintained circuits shall be switched

over to the emergency supply via appropriately rated contactors. As shown on the drawings, local lighting circuits on each floor shall be monitored by sub-circuit changeover relays located in the emergency EMDB's. In case of failure of any circuit due to trip or loss of power in a particular area, the emergency luminaries for only that area shall be switched on.

2) SELF CONTAINED

Luminaries providing maintained or non-maintained emergency lighting in which all elements, such as battery, the lamp the control unit and the test and monitoring facilities, where provided are contained within the luminaire. The self contained units shall constantly monitor their mains charging input for failure. In the case of maintained luminaries, these shall be fed via a local power supply in during mains healthy condition. In the event of mains failure, both the maintained and non-maintained circuits shall be switched over to the self battery found on each luminaire.

1.4 DESCRIPTION OF COMPONENTS

The light fixtures covered by the present specification refer to all expansion project buildings and include the following types:

- Safety lighting fixtures equipped with fluorescent and halogen lamps
- They and their various components must comply with the following standards:
- IEC.529 : degrees of protection provided by enclosures,
- IEC.536 : protection against electric shock: safety rules;
- IEC.695.2.1 : glow wire test – flammability and extinguishability,
- IEC.79.0 and 79.12: electrical apparatus for explosive atmospheres;
- IEC.400 : lamp- and starter-holders for tubular fluorescent lamps;
- IEC.64: tungsten filament lamps for general lighting purposes.
- The lighting fixtures and their various components must also comply with the Fire Safety Regulations.
- The supply voltage is 220 volts AC for all the fittings.
- All the fittings are equipped with their lamp(s).
- The caps and terminal blocks are easily replaceable.

1.5 CENTRAL BATTERY UNIT

The central battery cabinet shall be aesthetically pleasing and made of zinc coated steel, paint finished in stove enamel epoxy gloss. The system shall be dual output to accommodate both maintained and non-maintained luminaries. The CBU shall incorporate deep-discharge protection to prevent permanent damage to the batteries by over discharge. The following minimum specifications are required:

Input Voltage	: 230 VAC +/- 6%, 50 Hz
Output (mains healthy)	: 110 VAC, 50Hz
Output (emergency)	: 110 VDC
Charger	: Thyristor controller electronic charger with electronic current limit and temperature compensation
Protection	: Short circuit, low battery voltage, and reverse battery connection
Recharge time	: 80% within 14 hours, full recharge within 24 hours
End of discharge	: Battery voltage not less than 85% of rated nominal (BS5266)
Fuses	: Industrial HRC fuses complying with BS88
Contactors	: Complying with BS5424
Indicators	: Mains healthy, charge fail, charge ammeter, battery voltmeter

Systems with 220 Volts DC output shall not be acceptable due to high risk of injury in case of accidental contact during installation or routine maintenance.

Batteries shall be Sealed Valve Regulated Lead Acid type with 10 years design life at 30°C ambient temperature. These shall be located in separate compartment inside the central battery cabinet.

The central battery units shall be located in well ventilated rooms maintained at required maximum temperature.

1.6 SELF CONTAINED

The self-contained luminaires emergency lighting systems shall be capable of maintaining the stipulated lighting levels for a period of not less than one hour (rated duration).

Input voltage	: 220VAC -230VAC 50Hz in normal case
	: 1.5VDC -3.6VDC 0HZ in failure case
Recharge time	: 80% within 14 hours, full recharge within 24 hours
Indicators	: Green LED for charging and main power on

1.7 EMERGENCY / EXIT LUMINAIRES

Emergency and Exit luminaires are to be provided as shown on the drawings. The type and IP rating of the luminaire shall be as per the lighting schedule. Additional luminaries shall be considered in order to fulfill the design requirements described in section 2 of these specifications.

1.8 CENTRAL MONITORING UNIT

Automatic testing of all emergency and converted luminaries shall be carried out via the Central Monitoring Unit (CMU). **Addressable modules shall be provided within every luminaire to be connected to the CMU via a separate monitoring loop.** The panel shall be programmable via a PC or via the front controls, and should be able to produce accurate test reports as and when required.

Central Monitoring unit shall monitor only the luminaries connected to that unit. The CMU's shall be networkable to allow complete monitoring and testing from a single location. In addition the following features are required as a minimum:

- **Work in both self contained or central battery system**
- Reporting of luminaire number, location, and nature of fault detected
- Constant communication with luminaires to prove system integrity
- Reporting of charger or mains supply failure within 60 seconds of the event
- Disabling of loops, zones and luminaries to carry out maintenance
- Fully automatic daily, weekly, quarterly, and annual testing
- Manual testing of entire system shall be possible at any time
- On-board power supply to maintain memory if power fails
- Compliant with requirements of PR/EN62034
- 4 programmable monitored alarm outputs
- 2 programmable volt-free contacts
- 20 test zone LED's and audible buzzer
- 2 lines by 40 characters backlit LCD
- 1 RS-232 port for connecting an external printer

1.9 WIRING

All wiring of the Emergency Lighting and Monitoring system shall be done using fire resistant cabling approved by the Civil Defense .The diameter of the cable shall be selected as per the estimated loading of each circuit.

SECTION 28 31 00 FIRE ALARM AND VOICE EVACUATION**1.1. DESCRIPTION**

The System shall comprise of a range of analogue addressable, microprocessor based fire alarm control equipment to offer flexibility in both design and operation. The System shall be of modular concept for easy tailoring of system design, to meet the full requirements of this Project.

The System shall be designed to provide early warning smoke detection, to quickly identify the location of fire and provide user definable text informing the occupants of the building of potential smoke spread. Simultaneously, the System shall alert and evacuate the occupants, and control all necessary auxiliary command functions such as elevator control, air handling shut down, gas shut off & damper control, as per the cause and effects requirements of the building. The complete facility shall be protected by a fully Analogue Addressable Fire Alarm System. A range of smoke, heat, multi-sensor detectors, manual call point and sounders shall be installed in the building.

1.2. QUALITY CONTROL

The specialist shall allow for an internal authorized and certified quality control inspector to ensure that all QA guidelines are being followed. Only ISO approved specialist will be considered for these works. A certified report shall be forwarded to the professional team.

The addressable type fire detection and alarm system shall be designed installed and commissioned with all the necessary equipments and accessories in accordance with **LPCB/UL/FM/CE/VDS standards**.

1.3. ONE YEAR MAINTENANCE

Complete maintenance and repair service for the fire alarm system shall be available from a factory trained authorized representative of the manufacturer of the major equipment for a period of five (5) years after expiration of the guaranty.

The specialist shall submit a proposal for complete and comprehensive service and maintenance for the entire system. The maintenance shall be at least quarterly and a complete detailed report shall be issued in triplicate following the visit. As a minimum, 25% of all devices shall be tested every quarter.

1.4. SUBMITTALS AND SHOP DRAWING

1. Two copies of all submittals shall be submitted to the Architect/Engineer for review.
2. Sufficient information shall be clearly presented and shall include
3. Manufacturer's name, model numbers, power requirements, equipment layout, device arrangement and complete wiring.
4. Sequence and description of operation.
5. Product Data for each type of equipment, initiating device, signal device, peripheral device and cable provided on the project.
6. Shop drawings shall include battery calculations, floor plans and wiring diagrams.

1.5. OPERATION MANUAL

Operation manual shall include:

1. Installation instructions for use by installing contractor.
2. Operational instructions or manual for use by building personnel, including name and phone number of service representative.
3. Maintenance instructions as required for use by building personnel.

4. Copy of approved shop drawings.

1.6. TRANSPORTATION, HANDLING AND STORAGE

All the components of fire alarm system shall be provided in manufacturer's original new and unopened packing bearing manufacturer's name and label.

Store materials, not in actual use, in covered and well ventilated area and protect them from dirt, dust, moisture, direct sunlight and extreme temperatures.

For further requirements follow manufacturers written instructions regarding storage and handling.

1.7. WARRANTY

Submit written guarantee signed by the contractor, manufacturer and installer of fire alarm system for the period of 1 year from the date of substantial completion. The guarantee shall cover the repair and replacement of material with manufacturing defects and workmanship as directed by the engineer.

1.8. SYSTEM DESCRIPTION

The fire detection and alarm system shall comprise of Automatic Soft Addressable Modular design main fire alarm control panels, optical smoke & heat, multi Sensor, manual call points, electronic wall mounted Alarm sounder/flasher/ combined devices, The system shall be fully programmed to accommodate fire alarm zones. The system shall be configured to allow on site modifications with the minimum of disruption using the PC based software to facilitate future changes or alterations to existing buildings/network on site.

The complete Analogue Addressable Fire Detection System shall be located in the Control Room on the Ground Floor. The Main Control Panel will be equipped with sufficient loop cards to meet the need of the entire project. There shall be sufficient spare capacity to cater for at least the full requirement plus 20% spare

The system will be capable of providing fire, fault, action, supervisory, disable and monitoring facilities

In the event of a fire alarm the FACP unit shall operate the alarm signal throughout the building in accordance with the cause and effects and any additional requirements laid down by the Civil Defense:

- Signal the local authority fire brigade via Auto Dialer
- Activate all control by event function related to the alarm
- Signal to graphic monitoring station
- Release door with hold open device shall be signal to release.
- Operate fire dampers, shutdown HVAC, and general exhaust fan.
- Signal to elevator

1.9. PRODUCTS

1.9.1. FIRE ALARM CONTROL PANEL

- a) The FACP shall incorporate a number of features to allow easy operation through a user-friendly menu. It shall have a Graphical LCD display and non-tactile membrane type switches/keypad in durable plastic. For reliability it shall have its own on-board circuit protection and multiprocessor architecture. The keypad shall include means to enter of two minimum level passwords to prevent unauthorized manual control. The system shall be on and off-site programmable by using the keypad and, also by connecting a PC to the communication port on the main processor/display board.

- b) The FACP shall be fully configurable and expandable at the owner's site without the need for handheld programmer. All the configurations shall be done by the control panel.
- c) The FACP shall be of modular construction, capable of monitoring 10 monitored detection loops and shall have repeated code facility to learn if the loop have duplicated address. Each loop shall be capable to accommodate a minimum of 242 addressable devices.

Each loop length maximum 1.2Km

Detectors (Smoke, heat, flame), manual call points, modules, sounder strobe shall be connected on the same two wire loop.

- a) Each device and zone shall have its own 40 character text message allocation; also each device shall be linked to a software zone. Can be edited the device detailed and type, sensitivity and address on the control panel.
- b) The FACP shall be capable of a minimum of 140 zone indications per panel and have 3 common 1-amp rated sounder circuits and 2 common 1-amps rated auxiliary relay contacts. The panel shall have an internal event printer; this shall be mounted within the same enclosure. The printer shall have the facility to be programmed to print alarms only; failure of the printer must not affect any other part of the system.
- c) The FACP shall have a one man walk test facility to periodically test the detectors without having to continually manually reset from the panel. The test feature shall have silent mode option.
- d) The FACP shall automatically program to switch the sounders to Day/Night mode whereby any alarm condition is recognized outside normal hours shall provide an evacuation signal throughout the entire premises.
- e) The FACP shall have internal memory lock to prevent unauthorized data entry and shall also have 2000-memory event log for system information and shall be stored in non-volatile memory.
- f) The FACP shall have the facility to networking on RS485 to other panels and repeaters, while communication to colour graphics shall be via RS232.
- g) Alarm and faults shall be indicated as for a fire alarm via the LCD along with an associated LED and printer. A buzzer shall sound in the event of an alarm, prewarning of fault.
- h) The FACP shall have verification facility to delay acceptance of alarms until they can be confirmed, and shall meet the applicable standard. All the devices in the zone can be configuring for verification through the panel. Manual call point or contact devices connected to a loop will be separated and will not be verified.
- i) The Batteries and charger shall maintain the entire system, plus 20% for a minimum period of 24 hours stand-by in the event of mains failure. After 24 hours has elapsed the system shall be capable of initiating a full global alert and evacuate for a minimum period of 30 minutes.
- j) The network shall be peer to peer with each panel generating command to other panel as required. The network shall be class A wiring up to 1.2Km between two panels.

- k) Each control panel shall operate independently by performing its own specific control, alarm supervision and history gathering. The system which requires master or slave is not acceptable.

1.9.2. REPEATER PANELS

Repeater panels shall be connected via panel RS 485 repeater output and installed at each fireman's entrance (2 No.). The repeater panel shall have basic control, and have a LCD large display. The repeater panels shall be installed in slim line, aesthetically pleasing enclosures.

1.9.3. NETWORK REPEATER PANELS

Network repeater panels shall be class A wiring with other control panel shall be installed on every level; these shall be mounted as shown on the drawings. The repeater panel shall show all alarms, action, supervisory and faults. The LCD Display Panels are required to have full control. The panel must be connected to a suitable supply that will in the event of a mains power failure, be capable of maintaining its self for 24 hours plus ½ hour in alarm condition. Should the PSU be in fault the condition must be automatically reported and the information displayed on the FACP

1.9.4. GRAPHICAL MONITORING CENTRE

The graphical monitoring centre shall pictorially represent Fire, Fault, Action and Disable events on a monitor display. The purpose of this facility shall be to provide the user with the additional visual information. The system shall operate on the Windows environment and required to have full control or monitor, up to 8 users with multi-level password control, user friendly, direct and complete user graphics view interface, switching between device layouts of different monitoring zones. Pop out layout of abnormal information automatically. Flash the abnormal device(s) in different colors' for alarm, action, fault and isolation.

1.9.5. INTELLIGENT MULTI SENSORS AND INTELLIGENT PHOTOELECTRIC SMOKE SENSORS

The Multi Smoke and Heat Sensor shall operate on a 'Fuzzy Logic Technology', by the use of a minimum of an 8 bit processor, to enable the sensors to store and analyze fire scenarios, thereby reducing susceptibility to false alarms and improve the sensitivity. The device shall comprise of highly reliable photoelectric sensors, together with a dual heat sensor providing rate of rise and fixed temperature capabilities. The sensitivity of the device shall be variable. The device shall compress with remote indicator output. The addressable code for the device shall be electronically programmed and stored in the sensor and be non-volatile. The programming of this code shall be facilitated by a digital electronic hand held device.

The detector shall be capable of the features: sounder /strobe base at a minimum of 85dBA or isolator base. The detector base shall have no electronic circuit and shall include a separate twist-lock base.

1.9.6. INTELLIGENT PHOTOELECTRIC SMOKE SENSOR

The Photoelectric Smoke Sensor shall operate on a 'Fuzzy Logic Technology', by the use of a minimum of an 8 bit processor, to enable the sensors to store and analyze fire scenarios, thereby reducing susceptibility to false alarms and improve the sensitivity. The sensitivity of the device shall be variable. The device shall compress with remote indicator output. The addressable code for the device shall be electronically programmed and stored in the sensor and be non-volatile. The programming of this code shall be facilitated by a digital electronic hand held device.

The detector shall be capable of the features: sounder /strobe base at a minimum of 85dBA or isolator base. The detector base shall have no electronic circuit and shall include a separate twist-lock base.

1.9.7. INTELLIGENT DUAL HEAT SENSOR

The Heat Sensor shall incorporate a highly linear thermistor circuit. A voltage is produced proportional to temperature which shall be scaled, linearised and transmitted as an analogue value.

The sensors software shall allow the device to operate as both a rate of rise sensor and or a fixed temperature sensor. It shall be possible to de-select the rate of rise for specific locations. The device shall compress with remote indicator output. The addressable code for the device shall be stored in the sensor and be non-volatile. The programming of this code shall be facilitated by a hand held device.

The detector shall be capable of the features: sounder /strobe base at a minimum of 85dBA or isolator base. The detector base shall have no electronic circuit and shall include a separate twist-lock base.

1.9.8. MANUAL CALL POINTS

The Manual Break Glass Call Point shall have response time of less than 1 second and shall have a very low standby current, incorporating a status LED, which flashes when, polled or is continuously lit when operated. An additional set of volt free change over contacts shall be incorporated for additional local signaling as required. The addressable code for the device shall be electronically programmed and stored in the sensor and be non-volatile. The programming of this code shall be facilitated by a digital electronic hand held device.

1.9.9. ADDRESSABLE MONITOR MODULE

The Monitor shall be designed to interface sprinkler flow switch and supervisory tamper switch. All field wiring to be fully monitored for alarm and fault condition. The interface shall be designed for 1st fix of back plate will all terminations and be free of all electronics. The intelligent interface shall be installed at the commissioning stage. The addressable code for the device shall be electronically programmed and stored in the sensor and be non-volatile. The programming of this code shall be facilitated by a digital electronic hand held device.

1.9.10. ADDRESSABLE INTERFACE MODULE

The Interface shall have a relay inputs with NO and NC volt free contacts, these shall be separately driven under the control of the fire alarm panel. The output relay shall be rated to switch up to 1A 120V AC. A single input shall be provided with an open and short circuit monitoring, which shall enable monitoring for confirmation signal that the plant under control has operated. If the relay has not operated within a programmed period of time a specific fault shall be generated.

The interface shall be designed for 1st fix of back plate will all terminations and be free of all electronics. The intelligent interface shall be installed at the commissioning stage. The addressable code for the device shall be electronically programmed and stored in the sensor and be non-volatile. The programming of this code shall be facilitated by a digital electronic hand held device.

1.9.11. CIRCUIT ISOLATOR

The short Circuit Isolator shall be installed at least every twenty devices and or each separate fire zone and between floors. The circuit isolator shall be used to protect the installation wiring in the event of a short in the cables. Should a fault occur the isolator shall automatically disconnect the loop either side of the short, the control panel shall recognize this disconnection and shall communicate in both directions. This process will enable a limited number of devices to be off line whilst the fault is being investigated and repaired. Once the

short is removed the isolators shall automatically reinstate the loop connection. The isolator shall incorporate an LED, which illuminates when the short is detected on the loop circuit.

1.9.12. ADDRESSABLE REMOTE SENSOR INDICATOR

Remote Indicators shall be installed as shown on the drawings. The remote indicators shall be addressable linked to any number of detectors within the given location. The remote indicator shall be installed in a clearly identifiable location. The remote indicator shall fit onto a standard electrical accessory box.

1.9.13. SOUNDER BASE

The sounder shall be loop powered low profile design and shall incorporate an individual loop address; the unit shall neatly be incorporated within the sensor to ensure aesthetics are maintained. The sounder shall be rated 89dBA at 1 meter and 30 units per loop. The addressable code for the device shall be stored in the sounder and be nonvolatile. The programming of this code shall be facilitated by a hand held device.

1.9.14. SOUNDER-STROBE BASE

The sounder shall be loop powered low profile design and shall incorporate an individual loop address; the unit shall neatly be incorporated within the sensor to ensure aesthetics are maintained. The sounder shall be rated 90dBA at 1 meter and 30 units per loop. The addressable code for the device shall be stored in the sounder and be nonvolatile. The programming of this code shall be facilitated by a hand held device.

1.9.15. SOUNDER AND STROBE LOOP POWERED

The sounder and strobe shall be sited on the loop; surface mounted and shall incorporate an individual address. The sounder shall be rated 97 dBA at 1 meter and 30 units per loop. The addressable code for the device shall be stored in the sensor and be non-volatile. Weather resistant version should be utilized for external applications.

1.9.16. ADDRESSABLE SOUNDER AND STROBE SEPARATE 24V

The sounder and strobe shall be surface mounted and shall incorporate an individual loop address. The addressable code for the device shall be stored in the sensor and be non-volatile. Weather resistant version should be utilized for external applications.

1.9.17. FLASHING BEACONS SEPARATE 24V

Flashing Beacons shall be installed as shown on the drawings. The flashing beacons shall be utilized to assist with the safe evacuation of the buildings. The Flashing Beacons shall be installed at the door way to all major exists. Flashing Beacons shall also be installed in Plant Rooms of high ambient noise levels.

1.9.18. INTELLIGENT POWER SUPPLIES

The specialist shall determine how many distributed power supplies shall be required for the complete project. Each power supply shall be fully monitored by the fire detection system for:-

- Mains Healthy
- Mains Fail
- Charger Mail Function (Low Voltage)
- Batteries Mail Function (Low Amperage)

The Batteries and charger shall maintain the full load of the connected load, plus 20% for a minimum period of 24 hours stand-by, in the event of mains failure. After 24 hours has elapsed the system shall be capable of initiating a full global alert and evacuate or operation of all equipment reliant on the PSU for its supply for a minimum period of 30 minutes.

1.9.19. DUCT PROBES

Duct Probes shall be installed in all extract ducts from each Air Handling Unit. The Duct Probe shall house a standard Photo Electric Smoke Sensor see 1.5. The Duct Probe shall be a polycarbonate box that is mounted to the side of the duct. The Box shall have adjustable tubes that penetrate the duct by at least 75% for the inlet and 25% for the outlet.

1.10. ELECTRICAL INSTALLATION

The Fire Alarm system shall installed used fire rated cable. The fire rated cable shall be a minimum of 1.5 mm and shall be red Low Smoke and Fume (LSF) outer sheaf. The Fire rated capable shall be clipped directly to the building fabric for single and double cable runs. The Clips shall be fire rated metal with a LSF Coated Red finish. The clips shall be fixed suitably and shall be no greater than 300mm between fixings. Where there are more than 2 cables, the cables shall be installed on suitable galvanized tray or in trunking. Where cables run below 2.4 meters from finished floor level then all cables shall be installed within galvanized conduit.

Correct corresponding termination devices and boxes shall be used. All devices shall be terminated using a compression lock gland colored red manufactured in LSF material. Where devices are fixed surface a suitable LSF shroud shall be used.

Before any devices are connected the contractor must fully demonstrate and prove the quality of the installation for loop integrity, continuity, resistance and capacitance.

The system specialist shall install all devices; this shall include 2nd fix and 3rd fix. Prior to completion of the project the system specialist must demonstrate the entire system to the approval of the Consultant. The entire system shall have been 100% tested independently prior to the demonstration to the Consultant. Should any items fail during the test a new and complete 100% re-test shall be carried. The specialist will be responsible for any costs involved with re-testing of the system.

The system specialist will be responsible for providing complete and detailed manuals for the entire system. The manuals shall be fully detailed and shall describe the operation and function of each part of the system.

1.11. SERVICES

1.11.1. SYSTEM PROGRAMMING

The specialist shall completely programme the entire integrated system to meet the full needs of the specification. This shall be done off site and proven to the Consultant.

1.11.2. SYSTEM PACKAGING

The specialist shall provide full packaging for the entire consignment. The packaging shall be weather resistant and shall be suitable to ensure all goods are in complete and perfect condition ready for installation.

1.11.3. 100% WITNESS TESTING

The specialist shall fully test the entire system and when satisfied offer the entire system to the Consultant for completes 100% witness testing. Should any failures occur then a complete 100% retest shall be carried out at the cost of the specialist. The test must for the entire integrated system; partial tests will not be acceptable.

1.12. INTENSIVE TRAINING

The specialist shall allow for on site and off site intensive training for a period of 1 week for a minimum of 4 persons. The training procedures shall be submitted to the consultant for approval before training will commence. The training shall be carried out by a professional and approved training manager.

1.13. EXTENDED 2 YEAR GUARANTEE

The specialist shall provide an offer for a fully inclusive and comprehensive agreement to include for all consumables and materials that will be required to keep the system 100% operational for 2 years. The specialist shall allow all air freight, delivery and labor charges within the price.

1.14. SPARE PARTS

The specialist shall provide an offer for fully inclusive spare parts to include for all consumables and materials that will be required to keep the system 100% operational for 2 years. The specialist shall allow all air freight, delivery and labor charges within the price.

1.15. ELECTRICAL INSTALLATION

The electrical installation will be as specified earlier for each system. The specialist is to ensure that the entire system is complete and installed in accordance with correct installation policies. Only approved installation teams shall be utilized with the installation of the complete integrated system.

Letter of Bid / Form of Bid

Title: _____

Bidder: _____

We, the undersigned Bidder, hereby submit the first part of our Bid, the Technical Proposal

In submitting our Bid we make the following declarations:

- (a) **No reservations:** We have examined and have no reservations to the bidding document, including addenda issued in accordance with Instructions to Bidders;
- (b) **Eligibility:** We meet the eligibility requirements and have no conflict of interest in accordance with Instructions to Bidders;
- (c) **Bid/Proposal-Securing Declaration:** We have not been suspended nor declared ineligible by the Procuring agency/Employer based on execution of a Bid Securing Declaration or Bid Securing Declaration in the Procuring agency/Employer's country in accordance with Instructions to Bidders;
- (d) **Conformity:** We offer to execute works in conformity with the bidding document and in accordance with the works requirements: **[327- Upgradation of Fire Fighting System at SBP Buildings, Head Office Karachi]**;
- (e) **Bid Validity Period:** Our Bid shall be valid for the period specified in **Instruction to Bidders** (as amended, if applicable) from the date fixed for the Bid submission deadline (as amended, if applicable), and it shall remain binding upon us, and may be accepted at any time before the expiration of that period;
- (f) **Performance Security:** If our Bid is accepted, we commit to obtain performance security in accordance with the bidding document;
- (g) **One Bid per Bidder:** We are not submitting any other Bid(s) as an individual Bidder, and we are not participating in any other bid(s) as a Joint Venture member or as a subcontractor, and meet the requirements;
- (h) **Suspension and Debarment:** We, along with any of our subcontractors, suppliers, consultants, manufacturers, or service providers for any part of the contract, are not subject to, and not controlled by any entity (NACTA) or individual that is subject to, a temporary suspension or a debarment imposed by the Procuring agency/Employer. Further, we are not ineligible under Pakistan laws;
- (i) **Binding Contract:** We understand that this Bid, together with your written acceptance thereof included in your Letter of Acceptance, shall constitute a binding contract between us, until a formal contract is prepared and executed;
- (j) **Not Bound to Accept:** We understand that you are not bound to accept the Most Advantageous Bid or any other Bid that you may receive; and
- (k) **Fraud and Corruption:** We hereby certify that we have taken steps to ensure that no person acting for us, or on our behalf, engages in any type of Fraud and Corruption.

Seal & Signature of Bidder: _____

Date: _____

Form of Bid Security (Bank Guarantee)

[The bank shall fill in this Bank Guarantee Form in accordance with the instructions indicated.]
[Guarantor letterhead or SWIFT identifier code]

Beneficiary: State Bank of Pakistan, Banking Services Corporation (SBP BSC)

No.: *[Purchaser to insert reference number for the Request for Bids]*

Date: *[Insert date of issue]*

Validity: 208 Days from the date of bid opening

BID GUARANTEE No.: *[Insert guarantee reference number]*

Guarantor: *[Insert name and address of place of issue, unless indicated in the letterhead]*

We have been informed that _____ *[insert name of the Bidder]* (hereinafter called "the Applicant") has submitted or will submit to the Beneficiary its Bid (hereinafter called "the Bid") for the execution of **327- Upgradation of Fire Fighting System at SBP Buildings, Head Office Karachi** under Invitation to Bid No. **ED/PROC-HOK/1223364/2026** ("the ITB").

Furthermore, we understand that, according to the Beneficiary's conditions, Bids must be supported by a Bid guarantee.

At the request of the Applicant, we, as Guarantor, hereby irrevocably undertake to pay the Beneficiary any sum or sums not exceeding in total an amount of **Rs.5,000,000/- (Five Million Rupees only)** upon receipt by us of the Beneficiary's complying demand, supported by the Beneficiary's statement, whether in the demand itself or a separate signed document accompanying or identifying the demand, stating that either the Applicant:

- (a) has withdrawn its Bid during the period of Bid validity set forth in the Applicant's Letter of Bid ("the Bid Validity Period"), or any extension thereto provided by the Applicant; or
- (b) having been notified of the acceptance of its Bid by the Beneficiary during the Bid Validity Period or any extension thereto provided by the Applicant, (i) has failed to sign the contract agreement, or (ii) has failed to furnish the performance security, in accordance with the Instructions to Bidders ("ITB") of the Beneficiary's bidding document.

This guarantee will expire: (a) if the Applicant is the successful Bidder, upon our receipt of copies of the Contract agreement signed by the Applicant and the performance security issued to the Beneficiary in relation to such Contract agreement; or (b) if the Applicant is not the successful Bidder, upon the earlier of (i) our receipt of a copy of the Beneficiary's notification to the Applicant of the results of the Bidding process; or (ii) twenty-eight days after the end of the Bid Validity Period.

Consequently, any demand for payment under this guarantee must be received by us at the office indicated above on or before that date.

[Signature(s)]

Note: All italicized text is for use in preparing this form and shall be deleted from the final product.

SUPPLIER ACCOUNT DETAILS – S2 FORM

		STATE BANK OF PAKISTAN BANKING SERVICES CORPORATION Supplier Bank Account (IBAN) Details Form				S-2	
1. For Office use:							
*Office/Deptt				*Supplier No.			WHT Rate
Supplier	New		Update				
*Supplier Type				*Liability A/C			
				*Prepayment A/C			
2. Supplier Information							
*Supplier Name							
*Supplier NTN				CNIC No.			
*Supplier Address							
					*Supplier City		
*Contact No.					Mobile		
*E-mail Address					Fax No.		
3. Bank Account Information							
*Bank Name							
*IBAN (24 Characters)						*Supplier Stamp & Signature	
*Branch Type	Islamic		Commercial				
*Title of Account							
(For Office use only)							
Forwarded By		Verified By		Entered By (Supplier Mgt User)			
(Procurement Function)							
				Date			
Field marked with * are mandatory. Information without complete Bank Account Details (IBAN) & NTN/ CNIC will not be accepted. Any changes in Supplier's particulars should be conveyed immediately to SBP BSC. SBP BSC will not be responsible for credit into wrong account of supplier due to change in bank account details not conveyed to SBP BSC or delay in settlement of supplier's claims.							

LIST OF DRAWINGS

S. NO	DRAWING NOS.	DESCRIPTION	SCALE	DATE	REVISION	REMARKS
01	FF-100	LIST OF DRAWINGS	N.T.S.	16/12/2014	0	
02	FF-101	LEGEND ,NOTES AND DETAILS OF FIRE FIGHTING SYSTEM	N.T.S.	16/12/2014	0	
03	FF-102	GENERAL SITE PLAN (FIRE FIGHTING SYSTEM)	N.T.S.	16/12/2014	0	
04	FF-103	LAYOUT OF EXISTING PUMP ROOM & UNDER GROUND TANK	3/32"=1'-0"	16/12/2014	0	
05	FF-104	LAYOUT OF PROPOSED PUMP ROOM & UNDER GROUND TANK	3/32"=1'-0"	16/12/2014	0	
06	FF-105	LAYOUT OF FIRE FIGHTING SYSTEM GROUND FLOOR VAULT AREA MAIN BUILDING	3/32"=1'-0"	16/12/2014	0	
07	FF-106	LAYOUT OF FIRE FIGHTING SYSTEM FIRST FLOOR FOR MAIN BUILDING	3/32"=1'-0"	16/12/2014	0	
08	FF-107	LAYOUT OF FIRE FIGHTING SYSTEM SECOND FLOOR FOR MAIN BUILDING	3/32"=1'-0"	16/12/2014	0	
09	FF-108	LAYOUT OF FIRE FIGHTING SYSTEM THIRD FLOOR MAIN BUILDING	3/32"=1'-0"	16/12/2014	0	
10	FF-109	LAYOUT OF FIRE FIGHTING SYSTEM, 4TH FLOOR MAIN BUILDING	3/32"=1'-0"	16/12/2014	0	
11	FF-110	LAYOUT OF FIRE FIGHTING SYSTEM 5TH FLOOR MAIN BUILDING	3/32"=1'-0"	16/12/2014	0	
12	FF-111	LAYOUT OF FIRE FIGHTING SYSTEM 6TH FLOOR MAIN BUILDING	3/32"=1'-0"	16/12/2014	0	
13	FF-112	LAYOUT OF FIRE FIGHTING SYSTEM 7TH FLOOR MAIN BUILDING	3/32"=1'-0"	16/12/2014	0	
14	FF-113	LAYOUT OF FIRE FIGHTING SYSTEM 8TH FLOOR MAIN BUILDING	3/32"=1'-0"	16/12/2014	0	
15	FF-114	LAYOUT OF FIRE FIGHTING SYSTEM 9TH FLOOR MAIN BUILDING	3/32"=1'-0"	16/12/2014	0	
16	FF-115	LAYOUT OF FIRE FIGHTING SYSTEM 10TH FLOOR MAIN BUILDING	3/32"=1'-0"	16/12/2014	0	
17	FF-116	LAYOUT OF FIRE FIGHTING SYSTEM & AT ROOF	3/32"=1'-0"	16/12/2014	0	
18	FF-117	LAYOUT OF FIRE FIGHTING SYSTEM GROUND FLOOR LRC BUILDING	3/32"=1'-0"	16/12/2014	0	
19	FF-118	LAYOUT OF FIRE FIGHTING SYSTEM AT FIRST FLOOR LRC BUILDING	3/32"=1'-0"	16/12/2014	0	
20	FF-119	LAYOUT OF FIRE FIGHTING SYSTEM AT SECOND FLOOR LRC BUILDING	3/32"=1'-0"	16/12/2014	0	
21	FF-120	LAYOUT OF FIRE FIGHTING SYSTEM AT THIRD FLOOR AUDITORIUM LRC BUILDING	3/32"=1'-0"	16/12/2014	0	
22	FF-121	LAYOUT OF FIRE FIGHTING SYSTEM AT ROOF LRC BUILDING	3/32"=1'-0"	16/12/2014	0	
23	FF-122	LAYOUT OF SITE PLAN FIRE FIGHTING SYSTEM CANTEEN BUILDING	3/32"=1'-0"	16/12/2014	0	
24	FF-123	LAYOUT OF FIRE FIGHTING SYSTEM AT GROUND FLOOR CANTEEN BUILDING	3/32"=1'-0"	16/12/2014	0	

S. NO	DRAWING NOS.	DESCRIPTION	SCALE	DATE	REVISION	REMARKS
24	FF-124	LAYOUT OF FIRE FIGHTING SYSTEM AT FIRST FLOOR CANTEEN BUILDING	3/32"=1'-0"	16/12/2014	0	
25	FF-125	LAYOUT OF FIRE FIGHTING SYSTEM SECOND FLOOR FOR CANTEEN BUILDING	3/32"=1'-0"	16/12/2014	0	
26	FF-126	LAYOUT OF FIRE FIGHTING SYSTEM AT THIRD FLOOR CANTEEN BUILDING	3/32"=1'-0"	16/12/2014	0	
27	FF-127	LAYOUT OF FIRE FIGHTING SYSTEM AT FOURTH FLOOR CANTEEN BUILDING	3/32"=1'-0"	16/12/2014	0	
28	FF-128	LAYOUT OF FIRE FIGHTING SYSTEM AT ROOF CANTEEN BUILDING	3/32"=1'-0"	16/12/2014	0	
29	FF-129	LAYOUT OF FIRE FIGHTING SYSTEM ART GALLERY MUSEUM DEPARTMENT AT FIRST FLOOR	3/32"=1'-0"	16/12/2014	0	
30	FF-130	LAYOUT OF FIRE FIGHTING SYSTEM ART GALLERY MUSEUM DEPARTMENT AT ROOF	3/32"=1'-0"	16/12/2014	0	
31	FF-131	LAYOUT OF FIRE FIGHTING PLANT ROOM, FOR MAIN BUILDING & LRC	3/32"=1'-0"	16/12/2014	0	
32	FF-132	LAYOUT OF FIRE FIGHTING PLANT ROOM AT ROOF, FOR MAIN BUILDING & LRC	3/32"=1'-0"	16/12/2014	0	
33	FF-133	LAYOUT OF FIRE FIGHTING SYSTEM, MOSQUE AT GROUND FLOOR	3/32"=1'-0"	16/12/2014	0	
34	FF-134	LAYOUT OF FIRE FIGHTING SYSTEM, MOSQUE AT ROOF	3/32"=1'-0"	16/12/2014	0	
35	FF-135	LAYOUT OF FIRE FIGHTING SYSTEM, DAY CARE AT GROUND FLOOR	3/32"=1'-0"	16/12/2014	0	
36	FF-136	LAYOUT OF FIRE FIGHTING SYSTEM, DAY CARE AT ROOF	3/32"=1'-0"	16/12/2014	0	

YOUSUF HASAN ASSOCIATES, CONSULTING ENGINEERS, KARACHI

TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	GENERAL NOTES • THE DRAWINGS MUST BE READ IN COMBINATION WITH ALL OTHER ARCHITECTURAL AND ENGINEERING DRAWINGS RELATED TO THE SAME SHEET AND ANY ITEM ON IT. • ALL DIMENSIONS MUST BE VERIFIED BY THE CONTRACTOR BEFORE PREPARATION OF SHOP DRAWINGS. • SHOP DRAWINGS OF MANUFACTURED ITEMS MUST BE APPROVED BY THE ARCHITECT-ENGINEER BEFORE WORK ON THESE ITEMS IS BEGUN. • THE ARCHITECT-ENGINEER MUST BE INFORMED ABOUT CONTRADICTION OF ANY KIND BEFORE CONSTRUCTION BEGINS. • DO NOT SCALE THIS DRAWING AS A BASIS FOR CONSTRUCTION WORK FROM FIGURED DIMENSIONS ONLY. • ALL THE MENTIONED DIMENSIONS ARE IN METER UNLESS OTHERWISE STATED. • ALL THE MENTIONED LEVELS ARE IN METER UNLESS OTHERWISE STATED.	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING DRAWING TITLE LIST OF DRAWINGS	DRAWING NO. FF-100		REVISION 0
								DRAWN BY NABELL	SERIAL NO. 00	DATE 24/02/2015
								PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE N.T.S.

LEGEND:

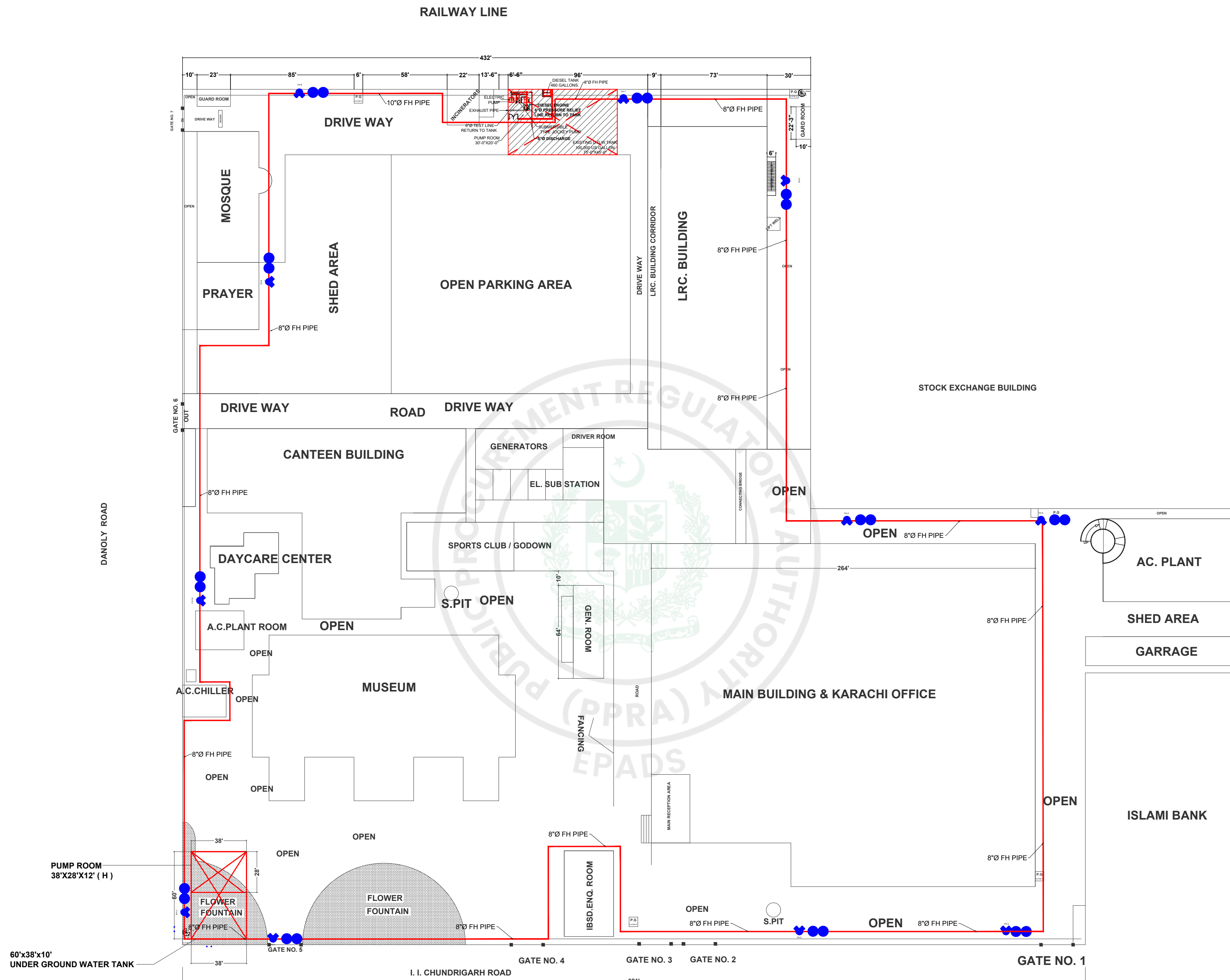
SYMBOL	DESCRIPTION
	FIRE SPRINKLER SUPPLY PIPE
	SPRINKLER
	FIRE HOSE CABINET (DOUBLE COMPARTMENT) INCLUDING : a- ONE AUTOMATIC HOSE REEL 25mm dia X 30m b- ø65mm LANDING VALVE WITH PRV WHEREVER REQUIRED & ø65mm COLLAPSABLE UL LISTED FIRE HOSE 30 MTRS. LONG WITH ADJUSTABLE 3-way BRASS NOZZLE. c - PORTABLE DRY POWDER FIRE EXTINGUISHER -6.0Kg d - PORTABLE CO2 FIRE EXTINGUISHER - 2.20 Kg.

GENERAL NOTES

1. THESE DRAWINGS ARE TO BE READ IN CONJUNCTION WITH THE TECHNICAL SPECIFICATION AND ALL RELEVANT ARCHITECTURAL, CIVIL AND STRUCTURAL DRAWINGS/DOCUMENTS AND ACTUAL SITE CONDITION.
2. ALL DIMENSIONS ARE TO BE VERIFIED ON SITE BY THIS CONTRACTOR BEFORE COMMENCEMENT OF WORK. ALL DISCREPANCIES MUST BE REPORTED IMMEDIATELY TO THE CONSULTING ENGINEER BEFORE PROCEEDING.
3. CONTRACTOR SHALL SUPPLY AND INSTALL ALL SECONDARY SUPPORT REQUIRED FOR SUSPENDING THEIR OWN SERVICES.
4. INSTALLATION HEIGHT OF TRAYS, TRUNKINGS, CONDUITS AND PIPING SHALL BE COORDINATED ON SITE WITH OTHER SERVICES AND TO THE SATISFACTION OF CLIENT/ CONSULTING ENGINEER.
5. ALL FIRE PROTECTION SYSTEM AND APPLIANCES SHALL CONFORM TO THE LATEST LOCAL CIVIL DEFENCE AND NFPA REQUIREMENTS AND CURRENT EDITIONS OF THE FOLLOWING STANDARDS:-
 - a) INSTALLATION OF SPRINKLER SYSTEM - NFPA 13
 - b) CLEAN AGENT FIRE EXTINGUISHING SYSTEMS - NFPA 2001
 - c) STANDPIPE AND HOSE SYSTEM - NFPA 14
 - d) PORTABLE FIRE EXTINGUISHER - NFPA 10
 - e) FIRE PUMP - NFPA 20
6. ALL BURIED AND CONCEALED PIPE WORK AND FITTINGS SHALL BE TESTED PRIOR TO MAKING GOOD.
7. ALL PIPE WORK SHALL BE SUITABLY SUPPORTED BY APPROVED BRACKETS AND HANGERS,SEISMIC SUPPORTS WHEREVER REQUIRED
8. ALL PIPE WORK SHALL BE PAINTED WITH ONE COAT OF RED ZINC CHROMATE PRIMER, ONE UNDERCOAT AND TWO COATS OF ENAMELED FINISHING PAINT.
9. THE CONTRACTOR SHALL CAREFULLY EXAMINE, COMPARE AND VERIFY THE DATA FURNISHED BY THE DRAWINGS AND SPECIFICATIONS. ANY DOUBT AS TO THE MEANING OF THE DRAWINGS (INCLUDING NOTES THERE IN) SHALL BE IMMEDIATELY REPORTED TO THE CONSULTING ENGINEERS WHO SHALL PROMPTLY CORRECT SUCH DISCREPANCIES, ERRORS AND OMISSIONS AFTER ITS DISCOVERY.
10. FINAL INSPECTION AND TESTING MUST BE CARRIED OUT BY THE CONTRACTOR TOGETHER WITH CONSULTING ENGINEERS BEFORE FIRE BRIGADE PERSONNEL ARE INVITED TO INSPECT AND TEST THE SYSTEM.
11. ALL PIPES PASSING THROUGH R.C. FLOOR SLAB, R.C. BEAM AND WALLS SHALL BE CONTAINED WITH PIPE SLEEVES OF ONE DIAMETER LARGER THAN THE PENETRATING PIPES AND CAULKED WITH APPROVED TYPE ELASTOMERIC FIRE RESISTANCE COMPOUND.
12. ALL ZONE CONTROL VALVES SHALL HAVE TEST DRAIN PIPES CONNECTED TO NEAREST DRAIN POINT.
13. ALL FIRE FIGHTING SYSTEMS AND EQUIPMENT SHALL BE ACCORDING TO NFPA CODES AND TO THE APPROVAL OF LOCAL CIVIL DEFENCE DEPARTMENT.
14. ALL HOLES, OPENINGS FOR THE CABLES TO BE FITTED WITH SMOKE / FLAME PROOF FIRE BARRIER.
15. PRESSURIZATION FANS PROVIDED FOR STAIR CASES, SERVICE LOBBIES, AND SHALL BE INTERLOCKED WITH FIRE ALARM SYSTEM.
16. THE CONTRACTOR SHALL SUBMIT COMPLETE HYDRAULIC CALCULATION TO CONSULTANT APPROVAL AND SELECT PUMP HEAD ACCORDINGLY.
17. THE CONTRACTOR SHALL PREPARE COMPLETE SHOP DRAWING AND TAKE APPROVAL FROM CIVIL DEFENSE BEFORE STARTING ANY WORK
18. THE CONTRACTOR SHALL PROVIDE AUXILIARY DRAIN TO COMPLY WITH NFPA 13.

TENDER DRAWING

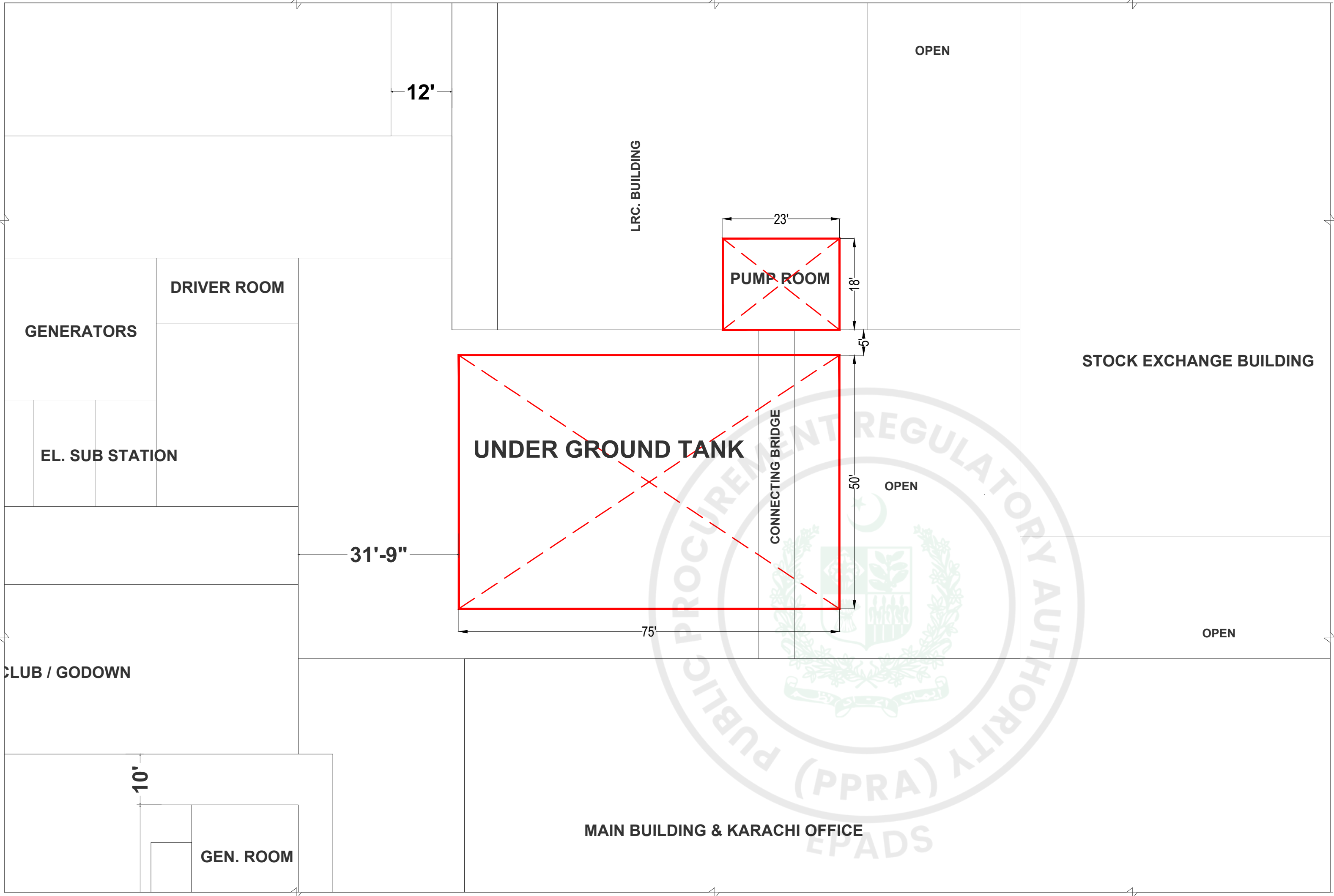
CLIENT STATE BANK OF PAKISTAN KARACHI	GENERAL NOTES • THE DRAWINGS MUST BE READ IN CONJUNCTION WITH ALL OTHER ARCHITECTURAL AND ENGINEERING DRAWINGS RELATED TO THE SAME SHEET AND ANY ITEM ON IT. • ALL DIMENSIONS MUST BE VERIFIED BY THE CONTRACTOR BEFORE PREPARATION OF SHOP DRAWINGS. • SHOP DRAWINGS OF MANUFACTURED ITEMS MUST BE APPROVED BY THE ARCHITECT-ENGINEER BEFORE WORK ON THESE ITEMS IS BEGUN. • THE ARCHITECT-ENGINEER MUST BE INFORMED ABOUT CONTRADICTION OF ANY KIND BEFORE CONSTRUCTION BEGINS. • DO NOT SCALE THIS DRAWING AS A BASIS FOR CONSTRUCTION WORK FROM PRINTED DIMENSIONS ONLY. • ALL THE MENTIONED DIMENSIONS ARE IN METER UNLESS OTHERWISE STATED. • ALL THE MENTIONED LEVELS ARE IN METER UNLESS OTHERWISE STATED.	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS  YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING	DRAWING NO. FF-101		REVISION 0
							DRAWING TITLE LEGEND ,NOTES AND DETAILS OF FIRE FIGHTING SYSTEM	DRAWN BY NABELL	SERIAL NO. 01	DATE 24/02/2015
								PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE N.T.S.



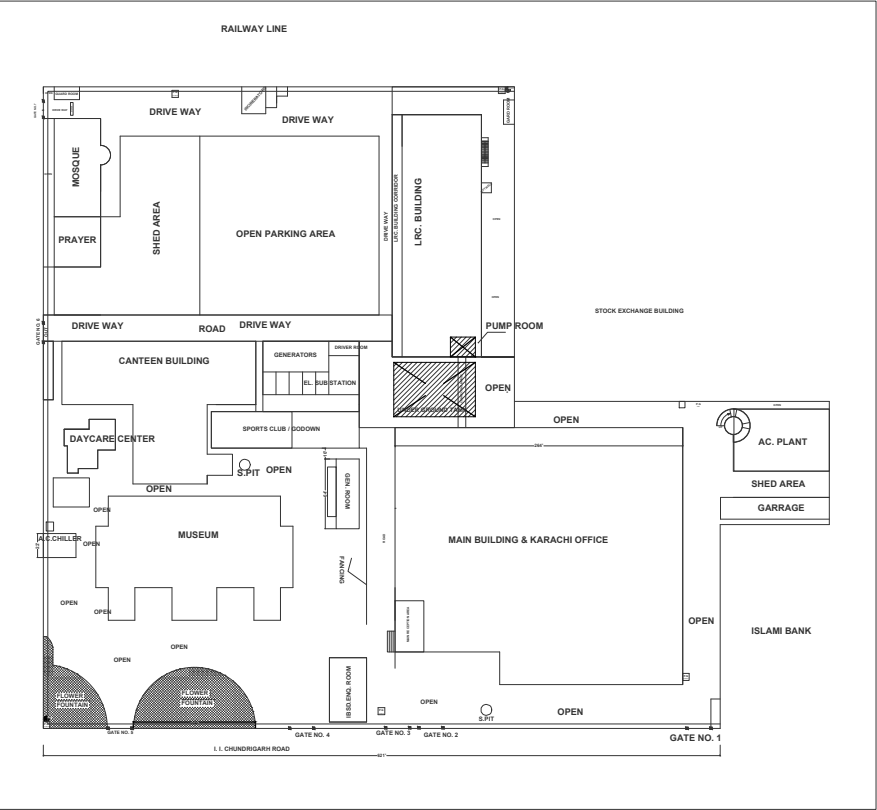
NEW SITE PLAN OF MAIN BANK BUILDING PREMISES

TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING DRAWING TITLE GENERAL SITE PLAN (FIRE FIGHTING SYSTEM)	DRAWING NO. FF-102		REVISION 0
							DRAWN BY SHEHARYAR	SERIAL NO. 02	DATE 24/02/2015
							PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE 1/64" = 1'-0"



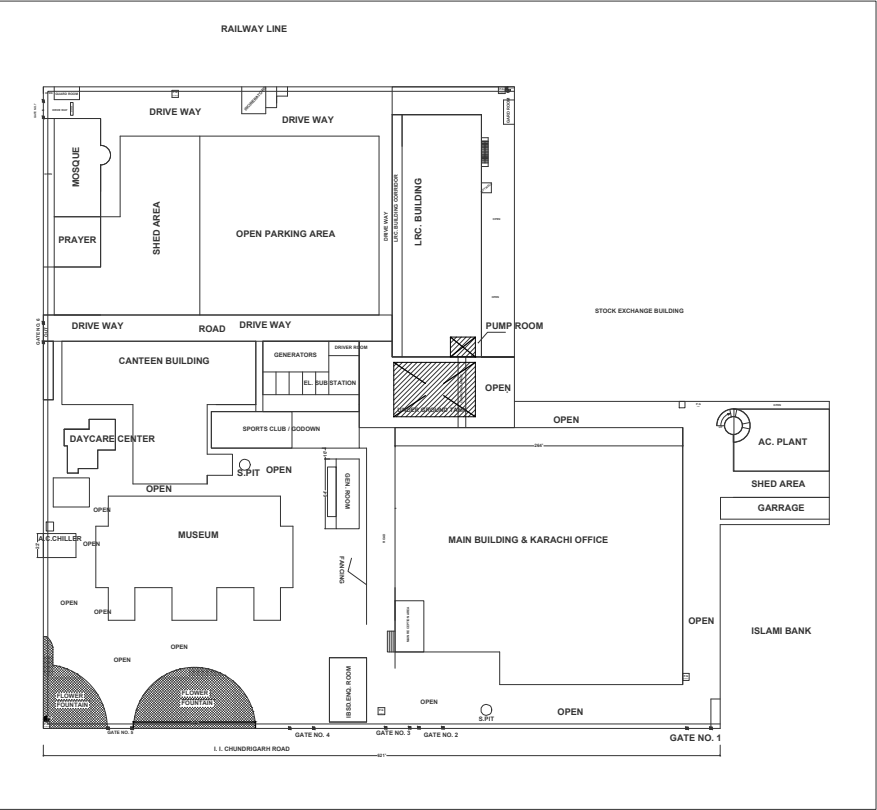
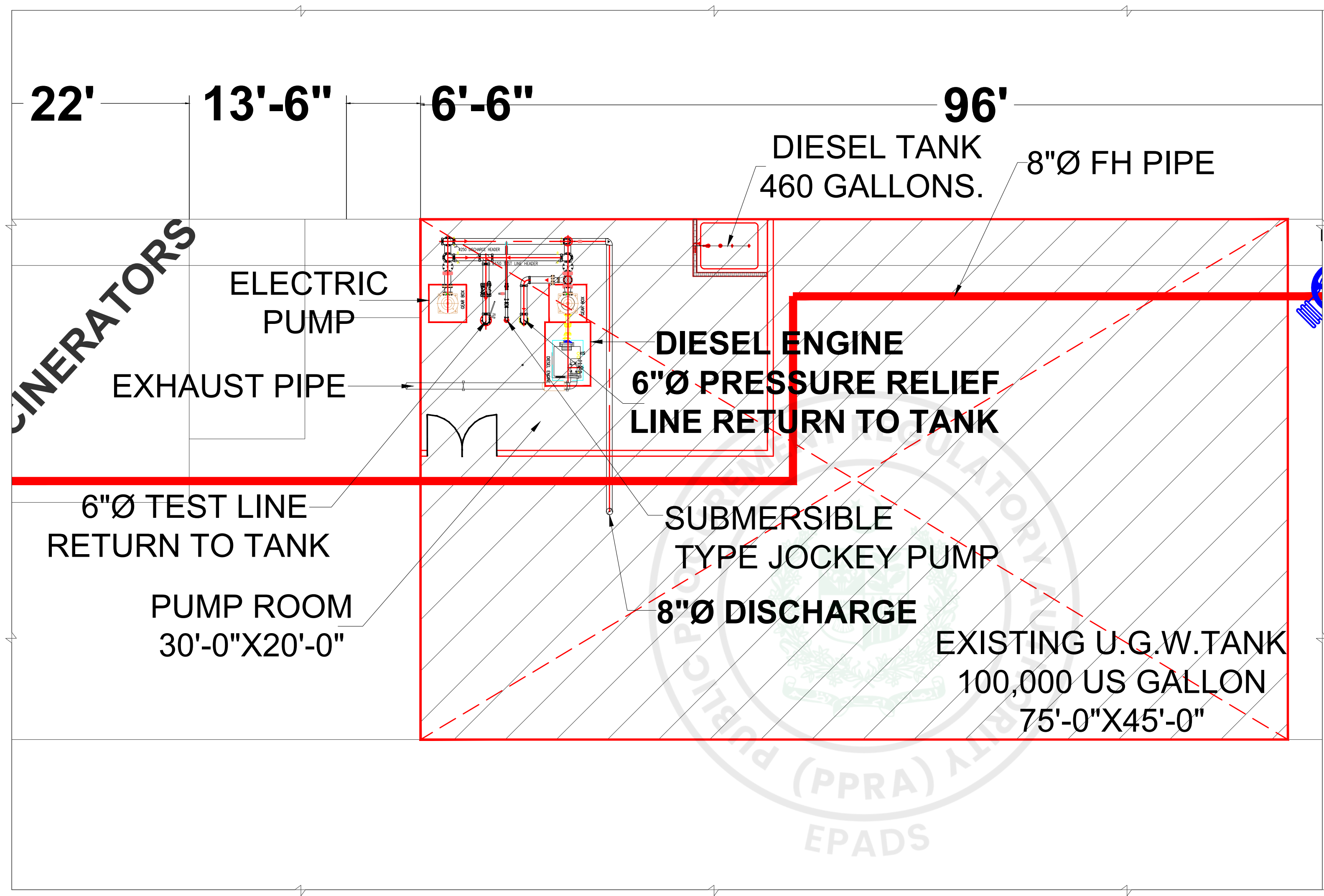
ENLARGE DETAIL OF EXISTING PUMP ROOM AND UNDER GROUND TANK



KEY PLAN

TENDER DRAWING

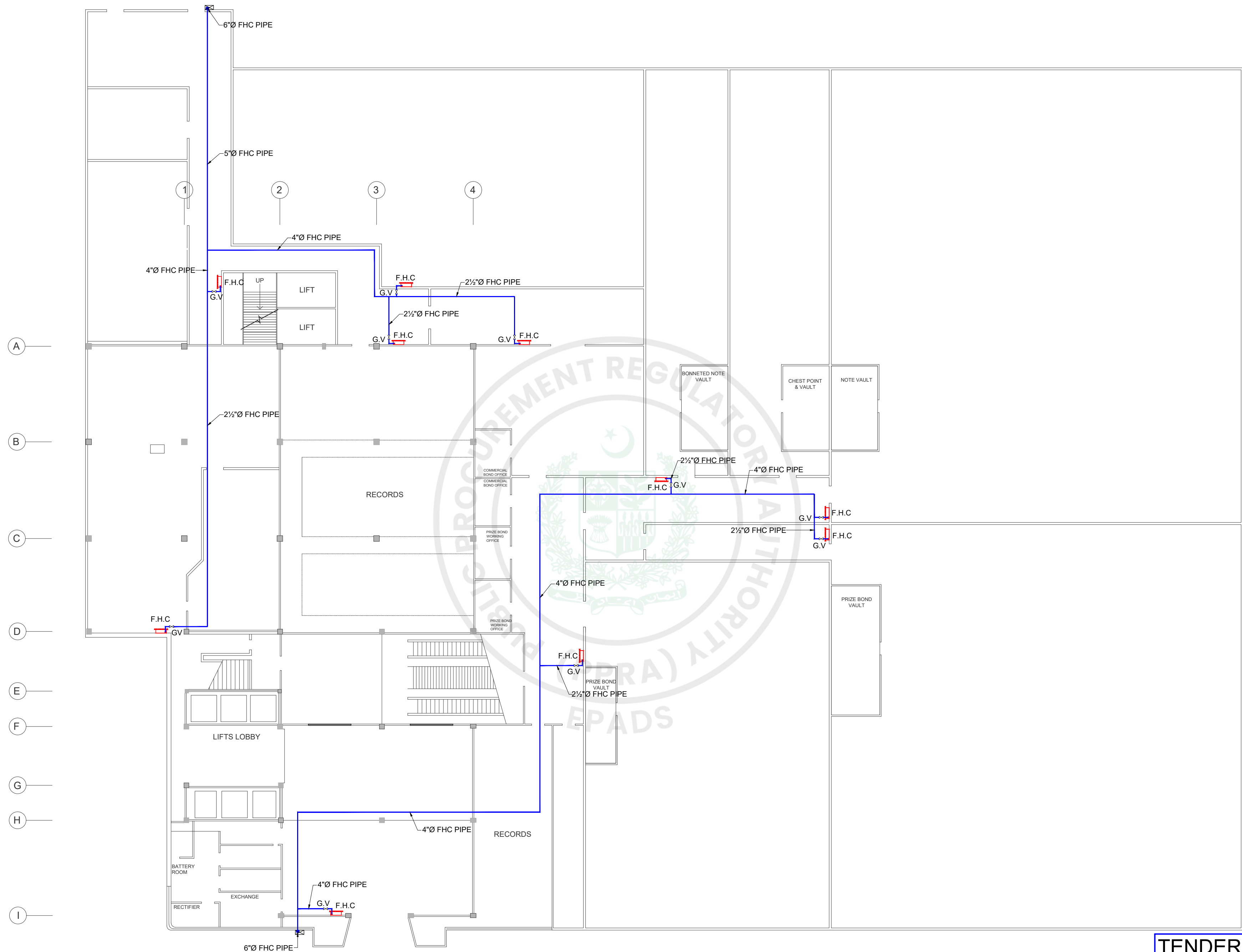
CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS  YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING		DRAWING NO. FF-103		REVISION 0
						DRAWING TITLE LAYOUT OF EXISTING PUMP ROOM & UNDER GROUND TANK		DRAWN BY JUNAID	SERIAL NO. 03	DATE 24/02/2015
								PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE 1/16"-1'-0"



KEY PLAN

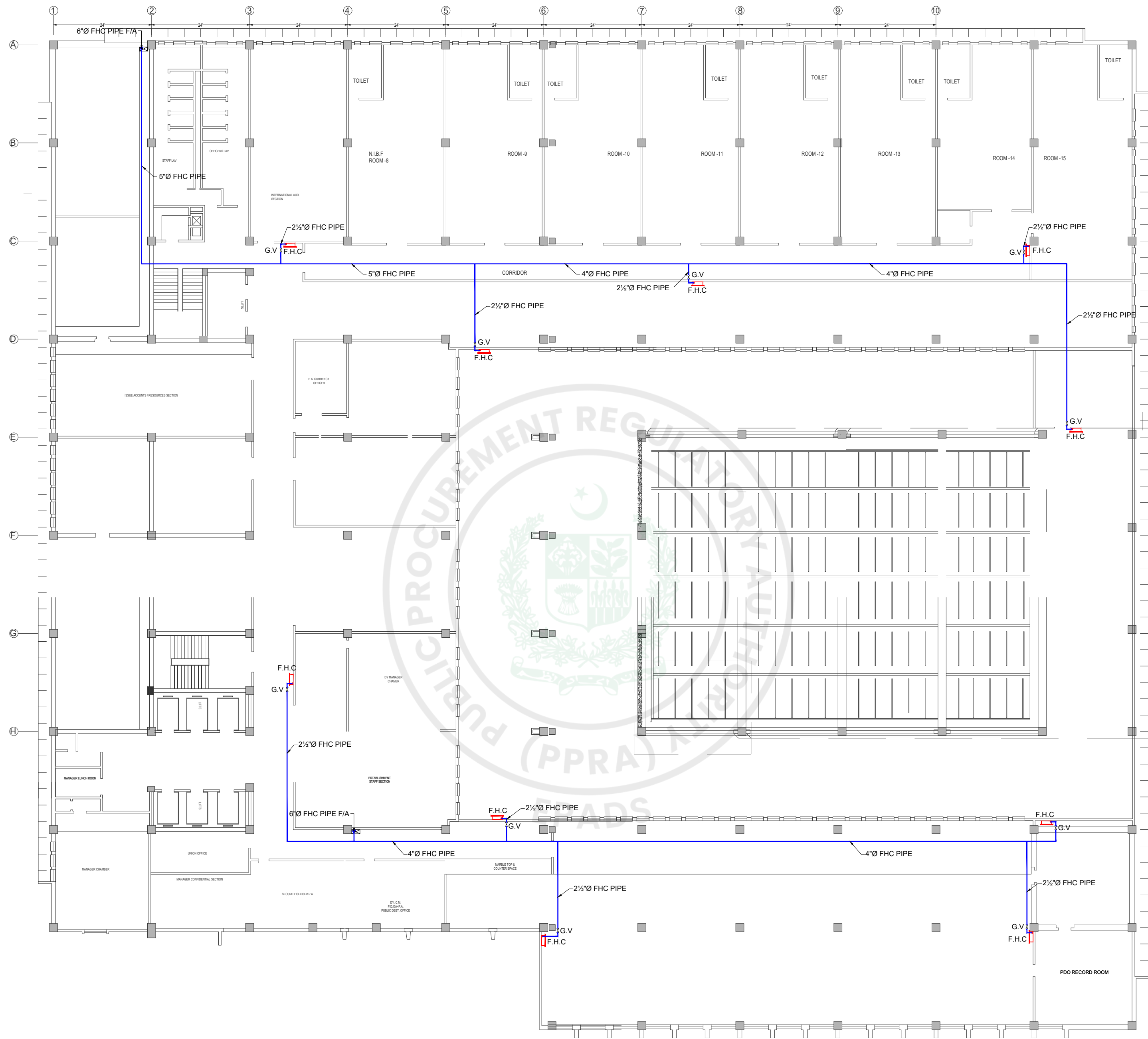
ENLARGE DETAIL OF PROPOSED PUMP ROOM AND UNDER GROUND TANK

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT HEAD OFFICE UP GRADATION (PHASE - 2) MAIN BLDG. & LRC BLDG.	REV.	DATE	DESCRIPTION	H.V.A.C & MECHANICAL CONSULTANTS YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING	DRAWING NO. FF-104		REVISION 0
						DRAWING TITLE LAYOUT OF PROPOSED PUMP ROOM & UNDER GROUND TANK	DRAWN BY AK	SERIAL NO. 04	DATE 16/12/2014
							PROJECT NO. 2009-04-SBP	APPROVED BY	SCALE 1/16"-1'-0"



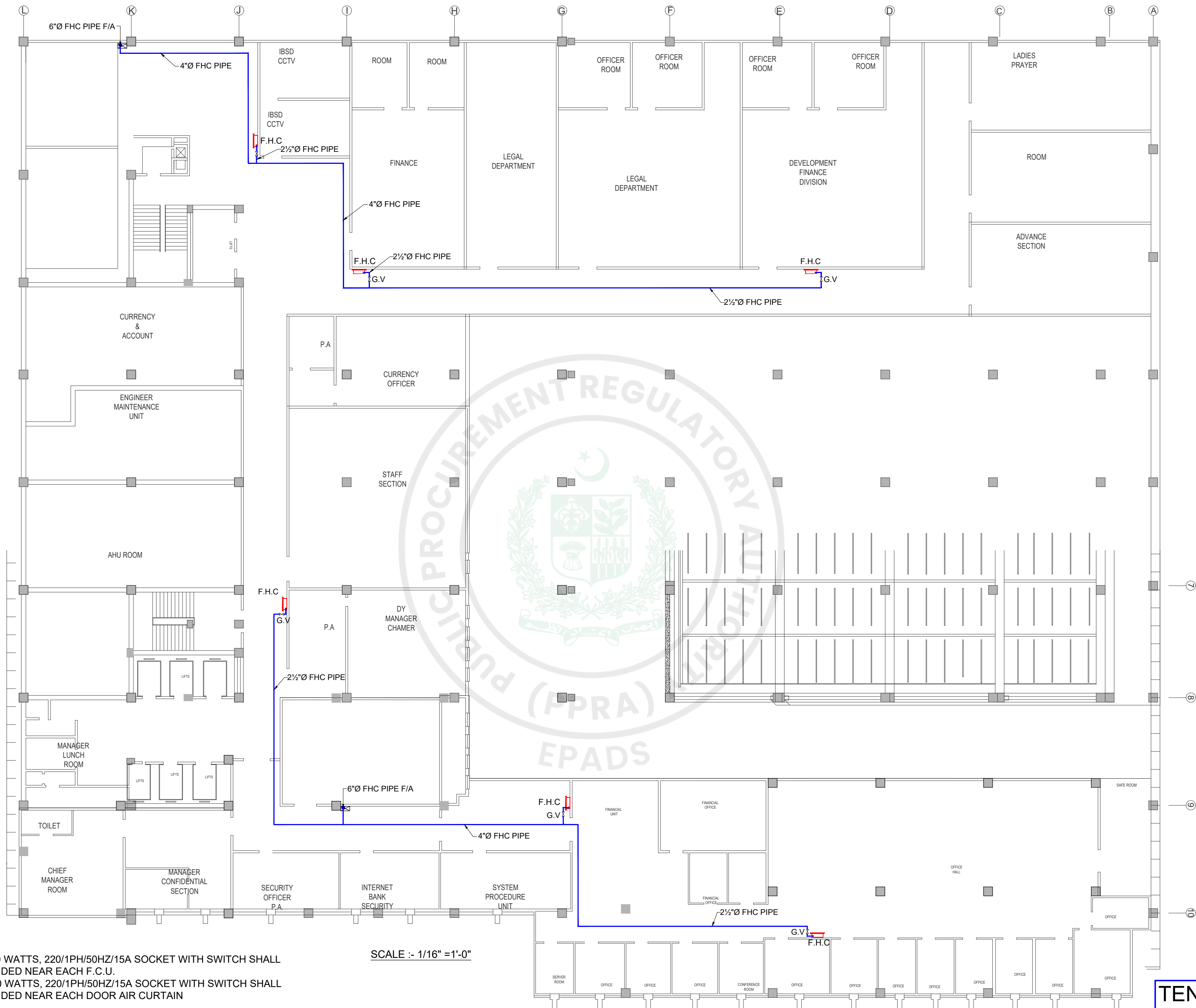
TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS  YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING		DRAWING NO. FF-105		REVISION A
		A	02-05-18	AS PER CLIENT COMMENTS		DRAWING TITLE		DRAWN BY	SERIAL NO.	DATE
						LAYOUT OF FIRE FIGHTING SYSTEM GROUND FLOOR VAULT AREA MAIN BUILDING		JUNAID	05	24/02/2015
								PROJECT NO.	APPROVED BY	SCALE
								2014-12-SBP		1/16"=1'-0"



TENDER DRAWING

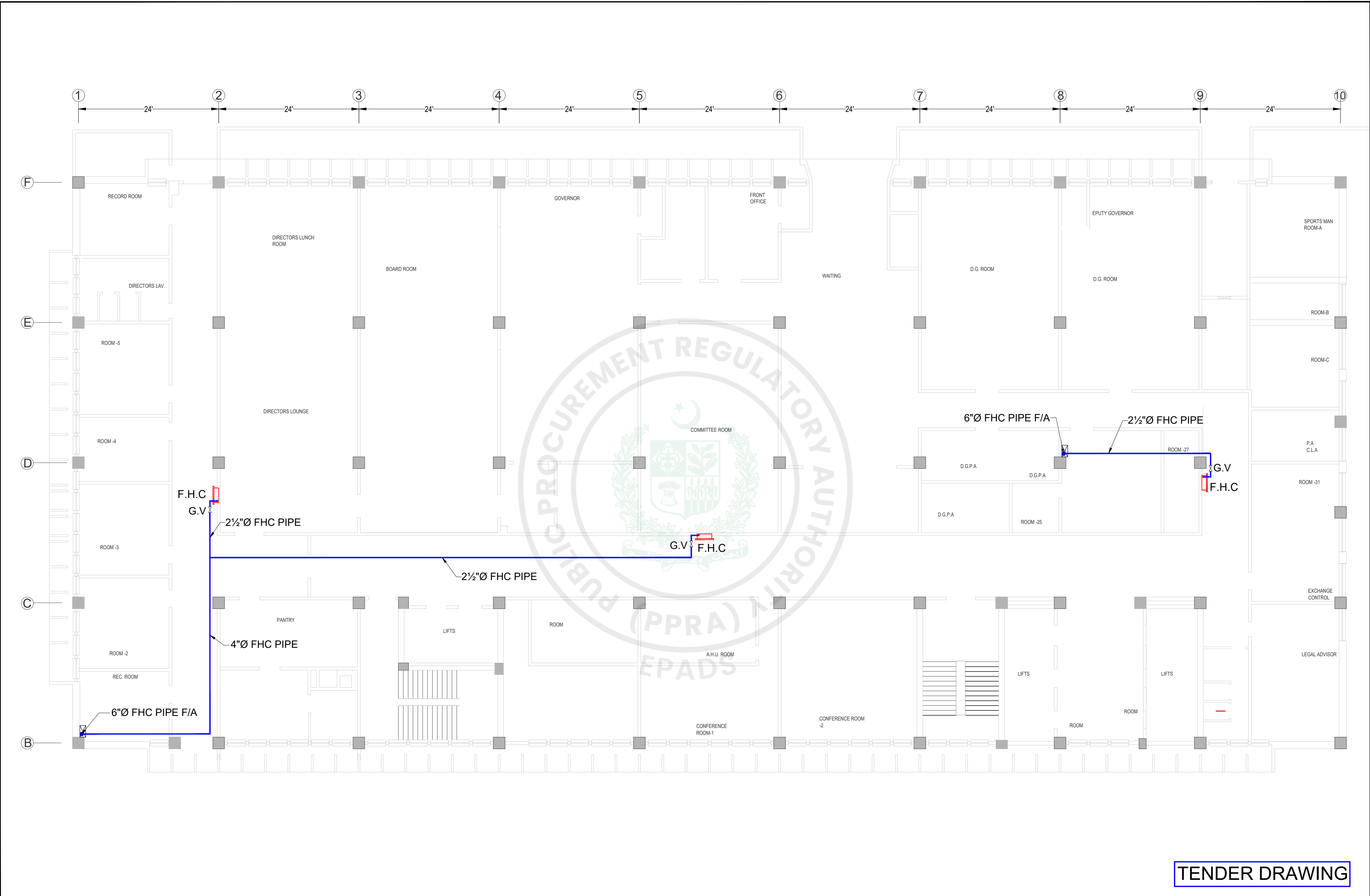
CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	H.V.A.C & MECHANICAL CONSULTANTS YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING	DRAWING NO. FF - 106		REVISION 0
						DRAWING TITLE LAYOUT OF FIRE FIGHTING SYSTEM FIRST FLOOR FOR MAIN BUILDING	DRAWN BY NABEEL	SERIAL NO. 06	DATE 16/12/2014
							PROJECT NO. 2009-04-SBP	APPROVED BY	SCALE 1/16"=1'-0"



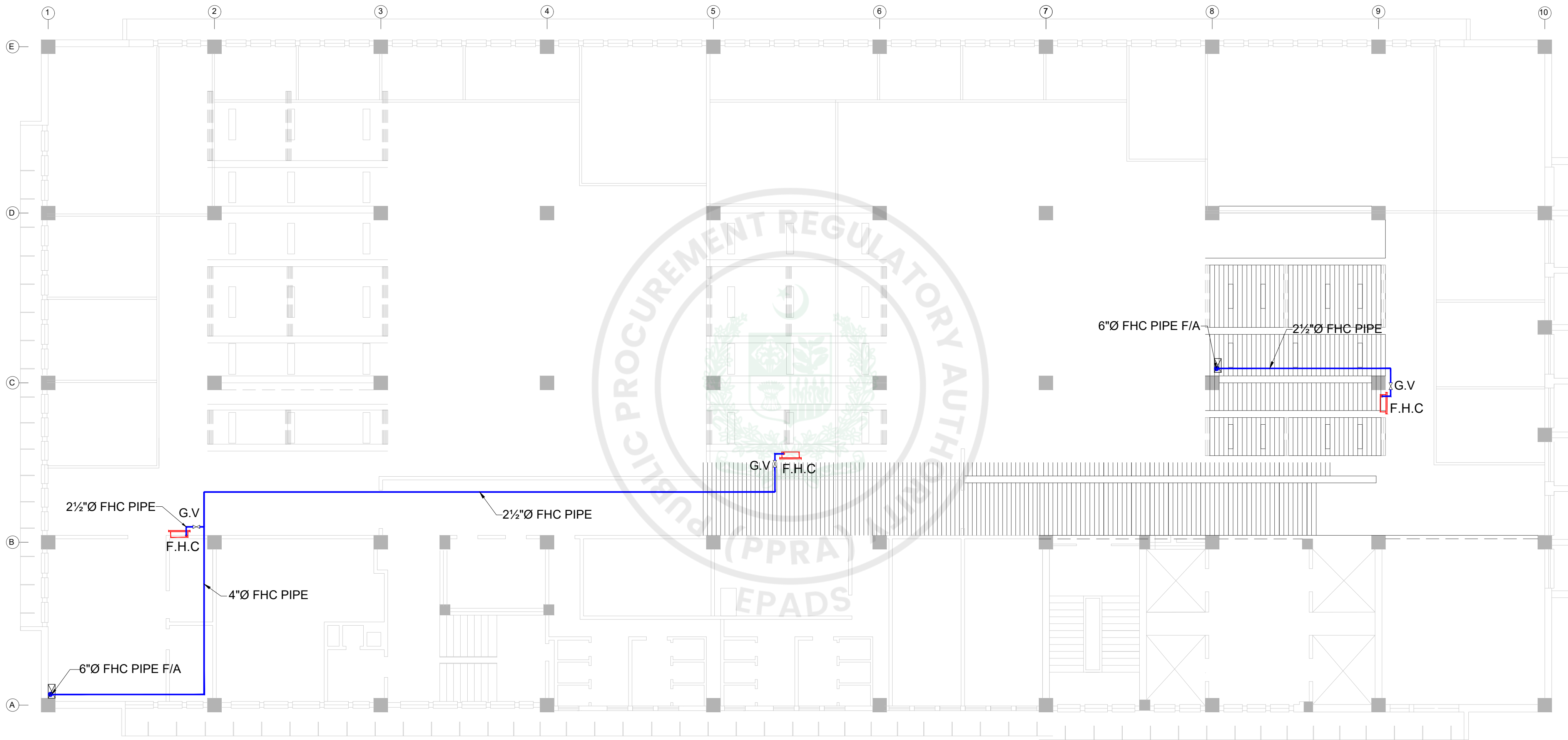
- NOTES:
- 1- E.S.P. 300 WATTS, 220/1PH/50HZ/15A SOCKET WITH SWITCH SHALL BE PROVIDED NEAR EACH F.C.U.
 - 2- E.S.P. 800 WATTS, 220/1PH/50HZ/15A SOCKET WITH SWITCH SHALL BE PROVIDED NEAR EACH DOOR AIR CURTAIN

SCALE :- 1/16" = 1'-0"

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING		DRAWING NO. FF - 107		REVISION 0	
						DRAWING TITLE LAYOUT OF FIRE FIGHTING SYSTEM SECOND FLOOR FOR MAIN BUILDING		DRAWN BY NABEEL	SERIAL NO. 07	DATE 24/02/2015	
								PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE 1/16"=1'-0"	

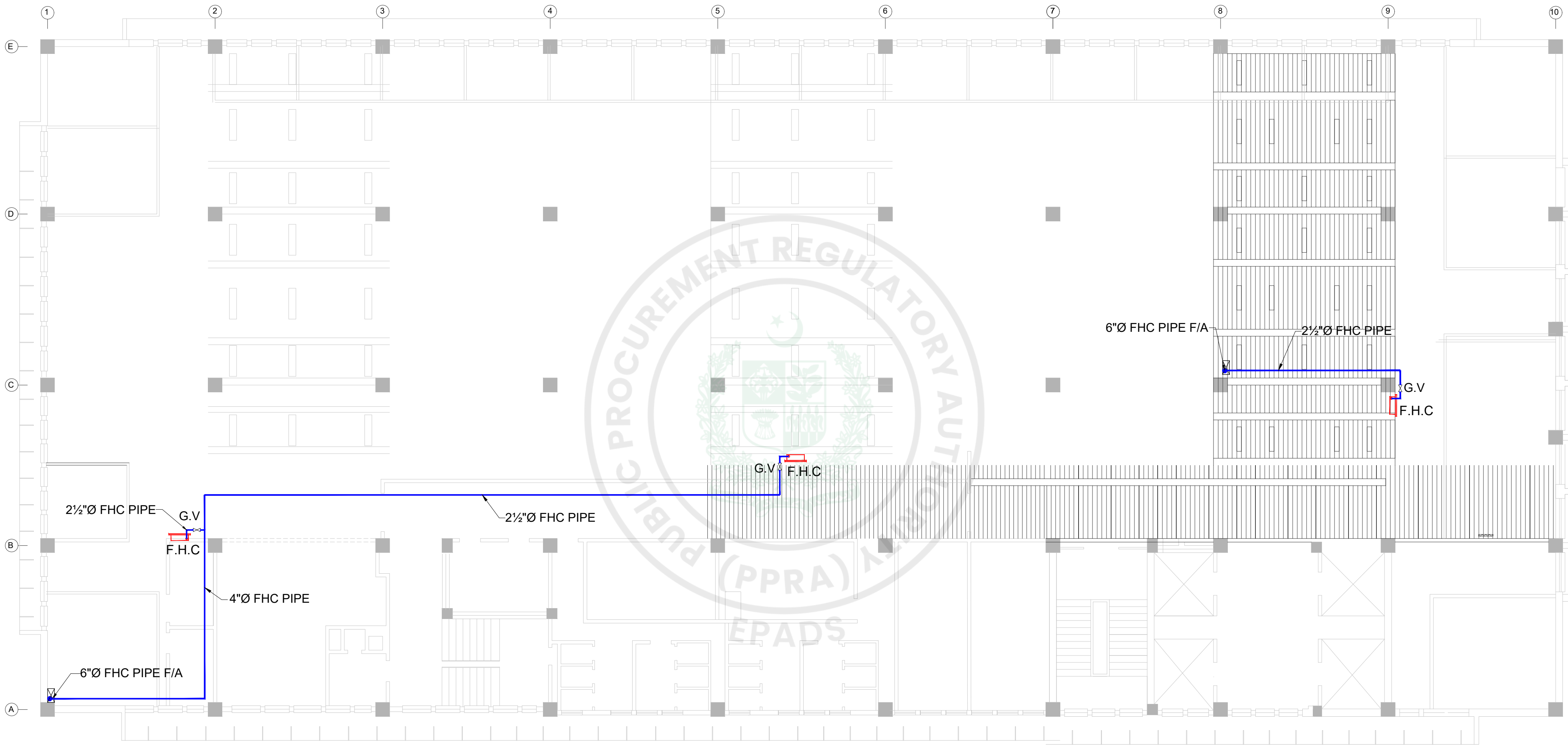


CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS YOU S U F H A S A N A S S O C I A T E S 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 432 1559 (921) 438 9931, FAX- 021-432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING		DRAWING NO. FF- 108		REVISION 0
						DRAWING TITLE LAYOUT OF FIRE FIGHTING SYSTEM THIRD FLOOR MAIN BUILDING		DRAWN BY SHEHARYAR	SERIAL NO. 08	DATE 24/02/2015
								PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE 3/32"=1'-0"



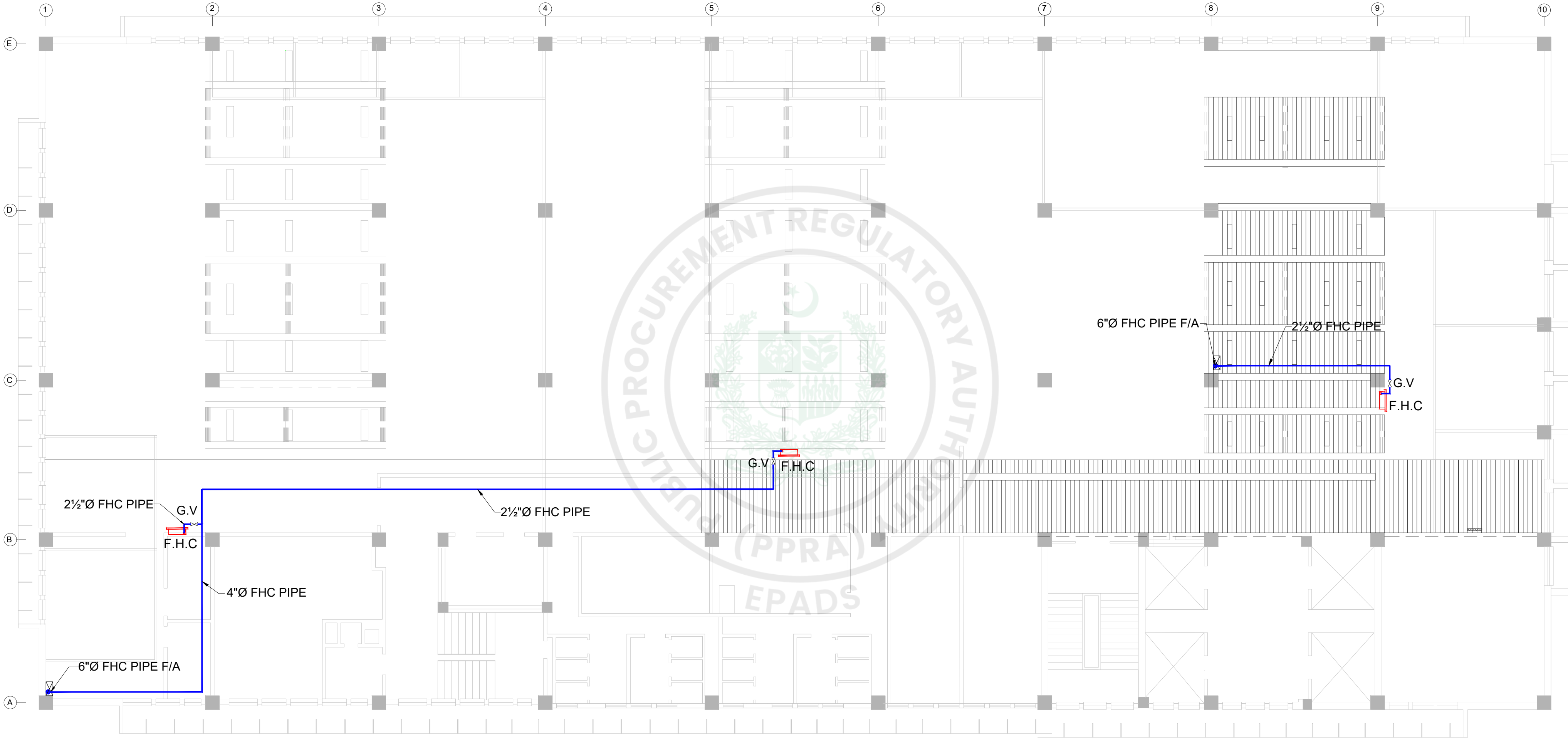
TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS  YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING DRAWING TITLE LAYOUT OF FIRE FIGHTING SYSTEM, 4TH FLOOR MAIN BUILDING	DRAWING NO. FF- 109		REVISION 0
							DRAWN BY NABEEL	SERIAL NO. 09	DATE 24/02/2015
							PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE 3/32"=1'-0"



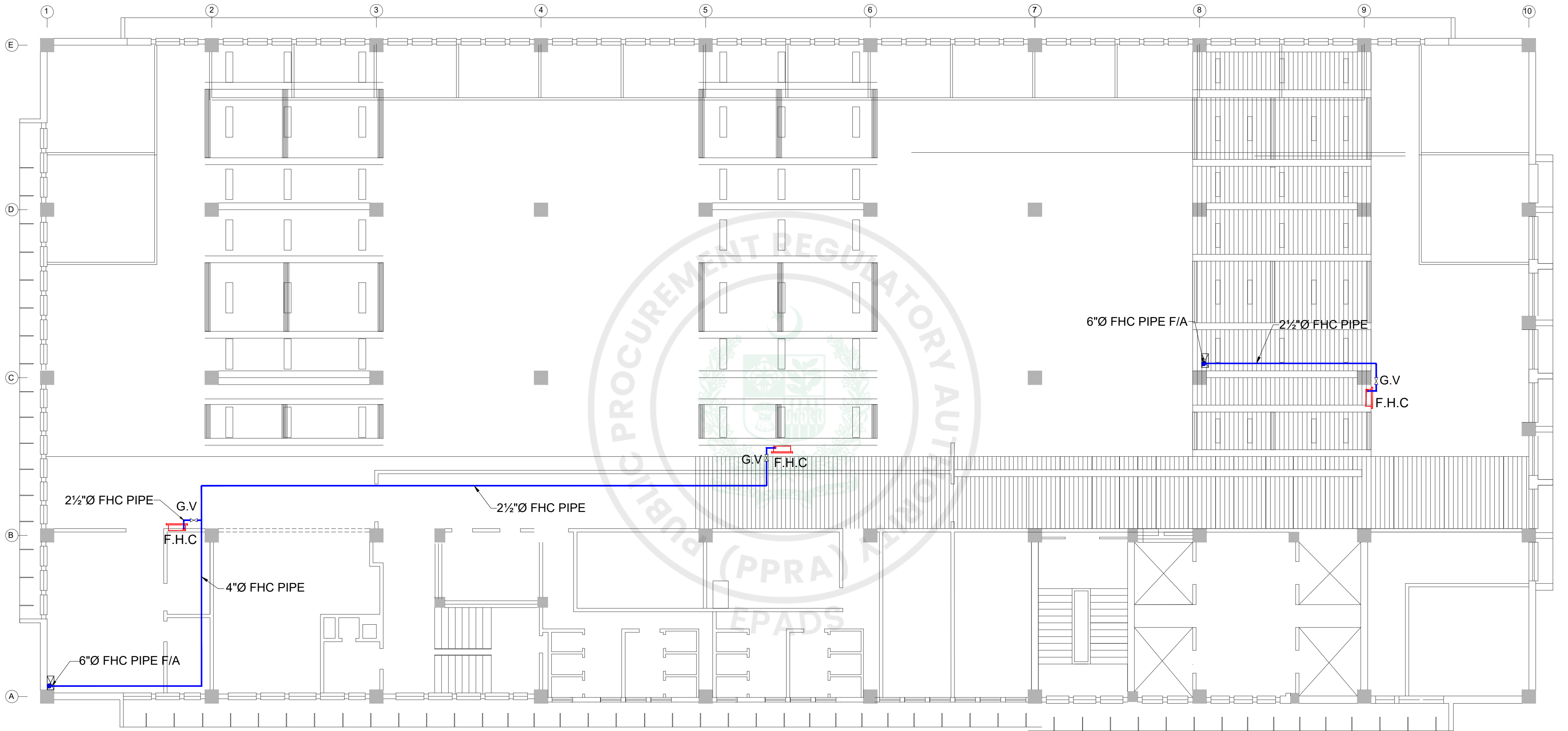
TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS  YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING		DRAWING NO. FF - 110		REVISION 0
						DRAWING TITLE LAYOUT OF FIRE FIGHTING SYSTEM 5TH FLOOR MAIN BUILDING		DRAWN BY NABEEL	SERIAL NO. 10	DATE 24/02/2015
								PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE 3/32"=1'-0"



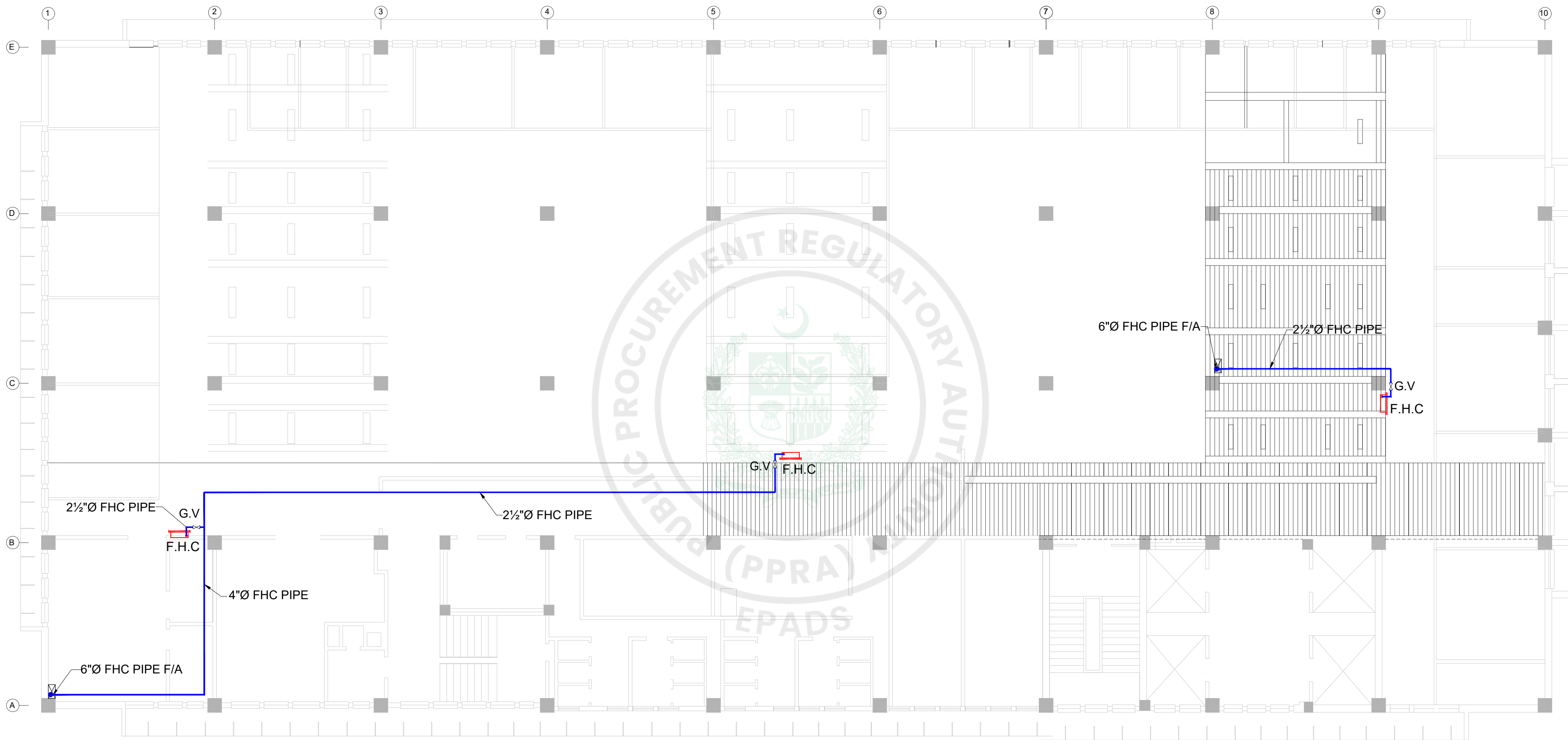
TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING		DRAWING NO. FF- 111		REVISION 0	
						DRAWING TITLE LAYOUT OF FIRE FIGHTING SYSTEM 6TH FLOOR MAIN BUILDING		DRAWN BY JUNAID	SERIAL NO. 11	DATE 24/02/2015	
								PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE 3/32"=1'-0'	



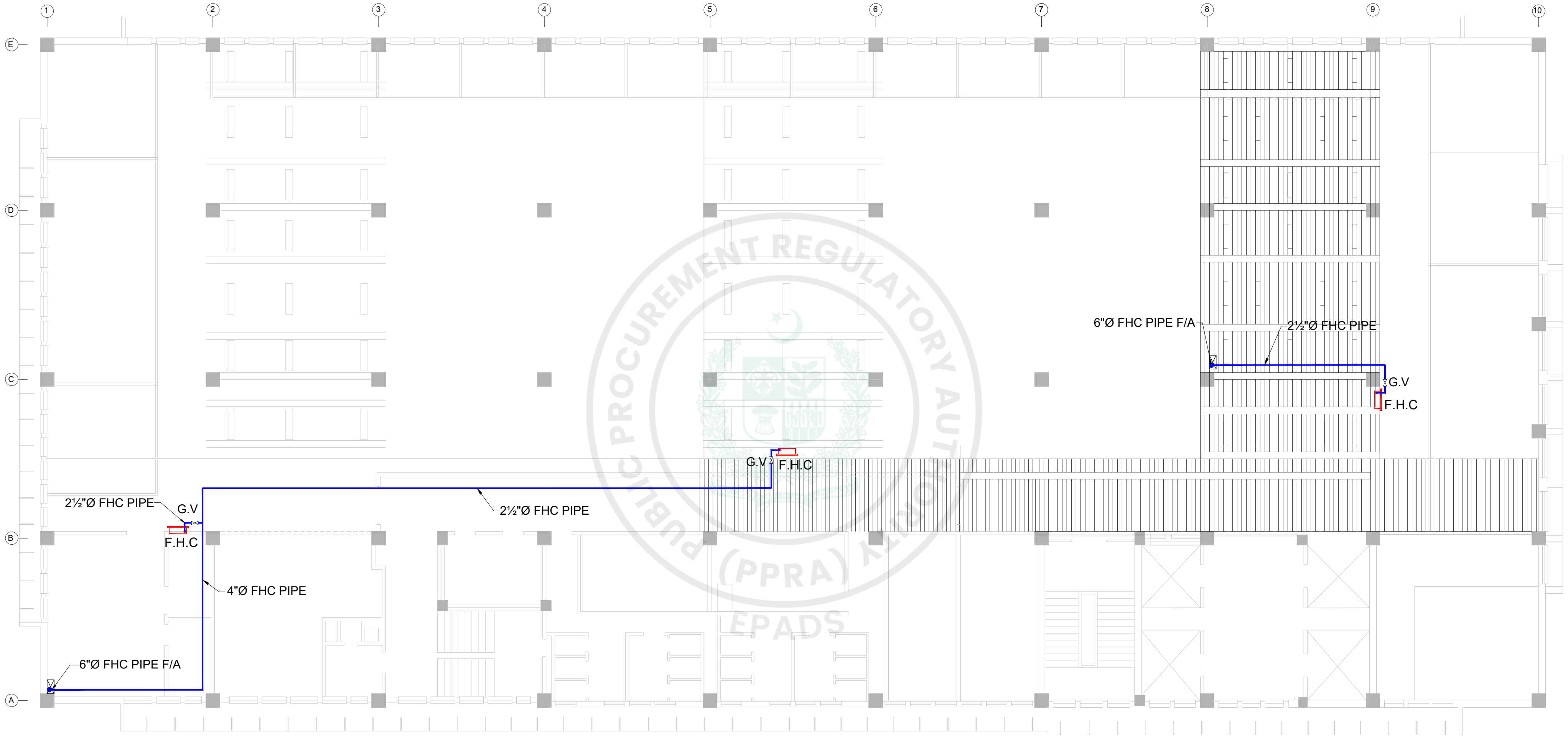
TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS  YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING		DRAWING NO. FF - 112		REVISION 0
						DRAWING TITLE LAYOUT OF FIRE FIGHTING SYSTEM 7TH FLOOR MAIN BUILDING		DRAWN BY JUNAID	SERIAL NO. 12	DATE 24/02/2015
								PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE 3/32"=1'-0"



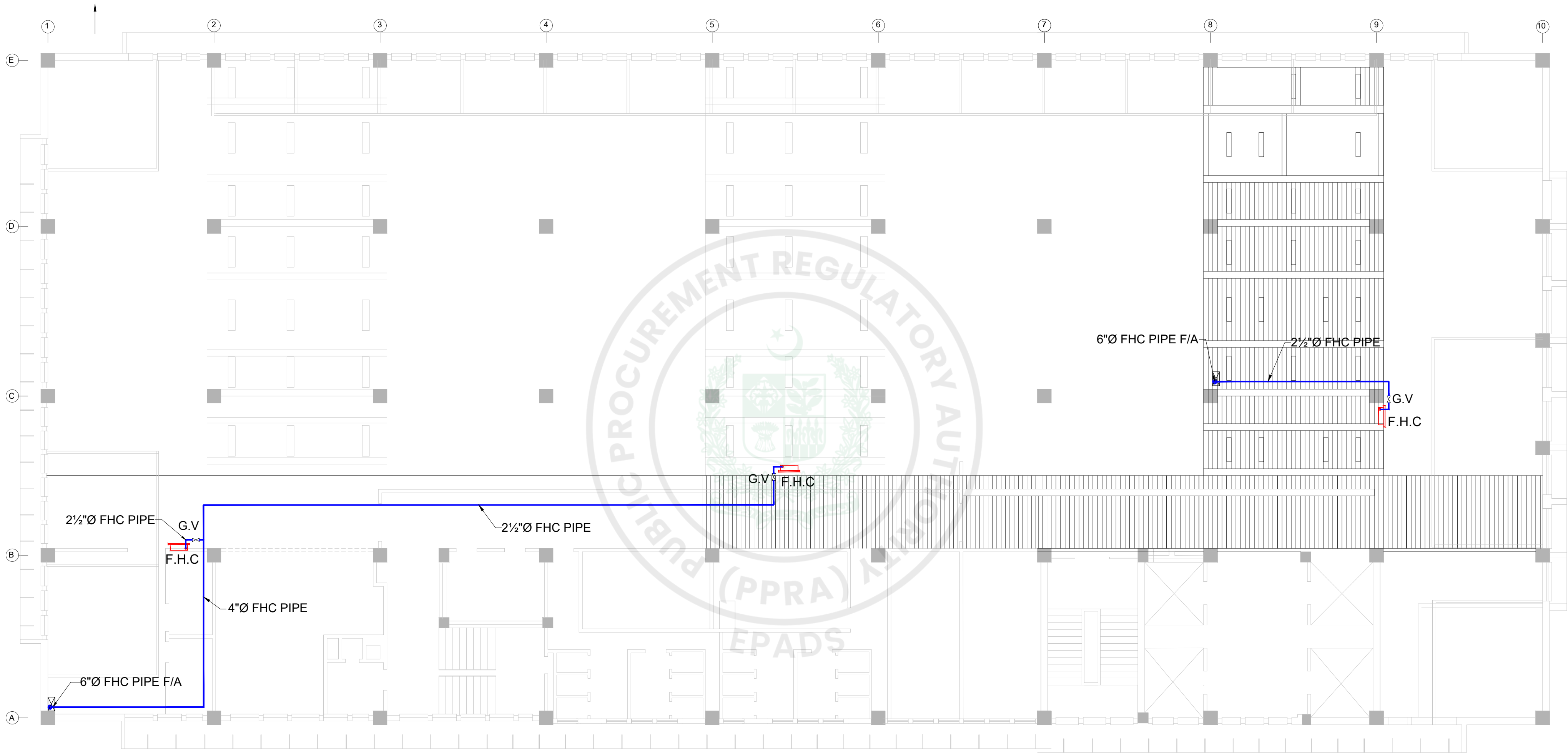
TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE AIRCONDITIONING		DRAWING NO. FF - 113		REVISION 0
						DRAWING TITLE LAYOUT OF FIRE FIGHTING SYSTEM 8TH FLOOR MAIN BUILDING		DRAWN BY NABEEL	SERIAL NO. 13	DATE 24/02/2015
								PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE 3/32"=1'-0"



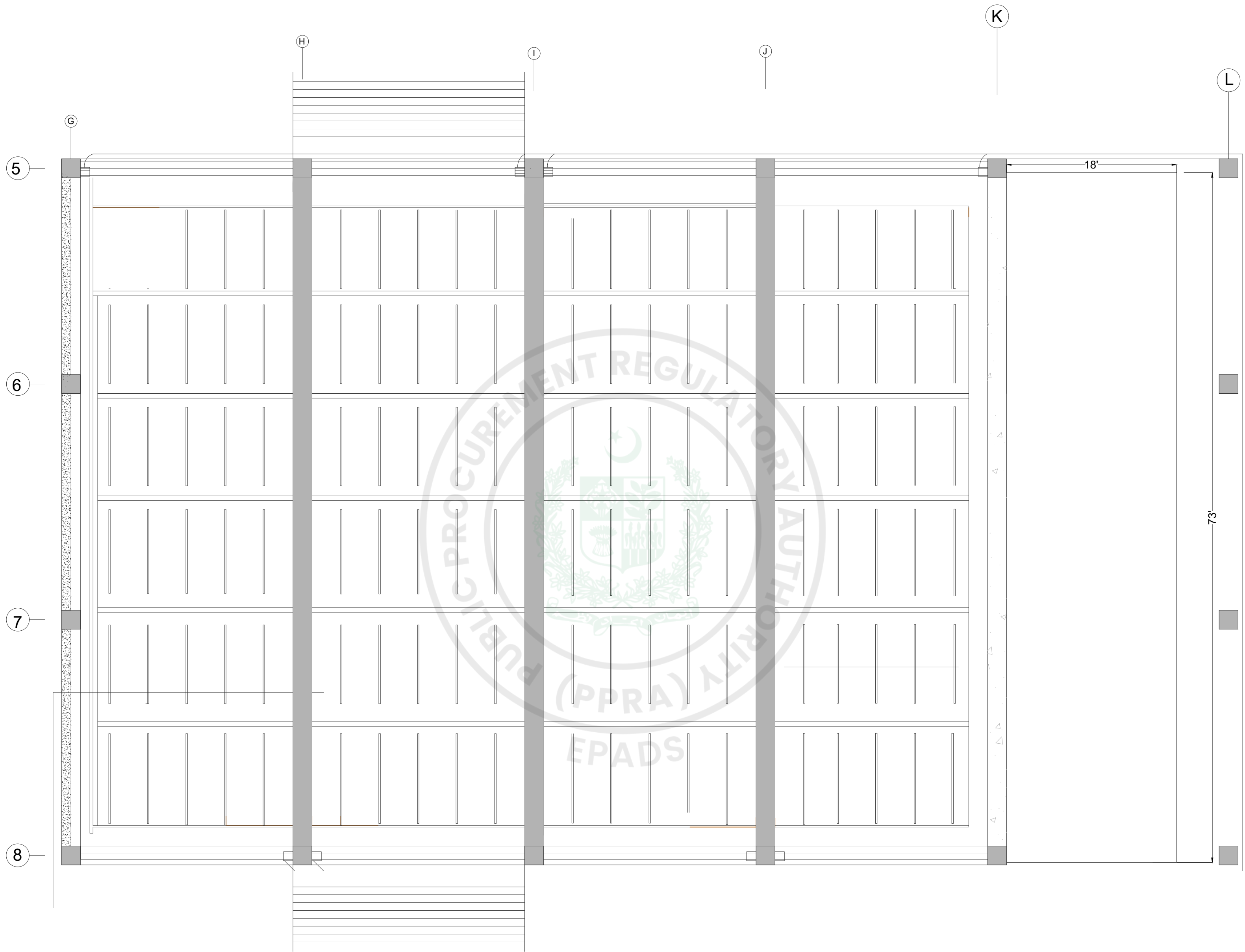
TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING DRAWING TITLE LAYOUT OF FIRE FIGHTING SYSTEM 9TH FLOOR MAIN BUILDING	DRAWING NO. FF - 114		REVISION 0
							DRAWN BY JUNAID	SERIAL NO. 14	DATE 24/02/2015
							PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE 3/32"=1'-0"



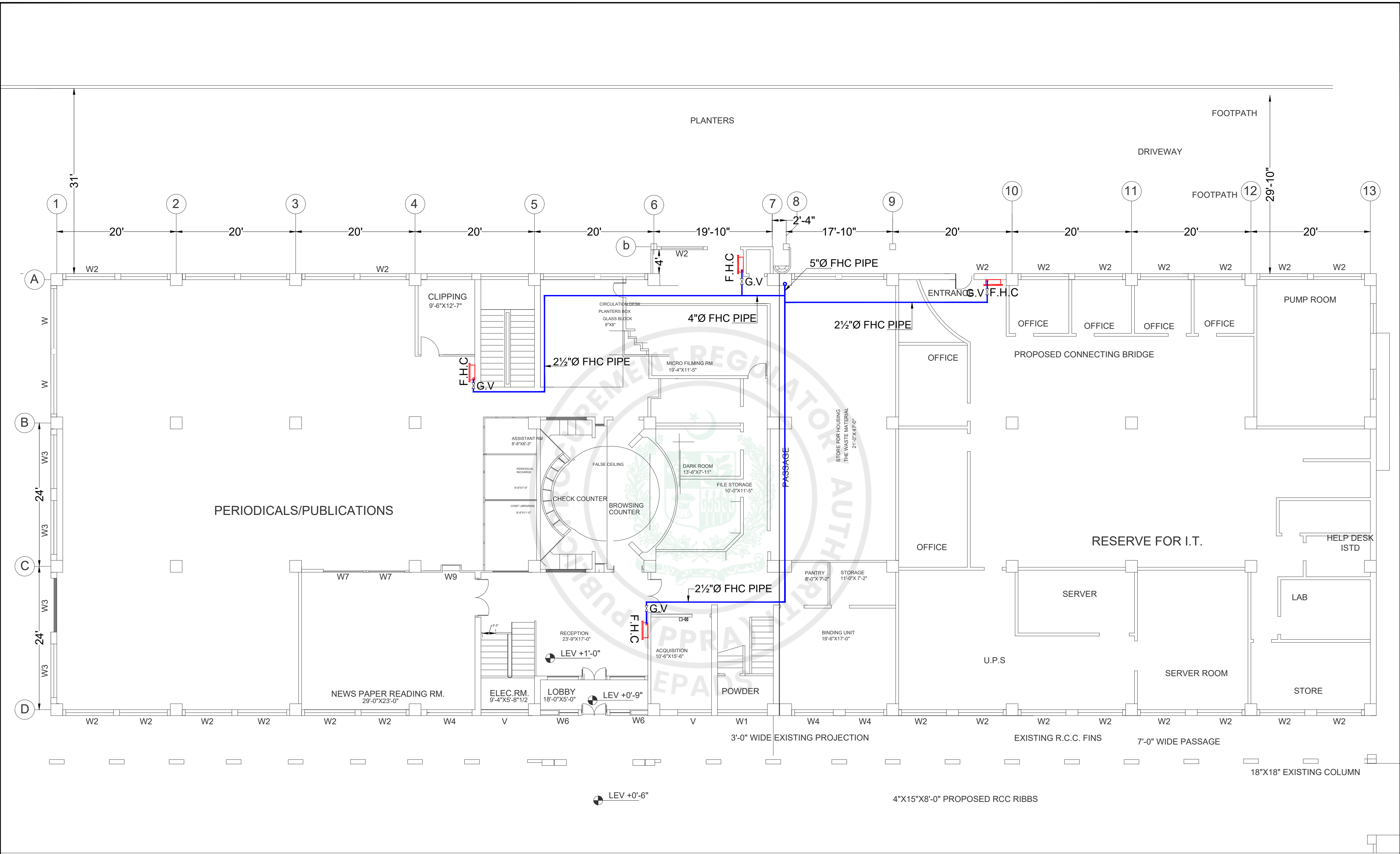
TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING DRAWING TITLE LAYOUT OF FIRE FIGHTING SYSTEM 10TH FLOOR MAIN BUILDING	DRAWING NO. FF - 115		REVISION 0
							DRAWN BY JUNAID	SERIAL NO. 15	DATE 24/02/2015
							PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE 3/32"=1'-0"



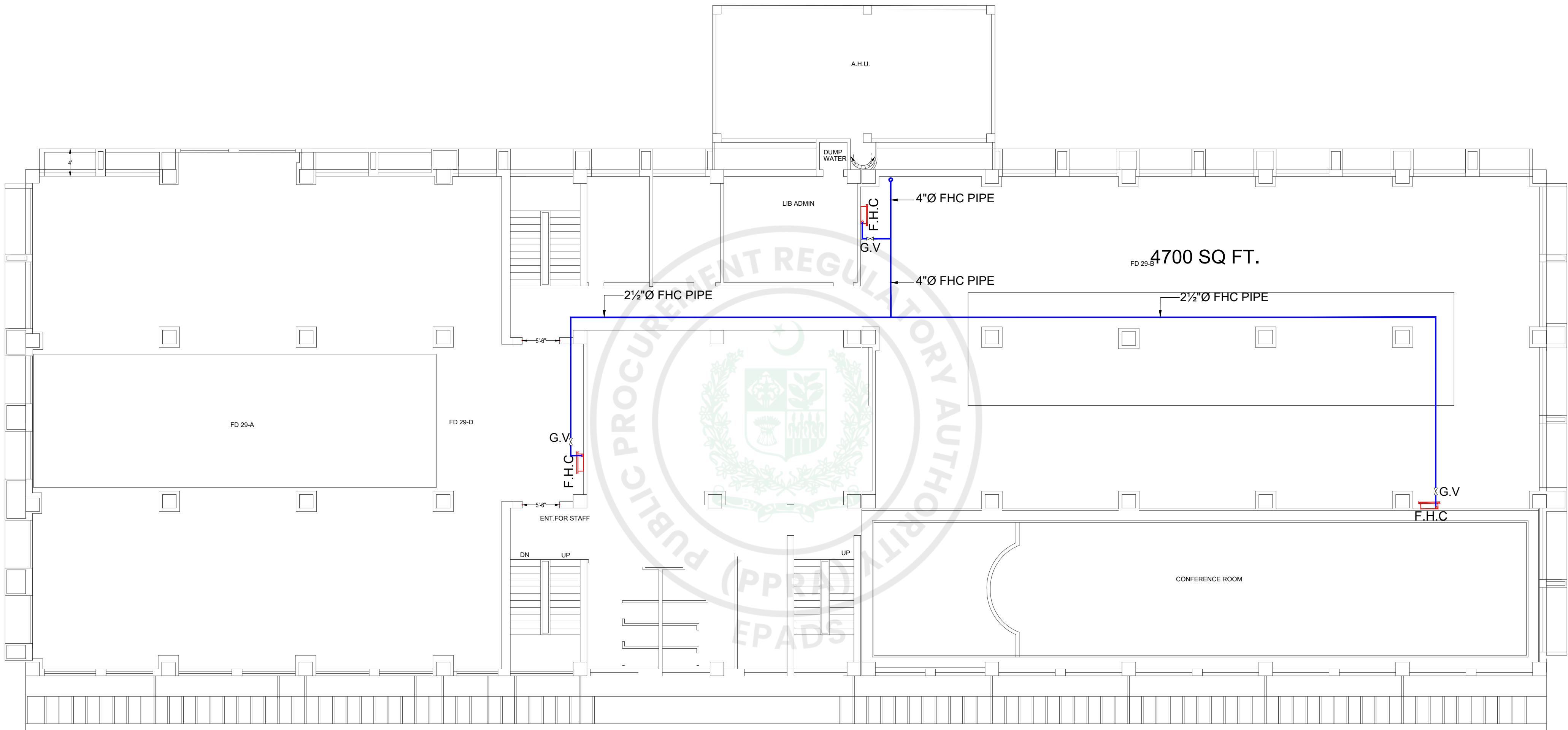
PROGRESS SET
PRELIMINARY DRAWINGS

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS YOUUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING		DRAWING NO. FF- 116		REVISION 0
						DRAWING TITLE LAYOUT OF FIRE FIGHTING SYSTEM & AT ROOF		DRAWN BY NABEEL	SERIAL NO. 16	DATE 24/02/2015
								PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE 1/8"=1'-0"



TENDER DRAWING

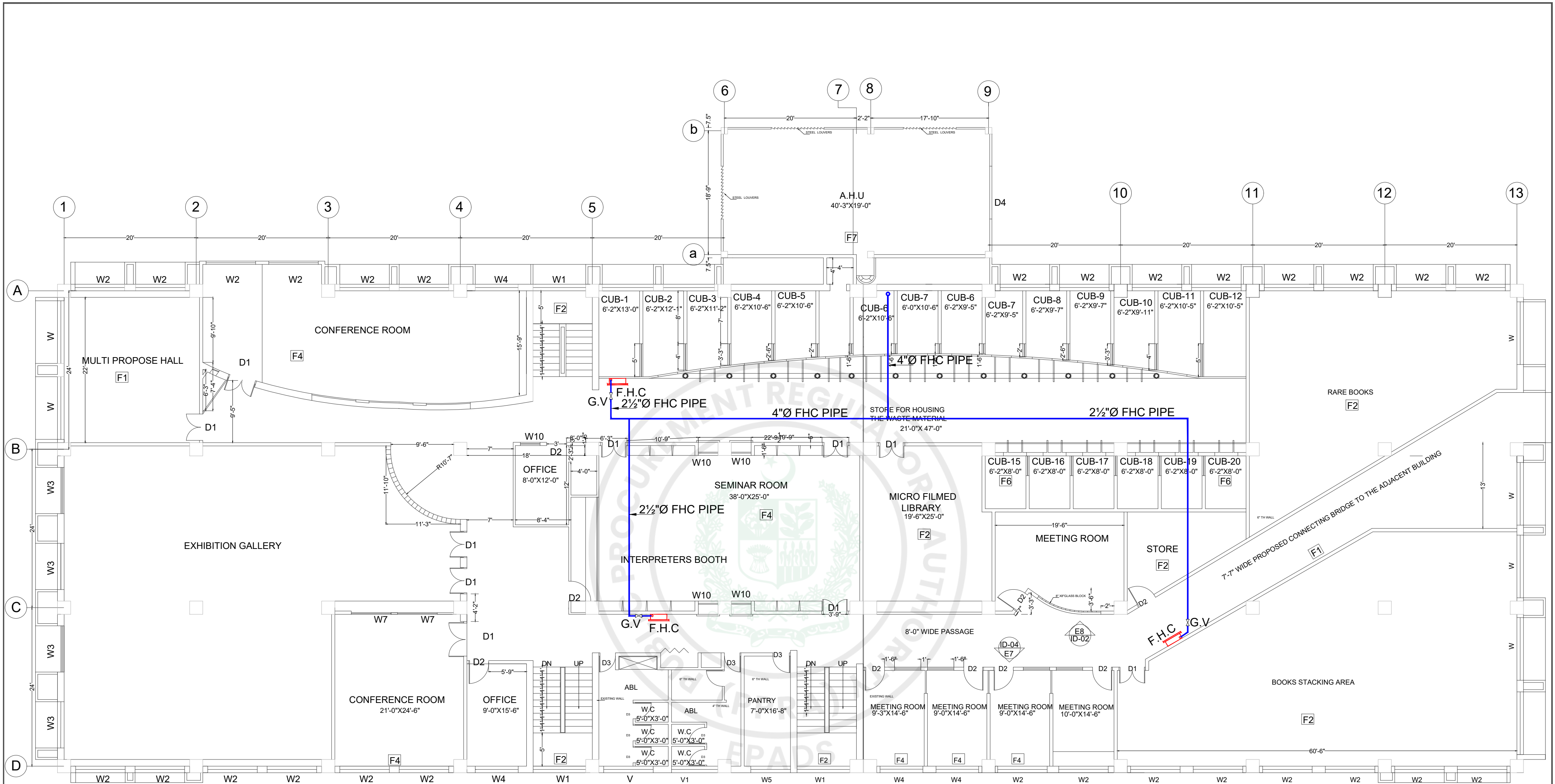
CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS YOU S U F H A S A N A S S O C I A T E S 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING DRAWING TITLE LAYOUT OF FIRE FIGHTING SYSTEM GROUND FLOOR LRC BUILDING	DRAWING NO. FF- 117		REVISION 0
							DRAWN BY SHAHERYAR	SERIAL NO. 17	DATE 24/02/2015
							PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE 3/32"=1'-0"



FIRST FLOOR PLAN (LRC)

TENDER DRAWING

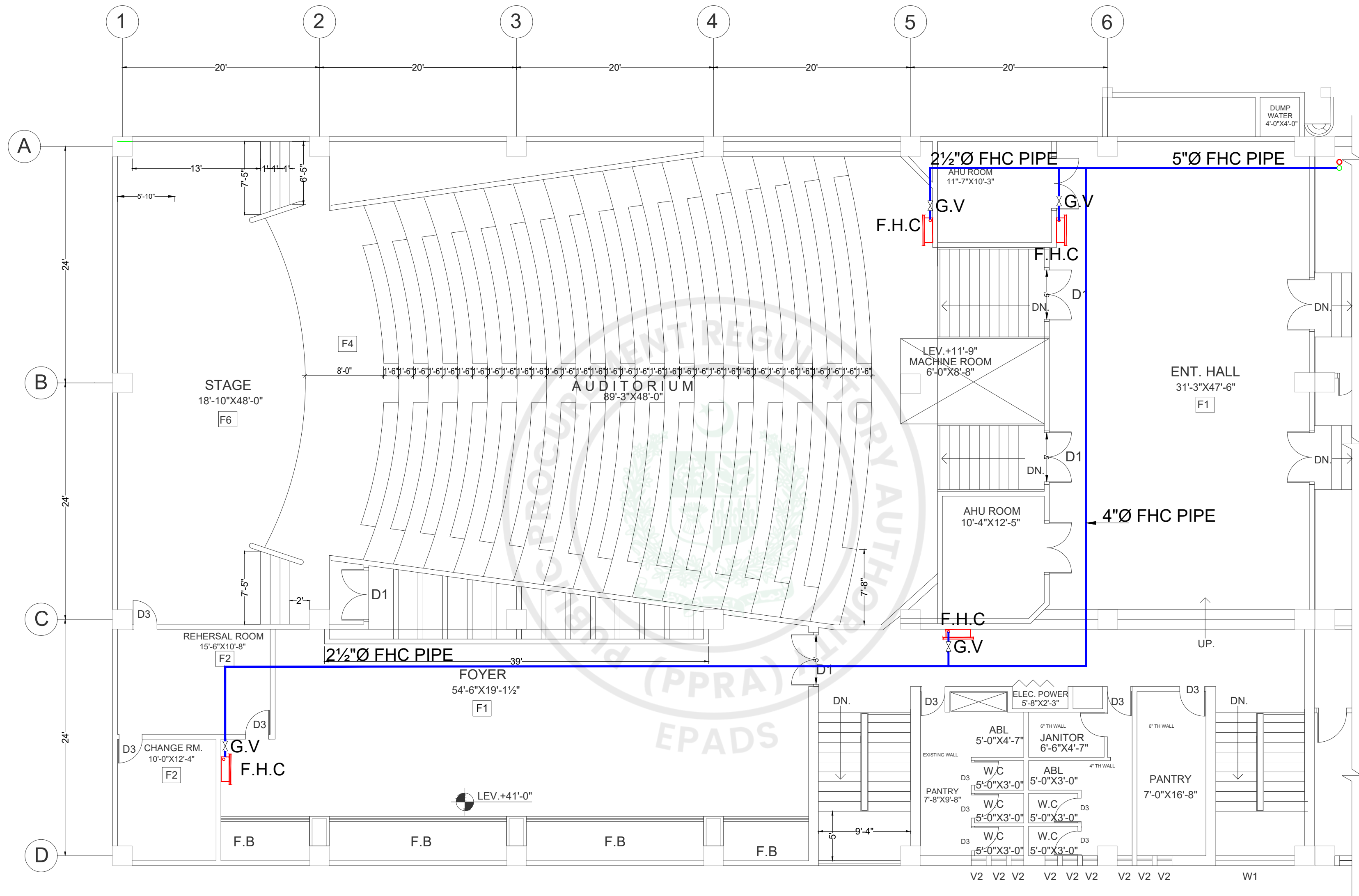
CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING	DRAWING NO. FF- 118		REVISION 0
							DRAWN BY JUNAID	SERIAL NO. 18	DATE 24/02/2015
						DRAWING TITLE LAYOUT OF FIRE FIGHTING SYSTEM AT FIRST FLOOR LRC BUILDING	PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE 3/32"=1'-0"



2ND FLOOR PLAN LRC
SCALE : 3/32"=1'-0"

TENDER DRAWING

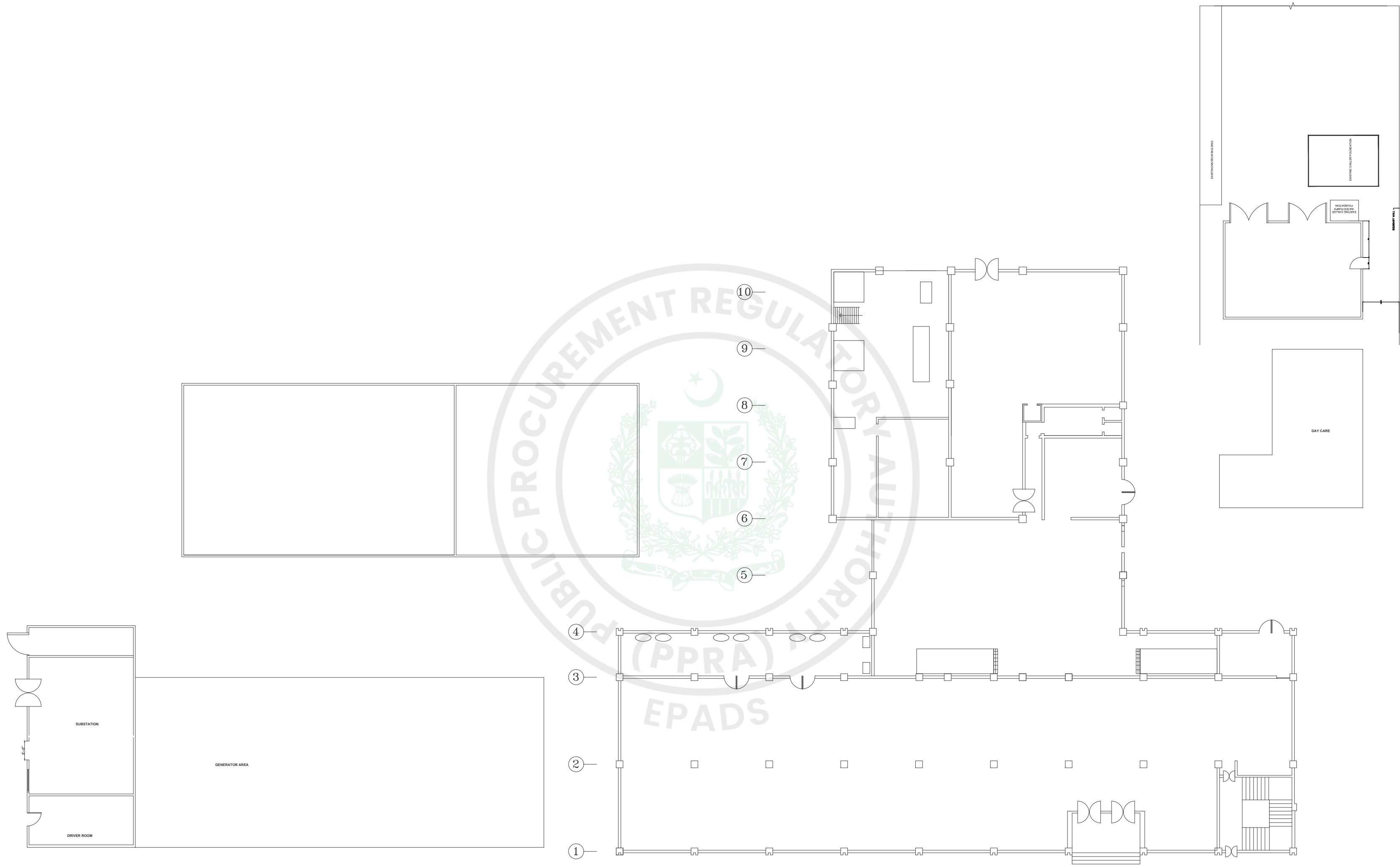
CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS YOUUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING		DRAWING NO. FF- 119		REVISION 0	
						DRAWING TITLE LAYOUT OF FIRE FIGHTING SYSTEM AT SECOND FLOOR LRC BUILDING		DRAWN BY JUNAID	SERIAL NO. 19	DATE 24/02/2015	
								PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE 3/32"=1'-0"	



FIRE H
GATE V
FIRE H
FIRE H
FIRE R

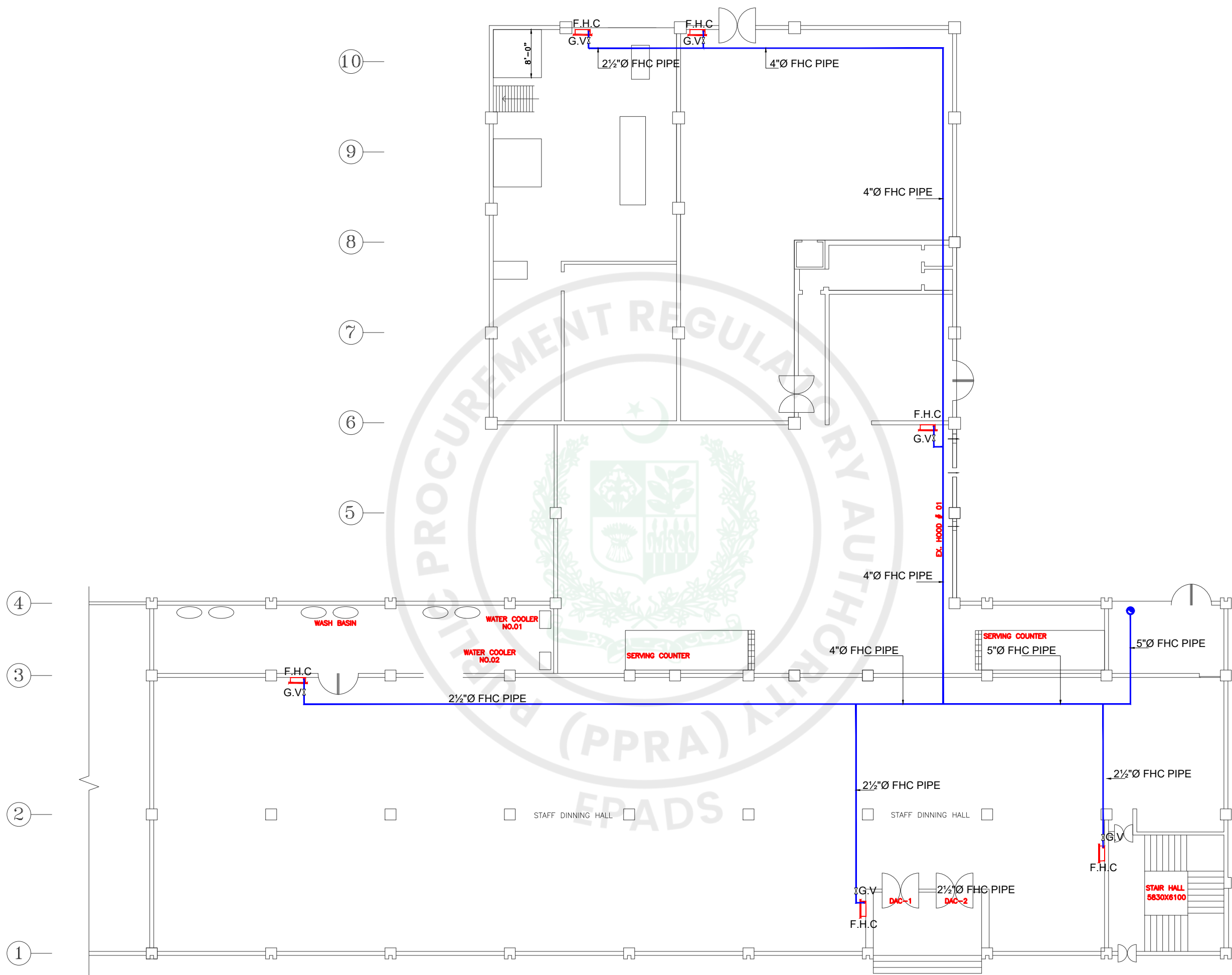
TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING		DRAWING NO. FF- 120		REVISION 0	
						DRAWING TITLE LAYOUT OF FIRE FIGHTING SYSTEM AT THIRD FLOOR AUDITORIUM LRC BUILDING		DRAWN BY JUNAID	SERIAL NO. 20	DATE 24/02/2015	
								PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE 1/8"=1'-0"	



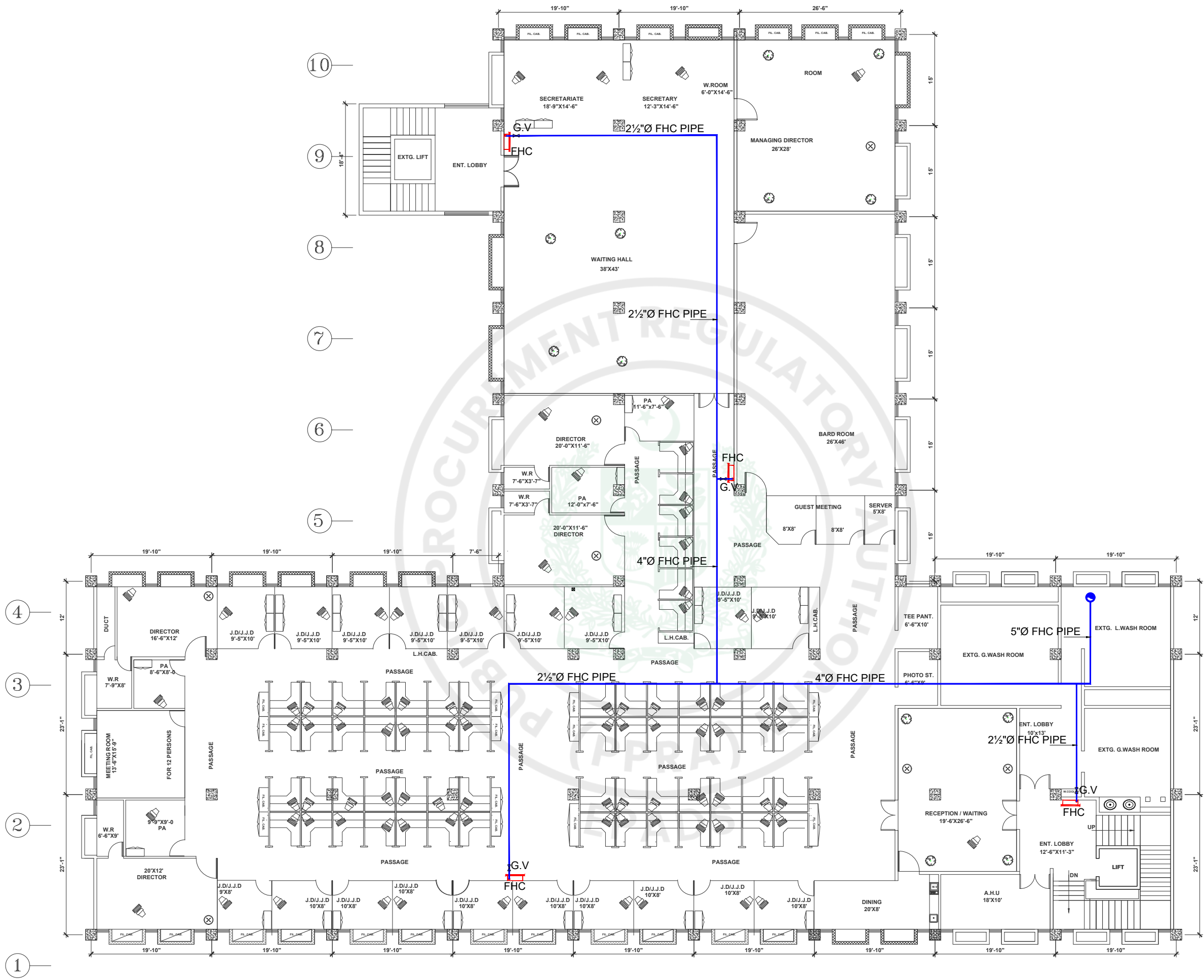
TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS	TITLE	DRAWING NO.		REVISION
	OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM				 YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	FIRE FIGHTING	FF-122	0	
		DRAWING TITLE	DRAWN BY	SERIAL NO.		DATE			
		LAYOUT OF SITE PLAN FIRE FIGHTING SYSTEM CANTEEN BUILDING	NABEEL	22		24/02/2015			
			PROJECT NO.	APPROVED BY		SCALE			
			2014-12-SBP			3/64"=1'-0"			



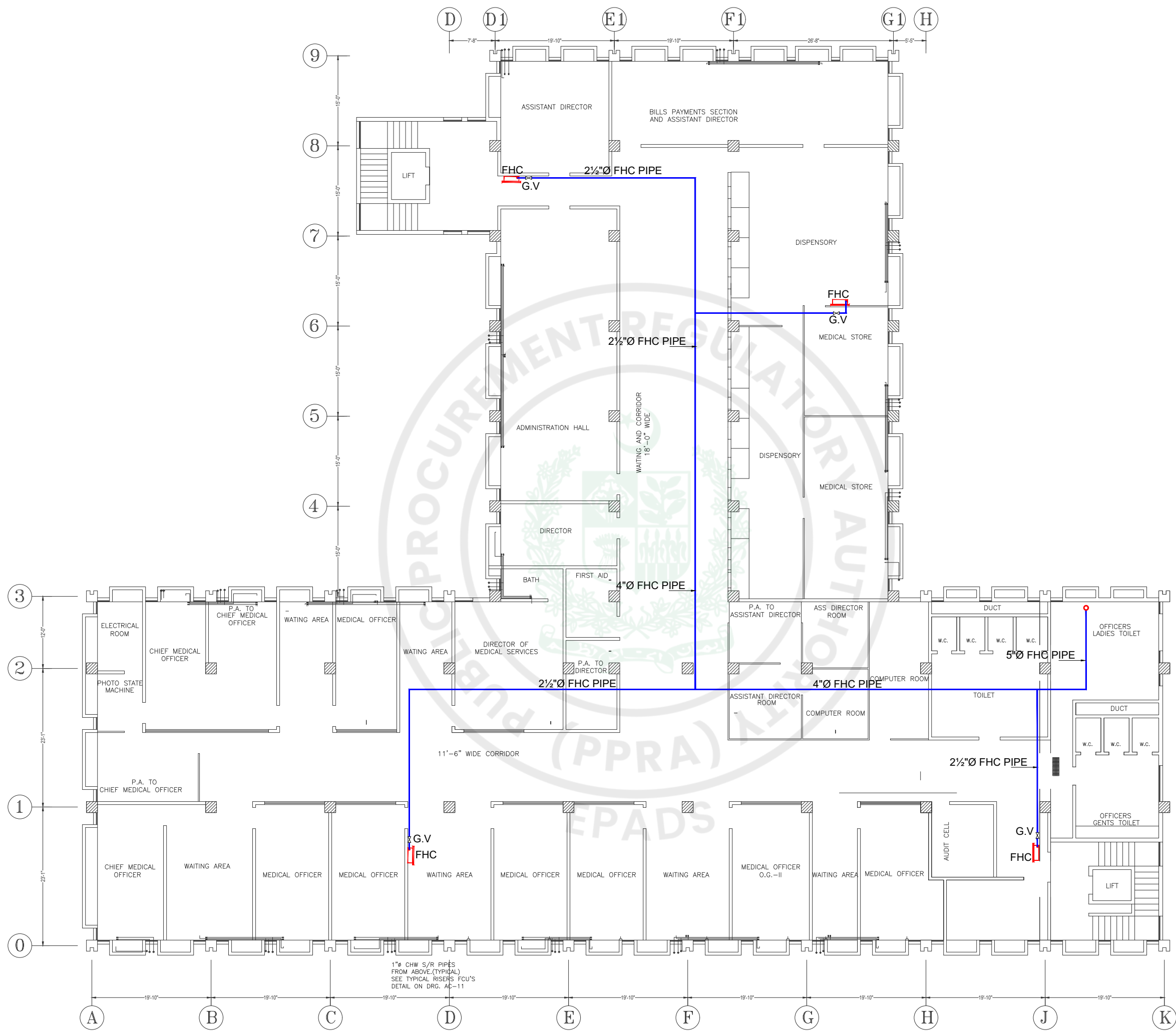
TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS	TITLE	DRAWING NO.		REVISION
	OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM				 YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	FIRE FIGHTING	FF-123	0	
						DRAWING TITLE	DRAWN BY	SERIAL NO.	DATE
						LAYOUT OF FIRE FIGHTING SYSTEM AT GROUND FLOOR CANTEN BUILDING	JUNAID	23	24/02/2015
							PROJECT NO.	APPROVED BY	SCALE
							2014-12-SBP		1/16"=1'-0"



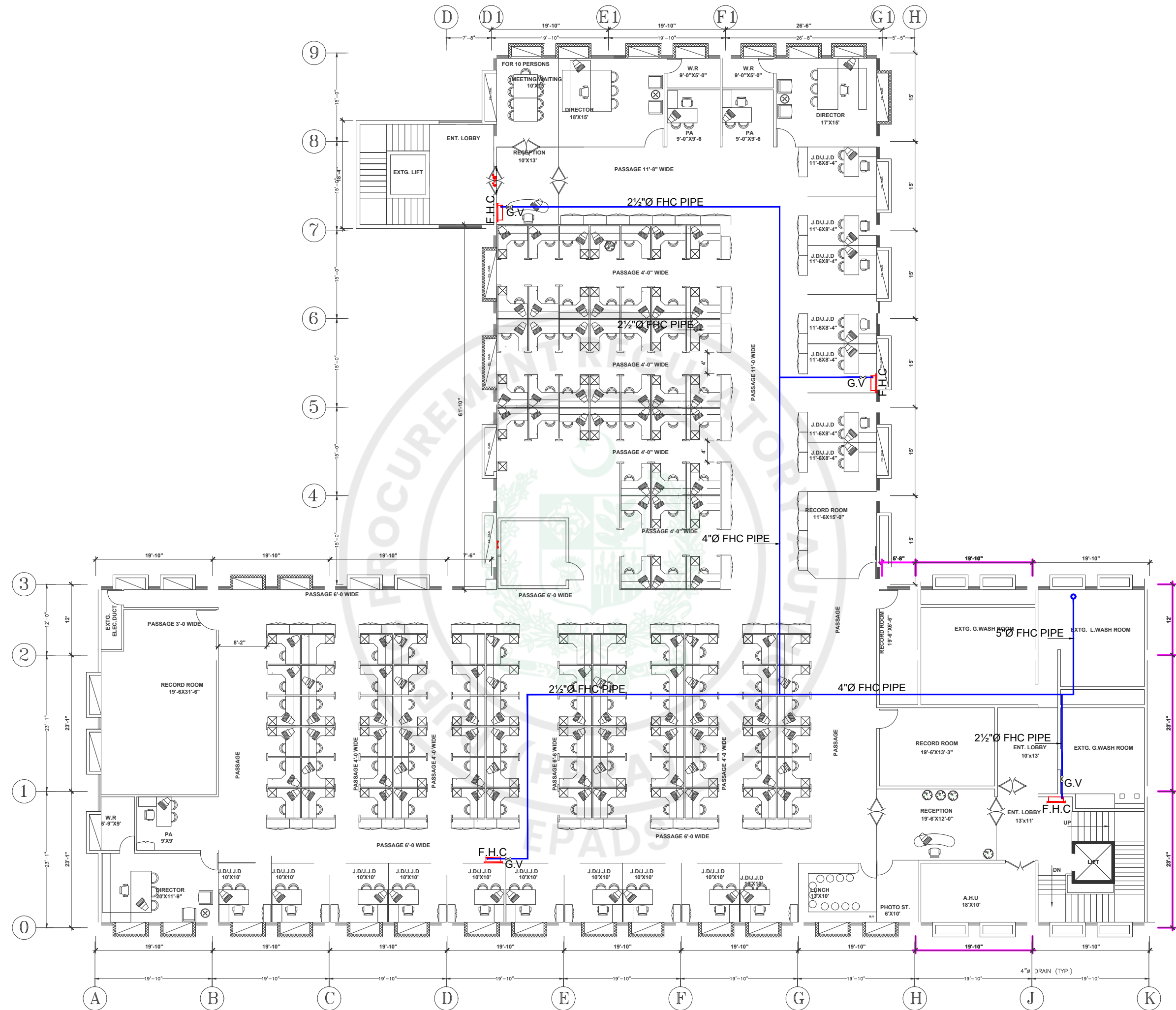
TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS Uh YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING		DRAWING NO. FF-124		REVISION 0
						DRAWING TITLE LAYOUT OF FIRE FIGHTING SYSTEM AT FIRST FLOOR CANTENEN BUILDING		DRAWN BY JUNAID	SERIAL NO. 24	DATE 24/02/2015
								PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE 1/16"=1'-0"



TENDER DRAWING

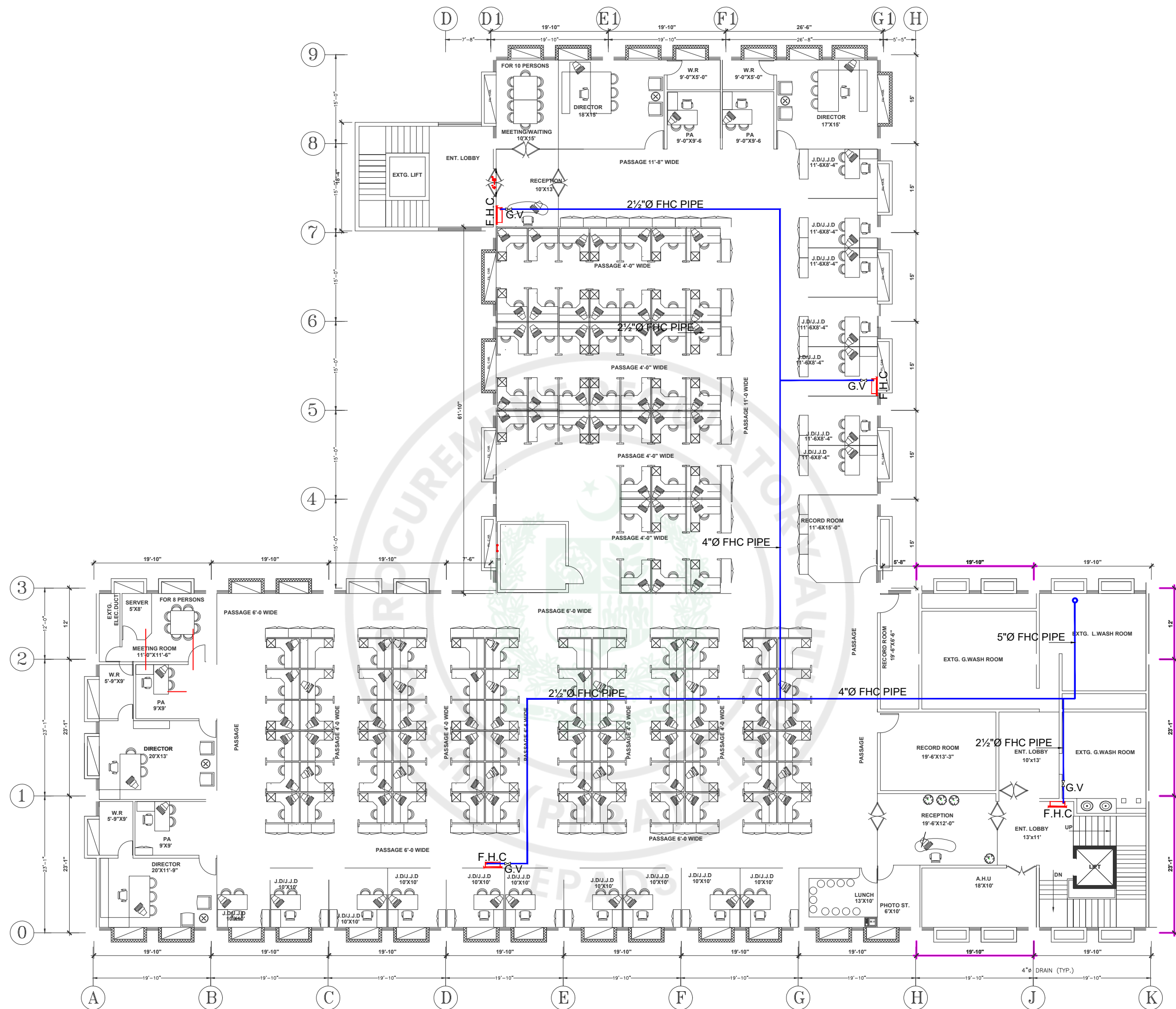
CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS YOU S U F H A S A N A S S O C I A T E S 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING	DRAWING NO. FF - 125		REVISION 0
						DRAWING TITLE LAYOUT OF FIRE FIGHTING SYSTEM SECOND FLOOR FOR CANTEEN BUILDING	DRAWN BY NABEEL	SERIAL NO. 25	DATE 24/02/2015
							PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE 1/16"=1'-0"



THIRD FLOOR

TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS	TITLE	DRAWING NO.		REVISION
	OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM				 YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	FIRE FIGHTING	FF-126	0	
					DRAWING TITLE	DRAWN BY	SERIAL NO.	DATE	
					LAYOUT OF FIRE FIGHTING SYSTEM AT THIRD FLOOR CANTEN BUILDING	SHEHARYAR	26	24/02/2015	
						PROJECT NO.	APPROVED BY	SCALE	
						2014-12-SBP		1/16"=1'-0"	

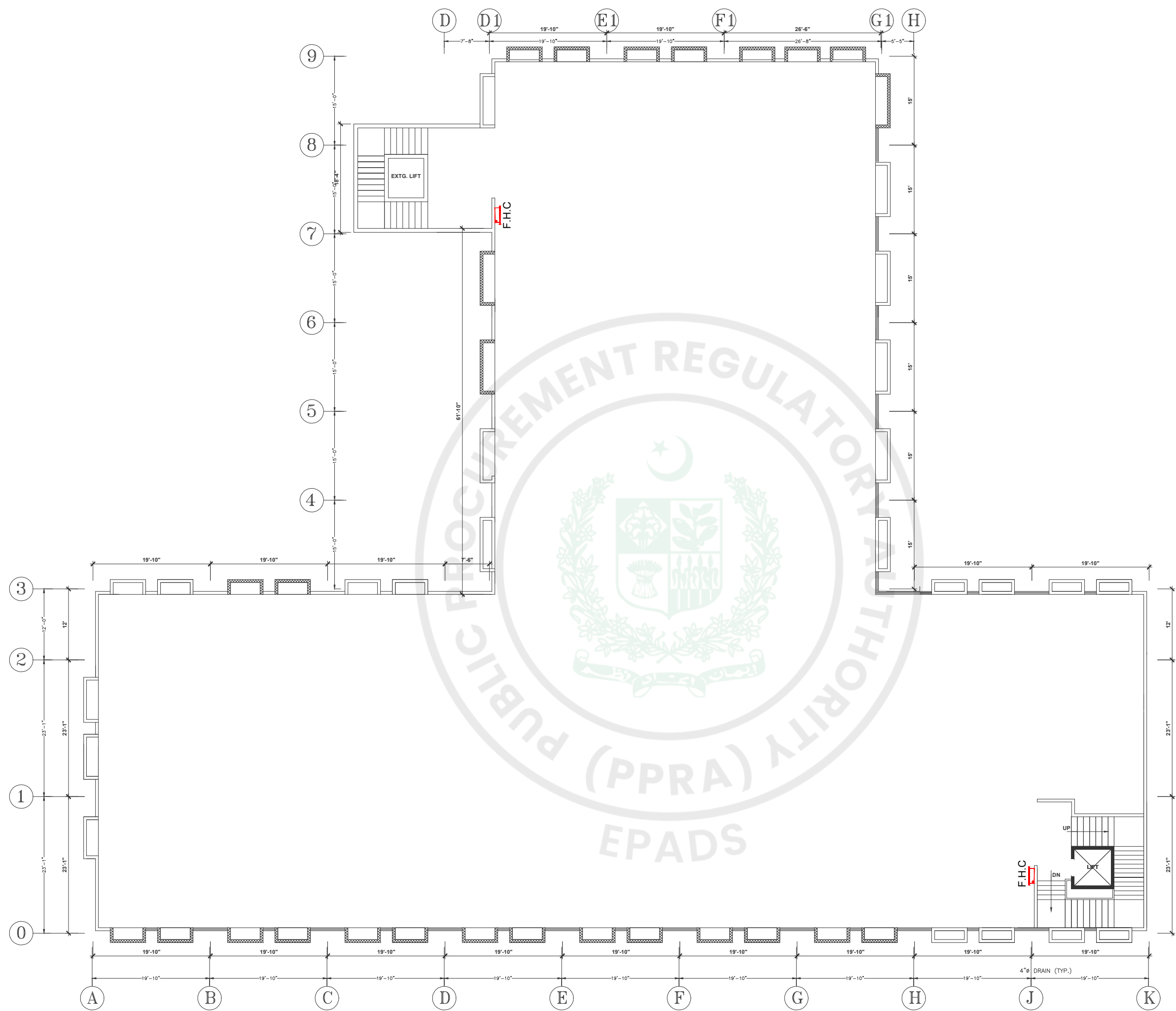


FOURTH FLOOR

CANTEEN BUILDING	
FIRE HOSE CABNET	22 NOS.
GATE VALVE 2 1/2"Ø	22 NOS.
FIRE HOSE CABINET PIPE 2 1/2"Ø	910 RFT
FIRE HOSE CABINET PIPE 4"Ø	550 RFT
FIRE RISER PIPE 5"Ø	200 RFT
MUSEUM	
FIRE HOSE CABNET	01 NOS.
DAY CARE	
FIRE HOSE CABNET	01 NOS.

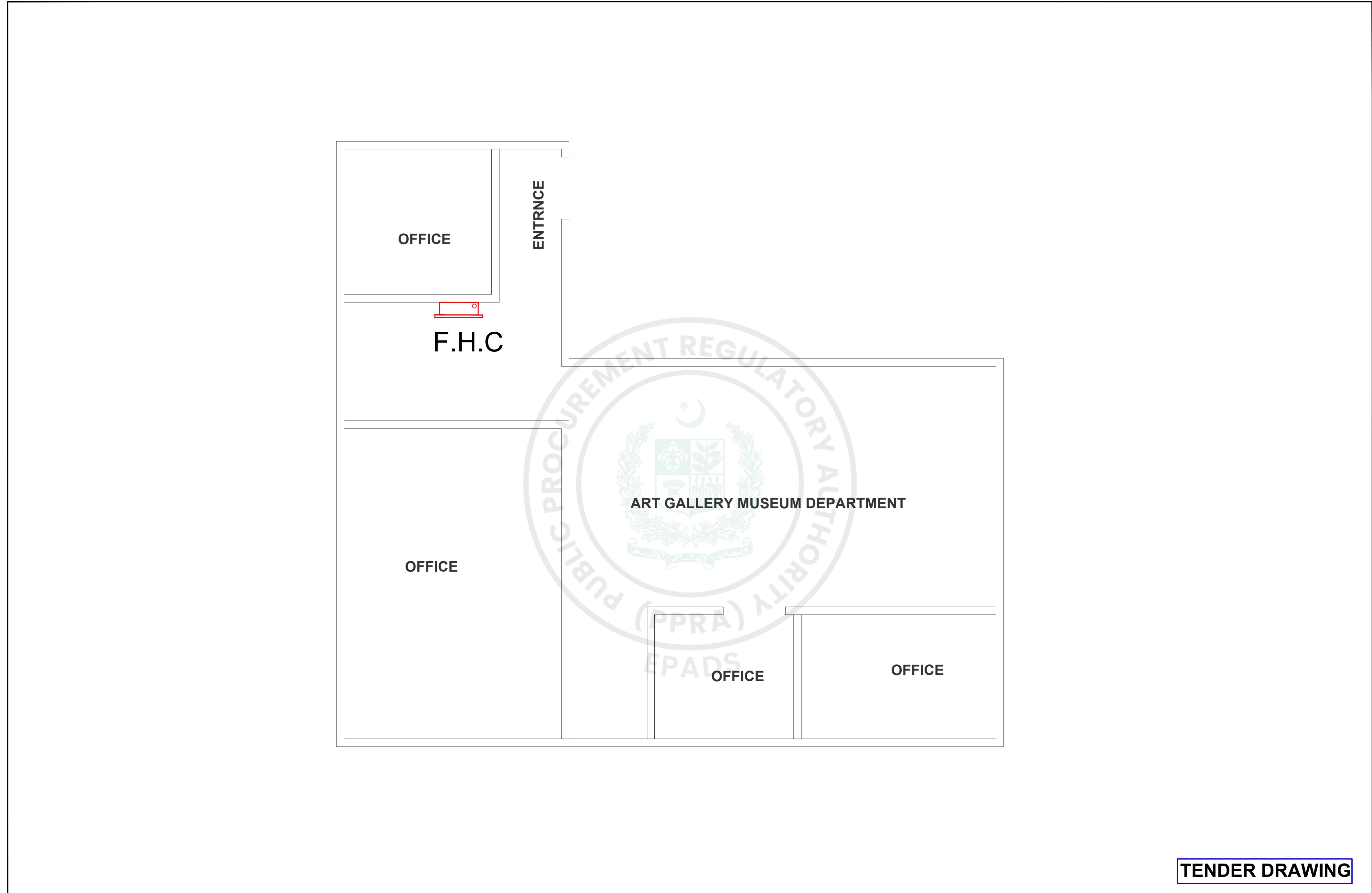
TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING DRAWING TITLE LAYOUT OF FIR FIGHTING SYSTEM AT FOURTH FLOOR CANTEEN BUILDING	DRAWING NO. FF-127		REVISION 0
							DRAWN BY NABEEL	SERIAL NO. 27	DATE 24/02/2015
							PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE 1/16"=1'-0"



TENDER DRAWING

CLIENT	PROJECT	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS	TITLE		DRAWING NO.		REVISION	
						FIRE FIGHTING		FF-128		0	
						DRAWING TITLE		DRAWN BY	SERIAL NO.	DATE	
STATE BANK OF PAKISTAN KARACHI	OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM				YOUUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	LAYOUT OF FIR FIGHTING SYSTEM AT ROOF CANTEN BUILDING		SHAHERYAR	28	24/02/2015	
								PROJECT NO.	APPROVED BY	SCALE	
								2014-12-SBP		1/16"=1'-0"	



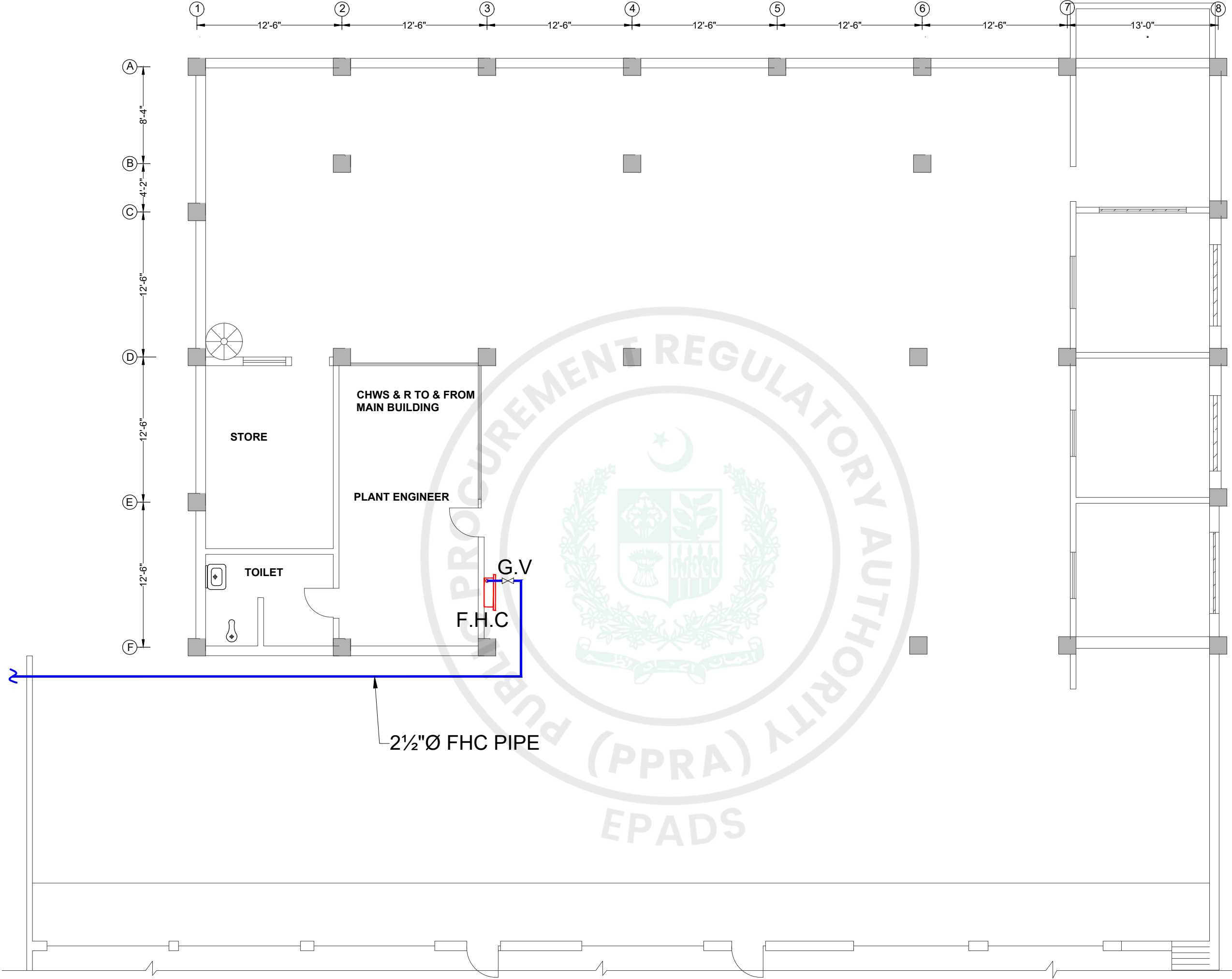
TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS  YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING		DRAWING NO. FF-129		REVISION 0
						DRAWING TITLE		DRAWN BY	SERIAL NO.	DATE
						LAYOUT OF A/C SYSTEM ART GALLERY MUSEUM DEPARTMENT AT FIRST FLOOR		JAUNAID	29	24/02/2015
								PROJECT NO.	APPROVED BY	SCALE
								2014-12-SBP		1/4"=1'-0"



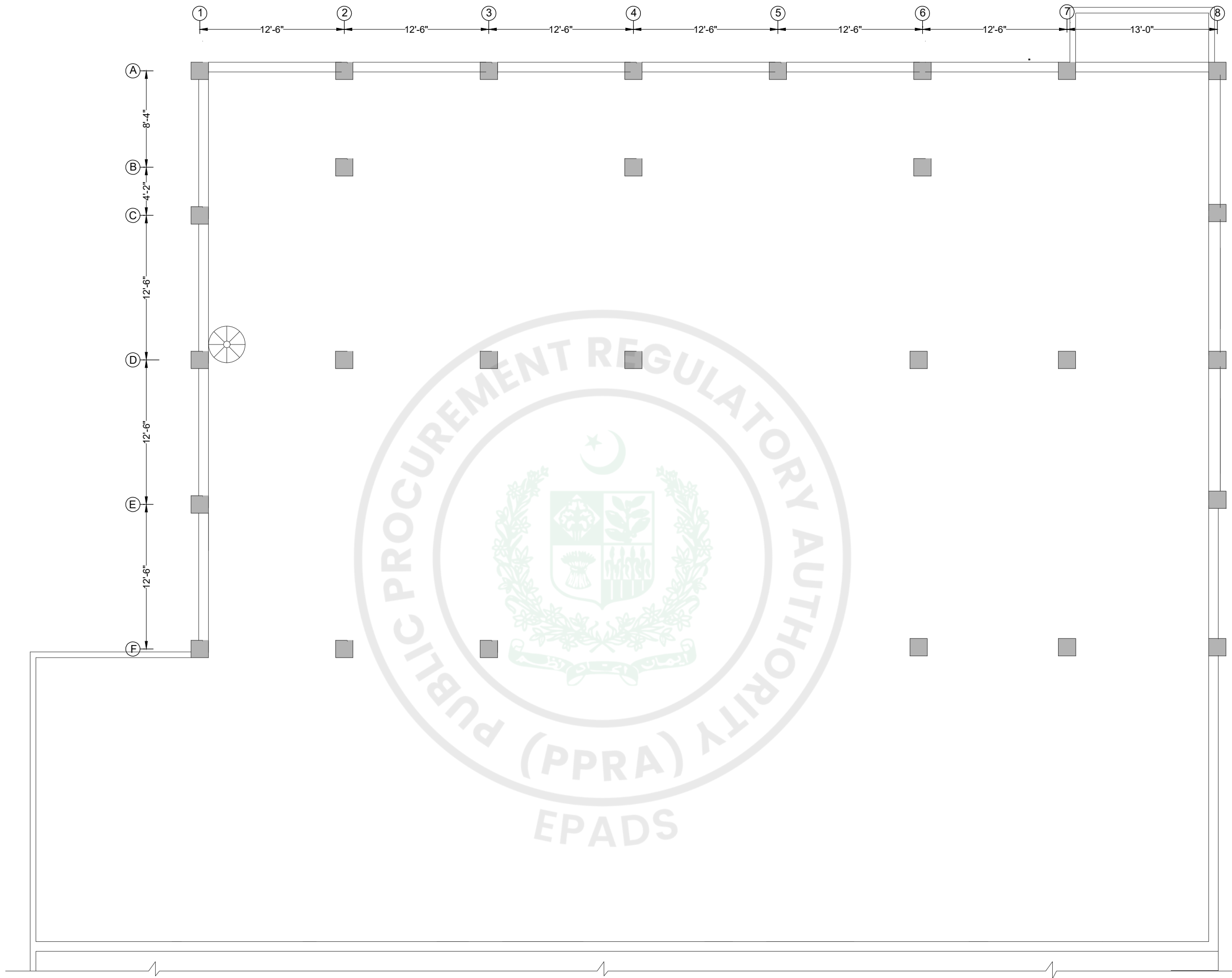
TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS  YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING		DRAWING NO. FF-130		REVISION 0
						DRAWING TITLE		DRAWN BY	SERIAL NO.	DATE
						LAYOUT OF A/C SYSTEM ART GALLERY MUSEUM DEPARTMENT AT ROOF		NABEEL	30	24/02/2015
								PROJECT NO.	APPROVED BY	SCALE
								2014-12-SBP		1/4"=1'-0"



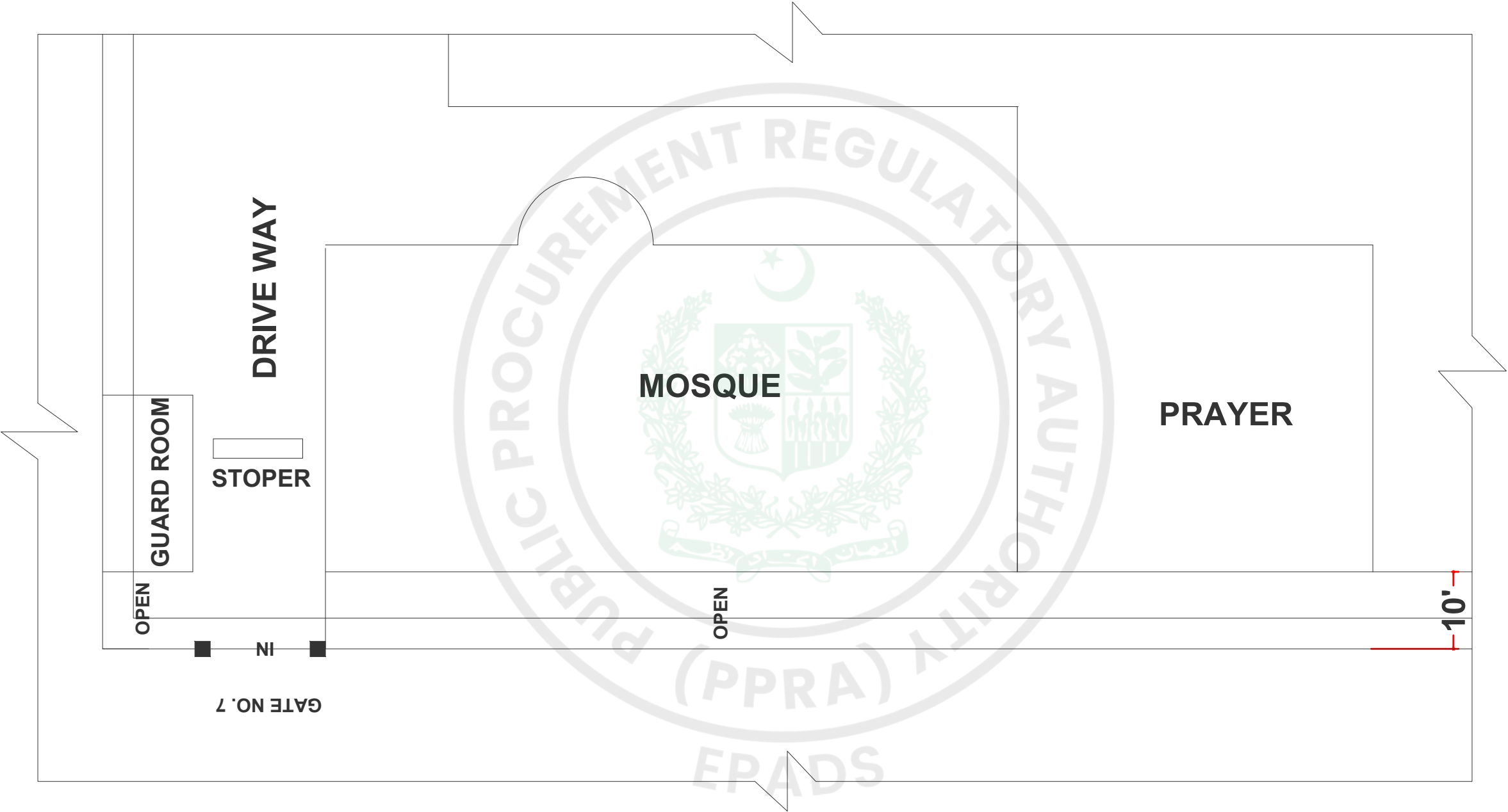
TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING		DRAWING NO. FF-131		REVISION 0
						DRAWING TITLE LAYOUT OF FIRE FIGHTING PLANT ROOM, FOR MAIN BUILDING & LRC		DRAWN BY SHAHERYAR	SERIAL NO. 31	DATE 24/02/2015
								PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE 1/8"=1'-0"



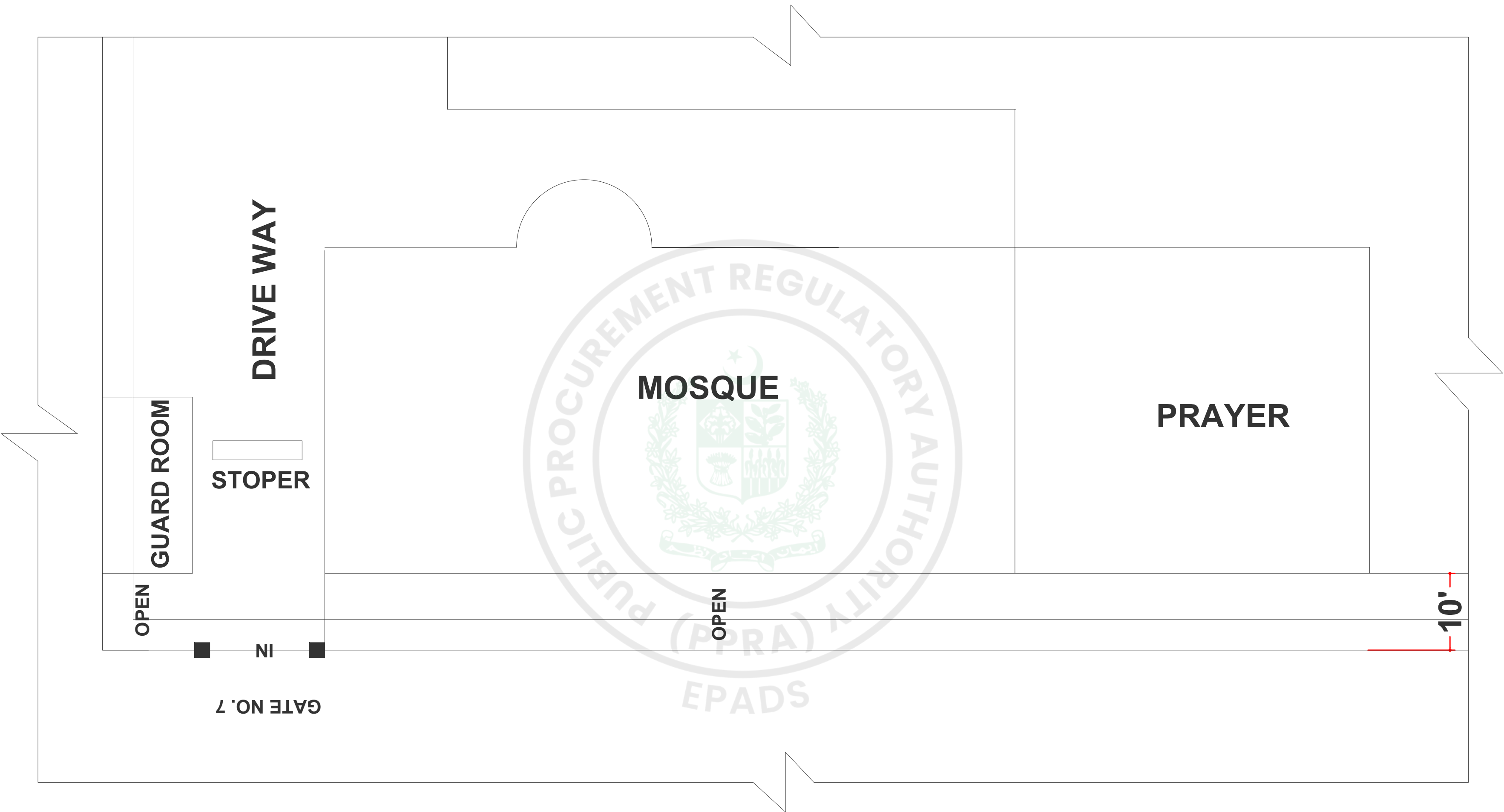
TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING		DRAWING NO. FF-132		REVISION 0
						DRAWING TITLE		DRAWN BY	SERIAL NO.	DATE
						LAYOUT OF FIRE FIGHTING PLANT ROOM, FOR MAIN BUILDING & LRC ROOF		JUNAID	32	24/02/2015
								PROJECT NO.	APPROVED BY	SCALE
								2014-12-SBP		1/8"=1'-0"



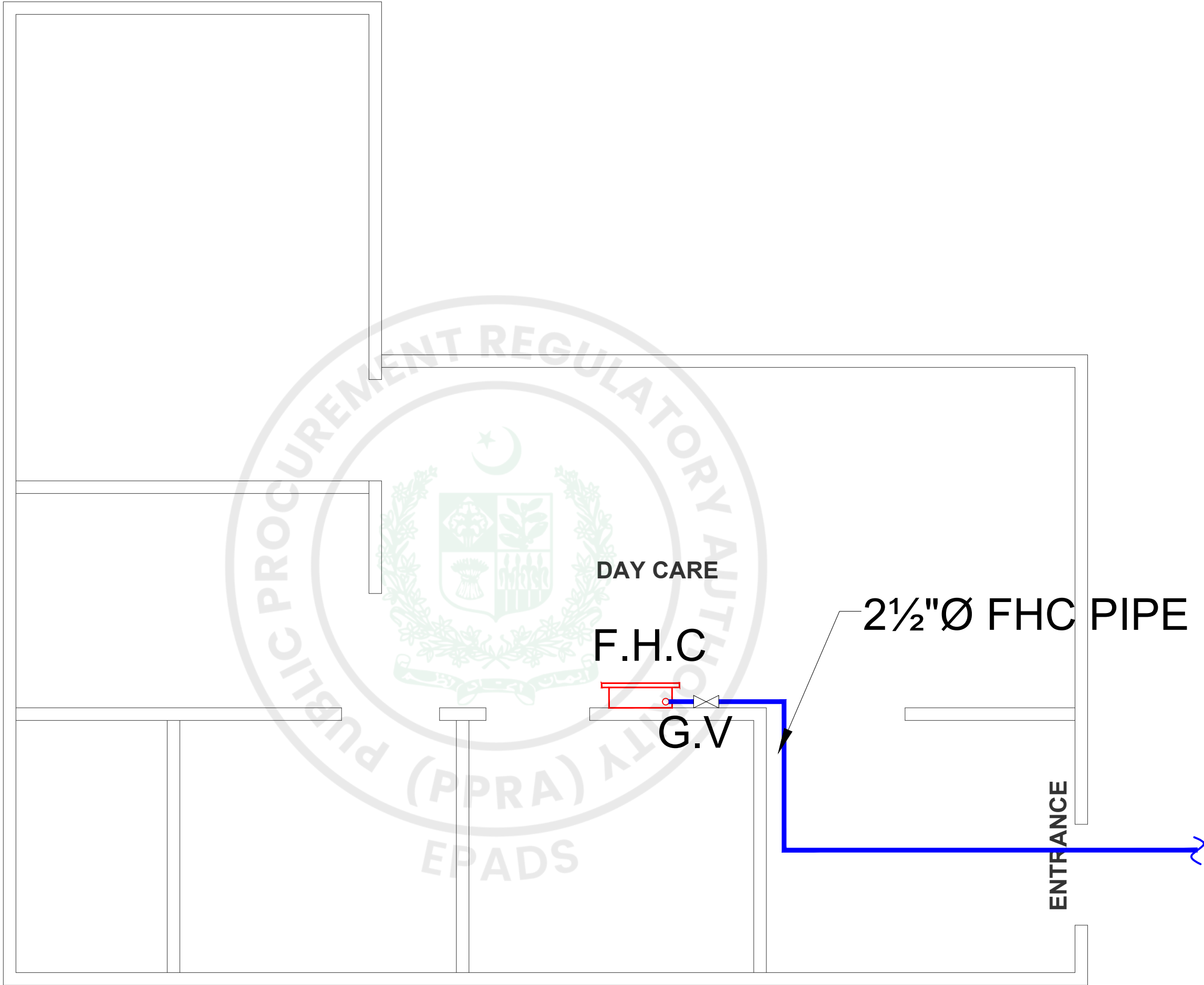
TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS  YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING		DRAWING NO. FF-133		REVISION 0
						DRAWING TITLE LAYOUT OF FIRE FIGHTING SYSTEM, MOSQUE AT GROUND FLOOR		DRAWN BY NABEEL	SERIAL NO. 33	DATE 24/02/2015
								PROJECT NO. 2014-12-SBP	APPROVED BY	SCALE 1/16" = 1'-0"



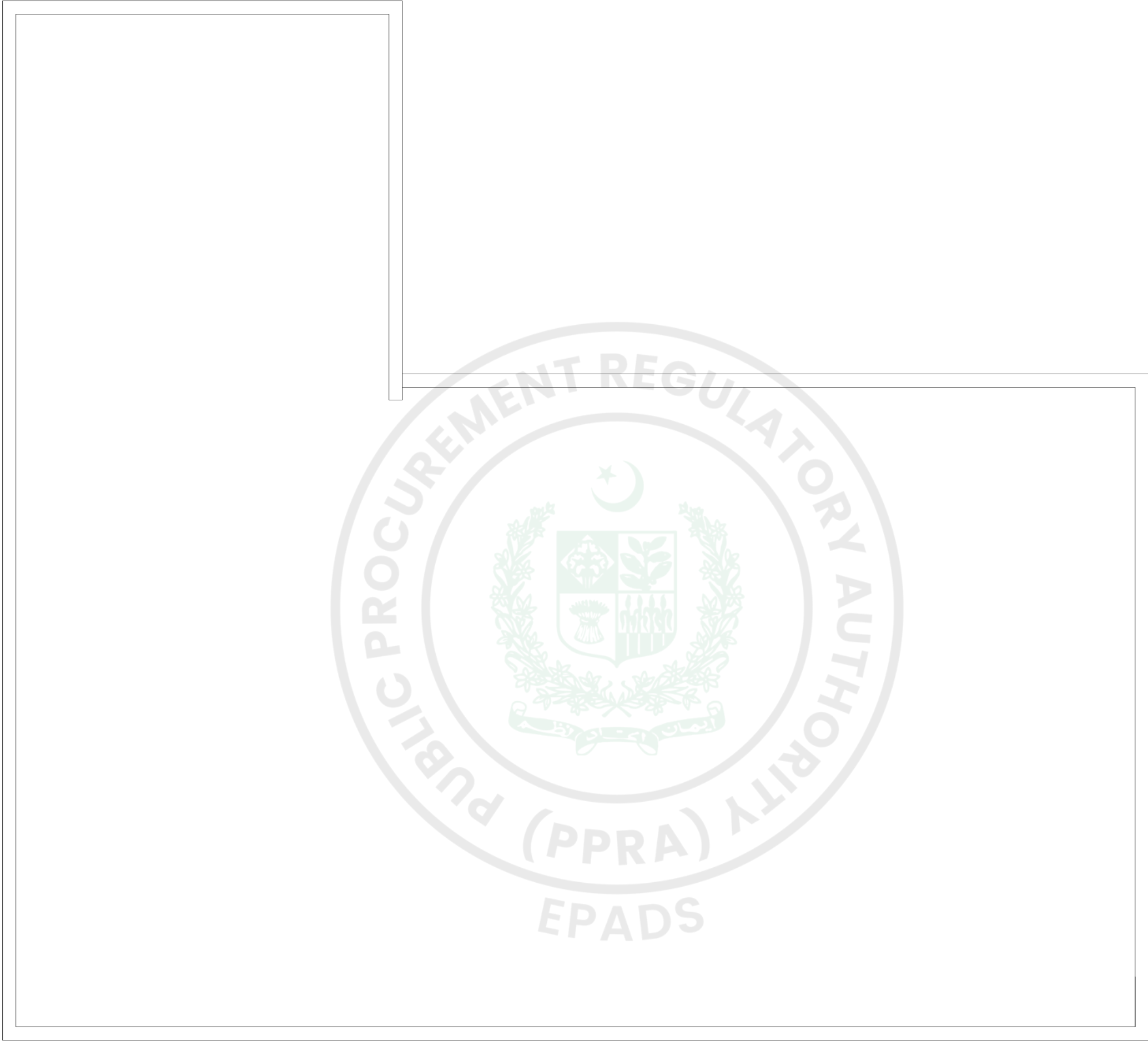
TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	CONSULTANTS YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING		DRAWING NO. FF-134		REVISION 0
						DRAWING TITLE		DRAWN BY	SERIAL NO.	DATE
						LAYOUT OF FIRE FIGHTING SYSTEM MOSQUE AT ROOF		JUNAID	34	24/02/2015
								PROJECT NO.	APPROVED BY	SCALE
								2014-12-SBP		3/32" = 1'-0"



TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS YOUSEF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING		DRAWING NO. FF-135		REVISION 0
						DRAWING TITLE		DRAWN BY	SERIAL NO.	DATE
						LAYOUT OF FIRE FIGHTING SYSTEM, DAY CARE AT GROUND FLOOR		NABEEL	35	24/02/2015
								PROJECT NO.	APPROVED BY	SCALE
								2014-12-SBF		1/4"=1'-0"



TENDER DRAWING

CLIENT STATE BANK OF PAKISTAN KARACHI	PROJECT OVERHAULING/REPAIRING OF FIRE FIGHTING SYSTEM	REV.	DATE	DESCRIPTION	FIRE FIGHTING CONSULTANTS YOUSUF HASAN ASSOCIATES 305, 3rd FLOOR, ANUM ESTATE SHAHRAH-E-FAISAL, KARACHI-75350, PAKISTAN PHONE NO (921) 3432 1559 (921) 3438 9931, FAX- 021-3432 1559 EMAIL ADDRESS: hasan@yhassociates.com	TITLE FIRE FIGHTING		DRAWING NO. FF-136		REVISION 0
						DRAWING TITLE		DRAWN BY	SERIAL NO.	DATE
						LAYOUT OF FIRE FIGHTING SYSTEM, DAY CARE AT ROOF		SHAHERYAR	36	24/02/2015
								PROJECT NO.	APPROVED BY	SCALE
								2014-12-SBP		1/4"=1'-0"